Data Science I with python

STAT 303-1-Sec20

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Table of contents

Preface

Welcome to **STAT 303-1 (Data Science with Python I), Section 20** at Northwestern University.

This book is designed to support your journey into data science, blending foundational concepts with hands-on practice. It builds on the original work of **Professor Arvind Krishna**, whose materials inspired the structure and spirit of this resource. The content has been thoughtfully updated to meet the evolving needs of students and the goals of the course.

In this first course, you will:

- Learn to use essential Python libraries for data science, including but not limited to **NumPy**, **Pandas**, and **Matplotlib**. These three are the main focus, but you will also be introduced to other useful libraries and tools as needed for real-world data analysis.
- Develop skills in **Exploratory Data Analysis (EDA)**, transforming messy, real-world datasets into meaningful insights through visualization, summarization, and pattern discovery

The book is organized to guide you step-by-step:

- **Chapters 1–3**: Set up your coding environment in VS Code, understand Python environments, and enhance your workflow
- **Chapters 4–5**: Review key concepts from STAT 201. If you need a refresher, visit the STAT 201 ebook.
- New material begins in **Chapter 6 – *Reading Data***

Throughout the quarter, this resource will be updated and refined to improve clarity, depth, and alignment with our teaching objectives.

As a **living document**, your feedback and suggestions are always welcome. Contributions from students, instructors, and the broader academic community help make this book stronger and more effective.

If you have ideas for improving the textbook, please share your feedback or suggestions using our dedicated Textbook Improvement Form. Your input is invaluable in making this resource better for everyone.

Thank you for joining us on this learning adventure. We hope this book becomes a valuable companion as you build your skills and confidence in data science with Python.

Part I

Getting started

1 Setting up your environment with VS Code

```
<IPython.core.display.Image object>
```

1.1 Learning Objectives

Setting up your Python data science environment is the foundation for all your work in this course. A well-configured setup will save you time, prevent errors, and make your workflow smoother. Whether you're new to VS Code or Python, these steps will help you build a reliable environment for coding, analysis, and reproducibility.

By the end of this chapter, you will be able to:

- Install and configure **Visual Studio Code (VS Code)** for Python and Jupyter Notebooks.
- Create and manage a **virtual environment** for this course.
- Install essential **Python libraries** for data science.
- Verify that your environment is working properly.

1.2 Why Choose Visual Studio Code (VS Code)?

Choosing the right editor can make a big difference in your productivity and learning experience. VS Code is a top choice for data science and Python development because it combines power, flexibility, and ease of use.

- **Free and Lightweight**
  VS Code is free, open-source, and runs efficiently on most systems without being resource-heavy.
- **Cross-Platform**
  Works consistently on Windows, macOS, and Linux.

- **Rich Python Support**
  With the official Python extension, you get:

  - Syntax highlighting and IntelliSense (auto-completion, code hints).

  - Built-in debugging tools.

  - Easy integration with virtual environments and Jupyter Notebooks.

- **Integrated Jupyter Notebook Support**
  You can open and run `.ipynb` notebooks directly in VS Code.
- **Customizable and Extensible**
  Large marketplace of extensions (e.g., formatting, linting, Git, Docker, AI tools).
- **Version Control Integration**
  Built-in Git and GitHub support for managing and sharing code.
- **Unified Workflow**
  One place for editing, running, debugging, documenting, and version-controlling Python code.

Whether you're just starting out or working on advanced projects, VS Code provides a modern, unified environment for all your coding needs.

1.3 Quick Start: Setting Up Python & VS Code

VS Code is a versatile editor. To use it for data science, you'll complete a few extra setup steps (e.g., Python & Jupyter extensions). Follow this step-by-step setup guide to get started quickly and avoid common pitfalls.

Step 1: Install Python

- Download and install the latest version from python.org.
- On Windows, check the box **"Add Python to PATH."** This makes Python available in your terminal.

Step 2: Install VS Code & Extensions

- Download VS Code from code.visualstudio.com.
- Open VS Code, go to the Extensions panel (`Ctrl+Shift+X`), and install:
  - **Python (Microsoft)**
  - **Jupyter (Microsoft)**

Step 3: Create a Course Folder

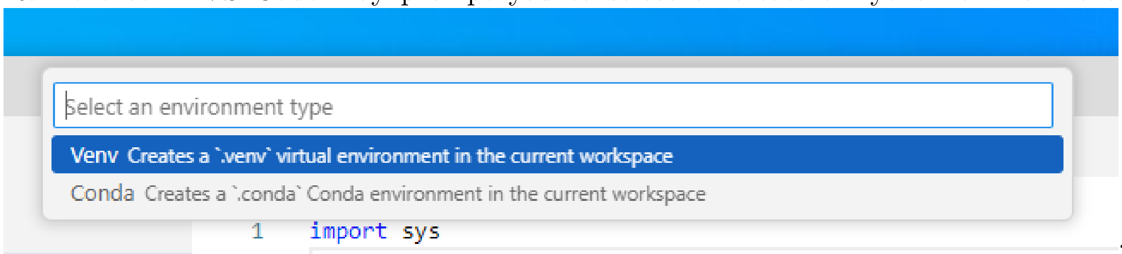
- Make a folder for your notebooks (e.g., `datasci` on Desktop or Documents).
- Open this folder in VS Code.

Step 4: Create Your First Notebook

- In VS Code, go to **File** → **New File** → **Save as** → `test.ipynb`.
- Add a code cell:

```
print("hello world")
```

- Run the cell. VS Code may prompt you to select or create a Python environment.



- If prompted, create a new environment using the GUI (recommended for beginners).
- Make sure the kernel (top right) shows **.venv (Python 3.xx.x)**.
- If you see `hello world` printed, your setup is working!

If everything works, you can skip the Full Setup below.

If you run into issues, continue with the detailed setup and troubleshooting steps in the next section.

This quick start helps you get coding fast, but the full setup ensures your environment is robust and reproducible for all course projects.

1.4 Full Setup: Reliable Python Environment in VS Code

If the quick start didn't work, or you want a more robust setup, follow these step-by-step instructions to ensure your Python environment is reproducible and ready for data science.

Step 1: Install Python

- Go to the official Python website and download the latest version for your operating system.
- **Important:** During installation, check the box **"Add Python to PATH"** so Python is available in your terminal.
- After installation, restart your terminal in VS Code.

- Verify installation by running:

```
python --version
```

  You should see the installed Python version.

Step 2: Install VS Code Extensions

- Open VS Code.
- Go to Extensions (`Ctrl+Shift+X`).
- Install:
  - **Python** (by Microsoft)
  - **Jupyter** (by Microsoft)

Step 3: Set Up Your Workspace

- Create a new folder for your course work (e.g., `STAT303-1` on Desktop or Documents).
- Open the folder in VS Code (`File > Open Folder`).

Step 4: Create a Notebook for Your Work

- In VS Code, go to `File > New File` and select `Jupyter Notebook`.
- Save the file as `your_notebook.ipynb`.
- Add a code cell:

```
print("hello world")
```

- Make sure to select the `.venv` kernel for your notebook (see next step).

Step 5A: Create a Virtual Environment (GUI Method)

- Run the cell. If prompted, create a Python environment using `venv` in the current workspace.
- Wait for Jupyter to start and install necessary components (e.g., ipykernel).
- After installation, ensure the selected kernel in the top right of the notebook says **.venv (Python 3.xx.x)**.
- If not, select the `.venv` kernel manually.

If you successfully created the environment using the GUI, skip Step 5B.

Step 5B: Create a Virtual Environment (Command Line Method)

- If not prompted to create your environment, use the terminal:
  - Open the terminal in VS Code.

- Run:

  ```
  python -m venv .venv
  ```

- Activate the environment:
  * On Windows:

    ```
    .\.venv\Scripts\activate
    ```

  * On macOS/Linux:

    ```
    source .venv/bin/activate
    ```

- You should see `(.venv)` at the start of your terminal prompt.
- Install essential packages:

  ```
  pip install numpy pandas matplotlib
  ```

Step 6: Verify Your Setup

- Run the code cell again. You should see `hello world` printed out.
- In the terminal, check your Python version and installed packages:

  ```
  python --version
  pip list
  ```

Troubleshooting Tips

- If the kernel does not show `.venv`, restart VS Code and ensure the environment is activated.
- Make sure the Python and Jupyter extensions are installed and enabled.
- If you see errors, check that your notebook is saved with the `.ipynb` extension and the correct kernel is selected.

A reliable environment is essential for reproducible data science. These steps will help you avoid common issues and set yourself up for success throughout the course.

1.5 Jupyter Notebooks in VS Code

After setting up your environment, you can take full advantage of Jupyter Notebooks directly in VS Code. This integration lets you:

- Create, edit, and run `.ipynb` notebooks without leaving VS Code
- Use interactive code cells for data analysis, visualization, and documentation
- Access rich outputs (plots, tables, markdown) inline with your code

- Switch kernels to use different Python environments for each notebook
- Debug, refactor, and version-control your notebooks with built-in tools

Getting Started:

- Open or create a notebook file (`.ipynb`) in VS Code
- Make sure the correct Python kernel/environment is selected (top right corner)
- Add code and markdown cells as needed
- Run cells individually or all at once

Learn More:

- Explore the official guide: [Jupyter Notebooks in VS Code]

Jupyter Notebooks in VS Code provide a powerful, flexible workflow for data science and Python development.

1.6 Rendering notebook as HTML using Quarto

Quarto is designed for high-quality, customizable, and publishable outputs, making it suitable for reports, blogs, or presentations. We'll use Quarto to print the **.ipynb* file as HTML for your assignment submission.

1.6.1 Installing and Verifying Quarto

- Download and install Quarto from the [official website].

- Follow the installation instructions for your operating system.

- Open your terminal within VS Code:

  – Go to `Terminal -> New Terminal`.

- Run the following command to verify that Quarto and its dependencies are correctly installed: `quarto --version`

- You should see the installed quarto version. If the command line doesn't work, you might see an error message like:

  ```
  - `quarto is not recognized`
  - `quarto command is not found`
  ```

This issue is often caused by Quarto not being added to the PATH environment variable. Similar to how you added the Python path to the environment variable above, you need to add the Quarto path to the system environment variable so that the command can be recognized by your operating system's shell.

On Windows, if you used the default installation path (without changing it), Quarto is installed in: `C:\Users\<USER>\AppData\Local\Programs\Quarto\bin`

Note:
Before moving forward, ensure that the command `quarto --version` successfully prints the version of your installed quarto. If it does not, you may need to troubleshoot your quarto installation or add it to the PATH environment variable.

1.6.2 Converting the Notebook to HTML

Check the procedure for rendering a notebook as HTML here. You have several options to format the file. Here are some points to remember when using Quarto to render your notebook as HTML:

1. The `Raw NBConvert` cell type is used to render different code formats into HTML or LaTeX. This information is stored in the notebook metadata and converted appropriately. **Use this cell type to put the desired formatting settings for the HTML file.**
2. In the formatting settings, remember to use the setting **`embed-resources: true`**. This will ensure that the rendered HTML file is self-contained, and is not dependent on other files. This is especially important when you are sending the HTML file to someone, or uploading it somewhere. If the file is self-contained, then you can send the file by itself without having to attach the dependent files with it.

Once you have entered the desired formatting setting in the `Raw NBConver` cell, you are ready to render the notebook to HTML. Open the terminal, navigate to the directory containing the notebook *(.ipynb file)*, and use the command: `quarto render filename.ipynb --to html`.

1.7 Independent Study

Goal: Get hands-on experience creating a reproducible Python environment using `pip` and `.venv` in VS Code.

1.7.1 Step-by-Step Instructions

Step 1: Create Your Workspace

- Make a folder named `STAT303_1` for your course work.
- Open this folder in VS Code.
- Create a new notebook: `setup_test.ipynb`.
- Add a code cell:

```
print("Hello, VS Code & .venv!")
```

- Run the cell. If prompted, create/select a Python environment (choose `.venv`).

Step 2: Create the `.venv` Environment

- If VS Code does not prompt you, open the terminal in VS Code.
- Run:

```
python -m venv .venv
```

- Activate the environment:

```
.\.venv\Scripts\activate
```

- Install essential packages:

```
pip install numpy pandas matplotlib
```

Step 3: Verify Environment Consistency

- In the terminal, check your Python version:

```
python --version
```

- In the notebook, run:

```
import sys
print(sys.executable)
```

- Make sure the Python path in the notebook matches the one in your terminal. This confirms you are using the same environment.

Step 4: Generate html from the notebook using Quarto

- In the terminal, make sure you are in the directory that contains your notebook

```
quarto render setup_test.ipynb  --to html
```

- Make sure the html file was generated in the same directory

Troubleshooting Tips

If you run into issues (e.g., kernel not showing `.venv`, code cell errors):

- Restart VS Code.
- Deactivate/reactivate `.venv`.
- Make sure **Python** & **Jupyter** extensions are installed.
- Save your notebook with the `.ipynb` extension.
- If the kernel is missing, use the kernel picker (top right) to select `.venv`.
- If you see an error like *"No such file or directory"* or *"No valid input files passed to render"*:

  1. Check your current location:
     - macOS / Linux: `pwd`

     - Windows: `cd`
  2. List files in the current directory:
     - macOS / Linux: `ls`

     - Windows: `dir`
  3. If the notebook isn't there:
     - Use `cd` to move into the correct folder, or

     - Provide the full (absolute) path to the file.

Extra:

- Try installing a package (e.g., `seaborn`) and import it in a new cell to test.
- Practice switching kernels to see how environments affect your code.
- Practice creating a `.conda` env and use `conda` to install the `seaborn` package

1.8 Reference

- Getting Started with VS Code
- Jupyter Notebooks in VS Code
- Install Python Packages
- Managing environments with `conda`

2 Python Environments and Package Management

`<IPython.core.display.Image object>`

2.1 Learning Objectives

Managing Python environments and packages is essential for reproducible, error-free data science. Understanding these concepts will help you avoid common pitfalls and collaborate more effectively. This chapter will guide you through the tools and best practices for setting up, sharing, and troubleshooting Python environments.

By the end of this chapter , you will be able to:

- Explain why virtual environments are important.
- Install essential data science packages in your Python environment.
- Create a `requirements.txt` file and use it to quickly recreate an environment.
- Recognize why dependency conflicts happen.
- Describe how tools like `conda` or `poetry` can help resolve dependency conflicts.

2.2 Data Science packages in Python

Python has a rich ecosystem of packages that make data analysis easier and more powerful. In this course, you will install and use several core packages:

- **NumPy** → numerical computing and array operations
- **pandas** → data manipulation and analysis

- **Matplotlib** → data visualization
- **Seaborn** → statistical data visualization (built on top of Matplotlib)

These packages give you the essential tools to load, clean, analyze, and visualize data.
Before using them, you need to install them in your Python environment.

In your previous VS Code setup lesson, you created a Python environment using `.venv` and selected it as the kernel to run your notebook. This created a folder called `.venv`, which contains: - A dedicated Python interpreter for your project - A `Lib` folder where all packages you install are stored

Any packages you install (e.g., with `pip install ...`) will only affect this environment, keeping your project isolated from others.

Using virtual environments is considered best practice for Python development, especially in data science projects. This isolation helps you: - Avoid version conflicts between packages - Keep project dependencies separate - Make your code more reproducible and shareable

Before we dive in, let's clarify what a Python environment is and why isolation matters.

2.3 Python Virtual Environments

A Python virtual environment is an **isolated workspace** with its own Python interpreter and its own set of installed packages.

This isolation makes it easy to manage dependencies for different projects and prevents package version conflicts.

Why use virtual environments?

- Keep each project's dependencies separate
- Avoid breaking code by accidentally upgrading/downgrading packages
- Share your code with others and ensure it works the same way on their machine

Virtual environments are a key tool for professional, reproducible Python workflows.

2.4 Install Data Science Packages Within Your Environment

Python's power for data science comes from its rich ecosystem of packages. These packages provide essential tools for scientific computing, data analysis, visualization, and machine learning.

Packages like `numpy`, `pandas`, and `matplotlib` are not included in the Python standard library—you need to install them in your environment before you can use them.

Test your setup: Add the following code to your `test.ipynb` notebook to check if your environment is ready:

```python
import numpy as np
import pandas as pd
print("Setup complete!")
```

If you see a `ModuleNotFoundError`, it means the package is not installed in your current environment. Use `pip install` to add missing packages.

Tip: Always install packages inside your active virtual environment to keep your project isolated and reproducible.

2.4.1 How to Install Packages

There are two main ways to install data science packages in your environment:

2.4.1.1 Installing from the Terminal

- **Open a New Terminal:** Check that your terminal is using the same environment as your notebook kernel.
  - If you see `(.venv)` at the start of the prompt, your virtual environment is active and matches the notebook kernel.
  - If you see `(base)`, it means the base conda environment is active (common if you have Anaconda installed).
  - To confirm which Python is being used, run:
    * On macOS/Linux: `which python`
    * On Windows: `where.exe python`

  Note: If you have both Anaconda and VS Code installed, environments can sometimes conflict. If the terminal and notebook use different environments, packages may be installed in the wrong place, causing import errors in your notebook.

- **Install packages with pip:**

  ```
  pip install numpy pandas
  ```

2.4.1.2 Installing from the Notebook

You can also install packages directly from a Jupyter Notebook cell using a magic command. This is convenient for quick installs without leaving the notebook interface.

- Add a new code cell and run:

  ```
  pip install numpy pandas
  ```

Best Practice: Always make sure you are installing packages into the correct environment. Double-check your kernel and terminal environment before running installation commands.

2.4.2 Backing Up and Sharing Your Environment

Reproducibility is a cornerstone of good data science. To ensure your code works for you, your collaborators, and on other machines, you need a way to recreate your Python environment exactly.

Backing up your environment makes it easy to share your work and avoid "it works on my machine" problems.

How to back up your environment:

Step 1: **Create a `requirements.txt` file**

- This file lists all installed packages and their versions, so anyone can recreate your environment.

Run the following command in your terminal to list all installed packages and their versions:

```
pip freeze
```

```
asttokens==2.4.1
colorama==0.4.6
comm==0.2.2
contourpy==1.3.0
cycler==0.12.1
debugpy==1.8.6
decorator==5.1.1
executing==2.1.0
fonttools==4.54.1
ipykernel==6.29.5
ipython==8.27.0
jedi==0.19.1
```

```
jupyter_client==8.6.3
jupyter_core==5.7.2
kiwisolver==1.4.7
matplotlib==3.9.2
matplotlib-inline==0.1.7
nest-asyncio==1.6.0
numpy==2.1.1
packaging==24.1
pandas==2.2.3
parso==0.8.4
pillow==10.4.0
platformdirs==4.3.6
prompt_toolkit==3.0.48
psutil==6.0.0
pure_eval==0.2.3
Pygments==2.18.0
pyparsing==3.1.4
python-dateutil==2.9.0.post0
pytz==2024.2
pywin32==306
pyzmq==26.2.0
six==1.16.0
stack-data==0.6.3
tornado==6.4.1
traitlets==5.14.3
tzdata==2024.2
wcwidth==0.2.13
Note: you may need to restart the kernel to use updated packages.
```

Using the redirection operator `>`, you can save the output of `pip freeze` to a `requirement.txt`. This file can be used to install the same versions of packages in a different environment.

```
pip freeze > requirement.txt
```

```
Note: you may need to restart the kernel to use updated packages.
```

Let's check whether the `requirement.txt` is in the current working directory

```
%ls
```

```
 Volume in drive C is Windows
 Volume Serial Number is A80C-7DEC

 Directory of c:\Users\lsi8012\OneDrive - Northwestern University\FA24\303-1\test_env

09/27/2024  02:25 PM    <DIR>          .
09/27/2024  02:25 PM    <DIR>          ..
09/27/2024  07:44 AM    <DIR>          .venv
09/27/2024  01:42 PM    <DIR>          images
09/27/2024  02:25 PM               695 requirement.txt
09/27/2024  02:25 PM            21,352 venv_setup.ipynb
               2 File(s)         22,047 bytes
               4 Dir(s)  166,334,562,304 bytes free
```

Step 2: **Share the file**

- Send the `requirements.txt` file to your collaborator, or save it for future use.

Step 3: **Recreate the environment elsewhere**

- On a new machine or environment, run:

  ```
  pip install -r requirements.txt
  ```

```
pip install -r requirement.txt
```

```
Requirement already satisfied: asttokens==2.4.1 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: colorama==0.4.6 in c:\users\lsi8012\onedrive - northwestern un
Requirement already satisfied: comm==0.2.2 in c:\users\lsi8012\onedrive - northwestern univer
Requirement already satisfied: contourpy==1.3.0 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: cycler==0.12.1 in c:\users\lsi8012\onedrive - northwestern uni
Requirement already satisfied: debugpy==1.8.6 in c:\users\lsi8012\onedrive - northwestern uni
Requirement already satisfied: decorator==5.1.1 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: executing==2.1.0 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: fonttools==4.54.1 in c:\users\lsi8012\onedrive - northwestern
Requirement already satisfied: ipykernel==6.29.5 in c:\users\lsi8012\onedrive - northwestern
Requirement already satisfied: ipython==8.27.0 in c:\users\lsi8012\onedrive - northwestern un
Requirement already satisfied: jedi==0.19.1 in c:\users\lsi8012\onedrive - northwestern unive
Requirement already satisfied: jupyter_client==8.6.3 in c:\users\lsi8012\onedrive - northwes
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Requirement already satisfied: matplotlib==3.9.2 in c:\users\lsi8012\onedrive - northwestern
Requirement already satisfied: matplotlib-inline==0.1.7 in c:\users\lsi8012\onedrive - northw
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```
Requirement already satisfied: nest-asyncio==1.6.0 in c:\users\lsi8012\onedrive - northweste
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Requirement already satisfied: packaging==24.1 in c:\users\lsi8012\onedrive - northwestern un
Requirement already satisfied: pandas==2.2.3 in c:\users\lsi8012\onedrive - northwestern univ
Requirement already satisfied: parso==0.8.4 in c:\users\lsi8012\onedrive - northwestern unive
Requirement already satisfied: pillow==10.4.0 in c:\users\lsi8012\onedrive - northwestern uni
Requirement already satisfied: platformdirs==4.3.6 in c:\users\lsi8012\onedrive - northweste
Requirement already satisfied: prompt_toolkit==3.0.48 in c:\users\lsi8012\onedrive - northwes
Requirement already satisfied: psutil==6.0.0 in c:\users\lsi8012\onedrive - northwestern univ
Requirement already satisfied: pure_eval==0.2.3 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: Pygments==2.18.0 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: pyparsing==3.1.4 in c:\users\lsi8012\onedrive - northwestern u
Requirement already satisfied: python-dateutil==2.9.0.post0 in c:\users\lsi8012\onedrive - no
Requirement already satisfied: pytz==2024.2 in c:\users\lsi8012\onedrive - northwestern unive
Requirement already satisfied: pywin32==306 in c:\users\lsi8012\onedrive - northwestern unive
Requirement already satisfied: pyzmq==26.2.0 in c:\users\lsi8012\onedrive - northwestern univ
Requirement already satisfied: six==1.16.0 in c:\users\lsi8012\onedrive - northwestern unive
Requirement already satisfied: stack-data==0.6.3 in c:\users\lsi8012\onedrive - northwestern
Requirement already satisfied: tornado==6.4.1 in c:\users\lsi8012\onedrive - northwestern uni
Requirement already satisfied: traitlets==5.14.3 in c:\users\lsi8012\onedrive - northwestern
Requirement already satisfied: tzdata==2024.2 in c:\users\lsi8012\onedrive - northwestern uni
Requirement already satisfied: wcwidth==0.2.13 in c:\users\lsi8012\onedrive - northwestern un
Note: you may need to restart the kernel to use updated packages.
```

Once you run the install command, all packages listed in your `requirements.txt` file will be installed, matching the exact versions you used in your original environment.

This ensures your code runs the same way everywhere—whether on your computer, a collaborator's machine, or a server.

Why is this important?

- Guarantees everyone is using the same package versions, reducing errors and "works on my machine" problems.
- Makes it easy to set up your project on a new computer, server, or cloud environment.
- Simplifies troubleshooting and collaboration—everyone starts from the same setup.

Pro Tip:

- Update your `requirements.txt` after installing or upgrading packages to keep it current.
- Consider version-controlling your `requirements.txt` file (e.g., with Git) so changes are tracked and shared with collaborators.

A well-maintained environment file is essential for reproducible, shareable, and professional data science projects.

2.5 Dependency Conflicts

A **dependency conflict** occurs when two or more packages in your Python environment require different or incompatible versions of the same library.
This can cause errors, unexpected behavior, or even break your code.

2.5.1 Why do dependency conflicts happen?

- Many Python packages rely on other packages (called *dependencies*) to work.
- If you install packages that depend on different versions of the same dependency, they may not work together.
- This is especially common in data science, where libraries evolve quickly and have complex interdependencies.

Example: Suppose you install two packages:

- `PackageA` requires `numpy==1.24.0`
- `PackageB` requires `numpy==2.2.0`

If you install both with `pip`, the **last specified version wins**. One of the packages may then fail because it cannot use the version of `numpy` that was actually installed.

2.5.2 How to avoid dependency conflicts

- Use **virtual environments** to isolate dependencies for each project.
- Check package requirements before installing new libraries.
- Use tools like **Poetry** or **Conda** to detect and resolve conflicts automatically.

2.6 How to Resolve Conflicts and Manage Environments

2.6.1 Use `conda` for Robust Environment Management

`conda` is a powerful tool for managing both Python environments and packages, especially in data science workflows. It helps you avoid and resolve dependency conflicts by handling package versions and system dependencies more effectively than `pip` alone.

Why use conda?

- Create fully isolated environments for different projects
- Install packages and all their dependencies from the Anaconda repository
- Easily switch between environments for different tasks
- Export and share environment configurations for reproducibility

How conda prevents dependency conflicts:

- When you install a package, conda automatically checks for compatible versions of all dependencies and installs them together.
- If a conflict is detected, conda will warn you, suggest solutions, or prevent incompatible installations.

Essential conda commands:

- Create a new environment:
  ```
  conda create --name myenv numpy pandas matplotlib
  ```
- Activate an environment:
  ```
  conda activate myenv
  ```
- Install a package in an environment:
  ```
  conda install seaborn
  ```
- List all environments:
  ```
  conda env list
  ```
- Export environment configuration:
  ```
  conda env export > environment.yml
  ```
- Recreate an environment from a file:
  ```
  conda env create -f environment.yml
  ```

Example: Suppose you need to work on two projects that require different versions of `scikit-learn`. You can create two separate environments:

```
conda create --name projectA scikit-learn=0.24
conda create --name projectB scikit-learn=1.2
```

Each environment will have its own compatible dependencies, so you avoid conflicts.

Summary: Using `conda` is highly recommended for managing complex dependencies, system-level packages, and reproducible environments in data science.

2.6.2 Use `poetry` for Modern Python Projects

`poetry` is a modern tool for managing Python dependencies and packaging. It streamlines environment setup, ensures reproducibility, and makes sharing your project easy.

Why use poetry?

- Automatically creates and manages virtual environments for each project
- Uses a `pyproject.toml` file to specify and lock project requirements
- Prevents dependency conflicts with a lock file (`poetry.lock`)
- Simplifies publishing packages to PyPI and sharing with others

How poetry helps with dependency management:

- Resolves and installs compatible versions of all dependencies automatically
- Keeps your environment reproducible and up-to-date
- Makes it easy to add, update, or remove packages

Essential poetry commands:

Command / File	Description
`poetry install`	**Installs all dependencies.** If a `poetry.lock` file exists, installs exact versions from it. If not, resolves dependencies, creates the `poetry.lock` file, and installs.
poetry add <package>	Adds a package, resolves the new dependency tree, updates the `pyproject.toml` and updates the `poetry.lock` file.
poetry remove <package>	Removes a package, resolves the new dependency tree, updates the `pyproject.toml` and updates the `poetry.lock` file.
`poetry show`	Lists all installed packages (and their versions) as recorded in the `poetry.lock` file.
`poetry show --tree`	Displays the dependency tree based on the resolved dependencies in the `poetry.lock` file.
`poetry show --outdated`	Compares versions in the `poetry.lock` file against the latest available on the remote repository.
poetry run <command>	Runs a command in the virtual environment whose packages are defined by the `poetry.lock` file.
`poetry.lock` (File)	**The most important file for reproducibility.** Pins exact versions of every package and dependency. **Always commit this to version control** to ensure all developers and deployments use identical dependencies.

Core Files to Know

- `pyproject.toml`:
  The *declarative* file where you define your project's metadata, dependencies, and build system.
  This file is meant to be edited by humans.
- `poetry.lock`:
  The *definitive record* of the exact versions of every package that makes your project work.
  This file is generated and managed by Poetry.
  You should commit this to version control to ensure everyone (and your deployments) uses identical dependencies.

Summary: `poetry` is highly recommended for modern Python projects where you want reliable dependency management, easy environment setup, and reproducible results.

2.6.3 Difference Between `conda` and `poetry`

Both `conda` and `poetry` are tools for managing Python environments and dependencies, but they have different strengths and use cases.

conda:

- Manages both Python environments and packages, including non-Python dependencies (e.g., C libraries, compilers)
- Works with multiple languages (Python, R, etc.)
- Uses the Anaconda repository, which includes many scientific and data science packages
- Great for data science workflows, especially when you need packages with compiled code or system-level dependencies
- Handles complex dependency resolution and environment isolation

poetry:

- Focuses on Python projects and packages
- Uses `pyproject.toml` and `poetry.lock` for reproducible builds
- Automatically creates and manages virtual environments
- Excellent for modern Python application development and publishing to PyPI
- Handles dependency resolution and version locking for Python packages only

Key differences:

- `conda` can install system-level and non-Python dependencies; `poetry` only manages Python packages
- `conda` environments can include packages from the Anaconda repository; `poetry` uses PyPI
- `poetry` is ideal for pure Python projects and reproducible builds; `conda` is better for scientific computing and mixed-language projects

When to use each:

- Use `conda` if you need scientific libraries, compiled code, or non-Python dependencies
- Use `poetry` for modern Python projects, apps, and libraries where you want easy dependency management and publishing

Summary Table:

Feature	conda	poetry
Language support	Python, R, more	Python only
Non-Python dependencies	Yes	No
Environment isolation	Yes	Yes
Dependency resolution	Excellent	Excellent
Reproducibility	Good	Excellent
Publishing to PyPI	No	Yes

Choose the tool that best fits your project needs!

2.6.4 Modern Python Package Managers: `pip`, `conda`, `poetry`, `uv`

Python package management has evolved rapidly, making it easier to install, update, and manage dependencies for any project.

- **pip** is the standard tool for installing Python packages from PyPI. It works well for most projects and is simple to use.
- **.venv** helps you create isolated environments so each project has its own dependencies.
- **conda** offers advanced environment and package management, especially for scientific computing and projects with non-Python dependencies.
- **poetry** provides modern dependency management, automatic environment creation, and reproducibility for Python projects.
- **uv** is a new, high-performance package manager written in Rust. It's designed for speed and supports modern workflows. Learn more: uv GitHub page

For this course:

- You'll use `pip` and `.venv` for installing packages and managing your environment.
- As you tackle larger or more complex projects, consider exploring `conda`, `poetry`, and `uv` for better performance, reproducibility, and ease of use.

Summary:

- Choose the tool that fits your workflow and project needs.

- Stay curious—new tools like `uv` are making Python development faster and more reliable.

A good package manager and environment strategy will save you time, prevent headaches, and make your code more reproducible and shareable.

2.7 Independent Study: Practice Environment Setup with `pip`

Objective: Build hands-on skills in creating, managing, and verifying Python environments and packages using `pip`.

2.7.1 Instructions

Step 1: Create Your Workspace

- Make a folder named `test_pip_env`.
- Open the folder in VS Code.

Step 2: Create a Virtual Environment

- create a `.venv` environment

Step 3: Install Required Packages

- With the environment active, install packages using `pip`:

  ```
  pip install numpy pandas
  ```

Step 4: Export Your Environment Configuration

- Save your environment's package list to a file:

  ```
  pip freeze > stat303_env.txt
  ```

Step 5: Deactivate and Remove the Environment

- Deactivate the environment:

  ```
  deactivate
  ```

- Remove the environment folder to clean up.

Step 6: Recreate the Environment from the Exported File

- Create a new environment and activate it.

- Install packages from your saved file:

```
pip install -r stat303_env.txt
```

- Verify that `numpy`, `pandas`, `matplotlib`, and (if installed) `scikit-learn` are present.

Step 7: Run a Jupyter Notebook

- Create a notebook in the recreated environment.
- Import `numpy`, `pandas`, `matplotlib`, and `scikit-learn`.
- Print the versions of these packages to confirm setup.
- Generate html file from the notebook using quarto

This exercise will help you master environment management and reproducibility—key skills for any data scientist.

2.8 Resources

- Tutorials in Python Packaging User Guide
- Poetry Documentation
- uv GitHub page

3 Enhancing Workflow in Jupyter Notebooks

```
<IPython.core.display.Image object>
```

3.1 Learning Objectives

In this lecture, we'll explore how to optimize your workflow in Jupyter notebooks by leveraging **magic commands** and **shell commands**, understanding **file paths**, and interacting with the **filesystem** using the `os` module.

By completing this lecture, you will be able to:

- Run shell commands directly within a notebook.
- Navigate and manipulate files and directories efficiently.
- Integrate external tools and scripts seamlessly into your workflow.

By mastering these techniques, you'll be able to work more efficiently and handle complex data science tasks with ease.

3.2 Magic Commands

3.2.1 What are Magic Commands?

Magic commands in Jupyter are shortcuts that extend the functionality of the notebook environment. There are two types of magic commands:

- **Line magics**: Commands that operate on a single line.
- **Cell magics**: Commands that operate on the entire cell.

You can access the full list of magic commands by typing:

```
# show all the avaiable magic commands on the system
%lsmagic
```

```
Available line magics:
%alias  %alias_magic  %autoawait  %autocall  %automagic  %autosave  %bookmark  %cd  %clear  %

Available cell magics:
%%!  %%HTML  %%SVG  %%bash  %%capture  %%cmd  %%code_wrap  %%debug  %%file  %%html  %%javasc

Automagic is ON, % prefix IS NOT needed for line magics.
```

3.2.2 Line Magic

In Jupyter notebooks, line magic commands are invoked by placing a single percentage sign (%) in front of the statement, allowing for quick, inline operations.

3.2.2.1 `%time`: Timing the execution of code

In data science, it is often crucial to evaluate the performance of specific code snippets or algorithms, and the `%time` magic command provides a simple and efficient way to measure the execution time of individual statements, helping you identify bottlenecks and optimize your code for better performance.

```python
def my_dot(a, b):
    """
   Compute the dot product of two vectors

    Args:
      a (ndarray (n,)):  input vector
      b (ndarray (n,)):  input vector with same dimension as a

    Returns:
      x (scalar):
    """
    x=0
    for i in range(a.shape[0]):
        x = x + a[i] * b[i]
    return x
```

```python
import numpy as np
np.random.seed(1)
a = np.random.rand(10000000)  # very large arrays
b = np.random.rand(10000000)
```

Let's use `%time` to measure the execution time of a single line of code.

```
# Example: Timing a list comprehension
%time np.dot(a, b)
```

```
CPU times: total: 15.6 ms
Wall time: 32.9 ms
```

```
2501072.5816813153
```

```
%time my_dot(a, b)
```

```
CPU times: total: 1.88 s
Wall time: 1.86 s
```

```
2501072.5816813707
```

To capture the output of `%time` (or `%timeit`), you cannot directly assign it to a variable as it's a magic command that prints the result to the notebook's output. However, you can use Python's built-in `time` module to manually time your code and assign the execution time to a variable.

Here's how you can do it using the `time` module:

```
import time
tic = time.time()  # capture start time
c = np.dot(a, b)
toc = time.time()  # capture end time

print(f"np.dot(a, b) =  {c:.4f}")
print(f"Vectorized version duration: {1000*(toc-tic):.4f} ms ")

tic = time.time()  # capture start time
c = my_dot(a,b)
toc = time.time()  # capture end time

print(f"my_dot(a, b) =  {c:.4f}")
print(f"loop version duration: {1000*(toc-tic):.4f} ms ")

del(a);del(b)  #remove these big arrays from memory
```

```
np.dot(a, b) =  2501072.5817
Vectorized version duration: 2.9922 ms
my_dot(a, b) =  2501072.5817
loop version duration: 2004.3225 ms
```

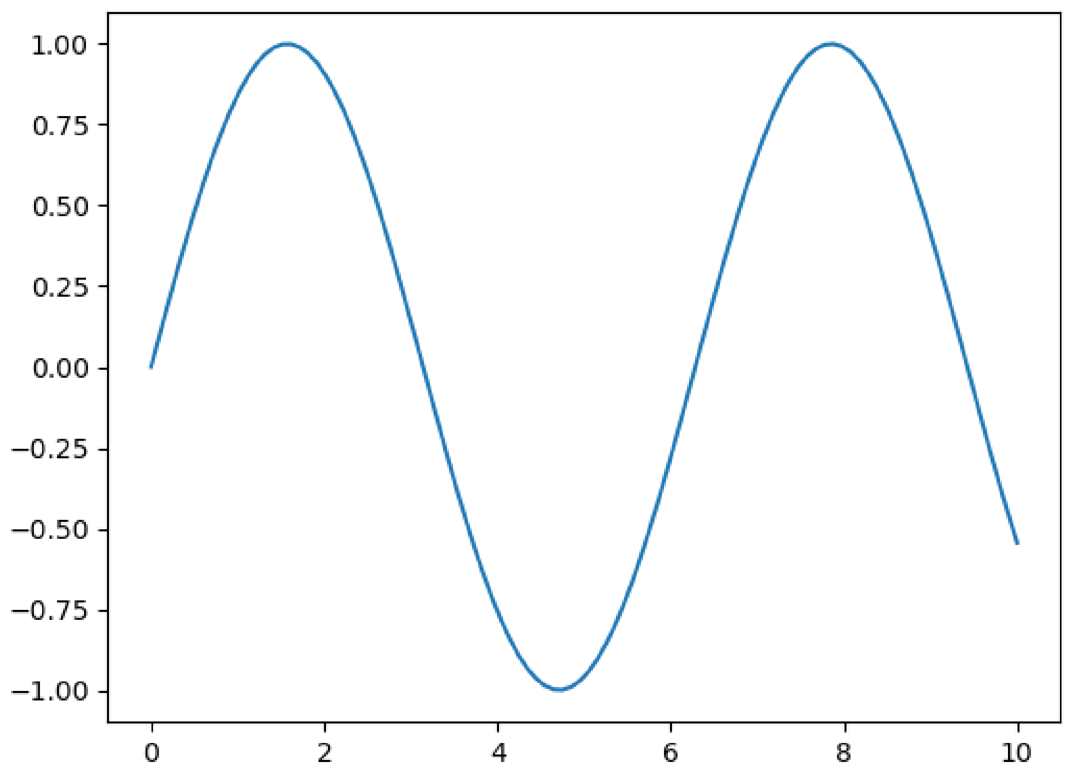
3.2.2.2 `%matplotlib inline`: Displaying plots inline

This command allows you to embed plots within the notebook.

```python
# Example: Using %matplotlib inline to display a plot
import matplotlib.pyplot as plt
import numpy as np

%matplotlib inline

x = np.linspace(0, 10, 100)
y = np.sin(x)
plt.plot(x, y);
```



3.2.3 Cell Magic Commands

While cell magic commands are denoted with double percentage signs (`%%`) at the beginning of the cell, enabling you to apply commands to the entire cell for more complex tasks. Note that a cell magic command in Jupyter notebook has to be the first line in the code cell!

3.2.3.1 `%%time`: Timing cell execution

This cell magic is useful for measuring the execution time of an entire cell.

```
%%time
# Example: Timing a cell with matrix multiplication

A = np.random.rand(1000, 1000)
B = np.random.rand(1000, 1000)
C = np.dot(A, B)
```

```
CPU times: total: 219 ms
Wall time: 23 ms
```

Question: there are several timing magic commands that can be confusing due to their similarities, they are `%time`,`%timeit`, `%%time`, and `%timeit`. Do your own research on the differences among them

3.3 Shell Commands in Jupyter Notebooks

What are Shell Commands?

Shell commands let you interact directly with your computer's operating system from within a Jupyter notebook. This means you can manage files, check your environment, and run system tools without leaving your notebook. To run a shell command in Jupyter, start the line with an exclamation mark (`!`). For example, `!ls` lists files on macOS/Linux, while `!dir` does the same on Windows.

How does it work?

- Python code is executed by the IPython kernel inside the notebook.
- Shell commands (lines starting with `!`) are sent to your computer's shell (like Bash, Command Prompt, or PowerShell), not to Python.
- This separation means you can use both Python and shell commands in the same notebook, but they run in different environments.

3.3.1 Using Shell Commands

3.3.1.1 Print the current working directory

To see where your notebook is running, use: - On macOS/Linux: `!pwd` - On Windows: `!cd`

This helps you understand where files will be saved or loaded from.

```
!cd
```

```
!cd
```

```
c:\Users\lsi8012\mydev\STAT303-1-class-notes
```

3.3.1.2 `!ls` (`!dir` on Windows): Listing files and directories

To view all files and folders in your current directory, use a shell command:

- On macOS/Linux: `!ls`
- On Windows: `!dir`

This is useful for quickly checking what data, scripts, or notebooks are available in your workspace. If you want to see hidden files (those starting with a dot), use `!ls -a` on macOS/Linux or `!dir /a` on Windows.

Tip: If you get an error, double-check that you are using the correct command for your operating system.

3.4 Installing Required Packages Within Your Notebook

When working in Jupyter notebooks, you often need to install new Python packages. There are two main ways to do this:

1. **Shell Command (not recommend):**
    - Use an exclamation mark (`!`) before the command:
        - `!pip install package_name`
    - This runs the command in your system shell, which may not always install the package in the same environment as your notebook kernel.
2. **Magic Command:**

- Use a percent sign (%) before the command:
  - `%pip install package_name`
- This is Jupyter-specific and ensures the package is installed in the environment where your notebook is running.

Example:

- `%pip install numpy` # Installs numpy in the notebook's environment
- `!pip install numpy` # Installs numpy using the system shell (may differ from notebook environment)

Unlike shell commands, the `%pip` magic command is designed specifically for Jupyter notebooks. It automatically installs packages into the exact environment your notebook kernel is using, reducing confusion and installation errors.

```
%pip install numpy
```

```
Requirement already satisfied: numpy in c:\users\lsi8012\appdata\local\anaconda3\lib\site-pa
Note: you may need to restart the kernel to use updated packages.
```

In most cases, you don't need to type `%` before `pip install numpy` because Jupyter's "automagic" feature is enabled by default. Automagic lets you use line magics without the `%` (unless their name conflicts with Python variables or commands).

How automagic works:

- If you type `pip install numpy`, Jupyter will interpret it as `%pip install numpy` automatically.

Best practice:

- Use `%pip install ...` for clarity and to avoid conflicts.
- Automagic is convenient, but explicit magic commands are safer in shared or complex notebooks.

Note: Cell magics still require the `%%` prefix.

3.5 File Paths in Python

Understanding how to specify file paths in Python is essential for loading and saving data. File paths tell Python where to find or store your files (e.g., datasets, results, or scripts).

There are two main types of file paths: **absolute paths** and **relative paths**.

- **Absolute Path:** Gives the complete location of a file or folder from the root of your file system. It always points to the same place, no matter where your code is running.
- **Relative Path:** Specifies the location of a file or folder in relation to your current working directory. It is shorter and more flexible, making your code easier to share and reuse.

Choosing the right type of path helps avoid errors and makes your code more portable across different computers and operating systems.

3.5.1 Absolute Path

An **absolute path** provides the full address to a file or directory, starting from the root of your system. It does **not** depend on where your code is running.

Example (Windows):

```
# Absolute path example (Windows)
file_path = r"C:\Users\Username\Documents\data.csv"
```

```
!conda env list
```

```
# conda environments:
#
base                  *  C:\Users\lsi8012\AppData\Local\anaconda3
```

The path associated with each conda env is absolute path.

3.5.2 Relative Path

A **relative path** describes the location of a file or folder in relation to your current working directory. It is shorter, more flexible, and makes your code easier to share and run on different computers.

Relative paths are especially useful in projects with organized folder structures, because they allow you to move your code and data together without changing file references.

To find your current working directory in your notebook, you can use a magic command:

- Magic command: `%pwd`

Knowing your current working directory helps you understand where Python will look for files when using relative paths.

```
# Magic command
%pwd
```

```
'c:\\Users\\lsi8012\\Documents\\Courses\\FA24\\DataScience_Intro_python_fa24_Sec20_21'
```

Example of a Relative Path:

```
# Example of relative path
file_path = "sample.txt"  # Relative to the current directory
```

The relative path `sample.txt` means that there is a file in the current working directory.

3.5.3 Methods to Specify File Paths in Windows

Specifying file paths correctly is crucial for avoiding errors and making your code portable. Windows uses backslashes (\) to separate folders, but in Python, a single backslash is an escape character (e.g., `\n` for newline), which can lead to mistakes if not handled properly.

Here are four reliable ways to specify file paths in Windows:

3.5.3.1 Method 1: Using Escaped Backslashes

Use double backslashes (\\) to prevent Python from interpreting them as escape characters.

```
file_path = "C:\\Users\\Username\\Documents\\data.csv"
```

3.5.3.2 Method 2: Using Raw Strings

Prefix the path with `r` to tell Python to treat backslashes as literal characters.

```
file_path = r"C:\Users\Username\Documents\data.csv"
```

3.5.3.3 Method 3: Using forward slashes (/)

Python accepts forward slashes (/) in file paths, even on Windows. This is often the simplest and most portable method.

```
file_path = "C:/Users/Username/Documents/data.csv"
```

3.5.3.4 Method 4: Using `os.path.join`

Use `os.path.join()` to build paths programmatically. This method automatically uses the correct separator for your operating system, making your code cross-platform.

Using these methods helps prevent bugs, makes your code easier to share, and ensures it works on different computers and operating systems.

```
import os
file_path = os.path.join("C:", "Users", "Username", "Documents", "data.csv")
```

This method works across different operating systems because `os.path.join` automatically uses the correct path separator (\ for Windows and /for Linux/Mac).

3.5.4 File paths in macOS and Linux

macOS and Linux use forward slashes (/) as path separators, which is exactly what Python expects by default. This means you can specify file paths directly, like `/Users/yourname/Documents/data.csv`, without worrying about escape characters or compatibility issues.

Why is this helpful? - Forward slashes work seamlessly in Python on macOS, Linux, and even Windows. - You avoid common errors caused by backslashes (which are escape characters in Python). - Code written with forward slashes is portable and recommended for all platforms.

3.5.4.1 Best Practices for File Paths in Data Science

- Prefer **relative paths** within your project folders—this makes your code portable and easy to share or move.
- Use **absolute paths** only for files outside your project or when you need a fixed location.
- Always check your **current working directory** before reading or writing files to avoid confusion and errors.

- Avoid hardcoding file paths directly in your code; use variables or configuration files for flexibility.
- For cross-platform compatibility, use **forward slashes (/)** in file paths or build paths programmatically with `os.path.join()`.
- Document your file structure and path conventions in your project README to help collaborators.
- When sharing code, test file paths on both Windows and macOS/Linux to ensure portability.

3.6 Interacting with the OS and Filesystem

In data science projects, you often work with data files (such as CSV, Excel, or JSON) stored in various folders. Managing these files efficiently is essential for reproducible workflows and organized projects.

The Python `os` module provides powerful tools to interact with your operating system and manage files and directories directly from your notebook. With `os`, you can: - Check your current working directory - List files and folders - Create, rename, or delete directories - Move between folders - Check if files or folders exist before using them

Let's import the `os` module and explore some of its most useful functions for data science tasks.

```
import os
```

We can get the location of the current working directory using the `os.getcwd` function.

```
os.getcwd()
```

```
'c:\\Users\\lsi8012\\mydev\\STAT303-1-class-notes'
```

The command `os.chdir('..')` in Python changes the current working directory to the parent directory of the current one.

Note that `..` as the path notation for the parent directory is universally true across all major operating systems, including Windows, macOS, and Linux. It allows you to move one level up in the directory hierarchy, which is very useful when navigating directories programmatically, especially in scripts where directory traversal is needed.

```
os.chdir('..')
```

```python
os.getcwd()
```

```
'c:\\Users\\lsi8012\\mydev'
```

`os.chdir()` is used to change the current working directory.

For example: `os.chdir('./week2')`

`./week2` is a relative path:

- `.` refers to the current directory.
- week2 is a folder inside the current directory.

The `os.listdir()` function in Python returns a list of all files and directories in the specified path. If no path is provided, it returns the contents of the current working directory.

```python
os.listdir()
```

```
['limonstar-web',
 'llm2005spring',
 'llm2005spring_back',
 'mcdc_2025su',
 'mcdc_2025su_new',
 'STAT201',
 'STAT303-1-class-notes',
 'STAT303-2-class-notes',
 'STAT303-3-class-notes',
 'stat359_su25',
 'stat362']
```

Check whether a specific folder/file exist in the current working directory

```python
'data' in os.listdir('.')
```

```
False
```

3.7 Navigate between directories and understand . and .. in paths

In Python, you can use the `os` module to work with directories and paths, and it's important to understand how relative paths work:

- `.` refers to the **current directory**
- `..` refers to the **parent directory**

This knowledge is essential for organizing files and writing portable code.

From the command line, you can use the `cd` (change directory) command to move between folders:

- `cd <folder_name>` moves **down** into a child folder
- `cd ..` moves **up** to the parent folder

By combining these commands, you can navigate to the desired location.

3.8 Independent Study

3.8.1 Setting Up Your Data Science Workspace

To reinforce and apply the skills from this lecture, complete the following hands-on tasks:

1. **Set Up Your Workspace**
   - Create a folder named `STAT303-1` to organize all course materials.
   - Set up a dedicated `.venv` environment for your coursework to keep dependencies isolated.
   - Organize your files into subfolders for datasets, assignments, project, quizzes, and lectures. The layout looks like below

   ```
   stat303-1/
     .venv/              # Python virtual environment
     datasets/
     project/
     homework assignments/
     small assignments/
     lectures/
   ```

   - Use the `os` module or shell commands in your notebook to create these directories programmatically.

2. **Practice Magic Commands**
   - Use `%timeit` to measure the execution time of a simple Python function in your notebook.
   - Explore `%lsmagic` to discover all available magic commands and experiment with a few.
3. **Run Shell Commands**
   - Use `!ls` (or `!dir` on Windows) to list the contents of the directories you created.
   - Use `!pwd` (or `!cd` on Windows) to print your current working directory.
4. **Navigate between directoires using `.` and `..`**

Tip: Document your process and any issues you encounter. This will help you troubleshoot and share your workflow with others.

3.9 References and Further Learning

- **IPython Magic Commands**
  IPython Documentation – Magic Commands
  A complete list of available line and cell magics, such as `%time`, `%pip`, and `%%bash`.
- **File and Directory Management in Python**
  Python `os` module documentation
  Covers functions like `os.getcwd()`, `os.chdir()`, and path handling.

Part II

Python refresher

4 Python Basics

```
<IPython.core.display.Image object>
```

Before diving into new material, this chapter provides a quick refresher on key Python concepts covered in STAT201, which is the prerequisite for this course. If you have not taken STAT201, or if you feel you need to strengthen your Python skills, please review chapters 3–8 in the STAT201 book. A solid understanding of Python basics will help you succeed in this course and make it easier to follow the advanced topics ahead.

4.1 Python Variables

Variables are fundamental building blocks in Python—they allow you to store, update, and reference data throughout your code. Choosing clear and descriptive variable names makes your code easier to read, debug, and share with others.

4.1.1 Rules for variable names

- Variable names must start with a letter (a–z, A–Z) or an underscore (`_`). They cannot begin with a number.
- Names can include letters, digits, and underscores (`a_variable`, `profit_margin`, `the_3_musketeers`).
- Variable names are case-sensitive: `score`, `Score`, and `SCORE` are all different variables.
- Avoid using Python reserved words (like `for`, `if`, `class`, etc.) as variable names.

Tip: Use descriptive names that reflect the purpose of the variable. This helps you and others understand your code at a glance.

Here are some examples of good variable names:

```
# Examples of valid variable names

a_variable = 23
is_today_Saturday = False
my_favorite_car = "Delorean"
the_3_musketeers = ["Athos", "Porthos", "Aramis"]
```

If you use an invalid variable name, Python will raise a `SyntaxError` and stop running your code. Always follow the naming rules to avoid these errors and keep your code readable.

4.1.2 Dynamic Typing in Python Variables

Python variables are dynamically typed, meaning you don't need to declare their type before using them. You can assign a value of any type to a variable, and even change its type later in your code:

```
x = 5      # x is an integer
x = "cat" # now x is a string
```

To check the type of a variable, use the built-in `type()` function:

```
type(x)
```

This flexibility makes Python easy to use, but it's important to keep track of your variable types to avoid confusion in your code.

```
# Print the variables defined above their values and their types
print("a_variable:", a_variable, "| type:", type(a_variable))
print("is_today_Saturday:", is_today_Saturday, "| type:", type(is_today_Saturday))
print("my_favorite_car:", my_favorite_car, "| type:", type(my_favorite_car))
print("the_3_musketeers:", the_3_musketeers, "| type:", type(the_3_musketeers))
```

```
a_variable: 23 | type: <class 'int'>
is_today_Saturday: False | type: <class 'bool'>
my_favorite_car: Delorean | type: <class 'str'>
the_3_musketeers: ['Athos', 'Porthos', 'Aramis'] | type: <class 'list'>
```

4.1.3 Multiple Variable Assignment in Python

Python allows you to assign values to several variables at once in a single line. This technique is especially useful for initializing related variables and can make your code cleaner and more efficient.

Example:

```
color1, color2, color3 = "red", "green", "blue"
```

After this assignment: - `color1` is "red" - `color2` is "green" - `color3` is "blue"

This approach works for any number of variables, as long as the number of values matches the number of variable names.

```
color1, color2, color3 = "red", "green", "blue"
print(color1)
print(color3)
```

```
red
blue
```

The same value can be assigned to multiple variables by chaining multiple assignment operations within a single statement.

```
color4 = color5 = color6 = "magenta"
print(color4)
print(color6)
```

```
magenta
magenta
```

4.2 Built-in data types

Python has several built-in data types for storing different kinds of information in variables.

```
<IPython.core.display.Image object>
```

Primitive Types
In Python, **integers, floats, booleans, and `None`** are often called *primitive data types* because they represent a single value.

Container (Data Structure) Types
Types such as **strings, lists, tuples, sets, and dictionaries** are *containers* because they can hold multiple values (characters in a string, items in a list, key–value pairs in a dictionary, etc.). We'll explore these container types in more detail in the next chapter.

Identifying Types
You can check the type of any object using Python's built-in `type()` function. For example:

```
print(type(42))          # int
print(type(3.14))        # float
print(type(True))        # bool
print(type(None))        # NoneType
print(type("hello"))     # str (a sequence / container of characters)
print(type([1, 2, 3]))   # list
```

```
<class 'int'>
<class 'float'>
<class 'bool'>
<class 'NoneType'>
<class 'str'>
<class 'list'>
```

4.3 Python Standard Library

The Python Standard Library provides a wide range of modules and built-in functions that support everyday programming tasks. These tools allow you to write code that is both efficient and readable without reinventing common functionality.

Examples of built-in functions: - `print()`: Displays output to the screen. - `len()`: Returns the length of an object (like a list or string). - `type()`: Shows the type of a variable. - `sum()`: Adds up all items in an iterable (like a list). - `range()`: Generates a sequence of numbers, often used in loops.

You can explore more built-in functions and modules in the official documentation. Using these tools makes your code more readable and powerful.

Example:

range(): The `range()` function returns a sequence of evenly-spaced integer values. It is commonly used in `for` loops to define the sequence of elements over which the iterations are performed.

Below is an example where the `range()` function is used to create a sequence of whole numbers upto 10:

```
print(list(range(1,10)))
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Date and Time:

Python includes a powerful built-in module called `datetime` for working with dates and times. This module lets you create, manipulate, and format date/time objects easily.

- You can get the current date and time.
- You can perform arithmetic with dates (e.g., add days, subtract dates).
- You can format dates and times for display or parsing.

This is essential for tasks like timestamping data, scheduling, or analyzing time series.

```
import datetime as dt
```

```
#Defining a date-time object
dt_object = dt.datetime(2022, 9, 20, 11,30,0)
```

Information about date and time can be accessed with the relevant attribute of the `datetime` object.

```
dt_object.day, dt_object.year
```

```
(20, 2022)
```

Formatting Dates and Times:

The `strftime` method in the `datetime` module lets you convert a `datetime` object into a readable string using custom formats. This is useful for displaying dates in a way that matches your needs (e.g., for reports, logs, or user interfaces).

Common format codes include: - `%Y`: 4-digit year (e.g., 2025) - `%m`: 2-digit month (01-12) - `%d`: 2-digit day (01-31) - `%H`: Hour (00-23) - `%M`: Minute (00-59) - `%S`: Second (00-59)

See the Python documentation for more formatting options.

```
dt_object.strftime('%m/%d/%Y')
```

```
'09/20/2022'
```

```
dt_object.strftime('%m/%d/%y %H:%M')
```

```
'09/20/22 11:30'
```

```python
dt_object.strftime('%h-%d-%Y')
```

```
'Sep-20-2022'
```

4.4 Third-Party Packages (Libraries)

While Python has many useful built-in functions like `print()`, `abs()`, `max()`, and `sum()`, these are often not enough for data analysis. Third-party libraries extend Python's capabilities and are essential for scientific computing and data science.

Popular libraries and their main uses:

1. **NumPy**: Efficient numerical operations, arrays, and mathematical functions. Essential for scientific and data analysis tasks.
2. **Pandas**: Powerful data manipulation and analysis. DataFrames and Series make reading, cleaning, and transforming data easy.
3. **Matplotlib & Seaborn**: Data visualization. Matplotlib creates a wide range of plots; Seaborn builds on Matplotlib for attractive statistical graphics.
4. **SciPy**: Advanced scientific computing—optimization, integration, statistics, and more.
5. **Scikit-learn**: Machine learning—tools for preprocessing, classification, regression, clustering, and model evaluation.
6. **Statsmodels**: Statistical modeling and inference (focuses on explanation, not just prediction).

How to use these libraries: 1. **Install the libraries** : Covered by Chapter 2 and 3 2. **Import the libraries** in your Python script or Jupyter notebook. 3. **Use their functions and classes** to analyze data, visualize results, and build models.

These libraries form the foundation of modern data science workflows in Python. You will gain plenty of hands-on experience with each of them throughout this course sequence.

4.4.1 Importing a Library

Use the `import` keyword to bring a library into your Python code.

Example:

```python
import numpy as np
```

Aliases like `np` make code shorter and easier to read.

```python
import numpy as np
np.arange(8)
```

```
array([0, 1, 2, 3, 4, 5, 6, 7])
```

Importing in Python: Key Styles

- Import a whole module:

  ```python
  import math
  ```

- Import specific items:

  ```python
  from random import randint
  ```

- Use an alias:

  ```python
  import pandas as pd
  ```

- Rename an imported item:

  ```python
  from os.path import join as join_path
  ```

Pick the style that fits your needs and keeps your code readable.

4.5 User-defined functions

A function is a reusable set of instructions that takes one or more inputs, performs some operations, and often returns an output. **Indeed, while python's standard library and ecosystem libraries offer a wealth of pre-defined functions for a wide range of tasks, there are situations where defining your own functions is not just beneficial but necessary.**

4.5.1 Creating and using functions

You can define a new function using the def keyword.

```python
def say_hello():
    print('Hello there!')
    print('How are you?')
```

Note the round brackets or parentheses () and colon : after the function's name. Both are essential parts of the syntax. The function's *body* contains an indented block of statements. The statements inside a function's body are not executed when the function is defined. To execute the statements, we need to *call* or *invoke* the function.

```
say_hello()
```

```
Hello there!
How are you?
```

```
def say_hello_to(name):
    print('Hello ', name)
    print('How are you?')
```

```
say_hello_to('Lizhen')
```

```
Hello  Lizhen
How are you?
```

```
name = input ('Please enter your name: ')
say_hello_to(name)
```

```
Please enter your name: George
Hello  George
How are you?
```

4.5.2 Variable scope: Local and global Variables

Local variable: When we declare variables inside a function, these variables will have a local scope (within the function). We cannot access them outside the function. These types of variables are called local variables. For example,

```
def greet():
    message = 'Hello'  # local variable
    print('Local', message)
greet()
```

```
Local Hello
```

```
# print(message) # try to access message variable outside greet() function, uncomment this li
```

As `message` was defined within the function `greet()`, it is local to the function, and cannot be called outside the function.

Global variable: Aa variable declared outside of the function or in global scope is known as a global variable. This means that a global variable can be accessed inside or outside of the function.

Let's see an example of how a global variable is created.

```python
message = 'Hello'  # declare global variable

def greet():
    print('Local', message)  # declare local variable

greet()
print('Global', message)
```

```
Local Hello
Global Hello
```

4.5.3 Function Arguments

4.5.3.1 Named Arguments

When calling functions with multiple arguments, using *named* arguments improves clarity and reduces mistakes. You can also split long function calls across multiple lines for readability.

Example:

```python
def greet(name, message):
    print(f"{message}, {name}!")

greet(name="Alice", message="Hello")
```

Named arguments make your code easier to understand and maintain.

Here is an example:

```python
def loan_emi(amount, duration, rate, down_payment=0):
    loan_amount = amount - down_payment
    emi = loan_amount * rate * ((1+rate)**duration) / (((1+rate)**duration)-1)
    return emi
```

```python
emi1 = loan_emi(
    amount=1260000,
    duration=8*12,
    rate=0.1/12,
    down_payment=3e5
)
```

```python
emi1
```

```
14567.19753389219
```

4.5.3.2 Optional Arguments

Functions with optional arguments offer more flexibility in how you can use them. You can call the function with or without the argument, and if there is no argument in the function call, then a default value is used.

```python
emi2 = loan_emi(
    amount=1260000,
    duration=8*12,
    rate=0.1/12)

emi2
```

```
19119.4467632335
```

4.5.3.3 `*args` and `**kwargs`

We can pass a variable number of arguments to a function using two special symbols in Python:
- `*args` for variable-length **positional arguments**
- `**kwargs` for variable-length **keyword arguments**

This is useful when you want your function to accept a variety of arguments.

```python
def myFun(*args,**kwargs):
    print("args: ", args)
    print("kwargs: ", kwargs)


# Now we can use both *args ,**kwargs
# to pass arguments to this function :
myFun('John',22,'cs',name="John",age=22,major="cs")
```

```
args:  ('John', 22, 'cs')
kwargs:  {'name': 'John', 'age': 22, 'major': 'cs'}
```

4.6 Control Flow

Control flow statements (like if, for, and while) let you decide how your code runs based on conditions and loops. Control flow in Python includes both conditional statements and iteration loops to manage program logic.

4.6.1 Conditional Statements

As in other languages, python has built-in keywords that provide conditional flow of control in the code.

4.6.1.1 Branching with `if`, `else` and `elif`

One of the most powerful features of programming languages is *branching*: the ability to make decisions and execute a different set of statements based on whether one or more conditions are true.

The `if` statement

In Python, branching is implemented using the `if` statement, which is written as follows:

```
if condition:
    statement1
    statement2
```

The `condition` can be a value, variable or expression. If the condition evaluates to `True`, then the statements within the *`if` block* are executed. Notice the four spaces before `statement1`, `statement2`, etc. The spaces inform Python that these statements are associated with the `if` statement above. This technique of structuring code by adding spaces is called *indentation*.

> **Indentation**: Python relies heavily on *indentation* (white space before a statement) to define code structure. This makes Python code easy to read and understand. You can run into problems if you don't use indentation properly. Indent your code by placing the cursor at the start of the line and pressing the `Tab` key once to add 4 spaces. Pressing `Tab` again will indent the code further by 4 more spaces, and press `Shift+Tab` will reduce the indentation by 4 spaces.

For example, let's write some code to check and print a message if a given number is even.

```
a_number = 34
```

```
if a_number % 2 == 0:
    print("We're inside an if block")
    print('The given number {} is even.'.format(a_number))
```

```
We're inside an if block
The given number 34 is even.
```

The `else` statement

We may want to print a different message if the number is not even in the above example. This can be done by adding the `else` statement. It is written as follows:

```
if condition:
    statement1
    statement2
else:
    statement4
    statement5
```

If `condition` evaluates to `True`, the statements in the `if` block are executed. If it evaluates to `False`, the statements in the `else` block are executed.

```
if a_number % 2 == 0:
    print('The given number {} is even.'.format(a_number))
else:
    print('The given number {} is odd.'.format(a_number))
```

```
The given number 34 is even.
```

The `elif` statement

Python also provides an `elif` statement (short for "else if") to chain a series of conditional blocks. The conditions are evaluated one by one. For the first condition that evaluates to `True`, the block of statements below it is executed. The remaining conditions and statements are not evaluated. So, in an `if`, `elif`, `elif`… chain, at most one block of statements is executed, the one corresponding to the first condition that evaluates to `True`.

```python
today = 'Wednesday'
```

```python
if today == 'Sunday':
    print("Today is the day of the sun.")
elif today == 'Monday':
    print("Today is the day of the moon.")
elif today == 'Tuesday':
    print("Today is the day of Tyr, the god of war.")
elif today == 'Wednesday':
    print("Today is the day of Odin, the supreme diety.")
elif today == 'Thursday':
    print("Today is the day of Thor, the god of thunder.")
elif today == 'Friday':
    print("Today is the day of Frigga, the goddess of beauty.")
elif today == 'Saturday':
    print("Today is the day of Saturn, the god of fun and feasting.")
```

```
Today is the day of Odin, the supreme diety.
```

In the above example, the first 3 conditions evaluate to `False`, so none of the first 3 messages are printed. The fourth condition evaluates to `True`, so the corresponding message is printed. The remaining conditions are skipped. Try changing the value of `today` above and re-executing the cells to print all the different messages.

Using `if`, `elif`, and `else` together

You can also include an `else` statement at the end of a chain of `if`, `elif`… statements. This code within the `else` block is evaluated when none of the conditions hold true.

```python
a_number = 49
```

```python
if a_number % 2 == 0:
    print('{} is divisible by 2'.format(a_number))
elif a_number % 3 == 0:
    print('{} is divisible by 3'.format(a_number))
elif a_number % 5 == 0:
    print('{} is divisible by 5'.format(a_number))
else:
    print('All checks failed!')
    print('{} is not divisible by 2, 3 or 5'.format(a_number))
```

```
All checks failed!
49 is not divisible by 2, 3 or 5
```

Non-Boolean Conditions

Note that conditions do not necessarily have to be booleans. In fact, a condition can be any value. The value is converted into a boolean automatically using the `bool` operator. Any value in Python can be converted to a Boolean using the `bool` function.

Only the following values evaluate to `False` (they are often called *falsy* values):

1. The value `False` itself
2. The integer `0`
3. The float `0.0`
4. The empty value `None`
5. The empty text `""`
6. The empty list `[]`
7. The empty tuple `()`
8. The empty dictionary `{}`
9. The empty set `set()`
10. The empty range `range(0)`

Everything else evaluates to `True` (a value that evaluates to `True` is often called a *truthy* value).

```python
if '':
    print('The condition evaluted to True')
else:
    print('The condition evaluted to False')
```

```
The condition evaluted to False
```

```python
if 'Hello':
    print('The condition evaluted to True')
else:
    print('The condition evaluted to False')
```

```
The condition evaluted to True
```

```python
if { 'a': 34 }:
    print('The condition evaluted to True')
else:
    print('The condition evaluted to False')
```

```
The condition evaluted to True
```

```python
if None:
    print('The condition evaluted to True')
else:
    print('The condition evaluted to False')
```

```
The condition evaluted to False
```

Nested conditional statements

The code inside an `if` block can also include an `if` statement inside it. This pattern is called `nesting` and is used to check for another condition after a particular condition holds true.

```python
a_number = 15
```

```python
if a_number % 2 == 0:
    print("{} is even".format(a_number))
    if a_number % 3 == 0:
        print("{} is also divisible by 3".format(a_number))
    else:
        print("{} is not divisibule by 3".format(a_number))
else:
    print("{} is odd".format(a_number))
    if a_number % 5 == 0:
        print("{} is also divisible by 5".format(a_number))
    else:
        print("{} is not divisibule by 5".format(a_number))
```

```
15 is odd
15 is also divisible by 5
```

Notice how the `print` statements are indented by 8 spaces to indicate that they are part of the inner `if`/`else` blocks.

> Nested `if`, `else` statements are often confusing to read and prone to human error. It's good to avoid nesting whenever possible, or limit the nesting to 1 or 2 levels.

Shorthand `if` conditional expression

A frequent use case of the `if` statement involves testing a condition and setting a variable's value based on the condition.

Python provides a shorter syntax, which allows writing such conditions in a single line of code. It is known as a *conditional expression*, sometimes also referred to as a *ternary operator*. It has the following syntax:

```
x = true_value if condition else false_value
```

It has the same behavior as the following `if-else` block:

```
if condition:
    x = true_value
else:
    x = false_value
```

Let's try it out for the example above.

```python
parity = 'even' if a_number % 2 == 0 else 'odd'
```

```python
print('The number {} is {}.'.format(a_number, parity))
```

```
The number 15 is odd.
```

The `pass` statement

`if` statements cannot be empty, there must be at least one statement in every `if` and `elif` block. We can use the `pass` statement to do nothing and avoid getting an error.

```python
a_number = 9
```

```python
# please uncomment the code below and see the error message
# if a_number % 2 == 0:

# elif a_number % 3 == 0:
#     print('{} is divisible by 3 but not divisible by 2')
```

As there must be at least one statement withihng the `if` block, the above code throws an error.

```python
if a_number % 2 == 0:
    pass
elif a_number % 3 == 0:
    print('{} is divisible by 3 but not divisible by 2'.format(a_number))
```

```
9 is divisible by 3 but not divisible by 2
```

4.6.2 Loops

4.6.2.1 `while` loops

Another powerful feature of programming languages, closely related to branching, is running one or more statements multiple times. This feature is often referred to as *iteration* on *looping*, and there are two ways to do this in Python: using `while` loops and `for` loops.

`while` loops have the following syntax:

```
while condition:
    statement(s)
```

Statements in the code block under `while` are executed repeatedly as long as the `condition` evaluates to `True`. Generally, one of the statements under `while` makes some change to a variable that causes the condition to evaluate to `False` after a certain number of iterations.

Let's try to calculate the factorial of `100` using a `while` loop. The factorial of a number `n` is the product (multiplication) of all the numbers from `1` to `n`, i.e., `1*2*3*...*(n-2)*(n-1)*n`.

```python
result = 1
i = 1

while i <= 10:
    result = result * i
    i = i+1

print('The factorial of 100 is: {}'.format(result))
```

```
The factorial of 100 is: 3628800
```

4.6.2.2 Infinite Loops

Suppose the condition in a `while` loop always holds true. In that case, Python repeatedly executes the code within the loop forever, and the execution of the code never completes. This situation is called an infinite loop. It generally indicates that you've made a mistake in your code. For example, you may have provided the wrong condition or forgotten to update a variable within the loop, eventually falsifying the condition.

If your code is *stuck* in an infinite loop during execution, just press the "Stop" button on the toolbar (next to "Run") or select "Kernel > Interrupt" from the menu bar. This will *interrupt* the execution of the code. The following two cells both lead to infinite loops and need to be interrupted.

```python
# INFINITE LOOP - INTERRUPT THIS CELL

result = 1
i = 1

while i <= 100:
    result = result * i
    # forgot to increment i
```

```python
# INFINITE LOOP - INTERRUPT THIS CELL

result = 1
i = 1

while i > 0 : # wrong condition
    result *= i
    i += 1
```

4.6.2.3 `break` and `continue` statements

In Python, `break` and `continue` statements can alter the flow of a normal loop.

We can use the `break` statement within the loop's body to immediately stop the execution and *break* out of the loop. with the continue statement. If the condition evaluates to `True`, then the loop will move to the next iteration.

```
i = 1
result = 1

while i <= 100:
    result *= i
    if i == 42:
        print('Magic number 42 reached! Stopping execution..')
        break
    i += 1

print('i:', i)
print('result:', result)
```

```
Magic number 42 reached! Stopping execution..
i: 42
result: 1405006117752879898543142606244511569936384000000000
```

```
i = 1
result = 1

while i < 8:
    i += 1
    if i % 2 == 0:
        print('Skipping {}'.format(i))
        continue
    print('Multiplying with {}'.format(i))
    result = result * i

print('i:', i)
print('result:', result)
```

```
Skipping 2
Multiplying with 3
Skipping 4
```

```
Multiplying with 5
Skipping 6
Multiplying with 7
Skipping 8
i: 8
result: 105
```

In the example above, the statement `result = result * i` inside the loop is skipped when `i` is even, as indicated by the messages printed during execution.

> **Logging**: The process of adding `print` statements at different points in the code *(often within loops and conditional statements)* for inspecting the values of variables at various stages of execution is called logging. As our programs get larger, they naturally become prone to human errors. Logging can help in verifying the program is working as expected. In many cases, `print` statements are added while writing & testing some code and are removed later.

Task: Guess the output and explain it.

```python
# Use of break statement inside the loop

for val in "string":
    if val == "i":
        break
    print(val)

print("The end")
```

```
s
t
r
The end
```

```python
# Program to show the use of continue statement inside loops

for val in "string":
    if val == "i":
        continue
    print(val)

print("The end")
```

```
s
t
r
n
g
The end
```

4.6.2.4 `for` loops

A `for` loop is used for iterating or looping over sequences, i.e., lists, tuples, dictionaries, strings, and *ranges*. For loops have the following syntax:

```
for value in sequence:
    statement(s)
```

The statements within the loop are executed once for each element in `sequence`. Here's an example that prints all the element of a list.

```python
days = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday']

for day in days:
    print(day)
```

```
Monday
Tuesday
Wednesday
Thursday
Friday
```

```python
# Looping over a string
for char in 'Monday':
    print(char)
```

```
M
o
n
d
a
y
```

```python
# Looping over a dictionary
person = {
    'name': 'John Doe',
    'sex': 'Male',
    'age': 32,
    'married': True
}

for key, value in person.items():
    print("Key:", key, ",", "Value:", value)
```

```
Key: name , Value: John Doe
Key: sex , Value: Male
Key: age , Value: 32
Key: married , Value: True
```

4.6.2.5 Iterating using `range`

The `range` function is used to create a sequence of numbers that can be iterated over using a `for` loop. It can be used in 3 ways:

- `range(n)` - Creates a sequence of numbers from `0` to `n-1`
- `range(a, b)` - Creates a sequence of numbers from `a` to `b-1`
- `range(a, b, step)` - Creates a sequence of numbers from `a` to `b-1` with increments of `step`

Let's try it out.

```python
for i in range(4):
    print(i)
```

```
0
1
2
3
```

```python
for i in range(3, 8):
    print(i)
```

```
3
4
5
6
7
```

```
for i in range(3, 14, 4):
    print(i)
```

```
3
7
11
```

Ranges are used for iterating over lists when you need to track the index of elements while iterating.

```
a_list = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday']

for i in range(len(a_list)):
    print('The value at position {} is {}.'.format(i, a_list[i]))
```

```
The value at position 0 is Monday.
The value at position 1 is Tuesday.
The value at position 2 is Wednesday.
The value at position 3 is Thursday.
The value at position 4 is Friday.
```

4.7 Object Oriented Programming

Python is an object-oriented programming language. In layman terms, it means that every number, string, data structure, function, class, module, etc., exists in the python interpreter as a python object. An object may have attributes and methods associated with it. For example, let us define a variable that stores an integer:

```
var = 2
```

The variable `var` is an object that has attributes and methods associated with it. For example a couple of its attributes are `real` and `imag`, which store the real and imaginary parts respectively, of the object `var`:

```
print("Real part of 'var': ",var.real)
print("Real part of 'var': ",var.imag)
```

```
Real part of 'var':  2
Real part of 'var':  0
```

Attribute: An attribute is a value associated with an object, defined within the class of the object.

Method: A method is a function associated with an object, defined within the class of the object, and has access to the attributes associated with the object.

For looking at attributes and methods associated with an object, say `obj`, press tab key after typing `obj.`.

Consider the example below of a class *example_class*:

```
class example_class:
    class_name = 'My Class'
    def my_method(self):
        print('Hello World!')

e = example_class()
```

In the above class, `class_name` is an attribute, while `my_method` is a method.

4.7.0.1 Call by Reference in Python

Python uses *call by object reference* (also called *call by sharing*). When you assign an object to a variable, the variable points to the object in memory—not a copy.

Changes made through one reference affect the original object. This is important to remember when working with mutable types like lists and dictionaries.

```
x = [5,3]
```

The variable name `x` is a reference to the memory location where the object `[5, 3]` is stored. Now, suppose we assign `x` to a new variable `y`:

```
y = x
```

In the above statement the variable name `y` now refers to the same object `[5,3]`. The object `[5,3]` does **not** get copied to a new memory location referred by `y`. To prove this, let us add an element to `y`:

```
y.append(4)
print(y)
```

```
[5, 3, 4]
```

```
print(x)
```

```
[5, 3, 4]
```

When we changed `y`, note that `x` also changed to the same object, showing that `x` and `y` refer to the same object, instead of referring to different copies of the same object.

5 Data Structures

<IPython.core.display.Image object>

In data science, you'll shape raw data **before modeling**. These built-ins are the foundation that explain why `Series`/`DataFrame` (pandas) and `ndarray` (NumPy) behave the way they do.

5.1 Primitives vs. Containers

- **Primitive (single value):** `int`, `float`, `bool`, `None`
- **Containers (hold multiple values):** `str`, `list`, `tuple`, `set`, `dict`

  Tip: Check any object's type with `type(x)`.

5.2 Core Built-in Data Structures

5.2.1 Sequences (ordered, indexable)

Types: `list`, `tuple`, `str`

```
# list (mutable)
nums = [10, 20, 20, 30]
nums.append(40)        # mutate in place
nums[0] = 11           # item assignment
nums_slice = nums[1:3] # slicing

# tuple (immutable)
pt = (42.0, -1.5)      # cannot reassign pt[0]

# string (immutable sequence of characters)
s = "data"
s2 = s.upper()         # returns a new string; s unchanged
```

When to use:

- `list`: grow/shrink, frequent edits, ordered data
- `tuple`: fixed-size records, function returns, hashable as dict keys
- `str`: text processing (we'll also use `nltk` later)

5.2.2 Sets (unordered, unique)

Types: `set`

```
a = {1, 2, 2, 3}        # {1, 2, 3}
b = {3, 4}
a | b                   # union -> {1, 2, 3, 4}
a & b                   # intersection -> {3}
a - b                   # difference -> {1, 2}
```

```
{1, 2}
```

When to use:

- Deduplication, membership tests, fast set algebra.

5.2.3 Mappings (key–value)

Types: `dict`

```
student = {"name": "Alex", "year": 3, "major": "Stats"}
student["year"] = 4              # update
student["gpa"] = 3.7             # insert
for k, v in student.items():     # iterate keys & values
    print(k, "->", v)
```

```
name -> Alex
year -> 4
major -> Stats
gpa -> 3.7
```

When to use:

- Labeled data, lookups, configuration/state.

5.3 Iterables vs. Iterators

In Python, an **iterable** is any object capable of returning its members one at a time, such as lists, tuples, strings, and dictionaries. You can loop over iterables using a `for` loop.

An **iterator** is an object that represents a stream of data; it produces the next value when you call `next()` on it. Iterators are created from iterables using the `iter()` function.

Key differences: - Iterables can be looped over, but do not remember their position. - Iterators remember their position and can only be advanced one item at a time.

Example:

```
my_list = [1, 2, 3]
my_iter = iter(my_list)
print(next(my_iter))  # 1
print(next(my_iter))  # 2
print(next(my_iter))  # 3
```

Understanding the difference helps you write efficient loops and work with data streams in Python.

5.4 Common, Efficient Operations

5.4.1 Comprehensions

Comprehensions are a concise way to create lists, sets, or dictionaries from iterables by applying an expression to each item in an iterable (such as a list, tuple, or range) and optionally filtering the items based on a condition. They are a powerful and efficient way to generate new collections without the need for explicit loops.

Basic Syntax:

```
new_list = [expression for item in iterable if condition]
```

- **expression:** What you want to include in the new list (or set/dict).
- **item:** Represents each element in the iterable as the comprehension iterates through it.
- **iterable:** The source of elements (list, tuple, range, etc.).
- **condition (optional):** A filter to control which items are included. If omitted, all items are included.

Why use comprehensions? - More readable and concise than loops. - Often faster than equivalent for-loops. - Preferred for simple transformations and filtering.

List comprehension example:

Create a list that has squares of natural numbers from 5 to 15.

```
sqrt_natural_no_5_15 = [(x**2) for x in range(5,16)]
print(sqrt_natural_no_5_15)
```

```
[25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225]
```

Create a list of tuples, where each tuple consists of a natural number and its square, for natural numbers ranging from 5 to 15.

```
sqrt_natural_no_5_15 = [(x,x**2) for x in range(5,16)]
print(sqrt_natural_no_5_15)
```

```
[(5, 25), (6, 36), (7, 49), (8, 64), (9, 81), (10, 100), (11, 121), (12, 144), (13, 169), (14
```

Creating a list of words that start with the letter 'a' in a given list of words.

```
words = ['apple', 'banana', 'avocado', 'grape', 'apricot']
a_words = [word for word in words if word.startswith('a')]
print(a_words)
```

```
['apple', 'avocado', 'apricot']
```

Set comprehension:

```
unique_lengths = {len(word) for word in ['cat', 'dog', 'mouse']}
uniq_initials = {name[0].upper() for name in ["amy","ann","bob","amy"]}
print(uniq_initials)
print(unique_lengths)
```

```
{'B', 'A'}
{3, 5}
```

Dictionary comprehension:

```python
word_lengths = {word: len(word) for word in ['cat', 'dog', 'mouse']}
print(word_lengths)
```

```
{'cat': 3, 'dog': 3, 'mouse': 5}
```

Comprehensions are preferred for simple transformations and filtering.

5.4.2 Membership & Lookups

Membership operations let you check if an item exists in a collection (like a list, set, or dictionary). Use `in` and `not in` for these checks. Lookups retrieve values by key or index, and are fastest in sets and dictionaries.

Examples:

- `3 in [1, 2, 3]` → `True` (checks if 3 is in the list)
- `'a' in {'a': 1, 'b': 2}` → `True` (checks if 'a' is a key in the dictionary)
- `5 not in {1, 2, 3}` → `True` (checks if 5 is not in the set)

Tip:

- Dictionary and set lookups are very fast (constant time).
- List and tuple membership checks are slower (linear time).
- For safe dictionary lookups, use `.get(key)` to avoid errors if the key is missing.

5.4.3 Unpacking

Unpacking lets you assign elements of a collection (like a list, tuple, or string) to multiple variables in a single step. This makes your code more readable and concise, especially when working with structured data.

Basic Syntax:

```python
pt = (3, 4)
x, y = pt  # x=3, y=4

a, b, c = [1, 2, 3]  # a=1, b=2, c=3
```

You can also use unpacking in loops and with function arguments.

Extended Unpacking: Python allows you to use the `*` operator to capture multiple elements:

```python
# Extended unpacking:
first, *rest = [10, 20, 30, 40]  # first=10, rest=[20, 30, 40]

a, *mid, z = [1,2,3,4]

print(rest)
print(mid)
```

```
[20, 30, 40]
[2, 3]
```

Unpacking in Loops:

```python
pairs = [(1, 'a'), (2, 'b'), (3, 'c')]
for num, char in pairs:
    print(num, char)
```

```
1 a
2 b
3 c
```

Unpacking with Dictionaries:

```python
student = {"name": "Alex", "year": 3}
for key, value in student.items():
    print(key, value)
```

```
name Alex
year 3
```

Why use unpacking?

- Makes code cleaner and more readable
- Quick variable assignment
- Useful for working with structured data (e.g., tuples, lists, dicts)
- Avoids manual indexing

Tip: Unpacking works with any iterable, including lists, tuples, strings, and even the results of functions that return multiple values.

5.4.4 Sorting in Python Iterables

Sorting is a common task when working with data. Python provides flexible ways to order lists, tuples, and other iterables.

5.4.4.1 `sorted()` (built-in function)

- Returns a **new sorted list** from any iterable.
- Works with lists, tuples, strings, sets, and more.
- Does **not modify** the original iterable.

```python
nums = [3, 1, 4, 1, 5]
print(sorted(nums))       # [1, 1, 3, 4, 5]
print(nums)               # [3, 1, 4, 1, 5] (unchanged)

word = "python"
print(sorted(word))       # ['h', 'n', 'o', 'p', 't', 'y']
```

```
[1, 1, 3, 4, 5]
[3, 1, 4, 1, 5]
['h', 'n', 'o', 'p', 't', 'y']
```

5.4.4.2 `.sort()` (list method)

- **In-place sort**: modifies the list itself.
- Only available for lists (not other iterables).
- Returns `None`.

```python
nums = [3, 1, 4, 1, 5]
nums.sort()
print(nums)    # [1, 1, 3, 4, 5]
```

```
[1, 1, 3, 4, 5]
```

5.4.4.3 Reverse Sorting

Both `sorted()` and `.sort()` accept a `reverse` argument.

```
nums = [3, 1, 4, 1, 5]
print(sorted(nums, reverse=True))
```

```
[5, 4, 3, 1, 1]
```

5.4.4.4 Sorting with a key

The `key` parameter lets you customize sorting logic.

```
# Sort by string length
words = ["pear", "apple", "banana", "kiwi"]
print(sorted(words, key=len))
# ['kiwi', 'pear', 'apple', 'banana']

# Sort by last character
print(sorted(words, key=lambda w: w[-1]))
# ['banana', 'pear', 'apple', 'kiwi']
```

```
['pear', 'kiwi', 'apple', 'banana']
['banana', 'apple', 'kiwi', 'pear']
```

5.4.4.5 Sorting Complex Data

For lists of tuples or dicts, use `key`.

```
# Sort by the second element of each tuple
pairs = [("a", 3), ("b", 1), ("c", 2)]
print(sorted(pairs, key=lambda t: t[1]))
# [('b', 1), ('c', 2), ('a', 3)]

# Sort list of dicts by a field
students = [
    {"name": "Alice", "grade": 85},
    {"name": "Bob", "grade": 92},
    {"name": "Chen", "grade": 78}
]
print(sorted(students, key=lambda s: s["grade"]))
```

```
[('b', 1), ('c', 2), ('a', 3)]
[{'name': 'Chen', 'grade': 78}, {'name': 'Alice', 'grade': 85}, {'name': 'Bob', 'grade': 92}]
```

Sorting strings alphabetically vs. numerically:

```python
nums = ["10", "2", "1"]
print(sorted(nums))
print(sorted(nums, key=int))
```

```
['1', '10', '2']
['1', '2', '10']
```

5.4.5 Lambda Functions in Python

Sometimes you only need a **tiny one-off function** for a specific task (like sorting by length or filtering items). Writing a full `def` feels heavy.
That's where **lambda functions** help.

Syntax

```python
lambda parameters: expression
```

- Creates an anonymous function (no name required).
- Must contain a single expression (not multiple statements).
- Automatically returns the value of the expression.

```python
square = lambda x: x**2
print(square(5))   # 25
```

```
25
```

Using Lambda Functions

- With `sorted()`

```python
words = ["pear", "apple", "banana", "kiwi"]

# Sort by word length
print(sorted(words, key=lambda w: len(w)))

# Sort by last character
print(sorted(words, key=lambda w: w[-1]))
```

```
['pear', 'kiwi', 'apple', 'banana']
['banana', 'apple', 'kiwi', 'pear']
```

- With `map()` and `filter()`

```python
nums = [1, 2, 3, 4, 5]

# Square each number
print(list(map(lambda x: x**2, nums)))

# Keep only even numbers
print(list(filter(lambda x: x % 2 == 0, nums)))
```

```
[1, 4, 9, 16, 25]
[2, 4]
```

- With `reduce()` (from `functools`)

```python
from functools import reduce

nums = [1, 2, 3, 4, 5]
product = reduce(lambda x, y: x * y, nums)
print(product)   # 120
```

```
120
```

Key Takeaways:

- `lambda` = quick, throwaway function for one-liners
- use `def` if the function
    - has multiple steps
    - is reused often

5.4.6 `enumerate()`

The `enumerate()` function is a built-in Python tool that adds a counter to any iterable, returning pairs of (index, item) as you loop. This is especially useful when you need both the item and its position in a loop.

Syntax:

```python
for index, value in enumerate(iterable, start=0):
    # use index and value
```

- `iterable`: Any sequence (list, tuple, string, etc.)
- `start`: Optional, sets the starting index (default is 0)

Why use `enumerate()`?

- Makes code cleaner and less error-prone
- Avoids manual index tracking with a separate variable
- Works with lists, tuples, strings, and more

Example:

```python
fruits = ['apple', 'banana', 'cherry']
for idx, fruit in enumerate(fruits):
    print(idx, fruit)
```

```
0 apple
1 banana
2 cherry
```

Tip: You can set the starting index with the `start` argument, e.g. `enumerate(my_list, start=1)`.

Use `enumerate()` for readable, efficient loops when you need both index and value.

5.4.7 zip()

The `zip()` function combines two or more iterables (like lists, tuples, or strings) into tuples, pairing elements by their position. This is useful for parallel iteration, creating pairs, or merging data from multiple sources.

Syntax:

```python
zip(iterable1, iterable2, ...)
```

- Each tuple contains one element from each iterable, matched by position.
- Stops at the shortest iterable.

Why use `zip()`? - Parallel iteration over multiple sequences - Pairing related data (e.g., names and scores) - Creating dictionaries from two lists

Example:

```
names = ['Alice', 'Bob', 'Chen']
scores = [85, 92, 78]
for name, score in zip(names, scores):
    print(name, score)
```

```
Alice 85
Bob 92
Chen 78
```

```
# Creating a dictionary from two lists:
gradebook = dict(zip(names, scores))
print(gradebook)
```

```
{'Alice': 85, 'Bob': 92, 'Chen': 78}
```

Tip: - You can use `zip(*zipped)` to unzip a list of tuples back into separate lists. - If the input iterables are different lengths, `zip()` stops at the shortest one.

5.4.8 Common Functions for Iterables

Python provides a variety of built-in functions to operate on iterables, making it easy to manipulate, process, and analyze collections like lists, tuples, strings, sets, and dictionaries. Below is a list of commonly used built-in functions specifically designed for iterables.

Function	Description	Example
`len()`	Returns the number of elements in an iterable.	`len([1, 2, 3])` → `3`
`min()`	Returns the smallest element in an iterable.	`min([3, 1, 4])` → `1`
`max()`	Returns the largest element in an iterable.	`max([3, 1, 4])` → `4`
`sum()`	Returns the sum of elements in an iterable (numeric types only).	`sum([1, 2, 3])` → `6`
`sorted()`	Returns a sorted list from an iterable (does **not** modify the original).	`sorted([3, 1, 2])` → `[1, 2, 3]`
`reversed()`	Returns an iterator that accesses the elements of an iterable in reverse.	`list(reversed([1, 2, 3]))` → `[3, 2, 1]`

Function	Description	Example
`enumerate()`	Returns an iterator of tuples containing indices and elements of the iterable.	`list(enumerate(['a', 'b', 'c']))` → `[(0, 'a'), (1, 'b'), (2, 'c')]`
`all()`	Returns `True` if all elements of the iterable are true (or if empty).	`all([True, 1, 'a'])` → `True`
`any()`	Returns `True` if **any** element of the iterable is true.	`any([False, 0, 'b'])` → `True`
`str.join(iterable)`	Joins elements of an iterable (e.g., list, tuple) into a single string, using the given string as a separator.	`''.join(['a', 'b', 'c'])` → `'abc'`

5.5 Independent Study

5.5.1 Sorting the data

- Sort [10, 2, 33, 25, 7] in descending order.
- Given words = ["data", "python", "AI", "science"], sort alphabetically ignoring case.
- Sort [("Ann", 22), ("Bob", 19), ("Chen", 22)] by age, preserving name order when ages match.

5.5.2 `lambda`

- Use `lambda` with `sorted()` to order the list:

```
nums = [-3, 1, -2, 5, 0]
```

- Use `lambda` with `filter()` to keep only names starting with a vowel:

```
names = ["Alice", "Bob", "Eve", "Uma", "Sam"]
```

- Use `lambda` with `map()` to convert Celsius to Fahrenheit:

```
temps_c = [0, 20, 37, 100]
```

5.5.3

You are given a list of strings representing student names. Some students appear more than once because they signed up for multiple activities.

Write a Python function that:

1. **Removes duplicate names** while *preserving the first occurrence order*.
2. **Returns a list of tuples `(index, name_length)`** for each **unique name**, where:
   - `index` is the original position of the name in the list (first occurrence).
   - `name_length` is the length of the name.

Finally, **sort the output list by `name_length` in descending order**.

```
names = ["Alice", "Bob", "Alice", "Charlie", "Bob", "Dave"]
```

5.5.4 Student data

```
# CELL 1: Student Data Setup
students = [
    (101, "Alice", "STAT303", 85),
    (102, "Bob", "STAT301", 92),
    (103, "Charlie", "STAT303", 78),
    (104, "Diana", "STAT301", 95),
    (105, "Eve", "STAT303", 88),
    (106, "Frank", "STAT304", 82),
    (107, "Grace", "STAT301", 90),
    (108, "Hank", "STAT304", 87),
    (109, "Ivy", "STAT303", 91),
    (101, "Alice", "STAT301", 88),
    (101, "Alice", "STAT304", 82),
    (101, "Alice", "STAT302", 86),
    (102, "Bob", "STAT303", 79),
    (102, "Bob", "STAT304", 85),
    (102, "Bob", "STAT302", 83),
    (103, "Charlie", "STAT301", 91),
    (103, "Charlie", "STAT304", 84),
    (103, "Charlie", "STAT302", 80),
    (104, "Diana", "STAT303", 89),
```

```
    (104, "Diana", "STAT304", 93),
    (104, "Diana", "STAT302", 96),
    (105, "Eve", "STAT301", 87),
    (105, "Eve", "STAT304", 90),
    (105, "Eve", "STAT302", 85),
    (106, "Frank", "STAT303", 76),
    (106, "Frank", "STAT301", 81),
    (106, "Frank", "STAT302", 79),
    (107, "Grace", "STAT303", 84),
    (107, "Grace", "STAT304", 88),
    (107, "Grace", "STAT302", 91),
    (108, "Hank", "STAT303", 80),
    (108, "Hank", "STAT301", 83),
    (108, "Hank", "STAT302", 82),
    (109, "Ivy", "STAT301", 94),
    (109, "Ivy", "STAT304", 89),
    (109, "Ivy", "STAT302", 92),
    (110, "Jack", "STAT303", 83),
    (110, "Jack", "STAT301", 86),
    (110, "Jack", "STAT304", 79),
    (110, "Jack", "STAT302", 81),
    (111, "Karen", "STAT303", 96),
    (111, "Karen", "STAT301", 92),
    (111, "Karen", "STAT304", 94),
    (111, "Karen", "STAT302", 98),
    (112, "Leo", "STAT303", 72),
    (112, "Leo", "STAT301", 75),
    (112, "Leo", "STAT304", 78),
    (112, "Leo", "STAT302", 74)
]
```

- What is the data type for this variable
- How many unique students in the dataset
- How many unique courses in the dataset
- what is the grade range?
- What is the average grade?

Part III

Core library foundations

6 Reading Data

<IPython.core.display.Image object>

6.1 Types of Data: Structured vs. Unstructured

Understanding the format of your data is crucial before you can analyze it effectively. Data exists in two primary formats:

6.1.1 Structured Data

Structured data is organized in a tabular format with clearly defined relationships between data elements. Key characteristics include:

- **Rows** represent individual **observations** (records or data points)
- **Columns** represent **variables** (attributes or features)
- Data follows a consistent schema or format
- Easy to search, query, and analyze using standard tools

Example: The dataset below contains movie information where each row represents one movie (observation) and each column contains specific attributes like title, rating, or budget (variables). Since all data in a row relates to the same movie, this is also called **relational data**.

	Title	US Gross	Production Budget	Release Date	Major Genre
0	The Shawshank Redemption	28241469	25000000	Sep 23 1994	Drama
1	Inception	285630280	160000000	Jul 16 2010	Horror/Thriller
2	One Flew Over the Cuckoo's Nest	108981275	4400000	Nov 19 1975	Comedy
3	The Dark Knight	533345358	185000000	Jul 18 2008	Action/Adventure
4	Schindler's List	96067179	25000000	Dec 15 1993	Drama

Common formats: CSV files, Excel spreadsheets, SQL databases

6.1.2 Unstructured Data

Unstructured data lacks a predefined organizational structure, making it more challenging to analyze with traditional methods.

Examples include: Text documents, images, audio files, videos, social media posts, sensor data

While harder to analyze directly, unstructured data can often be converted into structured formats for analysis (e.g., extracting features from images or sentiment scores from text).

In this sequence, we will focus on analyzing structured data.

6.2 Introduction to Pandas

Pandas is Python's premier library for data manipulation and analysis. Think of it as Excel for Python - it provides powerful tools for working with tabular data (like spreadsheets) and supports reading from various file formats:

- **CSV** (Comma-Separated Values)
- **Excel** spreadsheets (.xlsx, .xls)
- **JSON** (JavaScript Object Notation)
- **HTML** tables
- **SQL** databases
- And many more!

6.3 Reading CSV Files with Pandas

6.3.1 Understanding CSV Files

CSV (Comma-Separated Values) is the most popular format for storing structured data. Here's what makes CSV files special:

- **Simple structure**: Each row represents one record/observation
- **Comma delimiter**: Values within a row are separated by commas
- **Plain text**: Human-readable and lightweight
- **Universal compatibility**: Supported by virtually all data tools

> **Pro Tip**: When you open a CSV file in Excel, the commas are automatically converted to column separators, so you won't see them visually!

6.3.2 CSV File Structure Example

Here's a sample CSV file containing daily COVID-19 data for Italy:

```
date,new_cases,new_deaths,new_tests
2020-04-21,2256.0,454.0,28095.0
2020-04-22,2729.0,534.0,44248.0
2020-04-23,3370.0,437.0,37083.0
2020-04-24,2646.0,464.0,95273.0
2020-04-25,3021.0,420.0,38676.0
2020-04-26,2357.0,415.0,24113.0
2020-04-27,2324.0,260.0,26678.0
2020-04-28,1739.0,333.0,37554.0
...
```

Structure breakdown: - **Header row**: Column names (`date`, `new_cases`, `new_deaths`, `new_tests`) - **Data rows**: Each row contains values for one day - **Delimiter**: Commas separate each value within a row

6.3.3 Getting Started: Importing Pandas

Before we can work with data, we need to import the pandas library. By convention, pandas is imported with the alias `pd` - this is a universal practice that makes your code more readable and concise.

Why use `pd`?

- **Shorter syntax**: `pd.read_csv()` instead of `pandas.read_csv()`
- **Industry standard**: Everyone in the data science community uses this alias
- **Cleaner code**: Makes your scripts more readable

```python
import pandas as pd
import os
```

6.3.4 Loading Data with `pd.read_csv()`

What is `pd.read_csv()`?

The `pd.read_csv()` function is your gateway to loading CSV data into Python. It converts CSV files into a pandas **DataFrame** - think of it as a powerful, interactive spreadsheet that you can manipulate with code.

Key Features:

- **Automatic data type detection**: Pandas intelligently guesses column types (numbers, text, dates)
- **Header recognition**: Automatically uses the first row as column names
- **Missing data handling**: Gracefully handles empty cells
- **Flexible parsing**: Can handle various CSV formats and delimiters

```
movie_ratings = pd.read_csv('./datasets/movie_ratings.csv')
movie_ratings.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
0	Opal Dreams	14443	14443	9000000	Nov 22 2006	PG/PG-13
1	Major Dundee	14873	14873	3800000	Apr 07 1965	PG/PG-13
2	The Informers	315000	315000	18000000	Apr 24 2009	R
3	Buffalo Soldiers	353743	353743	15000000	Jul 25 2003	R
4	The Last Sin Eater	388390	388390	2200000	Feb 09 2007	PG/PG-13

The built-in python function `type` can be used to check the dataype of an object:

```
type(movie_ratings)
```

```
pandas.core.frame.DataFrame
```

The Result: A DataFrame

6.3.5 Understanding Pandas Data Structures

Pandas is built around **two fundamental data structures** that work together to handle different types of data:

6.3.5.1 Series: One-Dimensional Data

A **Series** is like a single column from a spreadsheet or a list with labels:

- **One-dimensional**: Contains data in a single column
- **Homogeneous**: All values are the same data type (numbers, text, etc.)
- **Indexed**: Each value has a label (index) for easy access
- **Example**: A list of movie titles, ratings, or prices

```python
movie_titles = movie_ratings['Title']
type(movie_titles)
```

```
pandas.core.series.Series
```

6.3.5.2 DataFrame: Two-Dimensional Data

A **DataFrame** is the star of pandas - it's like a complete spreadsheet or database table:

Key Characteristics:

- **Two-dimensional**: Organized in rows and columns
- **Heterogeneous**: Different columns can have different data types
- **Labeled axes**: Both rows (index) and columns have names/labels
- **Flexible**: Easy to filter, sort, group, and analyze

Structure:

- **Rows (Index)**: Each row represents one **observation** (e.g., one movie, customer, transaction)
- **Columns**: Each column represents one **variable/feature** (e.g., title, price, rating, date)
- **Cells**: Individual data points where rows and columns intersect

Default Behavior:

- **Row indices**: Automatically numbered 0, 1, 2, 3… (but customizable)
- **Column names**: Taken from the first row of your CSV file (the header)
- **Data types**: Automatically detected (numbers, text, dates, etc.)

6.3.6 Attributes and Methods of a Pandas DataFrame

Understanding DataFrame attributes and methods is essential for data exploration and analysis. These tools help you **inspect, understand, and summarize** your dataset efficiently.

What are Attributes vs Methods?

- **Attributes**: Properties of the DataFrame (accessed without parentheses)
- **Methods**: Functions that perform operations (accessed with parentheses)

> **Pro Tip**: All attributes and methods of a DataFrame can be viewed using Python's built-in `dir()` function: `dir(your_dataframe)`

6.3.6.1 Essential DataFrame Attributes

6.3.6.1.1 `dtypes`

- **Purpose**: Shows the data type of each column
- **Returns**: A Series containing the data type for each column

Why it matters: Understanding data types helps you:

- Identify if numbers are stored as text (and need conversion)
- Know which operations are valid for each column
- Optimize memory usage and performance

```
movie_ratings.dtypes
```

```
Title                  object
US Gross                int64
Worldwide Gross         int64
Production Budget       int64
Release Date           object
MPAA Rating            object
Source                 object
Major Genre            object
Creative Type          object
IMDB Rating           float64
IMDB Votes              int64
dtype: object
```

The table below describes the datatypes of columns in a Pandas DataFrame.

Pandas Type	Native Python Type	Description
object	string	The most general dtype. This datatype is assigned to a column if the column has mixed types (numbers and strings)
int64	int	This datatype is for integers from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 or for integers having a maximum size of 64 bits

Pandas Type	Native Python Type	Description
float64	float	This datatype is for real numbers. If a column contains integers and NaNs, Pandas will default to float64. This is because the missing values may be a real number

6.3.6.1.2 `shape`

Purpose: Returns the dimensions of the DataFrame as a tuple (rows, columns)

Why useful: Quickly understand the size of your dataset

```
# Check the dimensions of our movie dataset
print(f"Dataset shape: {movie_ratings.shape}")
print(f"Number of rows (movies): {movie_ratings.shape[0]}")
print(f"Number of columns (features): {movie_ratings.shape[1]}")
```

```
Dataset shape: (2228, 11)
Number of rows (movies): 2228
Number of columns (features): 11
```

6.3.6.1.3 `columns`

Purpose: Returns the column names as an Index object

Why useful: See what variables/features are available in your dataset

```
# View all column names
print("Available columns:")
for i, col in enumerate(movie_ratings.columns, 1):
    print(f"{i}. {col}")
```

```
Available columns:
1. Title
2. US Gross
3. Worldwide Gross
4. Production Budget
5. Release Date
6. MPAA Rating
```

```
7. Source
8. Major Genre
9. Creative Type
10. IMDB Rating
11. IMDB Votes
```

6.3.6.1.4 `index`

Purpose: Returns the row labels (index) of the DataFrame

Why useful: Understand how rows are labeled and indexed

```python
# Examine the index
print(f"Index type: {type(movie_ratings.index)}")
print(f"Index range: {movie_ratings.index[0]} to {movie_ratings.index[-1]}")
print(f"First 10 index values: {movie_ratings.index[:10].tolist()}")
```

```
Index type: <class 'pandas.core.indexes.range.RangeIndex'>
Index range: 0 to 2227
First 10 index values: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

6.3.6.2 Essential DataFrame Methods

6.3.6.2.1 `info()`

Purpose: Provides a comprehensive summary of the DataFrame

Returns: Information about data types, non-null counts, and memory usage

```python
# Get comprehensive dataset information
movie_ratings.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2228 entries, 0 to 2227
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Title              2228 non-null   object
 1   US Gross           2228 non-null   int64
 2   Worldwide Gross    2228 non-null   int64
 3   Production Budget  2228 non-null   int64
 4   Release Date       2228 non-null   object
```

```
 5   MPAA Rating        2228 non-null   object
 6   Source             2228 non-null   object
 7   Major Genre        2228 non-null   object
 8   Creative Type      2228 non-null   object
 9   IMDB Rating        2228 non-null   float64
 10  IMDB Votes         2228 non-null   int64
dtypes: float64(1), int64(4), object(6)
memory usage: 191.6+ KB
```

What `info()` tells you:

- **Data types** for each column
- **Non-null count** (helps identify missing data)
- **Memory usage** (useful for large datasets)
- **Total entries** and columns

6.3.6.2.2 `describe()`

Purpose: Generate descriptive statistics for numerical columns

Returns: Count, mean, std, min, 25%, 50%, 75%, max for each numeric column

```
# Get statistical summary of numerical columns
movie_ratings.describe()
```

	US Gross	Worldwide Gross	Production Budget	IMDB Rating	IMDB Votes
count	2.228000e+03	2.228000e+03	2.228000e+03	2228.000000	2228.000000
mean	5.076370e+07	1.019370e+08	3.816055e+07	6.239004	33585.154847
std	6.643081e+07	1.648589e+08	3.782604e+07	1.243285	47325.651561
min	0.000000e+00	8.840000e+02	2.180000e+02	1.400000	18.000000
25%	9.646188e+06	1.320737e+07	1.200000e+07	5.500000	6659.250000
50%	2.838649e+07	4.266892e+07	2.600000e+07	6.400000	18169.000000
75%	6.453140e+07	1.200000e+08	5.300000e+07	7.100000	40092.750000
max	7.601676e+08	2.767891e+09	3.000000e+08	9.200000	519541.000000

Key Statistics Explained:

- **count**: Number of non-missing values
- **mean**: Average value
- **std**: Standard deviation (measure of spread)
- **min/max**: Smallest and largest values

- **25%, 50%, 75%**: Quartiles (percentiles)

  Tip: Use `describe(include='all')` to include non-numeric columns in the summary!

6.3.6.2.3 `head()` and `tail()`

Purpose: View the first or last few rows of your dataset

Default: Shows 5 rows, but you can specify any number

```
# View first 3 rows
print("First 3 movies:")
movie_ratings.head(3)
```

```
First 3 movies:
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating	Sour
0	Opal Dreams	14443	14443	9000000	Nov 22 2006	PG/PG-13	Adap
1	Major Dundee	14873	14873	3800000	Apr 07 1965	PG/PG-13	Adap
2	The Informers	315000	315000	18000000	Apr 24 2009	R	Adap

```
# View last 3 rows
print("Last 3 movies:")
movie_ratings.tail(3)
```

```
Last 3 movies:
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date
2225	Robin Hood	105269730	310885538	210000000	May 14 2010
2226	Robin Hood: Prince of Thieves	165493908	390500000	50000000	Jun 14 1991
2227	Spiceworld	29342592	56042592	25000000	Jan 23 1998

6.3.6.2.4 `sample()`

Purpose: Get a random sample of rows from your DataFrame **Why useful**: Explore data patterns without bias toward top/bottom rows

```
# Get 5 random movies from the dataset
print("Random sample of 5 movies:")
movie_ratings.sample(5)
```

6.3.6.3 Quick Reference: Most Used Attributes & Methods

Attribute/Method	Purpose	Example Usage
`.shape`	Get dimensions (rows, cols)	`df.shape`
`.dtypes`	Check data types	`df.dtypes`
`.columns`	View column names	`df.columns`
`.info()`	Comprehensive overview	`df.info()`
`.describe()`	Statistical summary	`df.describe()`
`.head(n)`	First n rows	`df.head(10)`
`.tail(n)`	Last n rows	`df.tail(3)`
`.sample(n)`	Random n rows	`df.sample(5)`

6.4 Writing Data to CSV Files

After cleaning, analyzing, or transforming your data, you'll often need to **save your results** for future use or sharing. Pandas makes this easy with the `to_csv()` method.

6.4.1 Basic CSV Export

The simplest way to export a DataFrame is using the `to_csv()` method with just a filename:

```
# Basic export - saves DataFrame to CSV file
movie_ratings.to_csv('./datasets/movie_ratings_exported.csv')
print("  Data exported successfully!")
```

```
  Data exported successfully!
```

```
# Verify the file was created
if './datasets/movie_ratings_exported.csv' in [f'./datasets/{f}' for f in os.listdir('./datas

    print("  File successfully exported!")
else:
    print("  Export failed")
```

```
File successfully exported!
```

6.4.2 Important CSV Export Parameters

By default, pandas includes row indices in the exported CSV. Often you don't want this:

The `to_csv()` method has many useful parameters to customize your output:

6.4.2.1 Index Control

```
# Export WITHOUT row indices (most common)
movie_ratings.to_csv('./datasets/movies_no_index.csv', index=False)

# Export WITH row indices (default behavior)
movie_ratings.to_csv('./datasets/movies_with_index.csv', index=True)
```

6.4.2.2 Column Selection

Export only specific columns:

```
# Export only selected columns
columns_to_export = ['Title', 'IMDB Rating', 'Worldwide Gross']
movie_ratings[columns_to_export].to_csv('./datasets/movies_selected_columns.csv', index=Fals
```

6.4.2.3 Custom Separators

Change the delimiter (useful for different regional standards):

```
# Use semicolon instead of comma (common in European locales)
movie_ratings.to_csv('./datasets/movies_semicolon.csv', sep=';', index=False)

# Use tab separator (creates TSV file)
movie_ratings.to_csv('./datasets/movies_tab.txt', sep='\t', index=False)
```

6.4.2.4 Handling Missing Values

Control how missing/null values are represented:

```
# Replace NaN values with custom text
movie_ratings.to_csv('./datasets/movies_custom_na.csv',
                     index=False,
                     na_rep='Missing')
```

6.4.2.5 Encoding Specification

Ensure proper character encoding (important for international data):

```
# Specify UTF-8 encoding (recommended for international characters)
movie_ratings.to_csv('./datasets/movies_utf8.csv',
                     index=False,
                     encoding='utf-8')
```

6.4.3 Best Practices for CSV Export

Practice	Why Important	Example
Always use `index=False`	Prevents unnecessary row numbers	`df.to_csv('file.csv', index=False)`
Specify encoding	Ensures international characters display correctly	`encoding='utf-8'`
Use descriptive filenames	Makes files easy to identify later	`'cleaned_movie_data_2024.csv'`
Check file paths	Avoid overwriting important files	Use relative or absolute paths carefully
Handle missing data	Define how NaN values should appear	`na_rep='N/A'` or `na_rep=''`

6.4.4 Complete Export Example

Here's a comprehensive example combining multiple parameters:

```python
# Professional CSV export with all best practices
export_filename = './datasets/movie_analysis_final.csv'

# Select relevant columns for export
columns_for_analysis = [
    'Title', 'IMDB Rating', 'IMDB Votes',
    'US Gross', 'Worldwide Gross', 'Production Budget'
]

# Export with professional settings
movie_ratings[columns_for_analysis].to_csv(
    export_filename,
    index=False,           # No row numbers
    encoding='utf-8',      # International characters
    na_rep='N/A',         # Clear missing value indicator
    float_format='%.2f'    # 2 decimal places for numbers
)

print(f"  Analysis data exported to: {export_filename}")
print(f"  Exported {len(columns_for_analysis)} columns for {len(movie_ratings)} movies")
```

```
 Analysis data exported to: ./datasets/movie_analysis_final.csv
 Exported 6 columns for 2228 movies
```

6.4.5 Verifying Your Export

Always verify your exported data looks correct:

```python
# Read back the exported file to verify
verification_df = pd.read_csv(export_filename)

print(f"\n Exported dimensions: {verification_df.shape}")

print(" Exported file preview:")
verification_df.head(3)
```

```
Exported dimensions: (2228, 6)
Exported file preview:
```

	Title	IMDB Rating	IMDB Votes	US Gross	Worldwide Gross	Production Budget
0	Opal Dreams	6.5	468	14443	14443	9000000
1	Major Dundee	6.7	2588	14873	14873	3800000
2	The Informers	5.2	7595	315000	315000	18000000

6.5 Reading Other Data Formats

While **CSV is the most popular format** for structured data, real-world data comes in many different formats. As a data analyst, you'll frequently encounter various file types and data sources that require different reading approaches.

6.5.1 Common Data Formats Overview

Format	Extension	Use Case	Pandas Function
CSV	`.csv`	Most common tabular data	`pd.read_csv()`
Text	`.txt`	Tab-separated or custom delimiters	`pd.read_csv()` with `sep` parameter
JSON	`.json`	API data, nested structures	`pd.read_json()`
Excel	`.xlsx`, `.xls`	Spreadsheet data	`pd.read_excel()`
URLs	Various	Live data from web APIs	Various functions

6.5.2 What You'll Learn

In this section, we'll explore how to:

- **Read text files** with different delimiters (tabs, semicolons, etc.)
- **Handle JSON data** from APIs and web services
- **Access live data** directly from URLs and APIs
- **Extract tables** from HTML web pages

> **Key Insight**: The same `pd.read_csv()` function can read many text-based formats by adjusting parameters!

6.5.3 Reading Text Files (.txt)

Text files offer maximum flexibility for storing tabular data with custom delimiters.

Key Differences from CSV:

- **CSV files**: Always use commas (`,`) as delimitersLet's read a file where values are separated by tab characters:

- **Text files**: Can use **any character** as a delimiter

6.5.3.1 Reading Tab-Separated Files

Common Delimiters in Text Files:

- **Space** () → Simple space-separated files
- **Tab character** (`\t`) → Creates TSV (Tab-Separated Values) files- **Pipe symbol** (`|`) → Used when data contains commas
- **Semicolon** (`;`) → Common in European data files

```
# Read tab-separated file using \t as separator
# Note: We still use pd.read_csv() but specify the separator
movie_ratings_txt = pd.read_csv('./datasets/movie_ratings.txt', sep='\t')

movie_ratings_txt.head()
```

	Unnamed: 0	Title	US Gross	Worldwide Gross	Production Budget	Release Date	M
0	0	Opal Dreams	14443	14443	9000000	Nov 22 2006	P
1	1	Major Dundee	14873	14873	3800000	Apr 07 1965	P
2	2	The Informers	315000	315000	18000000	Apr 24 2009	R
3	3	Buffalo Soldiers	353743	353743	15000000	Jul 25 2003	R
4	4	The Last Sin Eater	388390	388390	2200000	Feb 09 2007	P

Key Points:

- Use the **same** **`pd.read_csv()`** function for text files
- Specify the delimiter with the **`sep`** parameter
- `\t` represents the tab character

6.5.3.2 Common Separator Examples

- The function automatically handles headers and data types

```
# Different separator examples (these are demonstrations)

# Semicolon-separated (common in Europe)
# df_semicolon = pd.read_csv('file.txt', sep=';')

# Pipe-separated (when data contains commas)
# df_pipe = pd.read_csv('file.txt', sep='|')

# Space-separated
# df_space = pd.read_csv('file.txt', sep=' ')

# Multiple spaces or mixed whitespace
# df_whitespace = pd.read_csv('file.txt', sep='\s+')

print("  Tip: Use sep='\\s+' for files with irregular spacing!")
```

```
  Tip: Use sep='\s+' for files with irregular spacing!
```

6.5.3.3 Automatic Delimiter Detection

Pandas can automatically detect delimiters using the Python engine:

```
# Let pandas automatically detect the delimiter
# This uses the 'python' engine with a 'sniffer' tool
try:
    auto_detected = pd.read_csv('./datasets/movie_ratings.txt',
                               sep=None,          # Auto-detect
                               engine='python')   # Use Python engine
    print(" Delimiter detected automatically!")
    print(f" Shape: {auto_detected.shape}")
except FileNotFoundError:
    print(" File not found - this is just a demonstration")
```

```
 Delimiter detected automatically!
 Shape: (2228, 12)
```

When to use automatic detection:

- When you're unsure about the delimiter
- For quick data exploration
- Avoid in production code (be explicit for reliability)

  Pro Tip: Always check the pandas documentation for parameter details!

6.5.4 Reading JSON Data

JSON (JavaScript Object Notation) is the **standard format for web APIs** and modern data exchange.

Why JSON is Everywhere:

- **Web APIs**: Most modern APIs return JSON data
- **Mobile Apps**: Standard format for app-server communication
- **Cloud Services**: AWS, Google Cloud, Azure all use JSON
- **IoT Devices**: Smart devices send data in JSON format

Key Advantages over CSV:

Feature	CSV	JSON
Structure	Flat tables only	**Nested / hierarchical**
Data Types	Text and numbers	**Objects, arrays, booleans**
Metadata	Must be stored separately or encoded in headers	**Supports metadata inline (keys + values in the same document)**
Complex / Nested Records	Must be flattened into multiple columns	**Native support for nested objects and arrays**

6.5.4.1 Reading JSON with Pandas

- **Data type inference** (numbers, dates, etc.)

`pd.read_json()` automatically converts JSON data into pandas DataFrames:

- **DataFrame conversion** when possible
- **Smart detection** of JSON structure

Automatic Processing:

- **Local files**: `pd.read_json('data.json')`
- **URLs**: `pd.read_json('https://api.example.com/data')`

- **Various orientations**: Records, index, values, etc.
- **JSON strings**: Direct JSON text

6.5.4.2 Real Example: TED Talks Dataset

Let's load a real JSON dataset containing TED Talks information:

```
# Load TED Talks data from GitHub (JSON format)

url = 'https://raw.githubusercontent.com/cwkenwaysun/TEDmap/master/data/TED_Talks.json'
print(" Loading TED Talks data from JSON...")
tedtalks_data = pd.read_json(url)
print(f" Loaded {tedtalks_data.shape[0]} TED Talks with {tedtalks_data.shape[1]} features!")
```

```
 Loading TED Talks data from JSON...
 Loaded 2474 TED Talks with 16 features!
```

```
# Explore the TED Talks dataset

print(f" Dataset Info:")
tedtalks_data.head(3)

print(f"Shape: {tedtalks_data.shape}")
print("\n First few TED Talks:")
print(f"Columns: {list(tedtalks_data.columns)}")
```

```
 Dataset Info:
Shape: (2474, 16)

 First few TED Talks:
Columns: ['id', 'speaker', 'headline', 'URL', 'description', 'transcript_URL', 'month_filmed
```

6.5.4.3 Reading Data from URLs and APIs

Modern data science often involves accessing live data from web APIs (Application Programming Interfaces). This allows you to get real-time information directly into your analysis pipeline.

Using the `requests` Library

The `requests` library is the standard tool for making HTTP requests in Python. You'll need to install it if not already available:

```
pip install requests
```

Basic API Request Pattern: 1. **Send request** to the API URL 2. **Check response** status (success/error) 3. **Parse JSON data** into usable format 4. **Convert to DataFrame** for analysis

Let's demonstrate with the **CoinGecko API**, which provides cryptocurrency market data:

```python
import requests

# Define the API endpoint
api_url = 'https://api.coingecko.com/api/v3/coins/markets'

# Add parameters to get USD prices for top 10 cryptocurrencies
params = {
    'vs_currency': 'usd',  # Price in US Dollars
    'order': 'market_cap_desc',  # Sort by market cap
    'per_page': 10,  # Get top 10 coins
    'page': 1,  # First page
    'sparkline': False  # Don't include price history
}

print(" Requesting cryptocurrency data from CoinGecko API...")

# Send GET request with parameters
response = requests.get(api_url, params=params)

# Check HTTP status codes
print(f" Response Status Code: {response.status_code}")

if response.status_code == 200:
    print("  Request successful!")

    # Parse JSON response
    crypto_data = response.json()

    # Display basic info about the response
    print(f"  Received data for {len(crypto_data)} cryptocurrencies")

    # Show raw JSON structure (first item only)
```

```python
    print(f"\n Sample JSON structure:")
    print(f"Keys available: {list(crypto_data[0].keys())}")

else:
    # Handle different error codes
    error_messages = {
        400: "Bad Request - Check your parameters",
        401: "Unauthorized - API key might be required",
        403: "Forbidden - Access denied",
        404: "Not Found - Invalid endpoint",
        429: "Rate Limited - Too many requests",
        500: "Server Error - Try again later"
    }

    error_msg = error_messages.get(response.status_code, "Unknown error")
    print(f" Request failed: {error_msg}")
    crypto_data = None
```

```
 Requesting cryptocurrency data from CoinGecko API...
 Response Status Code: 200
  Request successful!
  Received data for 10 cryptocurrencies

 Sample JSON structure:
Keys available: ['id', 'symbol', 'name', 'image', 'current_price', 'market_cap', 'market_cap
```

```python
# Convert JSON data to pandas DataFrame (if successful)
if crypto_data:
    # Method 1: Convert JSON directly to DataFrame
    crypto_df = pd.DataFrame(crypto_data)

    print(f"\n  Cryptocurrency Market Data:")
    print(f"Dataset shape: {crypto_df.shape}")

    # Select and display key columns
    key_columns = ['name', 'symbol', 'current_price', 'market_cap', 'price_change_percentage_
    display_df = crypto_df[key_columns].copy()

    # Format for better readability
    display_df['current_price'] = display_df['current_price'].apply(lambda x: f"${x:,.2f}")
    display_df['market_cap'] = display_df['market_cap'].apply(lambda x: f"${x:,.0f}")
```

```
    display_df['price_change_percentage_24h'] = display_df['price_change_percentage_24h'].app

    # Rename columns for display
    display_df.columns = ['Name', 'Symbol', 'Current Price', 'Market Cap', '24h Change %']

    print(f"\n  Top 10 Cryptocurrencies by Market Cap:")
    display(display_df)

    # Method 2: Manual processing for learning
    print(f"\n  Price Summary:")
    for coin in crypto_data[:5]:  # Show first 5
        name = coin['name']
        symbol = coin['symbol'].upper()
        price = coin['current_price']
        change_24h = coin['price_change_percentage_24h']

        # Format change with color indicator
        change_indicator = " " if change_24h > 0 else " "
        print(f"{change_indicator} {name} ({symbol}): ${price:,.2f} | 24h: {change_24h:.2f}%

else:
    print("  No data to process due to API request failure")
```

```
  Cryptocurrency Market Data:
Dataset shape: (10, 26)

  Top 10 Cryptocurrencies by Market Cap:
```

	Name	Symbol	Current Price	Market Cap	24h Change %
0	Bitcoin	btc	$114,167.00	$2,275,903,872,681	0.13%
1	Ethereum	eth	$4,133.97	$499,142,371,776	-1.06%
2	Tether	usdt	$1.00	$174,703,594,573	-0.04%
3	XRP	xrp	$2.86	$170,592,210,519	-1.02%
4	BNB	bnb	$1,006.92	$140,084,147,935	-1.12%
5	Solana	sol	$208.39	$113,247,264,117	-1.54%
6	USDC	usdc	$1.00	$73,396,062,741	0.00%
7	Lido Staked Ether	steth	$4,133.28	$35,272,520,571	-0.90%
8	Dogecoin	doge	$0.23	$34,944,052,528	-0.81%
9	TRON	trx	$0.33	$31,555,873,939	-0.92%

```
Price Summary:
Bitcoin (BTC): $114,167.00 | 24h: 0.13%
Ethereum (ETH): $4,133.97 | 24h: -1.06%
Tether (USDT): $1.00 | 24h: -0.04%
XRP (XRP): $2.86 | 24h: -1.02%
BNB (BNB): $1,006.92 | 24h: -1.12%
```

6.5.4.4 Read data from wikipedia

```
api_url = 'https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_per_capita'
# Use pandas to read all tables from the webpage
wiki_response = requests.get(api_url)
# Check HTTP status codes
print(f" Response Status Code: {wiki_response.status_code}")
if wiki_response.status_code == 200:
    print("  Request successful!")
else:
    # output error message based on the status code
    error_messages = {
        400: "Bad Request - Check your parameters",
        401: "Unauthorized - API key might be required",
        403: "Forbidden - Access denied",
        404: "Not Found - Invalid endpoint",
        429: "Rate Limited - Too many requests",
        500: "Server Error - Try again later"
    }
    error_msg = error_messages.get(wiki_response.status_code, "Unknown error")
    print(f"  Request failed: {error_msg}")
```

```
 Response Status Code: 403
  Request failed: Forbidden - Access denied
```

6.5.4.4.1 Understanding HTTP 403 Errors with Wikipedia

Why Wikipedia Returns 403 "Forbidden" Errors:

Wikipedia and many other websites block requests that appear to be automated bots or scrapers. When you make a request without proper headers, the server sees it as suspicious and blocks it with a 403 error.

Common Causes of 403 Errors:

- **Missing User-Agent**: No browser identification
- **Suspicious headers**: Looks like automated scraping

- **Rate limiting**: Too many requests too quickly
- **IP blocking**: Your IP has been flagged

6.5.4.4.2 Solution: Add Proper Headers

The key is to make your request look like it's coming from a real web browser by adding appropriate headers:

```
headers = {
    'User-Agent': 'My Data Analysis Project (your_email@example.com)'
}
url = 'https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)'

# Pass the headers to read_html
tables = pd.read_html(url, header=0, attrs=headers)
```

Let's define a helper function for this purpose

```
def read_html_tables_from_url(url, match=None, user_agent=None, timeout=10, **kwargs):
    """
    Read HTML tables from a website URL with error handling and customizable options.

    Parameters:
    -----------
    url : str
        The website URL to scrape tables from
    match : str, optional
        String or regex pattern to match tables (passed to pd.read_html)
    user_agent : str, optional
        Custom User-Agent header. If None, uses a default Chrome User-Agent
    timeout : int, default 10
        Request timeout in seconds
    **kwargs : dict
        Additional arguments passed to pd.read_html()

    Returns:
    --------
```

```
    list of DataFrames or None
        List of pandas DataFrames (one per table found), or None if error occurs

    Examples:
    ---------
    # Basic usage
    tables = read_html_tables_from_url("https://en.wikipedia.org/wiki/List_of_countries_by_GI

    # Filter tables containing specific text
    tables = read_html_tables_from_url(url, match="Country")

    # With custom user agent
    tables = read_html_tables_from_url(url, user_agent="MyBot/1.0")
    """
    import pandas as pd
    import requests

    # Default User-Agent if none provided
    if user_agent is None:
        user_agent = 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, li

    headers = {'User-Agent': user_agent}

    try:
        print(f" Fetching data from: {url}")

        # Make the HTTP request
        response = requests.get(url, headers=headers, timeout=timeout)
        response.raise_for_status()  # Raise an exception for bad status codes (4xx or 5xx)

        print(f" Successfully retrieved webpage (Status: {response.status_code})")

        # Parse HTML tables
        read_html_kwargs = {'match': match} if match else {}
        read_html_kwargs.update(kwargs)  # Add any additional arguments

        tables = pd.read_html(response.content, **read_html_kwargs)

        print(f" Found {len(tables)} table(s) on the page")

        # Display table information
        if tables:
```

```python
            print(f"\n  Table Overview:")
            for i, table in enumerate(tables):
                print(f"  Table {i}: {table.shape} (rows × columns)")
                if i >= 4:  # Limit output for readability
                    print(f"  ... and {len(tables) - 5} more tables")
                    break

        return tables

    except requests.exceptions.HTTPError as err:
        print(f"  HTTP Error: {err}")
        return None
    except requests.exceptions.ConnectionError as err:
        print(f"  Connection Error: {err}")
        return None
    except requests.exceptions.Timeout as err:
        print(f"  Timeout Error: Request took longer than {timeout} seconds")
        return None
    except requests.exceptions.RequestException as err:
        print(f"  Request Error: {err}")
        return None
    except ValueError as err:
        print(f"  No tables found on the page or parsing error: {err}")
        return None
    except Exception as e:
        print(f"  An unexpected error occurred: {e}")
        return None
```

Now let's demonstrate how to use our new function with different options:

```python
# Example 1: Basic usage - Get all tables from GDP Wikipedia page
url = 'https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_per_capita'
tables = read_html_tables_from_url(url)

if tables:
    print(f"\n Successfully retrieved {len(tables)} tables!")

    # Display info about the largest table
    largest_table = max(tables, key=len)
    print(f"\n  Largest table has {largest_table.shape[0]} rows and {largest_table.shape[1]}
    print(f"Columns: {list(largest_table.columns)}")
else:
```

```python
    print(" Failed to retrieve tables")
```

```
 Fetching data from: https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_per_ca
 Successfully retrieved webpage (Status: 200)
 Found 6 table(s) on the page

 Table Overview:
  Table 0: (1, 2) (rows × columns)
  Table 1: (224, 4) (rows × columns)
  Table 2: (9, 2) (rows × columns)
  Table 3: (13, 2) (rows × columns)
  Table 4: (2, 2) (rows × columns)
  ... and 1 more tables

 Successfully retrieved 6 tables!

 Largest table has 224 rows and 4 columns
Columns: ['Country/Territory', 'IMF (2025)[a][5]', 'World Bank (2022-24)[6]', 'United Nations
 Successfully retrieved webpage (Status: 200)
 Found 6 table(s) on the page

 Table Overview:
  Table 0: (1, 2) (rows × columns)
  Table 1: (224, 4) (rows × columns)
  Table 2: (9, 2) (rows × columns)
  Table 3: (13, 2) (rows × columns)
  Table 4: (2, 2) (rows × columns)
  ... and 1 more tables

 Successfully retrieved 6 tables!

 Largest table has 224 rows and 4 columns
Columns: ['Country/Territory', 'IMF (2025)[a][5]', 'World Bank (2022-24)[6]', 'United Nations
```

```python
# Example 2: Filtered search - Only get tables containing 'Country'
filtered_tables = read_html_tables_from_url(url, match='Country')

if filtered_tables:
    print(f"\n Found {len(filtered_tables)} table(s) containing 'Country'")

    # Examine the first filtered table
```

```python
    gdp_table = filtered_tables[0]
    print(f"\n  GDP Table Details:")
    print(f"Shape: {gdp_table.shape}")
    print(f"Columns: {list(gdp_table.columns)}")
    print("\n  Sample data:")
    display(gdp_table.head(3))
else:
    print("  No tables found containing 'Country'")
```

```
 Fetching data from: https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_per_ca
 Successfully retrieved webpage (Status: 200)
 Found 1 table(s) on the page

 Table Overview:
  Table 0: (224, 4) (rows × columns)

 Found 1 table(s) containing 'Country'

 GDP Table Details:
Shape: (224, 4)
Columns: ['Country/Territory', 'IMF (2025)[a][5]', 'World Bank (2022-24)[6]', 'United Nations

 Sample data:
 Successfully retrieved webpage (Status: 200)
 Found 1 table(s) on the page

 Table Overview:
  Table 0: (224, 4) (rows × columns)

 Found 1 table(s) containing 'Country'

 GDP Table Details:
Shape: (224, 4)
Columns: ['Country/Territory', 'IMF (2025)[a][5]', 'World Bank (2022-24)[6]', 'United Nations

 Sample data:
```

	Country/Territory	IMF (2025)[a][5]	World Bank (2022–24)[6]	United Nations (2023)[7]
0	Monaco	—	256581	256581
1	Liechtenstein	—	207973	207150

	Country/Territory	IMF (2025)[a][5]	World Bank (2022–24)[6]	United Nations (2023)[7]
2	Luxembourg	140941	137516	128936

6.6 Independent Study

6.6.1 Practice exercise 1: Reading `.csv` data

Read the file *Top 10 Albums By Year.csv*. This file contains the top 10 albums for each year from 1990 to 2021. Each row corresponds to a unique album.

6.6.1.1

Print the first 5 rows of the data.

6.6.1.2

How many rows and columns are there in the data?

6.6.1.3

Print the summary statistics of the data, and answer the following questions:

1. What proportion of albums have 15 or lesser tracks? Mention a range for the proportion.
2. What is the mean length of a track (in minutes)?

6.6.2 Practice exercise 2: Reading `.txt` data

Read the file *bestseller_books.txt*. It contains top 50 best-selling books on amazon from 2009 to 2019. Identify the delimiter without opening the file with Notepad or a text-editing software. How many rows and columns are there in the dataset?

Alternatively, you can use the argument `sep = None`, and `engine = 'python'`. The default engine is C. However, the 'python' engine has a 'sniffer' tool which may identify the delimiter automatically.

6.6.3 Practice exercise 3: Reading HTML data

Read the table(s) consisting of attendance of spectators in FIFA worlds cup from this page. Read only those table(s) that have the word *'attendance'* in them. How many rows and columns are there in the table(s)?

6.6.4 Practice exercise 4: Reading JSON data

Read the movies dataset from here. How many rows and columns are there in the data?

7 Pandas Fundamentals

7.1 Introduction

Pandas is an essential tool in the data scientist or data analyst's toolkit due to its ability to handle and transform data with ease and efficiency.

Built on top of the NumPy package, Pandas extends the capabilities of NumPy by offering support for working with tabular or heterogeneous data. While NumPy is optimized for working with homogeneous numerical arrays, Pandas is specifically designed for handling structured, labeled data, commonly represented in two-dimensional tables known as DataFrames. There are some similarities between the two libraries. Like NumPy, Pandas provides basic mathematical functionalities such as addition, subtraction, conditional operations, and broadcasting. However, while NumPy focuses on multi-dimensional arrays, Pandas excels in offering the powerful 2D `DataFrame` object for data manipulation.

Data in Pandas is often used to feed statistical analyses in SciPy, visualization functions in Matplotlib, and machine learning algorithms in Scikit-learn.

Typically, the Pandas library is used for:

- Data reading/writing from different sources (CSV, Excel, SQL, etc.).
- Data manipulation, cleaning, and transformation.
- Computing data distribution and summary statistics
- Grouping, aggregation, and pivoting for advanced data analysis.
- Merging, concatenating, and reshaping data.
- Handling time series and datetime data.
- Data visualization and plotting.

Let's import the Pandas library to use its methods and functions.

```
import pandas as pd
```

7.2 Creating a Pandas Series / DataFrame

- From a Python List or Dictionary
- By Reading Data from a file

7.2.1 Creating a Series/DataFrame from Python List or Dictionary

7.2.1.1 Creating a series:

- **From a Python List**: You can create a pandas Series by passing a list of values.

```
#Defining a Pandas Series
series_example = pd.Series(['these','are','english','words'])
series_example
```

```
0      these
1        are
2    english
3      words
dtype: object
```

Note that the default row indices are integers starting from 0. However, the index can be specified with the `index` argument if desired by the user:

```
#Defining a Pandas Series with custom row labels
series_example = pd.Series(['these','are','english','words'], index = range(101,105))
series_example
```

```
101      these
102        are
103    english
104      words
dtype: object
```

- **From a Dictionary**: A dictionary can also be used to create a Series, where keys become index labels.

```
#Dictionary consisting of the GDP per capita of the US from 1960 to 2021 with some missing va
GDP_per_capita_dict = {'1960':3007,'1961':3067,'1962':3244,'1963':3375,'1964':3574,'1965':382
```

```
#Example 2: Creating a Pandas Series from a Dictionary
GDP_per_capita_series = pd.Series(GDP_per_capita_dict)
GDP_per_capita_series.head()
```

```
1960    3007
1961    3067
1962    3244
1963    3375
1964    3574
dtype: int64
```

7.2.1.2 Creating a DataFrame

- **From a list of Python Dictionary**: You can create a DataFrame where keys are column names and values are lists representing column data.

```
#List of dictionary consisting of 52 playing cards of the deck
deck_list_of_dictionaries = [{'value':i, 'suit':c}
for c in ['spades', 'clubs', 'hearts', 'diamonds']
for i in range(2,15)]
```

```
#Example 3: Creating a Pandas DataFrame from a List of dictionaries
deck_df = pd.DataFrame(deck_list_of_dictionaries)
deck_df.head()
```

	value	suit
0	2	spades
1	3	spades
2	4	spades
3	5	spades
4	6	spades

- **From a Python Dictionary**: You can create a DataFrame where keys are column names and values are lists representing column data.

```
#Example 4: Creating a Pandas DataFrame from a Dictionary
dict_data = {'A': [1, 2, 3, 4, 5],
        'B': [20, 10, 50, 40, 30],
        'C': [100, 200, 300, 400, 500]}

dict_df = pd.DataFrame(dict_data)
dict_df
```

	A	B	C
0	1	20	100
1	2	10	200
2	3	50	300
3	4	40	400
4	5	30	500

7.3 Creating a Series/DataFrame by Reading Data from a File

In the real world, a Pandas DataFrame will typically be created by loading the datasets from existing storage such as SQL Database, CSV file, Excel file, text files, HTML files, etc., as we learned in the perious chapter of the book on Reading data.

- **Using `read_csv()`**: This is one of the most common methods to read data from a CSV file into a pandas DataFrame.
- **Using `read_excel()`**: You can also read data from Excel files.
- **Using `read_json()`**: You can also read data from json files.

7.4 Pandas data structures - Series and DataFrame

There are two core components of the Pandas library - Series and DataFrame.

A DataFrame is a two-dimensional object - comprising of tabular data organized in rows and columns, where individual columns can be of different value types (numeric / string / boolean etc.). A DataFrame has row labels (also called row indices) which refer to individual rows, and column labels (also called column names) that refer to individual columns. By default, the row indices are integers starting from zero. However, both the row indices and column names can be customized by the user.

Let us read the spotify data - *spotify_data.csv*, using the Pandas function `read_csv()`.

```
movie_ratings = pd.read_csv('./datasets/movie_ratings.csv')
movie_ratings.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
0	Opal Dreams	14443	14443	9000000	Nov 22 2006	PG/PG-13
1	Major Dundee	14873	14873	3800000	Apr 07 1965	PG/PG-13
2	The Informers	315000	315000	18000000	Apr 24 2009	R
3	Buffalo Soldiers	353743	353743	15000000	Jul 25 2003	R

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
4	The Last Sin Eater	388390	388390	2200000	Feb 09 2007	PG/PG-13

The object `spotify_data` is a pandas DataFrame:

```
type(movie_ratings)
```

```
pandas.core.frame.DataFrame
```

A Series is a one-dimensional array-like object in pandas that contains a sequence of values, where each value is associated with an index. All the values in a Series must have the same data type. Each column of a DataFrame is Series as shown in the example below.

```
#Extracting song titles from the spotify_songs DataFrame
movie_title = movie_ratings['Title']
movie_title
```

```
0                          Opal Dreams
1                         Major Dundee
2                        The Informers
3                     Buffalo Soldiers
4                   The Last Sin Eater
                     ...
2223                       King Arthur
2224                             Mulan
2225                        Robin Hood
2226    Robin Hood: Prince of Thieves
2227                        Spiceworld
Name: Title, Length: 2228, dtype: object
```

```
#The object spotify_songs is a Series
type(movie_title)
```

```
pandas.core.series.Series
```

A Series is essentially a column, and a DataFrame is a two-dimensional table made up of a collection of Series

7.5 Data Selection and Filtering

7.5.1 Extracting Column(s)

The first step when working with a DataFrame is often to extract one or more columns. To do this effectively, it's helpful to understand the internal structure of a DataFrame. Conceptually, you can think of a DataFrame as a dictionary of lists, where the keys are column names, and the values are lists or arrays containing data for the respective columns.

```
# Pandas format is simliar to this
movie_ratings_dict = {
    'Title':  ['Opal Dreams', 'Major Dundee', 'The Informers', 'Buffalo Soldiers', 'The Last
    'US Gross':  [14443, 14873, 315000, 353743, 388390],
    'Worldwide Gross': [14443, 14873, 315000, 353743, 388390],
    'Production Budget': [9000000, 3800000, 18000000, 15000000, 2200000]
}
```

For dictionary, we use key to retrive its values

```
movie_ratings_dict['Title']
```

```
['Opal Dreams',
 'Major Dundee',
 'The Informers',
 'Buffalo Soldiers',
 'The Last Sin Eater']
```

Similar like dictionary, we can extract a column by its column name

```
movie_ratings['Title']
```

```
0                          Opal Dreams
1                         Major Dundee
2                        The Informers
3                     Buffalo Soldiers
4                   The Last Sin Eater
                         ...
2223                       King Arthur
2224                             Mulan
2225                        Robin Hood
```

```
2226    Robin Hood: Prince of Thieves
2227                       Spiceworld
Name: Title, Length: 2228, dtype: object
```

Each column is a feature of the dataframe, we can also use `.` operator to extract a single column

```
movie_ratings.Title
```

```
0                         Opal Dreams
1                        Major Dundee
2                       The Informers
3                    Buffalo Soldiers
4                   The Last Sin Eater
                    ...
2223                      King Arthur
2224                            Mulan
2225                       Robin Hood
2226    Robin Hood: Prince of Thieves
2227                       Spiceworld
Name: Title, Length: 2228, dtype: object
```

When extracting multiple columns, you need to place the column names inside a list.

```
movie_ratings[['Title', 'US Gross', 'Worldwide Gross' ]]
```

	Title	US Gross	Worldwide Gross
0	Opal Dreams	14443	14443
1	Major Dundee	14873	14873
2	The Informers	315000	315000
3	Buffalo Soldiers	353743	353743
4	The Last Sin Eater	388390	388390
...	...	...	...
2223	King Arthur	51877963	203877963
2224	Mulan	120620254	303500000
2225	Robin Hood	105269730	310885538
2226	Robin Hood: Prince of Thieves	165493908	390500000
2227	Spiceworld	29342592	56042592

7.5.2 Extracting Row(s)

7.5.2.1 Extracting based on a Single Condition or Multiple Conditions

In many cases, we need to filter rows based on specific conditions or a combination of multiple conditions. Next, let's explore how to use these conditions effectively to extract rows that meet our criteria, whether it's a single condition or multiple conditions combined

```
# extracting the rows that have IMDB Rating greater than 8
movie_ratings[movie_ratings['IMDB Rating'] > 8]
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Ra
21	Gandhi, My Father	240425	1375194	5000000	Aug 03 2007	Other
56	Ed Wood	5828466	5828466	18000000	Sep 30 1994	R
67	Requiem for a Dream	3635482	7390108	4500000	Oct 06 2000	Other
164	Trainspotting	16501785	24000785	3100000	Jul 19 1996	R
181	The Wizard of Oz	28202232	28202232	2777000	Aug 25 2039	G
...	...	...	...	...	...	...
2090	Finding Nemo	339714978	867894287	94000000	May 30 2003	G
2092	Toy Story 3	410640665	1046340665	200000000	Jun 18 2010	G
2094	Avatar	760167650	2767891499	237000000	Dec 18 2009	PG/PG-1
2130	Scarface	44942821	44942821	25000000	Dec 09 1983	Other
2194	The Departed	133311000	290539042	90000000	Oct 06 2006	R

To combine multiple conditions in pandas, you need to use the `&` (AND) and `|` (OR) operators. Make sure to enclose each condition in parentheses () for clarity and to ensure proper evaluation order.

```
# extracting the rows that have IMDB Rating greater than 8 and US Gross less than 1000000
movie_ratings[(movie_ratings['IMDB Rating'] > 8) & (movie_ratings['US Gross'] < 1000000)]
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
21	Gandhi, My Father	240425	1375194	5000000	Aug 03 2007	Other
636	Lake of Fire	25317	25317	6000000	Oct 03 2007	Other

7.5.2.2 Negating Conditions with the ~ Operator

The tilde (`~`) is used for negating boolean conditions in pandas, making it a useful tool for excluding specific values or rows.

In some cases, we may want to extract rows that do not meet a specific condition. To accomplish this, we can use the ~ operator, which negates a condition. This operator allows us to filter data by excluding rows that satisfy a particular condition, making it a powerful tool for refining our queries. Let's explore how to use the ~ operator to negate conditions and select rows that do not meet our criteria

```
# Excluding the rows that have IMDB Rating that equals 8 using the tilde ~
movie_ratings[~(movie_ratings['IMDB Rating'] == 8)]
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date
0	Opal Dreams	14443	14443	9000000	Nov 22 2006
1	Major Dundee	14873	14873	3800000	Apr 07 1965
2	The Informers	315000	315000	18000000	Apr 24 2009
3	Buffalo Soldiers	353743	353743	15000000	Jul 25 2003
4	The Last Sin Eater	388390	388390	2200000	Feb 09 2007
...	...	...	...	...	...
2223	King Arthur	51877963	203877963	90000000	Jul 07 2004
2224	Mulan	120620254	303500000	90000000	Jun 19 1998
2225	Robin Hood	105269730	310885538	210000000	May 14 2010
2226	Robin Hood: Prince of Thieves	165493908	390500000	50000000	Jun 14 1991
2227	Spiceworld	29342592	56042592	25000000	Jan 23 1998

Another way to exclude rows where a column equals a certain value, you can use `!=` to create the condition.

```
# using the != operator
movie_ratings[movie_ratings['IMDB Rating'] != 8]
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date
0	Opal Dreams	14443	14443	9000000	Nov 22 2006
1	Major Dundee	14873	14873	3800000	Apr 07 1965
2	The Informers	315000	315000	18000000	Apr 24 2009
3	Buffalo Soldiers	353743	353743	15000000	Jul 25 2003
4	The Last Sin Eater	388390	388390	2200000	Feb 09 2007
...	...	...	...	...	...
2223	King Arthur	51877963	203877963	90000000	Jul 07 2004
2224	Mulan	120620254	303500000	90000000	Jun 19 1998
2225	Robin Hood	105269730	310885538	210000000	May 14 2010
2226	Robin Hood: Prince of Thieves	165493908	390500000	50000000	Jun 14 1991

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	M
2227	Spiceworld	29342592	56042592	25000000	Jan 23 1998	P

7.5.3 Extracting Subsets of Rows and Columns

Sometimes we may be interested in working with a subset of rows and columns of the data, instead of working with the entire dataset. The indexing operators loc and iloc provide a convenient way of selecting a subset of desired rows and columns.

Let us first sort the `movie_ratings` data frame by `IMDB Rating`.

```
movie_ratings_sorted = movie_ratings.sort_values(by = 'IMDB Rating', ascending = False)
movie_ratings_sorted.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MP
182	The Shawshank Redemption	28241469	28241469	25000000	Sep 23 1994	R
2084	Inception	285630280	753830280	160000000	Jul 16 2010	PG
790	Schindler's List	96067179	321200000	25000000	Dec 15 1993	R
1962	Pulp Fiction	107928762	212928762	8000000	Oct 14 1994	R
561	The Dark Knight	533345358	1022345358	185000000	Jul 18 2008	PG

7.5.3.1 Subsetting the DataFrame by `loc`

The operator `loc` uses axis labels (row indices and column names) to subset the data.

Let's subset the `title`, `worldwide gross`, `production budget`, and `IMDB raring` of top 3 movies.

```
# Subsetting the DataFrame by loc - using axis labels
movies_subset = movie_ratings_sorted.loc[[182,2084, 2092],[ 'Title', 'IMDB Rating', 'US Gros
movies_subset
```

	Title	IMDB Rating	US Gross	Worldwide Gross	Production Budget
182	The Shawshank Redemption	9.2	28241469	28241469	25000000
2084	Inception	9.1	285630280	753830280	160000000
2092	Toy Story 3	8.9	410640665	1046340665	200000000

The : symbol in `.loc` is a slicing operator that represents a range or all elements in the specified dimension (rows or columns). Use : alone to select all rows/columns, or with start/end points to slice specific parts of the DataFrame.

```
# Subsetting the DataFrame by loc - using axis labels. the colon is used to select all rows
movies_subset = movie_ratings_sorted.loc[:,['Title','Worldwide Gross','Production Budget','I
movies_subset
```

	Title	Worldwide Gross	Production Budget	IMDB Rating
182	The Shawshank Redemption	28241469	25000000	9.2
2084	Inception	753830280	160000000	9.1
790	Schindler's List	321200000	25000000	8.9
1962	Pulp Fiction	212928762	8000000	8.9
561	The Dark Knight	1022345358	185000000	8.9
...	...	...	...	...
1051	Glitter	4273372	8500000	2.0
1495	Disaster Movie	34690901	20000000	1.7
1116	Crossover	7009668	5600000	1.7
805	From Justin to Kelly	4922166	12000000	1.6
1147	Super Babies: Baby Geniuses 2	9109322	20000000	1.4

```
# Subsetting the DataFrame by loc - using axis labels. the colon is used to select a range o
movies_subset = movie_ratings_sorted.loc[182:561,['Title','Worldwide Gross','Production Budg
movies_subset
```

	Title	Worldwide Gross	Production Budget	IMDB Rating
182	The Shawshank Redemption	28241469	25000000	9.2
2084	Inception	753830280	160000000	9.1
790	Schindler's List	321200000	25000000	8.9
1962	Pulp Fiction	212928762	8000000	8.9
561	The Dark Knight	1022345358	185000000	8.9

Combining `.loc` with condition(s) to extract specific rows and columns based on criteria

```
# extracting the rows that have IMDB Rating greater than 8 or US Gross less than 1000000, onl
movie_ratings[(movie_ratings['IMDB Rating'] > 8) & (movie_ratings['US Gross'] < 1000000)][['

#using loc to extract the rows that have IMDB Rating greater than 8 or US Gross less than 100
movie_ratings.loc[(movie_ratings['IMDB Rating'] > 8) & (movie_ratings['US Gross'] < 1000000)
```

	Title	IMDB Rating
21	Gandhi, My Father	8.1
636	Lake of Fire	8.4

7.5.3.2 Subsetting the DataFrame by `iloc`

while `iloc` uses the position of rows or columns, where position has values 0,1,2,3,…and so on, for rows from top to bottom and columns from left to right. In other words, the first row has position 0, the second row has position 1, the third row has position 2, and so on. Similarly, the first column from left has position 0, the second column from left has position 1, the third column from left has position 2, and so on.

`<IPython.core.display.Image object>`

```
# let's check  the movie_ratings_sorted DataFrame
movie_ratings_sorted.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MP
182	The Shawshank Redemption	28241469	28241469	25000000	Sep 23 1994	R
2084	Inception	285630280	753830280	160000000	Jul 16 2010	PG
790	Schindler's List	96067179	321200000	25000000	Dec 15 1993	R
1962	Pulp Fiction	107928762	212928762	8000000	Oct 14 1994	R
561	The Dark Knight	533345358	1022345358	185000000	Jul 18 2008	PG

After sorting, the position-based index changes, while the label-based index remains unchanged. Let's pass the position-based index to `iloc` to retrieve the top 2 rows from the `movie_ratings_sorted` DataFrame.

```
movie_ratings_sorted.iloc[0:2,:]
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MP
182	The Shawshank Redemption	28241469	28241469	25000000	Sep 23 1994	R
2084	Inception	285630280	753830280	160000000	Jul 16 2010	PG

It is important to note that the endpoint is excluded in an `iloc` slice.

For comparison, let's pass the same argument to loc and see what it returns.

```
movie_ratings_sorted.loc[0:2,:]
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA R
0	Opal Dreams	14443	14443	9000000	Nov 22 2006	PG/PG-1
851	Star Trek: Generations	75671262	120000000	38000000	Nov 18 1994	PG/PG-1
140	Tuck Everlasting	19161999	19344615	15000000	Oct 11 2002	PG/PG-1
708	De-Lovely	13337299	18396382	4000000	Jun 25 2004	PG/PG-1
705	Flyboys	13090630	14816379	60000000	Sep 22 2006	PG/PG-1
...	...	...	...	...	...	...
955	The Brothers Solomon	900926	900926	10000000	Sep 07 2007	R
1637	Drumline	56398162	56398162	20000000	Dec 13 2002	PG/PG-1
1610	Hollywood Homicide	30207785	51107785	75000000	Jun 13 2003	PG/PG-1
569	Doom	28212337	54612337	70000000	Oct 21 2005	R
2	The Informers	315000	315000	18000000	Apr 24 2009	R

As you can see, `:` is used in the same way as in `loc` to denote a range of the index. All rows between label indices 0 and 2, inclusive of both ends, are returned.

```
# Subsetting the DataFrame by iloc - using index of the position of rows and columns
movies_iloc_subset1 = movie_ratings_sorted.iloc[0:10,[0,2,3,9]]
movies_iloc_subset1
```

	Title	Worldwide Gross	Production Budget	IMDB R
182	The Shawshank Redemption	28241469	25000000	9.2
2084	Inception	753830280	160000000	9.1
2092	Toy Story 3	1046340665	200000000	8.9
1962	Pulp Fiction	212928762	8000000	8.9
790	Schindler's List	321200000	25000000	8.9
561	The Dark Knight	1022345358	185000000	8.9
184	Cidade de Deus	28763397	3300000	8.8
487	The Lord of the Rings: The Fellowship of the Ring	868621686	109000000	8.8
497	The Lord of the Rings: The Return of the King	1133027325	94000000	8.8
1081	C'era una volta il West	5321508	5000000	8.8

Can you use `.iloc` for conditional filtering, why or why not?

To recap, here are the **Key differences between`loc` and `iloc` in pandas**:

- **Indexing Type:**

  - loc uses labels (names) for indexing.
  - iloc uses integer positions for indexing.

- **Inclusion of Endpoints:**
  - In a loc slice, both endpoints are included.
  - In an iloc slice, the endpoint is excluded.

7.5.4 Finding minimum/maximum of a column

When working with pandas, there are two main options for locating the minimum or maximum values in a DataFrame column:

- `idxmin()` and `idxmax()`: return the **index label** of the first occurrence of the maximum or minimum value in a specified column.

```
# movie_ratings_sorted.iloc[position_max_wgross,:]
max_index = movie_ratings_sorted['Worldwide Gross'].idxmax()
min_index = movie_ratings_sorted['Worldwide Gross'].idxmin()
print("Max index: ", max_index)
print("Min index: ", min_index)
```

```
Max index:  2094
Min index:  896
```

`idxmin()` and `idxmax()` return the index label of the minimum or maximum value in a column. You can use these returned index labels with `.loc` to extract the corresponding row.

```
print(movie_ratings_sorted.loc[max_index,'Worldwide Gross'])
print(movie_ratings_sorted.loc[min_index,'Worldwide Gross'])
```

```
2767891499
884
```

- `argmax()` and `argmin()`: Return the **integer position** of the first occurrence of the maximum or minimum value in a column. You can use these integer positions with `.iloc` to extract the corresponding row

```
# using argmax and argmin, which return the index of the maximum and minimum values
max_position = movie_ratings_sorted['Worldwide Gross'].argmax()
min_position = movie_ratings_sorted['Worldwide Gross'].argmin()
print("max position:", max_position)
print("min position:", min_position)

# using iloc to get the row with the maximum and minimum values
print(movie_ratings_sorted.iloc[max_position, 2])
print(movie_ratings_sorted.iloc[min_position, 2])
```

```
max position: 48
min position: 2149
2767891499
884
```

Tips:

- When working with non-unique or custom indices, it is recommended to use `idxmax()` and `idxmin()` to retrieve index labels, as `argmax()` may be less intuitive in such scenarios.
- For DataFrames, use `.idxmax(axis=1)` or `.idxmin(axis=1)` to find the index labels corresponding to the maximum or minimum values across rows, rather than columns.

7.5.5 Finding the top *n* minimum/maximum values of a column

To find the top n minimum or maximum values of a column in a pandas DataFrame, you can use the `nsmallest()` and `nlargest()` methods. Here's how you can do it:

```
# find the top 3 movies with the highest worldwide gross
top_3_movies = movie_ratings.nlargest(3, 'Worldwide Gross')
top_3_movies
```

	Title	US Gross	Worldwide Gross	Production Budget
2094	Avatar	760167650	2767891499	237000000
2093	Titanic	600788188	1842879955	200000000
497	The Lord of the Rings: The Return of the King	377027325	1133027325	94000000

Let's double check the result using the `movie_ratings_sorted` dataframe

```python
movie_ratings_sorted['Worldwide Gross'].nlargest(3)
```

```
2094    2767891499
2093    1842879955
497     1133027325
Name: Worldwide Gross, dtype: int64
```

Let's find the 3 movies with the smallest IMDb votes using the `nsmallest` method

```python
bottom_3_movies = movie_ratings.nsmallest(3, 'IMDB Votes')
bottom_3_movies
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating	Source
1000	Teeth	347578	2300349	2000000	Jan 18 2008	R	Original
1261	Birth	5005899	14603001	20000000	Oct 29 2004	R	Original
1298	CachÈ	3647381	17147381	8000000	Dec 23 2005	R	Original

Let's check the result using `sort_values` method

```python
# using the sort_values method
movie_ratings_sorted['IMDB Votes'].nsmallest(3)
```

```
1000    18
1261    25
1298    26
Name: IMDB Votes, dtype: int64
```

7.5.6 Sub-setting data

In the chapter on reading data, we learned about different ways for subsetting data, specifically:

- Selecting columns: `df['column_name']`, `df[['col1', 'col2']]`
- Selecting rows by label: `df.loc[]`
- Selecting rows by integer position: `df.iloc[]`
- Selecting by condition (Boolean indexing): `df[df['column_name'] > value]`
- Selecting by multiple conditions: `df[(df['column_name'] > value) & (df['other_column'] == another_value)]`

7.6 Other Manipulation

7.6.1 Setting and Resetting indices

In pandas, the `index` of a dataframe can be accessed by using `index` attribute, it represents the row labels of a DataFrame. When you create a DataFrame without specifying an index, pandas automatically assigns a RangeIndex, which is a sequence of integers starting from 0.

```
# read the txt data
bestseller_data = pd.read_csv('../Data/bestseller_books.txt', sep=';')
bestseller_data.head()
```

	Unnamed: 0.1	Unnamed: 0	Name	Author
0	0	0	10-Day Green Smoothie Cleanse	JJ Smith
1	1	1	11/22/63: A Novel	Stephen King
2	2	2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson
3	3	3	1984 (Signet Classics)	George Orwell
4	4	4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic

```
# accessing the index of the DataFrame
bestseller_data.index
```

```
RangeIndex(start=0, stop=550, step=1)
```

Alternatively, we can use the `index_col` parameter in the `pd.read_csv()` function to set a specific column as the index when reading a CSV file in pandas. This allows us to directly specify which column should be used as the index of the resulting DataFrame.

```
bestseller_index_data = pd.read_csv('../Data/bestseller_books.txt', sep=';', index_col='Unnam
bestseller_index_data.head()
```

	Unnamed: 0	Name	Author	User Rati
0	0	10-Day Green Smoothie Cleanse	JJ Smith	4.7
1	1	11/22/63: A Novel	Stephen King	4.6
2	2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7
3	3	1984 (Signet Classics)	George Orwell	4.7
4	4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8

```
bestseller_index_data.index
```

```
Index([  0,   1,   2,   3,   4,   5,   6,   7,   8,   9,
       ...
       540, 541, 542, 543, 544, 545, 546, 547, 548, 549],
      dtype='int64', length=550)
```

The `index` is crucial when subsetting your dataset using methods like `loc[]`, as it identifies the rows you want to select based on their labels.

```
# get the book that User rating is greater than 4.5
bestseller_index_data.loc[bestseller_index_data['User Rating']== 4.9].head()
```

	Unnamed: 0	Name	Author	User Rating	Rev
40	40	Brown Bear, Brown Bear, What Do You See?	Bill Martin Jr.	4.9	143
41	41	Brown Bear, Brown Bear, What Do You See?	Bill Martin Jr.	4.9	143
81	81	Dog Man and Cat Kid: From the Creator of Capta...	Dav Pilkey	4.9	506
82	82	Dog Man: A Tale of Two Kitties: From the Creat...	Dav Pilkey	4.9	478
83	83	Dog Man: Brawl of the Wild: From the Creator o...	Dav Pilkey	4.9	723

Additionally, you can modify the index at any point after the DataFrame is created by using the set_index() function to re-index the rows based on existing column(s) of the DataFrame.

```
bestseller_data_reindexed = bestseller_data.set_index('Unnamed: 0.1')
bestseller_data_reindexed.index
```

```
Index([  0,   1,   2,   3,   4,   5,   6,   7,   8,   9,
       ...
       540, 541, 542, 543, 544, 545, 546, 547, 548, 549],
      dtype='int64', name='Unnamed: 0.1', length=550)
```

Note that, the index column does not have to uniquely identify each row. For example, if we need to subset data by `Author` frequently in our analysis, we can set it as our index

```
bestseller_author_index = bestseller_data.set_index('Author')
bestseller_author_index.loc['Jen Sincero']
```

	Unnamed: 0.1	Unnamed: 0	Name	User Rati
Author				
Jen Sincero	546	546	You Are a Badass: How to Stop Doubting Your Gr...	4.7
Jen Sincero	547	547	You Are a Badass: How to Stop Doubting Your Gr...	4.7
Jen Sincero	548	548	You Are a Badass: How to Stop Doubting Your Gr...	4.7
Jen Sincero	549	549	You Are a Badass: How to Stop Doubting Your Gr...	4.7

```
bestseller_author_index.index
```

```
Index(['JJ Smith', 'Stephen King', 'Jordan B. Peterson', 'George Orwell',
       'National Geographic Kids', 'George R. R. Martin',
       'George R. R. Martin', 'Amor Towles', 'James Comey', 'Fredrik Backman',
       ...
       'R. J. Palacio', 'R. J. Palacio', 'R. J. Palacio', 'R. J. Palacio',
       'R. J. Palacio', 'Jeff Kinney', 'Jen Sincero', 'Jen Sincero',
       'Jen Sincero', 'Jen Sincero'],
      dtype='object', name='Author', length=550)
```

You can revert it back to a default index using `reset_index()`. It converts the current index into a regular column and resets the index to a default integer sequence.

```
bestseller_author_index.reset_index()
```

	Author	Unnamed: 0.1	Unnamed: 0	Name
0	JJ Smith	0	0	10-Day Green Smoothie Cleanse
1	Stephen King	1	1	11/22/63: A Novel
2	Jordan B. Peterson	2	2	12 Rules for Life: An Antidote to Chaos
3	George Orwell	3	3	1984 (Signet Classics)
4	National Geographic Kids	4	4	5,000 Awesome Facts (About Everything!) (N
...	...	...	...	...
545	Jeff Kinney	545	545	Wrecking Ball (Diary of a Wimpy Kid Book
546	Jen Sincero	546	546	You Are a Badass: How to Stop Doubting Yo
547	Jen Sincero	547	547	You Are a Badass: How to Stop Doubting Yo
548	Jen Sincero	548	548	You Are a Badass: How to Stop Doubting Yo
549	Jen Sincero	549	549	You Are a Badass: How to Stop Doubting Yo

In summary, the index is a powerful feature of pandas for organizing and accessing data, and it can be set, changed, or reset as needed to suit your data analysis.

Later, you will learn about **hierarchical indexing**, which allows you to create multi-level indices by passing multiple columns.

7.6.2 Dropping a column

In pandas, you can drop a column from a DataFrame using the drop() method. This is useful when you want to remove columns that are redundant or when you want to simplify your DataFrame by excluding unnecessary data.

```
df.drop(columns='column_name')
df.drop(columns= ['col_1', 'col_2'])
```

```
# drop the repeated rows in the bestseller dataframe
bestseller_df = bestseller_index_data.drop(columns = 'Unnamed: 0')
bestseller_df.head()
```

	Name	Author	User Rating	Reviews
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350
1	11/22/63: A Novel	Stephen King	4.6	2052
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979
3	1984 (Signet Classics)	George Orwell	4.7	21424
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665

```
# another way to drop the column
bestseller_index_data.drop('Unnamed: 0', axis=1).head()
```

	Name	Author	User Rating	Reviews
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350
1	11/22/63: A Novel	Stephen King	4.6	2052
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979
3	1984 (Signet Classics)	George Orwell	4.7	21424
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665

Note: The `df.drop()` function can be used to remove rows by specifying row labels, index values, or based on conditions. It includes an **`axis`** parameter, which defaults to dropping rows (`axis=0`). Feel free to explore its additional options.

7.6.3 Adding a column

In a DataFrame, it's common to create new columns based on existing columns for analysis or prediction purposes. If the column name you specify does not already exist, a new column will be added to the DataFrame. If the column name already exists, the new values will overwrite the existing column.

Let's classify books based on their user rating

```
def classify_rating(rating):
    if rating >= 4.5:
        return 'Highly Rated'
    elif 3.0 <= rating < 4.5:
        return 'Moderately Rated'
    else:
        return 'Low Rated'
```

```
bestseller_df['Rating_Class'] = bestseller_df['User Rating'].apply(classify_rating)

bestseller_df.head()
```

	Name	Author	User Rating	Reviews
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350
1	11/22/63: A Novel	Stephen King	4.6	2052
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979
3	1984 (Signet Classics)	George Orwell	4.7	21424
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665

7.6.4 Renaming a column

To rename a column in a pandas DataFrame, you can use the `rename()` method. .

- Renaming a single column

```
df.rename(columns={'old_name': 'new_name'}, inplace=False)
```

```
# rename Year to Publication Year
bestseller_df.rename(columns={'Year':'Publication Year'}, inplace=True)
bestseller_df.head()
```

	Name	Author	User Rating	Reviews
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350
1	11/22/63: A Novel	Stephen King	4.6	2052
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979
3	1984 (Signet Classics)	George Orwell	4.7	21424
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665

- Renameing multiple columns

Specifying a dictionary that maps old column names to new one

```
df_renamed_multiple = df.rename(columns={'col1': 'new_col1', 'col2': 'new_col2'})
```

```
# rename Name to Book Name, and Author to Writer
bestseller_df.rename(columns={'Name':'Book Name', 'Author':'Writer'}, inplace=True)
bestseller_df.head()
```

	Book Name	Writer	User Rating	Reviews
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350
1	11/22/63: A Novel	Stephen King	4.6	2052
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979
3	1984 (Signet Classics)	George Orwell	4.7	21424
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665

7.6.5 Sorting data

Sorting dataset is a very common operation. The sort_values() function of Pandas can be used to sort a Pandas DataFrame or Series. Let us sort the spotify data in decreasing order of *track_popularity*:

```
bestseller_sorted = bestseller_index_data.sort_values(by = 'User Rating', ascending = False)
bestseller_sorted.head()
```

	Unnamed: 0	Name	Author	User Rating	Reviews	Price
521	521	Unfreedom of the Press	Mark R. Levin	4.9	5956	11
420	420	The Legend of Zelda: Hyrule Historia	Patrick Thorpe	4.9	5396	20
248	248	Oh, the Places You'll Go!	Dr. Seuss	4.9	21834	8

	Unnamed: 0	Name	Author	User Rating	Reviews	Price
247	247	Oh, the Places You'll Go!	Dr. Seuss	4.9	21834	8
246	246	Oh, the Places You'll Go!	Dr. Seuss	4.9	21834	8

7.6.6 Ranking data

With the rank() function, we can rank the observations.

For example, let us add a new column to the bestseller data that provides the rank of the `User Rating` column:

```
bestseller_ranked = bestseller_df.copy()
bestseller_ranked['Rating_rank']=bestseller_ranked['User Rating'].rank()
bestseller_ranked.head()
```

	Book Name	Writer	User Rating	Reviews
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350
1	11/22/63: A Novel	Stephen King	4.6	2052
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979
3	1984 (Signet Classics)	George Orwell	4.7	21424
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665

Note the column `Rating_rank`. Why does it contain floating point numbers? Check the `rank()` documentation to find out!

7.7 Arithematic operations within and between DataFrames

Pandas is built on top of NumPy, and for arithmetic operations, it inherits many of NumPy's powerful features. Pandas provides an intuitive and efficient way to perform arithmetic operations both within a DataFrame (on columns or rows) and between multiple DataFrames. These operations are element-wise, meaning they are applied to corresponding elements of the DataFrame or Series.

7.7.1 Arithematic operations within a DataFrame

You can easily perform arithmetic operations on columns or rows within a DataFrame

- Arithmetic Between Columns

```
# Create a DataFrame
data = {'A': [1, 2, 3],
        'B': [4, 5, 6],
        'C': [7, 8, 9]}
df = pd.DataFrame(data)

# Adding two columns
df['A_plus_B'] = df['A'] + df['B']

# Multiplying two columns
df['A_times_C'] = df['A'] * df['C']

df
```

	A	B	C	A_plus_B	A_times_C
0	1	4	7	5	7
1	2	5	8	7	16
2	3	6	9	9	27

- Arithmetic on Entire Rows

```
# Summing values across columns for each row
df['row_sum'] = df.sum(axis=1)
df
```

	A	B	C	A_plus_B	A_times_C	row_sum
0	1	4	7	5	7	24
1	2	5	8	7	16	38
2	3	6	9	9	27	54

7.7.2 Arithematic operations between DataFrames

When performing operations between two DataFrames, pandas aligns the data based on both the `index` and `column labels`. If the indices or columns do not match, pandas automatically fills the missing values with NaN (not-a-number) unless a fill value is provided.

Let us create two toy DataFrames:

```
#Creating two toy DataFrames
toy_df1 = pd.DataFrame([(1,2),(3,4),(5,6)], columns=['a','b'])
toy_df2 = pd.DataFrame([(100,200),(300,400),(500,600)], columns=['a','b'])
```

```
#DataFrame 1
toy_df1
```

	a	b
0	1	2
1	3	4
2	5	6

```
#DataFrame 2
toy_df2
```

	a	b
0	100	200
1	300	400
2	500	600

Element by element operations between two DataFrames can be performed with the operators `+`, `-`, `*`,`/`,`**`, and `%`. Below is an example of element-by-element addition of two DataFrames:

```
# Element-by-element arithmetic addition of the two DataFrames
toy_df1 + toy_df2
```

	a	b
0	101	202
1	303	404

	a	b
2	505	606

Note that these operations create problems when the row indices and/or column names of the two DataFrames do not match. See the example below:

```
#Creating another toy example of a DataFrame
toy_df3 = pd.DataFrame([(100,200),(300,400),(500,600)], columns=['a','b'], index=[1,2,3])
toy_df3
```

	a	b
1	100	200
2	300	400
3	500	600

```
#Adding DataFrames with some unmatching row indices
toy_df1 + toy_df3
```

	a	b
0	NaN	NaN
1	103.0	204.0
2	305.0	406.0
3	NaN	NaN

Note that the rows whose indices match between the two DataFrames are added up. The rest of the values are missing (or `NaN`) because only one of the DataFrames has that index.

As in the case of row indices, missing values will also appear in the case of unmatching column names, as shown in the example below.

```
toy_df4 = pd.DataFrame([(100,200),(300,400),(500,600)], columns=['b','c'])
toy_df4
```

	b	c
0	100	200
1	300	400

	b	c
2	500	600

```
#Adding DataFrames with some unmatching column names
toy_df1 + toy_df4
```

	a	b	c
0	NaN	102	NaN
1	NaN	304	NaN
2	NaN	506	NaN

7.7.3 Arithmetic Between DataFrame and Series (Broadcasting)

Pandas supports **broadcasting**, where a Series can be broadcasted to match the dimensions of a DataFrame during an arithmetic operation.

```
# Broadcasting: The row [1,2] (a Series) is added on every row in df2
toy_df1.loc[0,:] + toy_df2
```

	a	b
0	101	202
1	301	402
2	501	602

Note that the `+` operator is used to add values of a Series to a DataFrame based on column names. For adding a Series to a DataFrame based on row indices, we cannot use the `+` operator. Instead, we'll need to use the `add()` function as explained below.

Broadcasting based on row/column labels: We can use the add() function to broadcast a Series to a DataFrame. By default the Series adds based on column names, as in the case of the `+` operator.

```
# Add the first row of df1 (a Series) to every row in df2
toy_df2.add(toy_df1.loc[0,:])
```

	a	b
0	101	202
1	301	402
2	501	602

For broadcasting based on row indices, we use the `axis` argument of the `add()` function.

```
# The second column of df1 (a Series) is added to every col in df2
toy_df2.add(toy_df1.loc[:,'b'],axis='index')
```

	a	b
0	102	202
1	304	404
2	506	606

7.8 Advanced DataFrame Operations

Pandas provides powerful methods for applying functions to DataFrame rows, columns, or individual elements. These methods allow for flexible data transformations, making them essential for advanced data manipulation.

7.8.1 `apply()`: Row-wise, or Column-wise

The `apply()` function in pandas is used to apply a function along an axis of a DataFrame.

- Row-wise: You can apply a function to each row using `axis=1`.
- Column-wise: You can apply a function to each column using `axis=0` (default behavior).

Let's create a DataFrame and apply a custom function to each row.

```
# Create a DataFrame
df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [4, 5, 6],
    'C': [7, 8, 9]
})
```

```
df
```

	A	B	C
0	1	4	7
1	2	5	8
2	3	6	9

```
# Function to compute the row sum
def row_sum(row):
    return row.sum()
```

```
# Apply the function row-wise (axis=1)
df['Row_Sum'] = df.apply(row_sum, axis=1)
df
```

	A	B	C	Row_Sum
0	1	4	7	12
1	2	5	8	15
2	3	6	9	18

Let's apply the same function to each column next

```
# Apply the function column-wise (axis=0)
column_sum = df.apply(sum, axis=0)
column_sum
```

```
A           6
B          15
C          24
Row_Sum    45
dtype: int64
```

7.8.2 `map()`: Element-wise

`map()` is typically used for element-wise transformations on a Series.

```python
# write the function to compute square of a number
def square(x):
    return x ** 2

# use the map() function to apply a function element-wise
df['A_squared'] = df['A'].map(square)
df
```

	A	B	C	Row_Sum	A_squared
0	1	4	7	12	1
1	2	5	8	15	4
2	3	6	9	18	9

Element-wise transformation on an entire DataFrame.

```python
df.map(square)
```

	A	B	C	Row_Sum	A_squared
0	1	16	49	144	1
1	4	25	64	225	16
2	9	36	81	324	81

When applying a function to an entire DataFrame, it's important to ensure that all columns have the same data type. Otherwise, you might encounter errors or unexpected behavior.

```python
# Create a DataFrame
df_str = pd.DataFrame({
    'A': [1, 2, 3],
    'B': ['I', 'major', 'in'],
    'C': [7, 8, 9]
})
# uncomment the following line to run the code
# df_str.map(square)
```

7.8.3 Key Differences Between `map()` and `apply()`

Feature	`map()`	`apply()`
Applies to	Both Series and DataFrame	Both Series and DataFrame
Functionality	Element-wise transformations	Can apply more complex functions (row-wise or column-wise for DataFrame)
Input	Function, dictionary, or another Series	Function
Use Case	Used for simpler element-wise replacements or mappings	Used for more complex operations that can act on rows, columns, or individual elements
Axis Option	N/A	Can specify axis in DataFrame (row-wise or column-wise)

7.8.3.1 Summary:

- `map()`: Primarily used for element-wise operations. It can take functions, dictionaries, or Series as input for substitution or transformation.
- `apply()`: Apply complex functions across rows or columns.

7.9 `Lambda` Functions in Pandas

A lambda function is a small, anonymous function defined using the lambda keyword in Python. It can take any number of arguments but can only have one expression. In pandas, lambda functions are often used for quick and concise operations on data.

7.9.1 Syntax of Lambda Functions:

```
lambda arguments: expression
```

7.9.2 Key Features

- **Anonymous**: Lambda functions are defined without a name.
- **Concise**: They are typically used for small, simple functions that are not reused elsewhere.
- **Single Expression**: They can only contain one expression, which is evaluated and returned.

Lambda functions are commonly used with `apply()` and `map()` to efficiently transform data in pandas, allowing for quick, inline function definitions without the need to explicitly define separate functions.

Let's use a lambda function to rewrite the `apply()` and `map()` operations from the previous example

```
df['A_squared_lamdba'] = df['A'].map(lambda x: x ** 2)
df
```

	A	B	C	Row_Sum	A_squared	A_squared_lamdba
0	1	4	7	12	1	1
1	2	5	8	15	4	4
2	3	6	9	18	9	9

```
df_squared = df.apply(lambda x: x ** 2)
df_squared
```

	A	B	C	Row_Sum	A_squared	A_squared_lamdba
0	1	16	49	144	1	1
1	4	25	64	225	16	16
2	9	36	81	324	81	81

Let's look at other examples using lambda

Example 1: adding a new column based on existing columns

```
# Create a DataFrame
df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [4, 5, 6]
})

# Use a lambda function to create a new column 'C' as the sum of 'A' and 'B'
df['C'] = df.apply(lambda row: row['A'] + row['B'], axis=1)
print(df)
```

```
   A  B  C
0  1  4  5
1  2  5  7
2  3  6  9
```

Example 2: Filter rows based on a condition

```
# Filter rows where values in column 'A' are greater than 1
filtered_df = df[df['A'].apply(lambda x: x > 1)]
print(filtered_df)
```

```
   A  B  C
1  2  5  7
2  3  6  9
```

7.10 Understanding `inplace=True` and `inplace=False`

Above, we covered multiple methods that allow you to modify a DataFrame or Series. The `inplace` parameter of these methods determines whether the operation modifies the original object or returns a new object.

- `inplace=True`: Modifies the original DataFrame or Series directly and returns None. The changes are made in place, meaning the original object is altered.
- `inplace=False`: Returns a new DataFrame or Series with the modifications, leaving the original object unchanged. This is the default behavior for most methods.

Note: The default value for the inplace parameter is False for most methods.

Usage Example:

```
# Create a DataFrame
inplace_df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [4, 5, 6]
})

# Drop a column without modifying the original DataFrame
df_dropped = inplace_df.drop('A', axis=1)
print(df_dropped)  # New DataFrame without column 'A'
print(inplace_df)          # Original DataFrame remains unchanged

# Drop a column in place
inplace_df.drop('A', axis=1, inplace=True)
print(inplace_df)          # Original DataFrame is now modified
```

```
   B
0  4
1  5
2  6
   A  B
0  1  4
1  2  5
2  3  6
   B
0  4
1  5
2  6
```

7.11 Case study

To see the application of arithematic operations on DataFrames, let us see the case study below.

Song recommendation: Spotify recommends songs based on songs listened by the user. Suppose you have listened to the song *drivers license*. Spotify intends to recommend you 5 songs that are *similar* to *drivers license*. Which songs should it recommend?

Let us see the available song information that can help us identify songs similar to *drivers license*. The columns attribute of DataFrame will display all the columns names. The description of some of the column names relating to audio features is here.

```
spotify_data.columns
```

```
Index(['artist_followers', 'genres', 'artist_name', 'artist_popularity',
       'track_name', 'track_popularity', 'duration_ms', 'explicit',
       'release_year', 'danceability', 'energy', 'key', 'loudness', 'mode',
       'speechiness', 'acousticness', 'instrumentalness', 'liveness',
       'valence', 'tempo', 'time_signature'],
      dtype='object')
```

Solution approach: We have several features of a song. Let us find songs similar to *drivers license* in terms of *danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, time_signature* and *tempo*. Note that we are considering only audio features for simplicity.

To find the songs most similar to *drivers license*, we need to define a measure that quantifies the similarity. Let us define similarity of a song with *drivers license* as the Euclidean distance of the song from *drivers license*, where the coordinates of a song are: (danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, time_signature, tempo). Thus, similarity can be formulated as:

$$Similarity_{DL-S} = \sqrt{(danceability_{DL} - danceability_S)^2 + (energy_{DL} - energy_S)^2 + \cdots + (tempo_{DL} - tempo_S)^2}$$

where the subscript DL stands for *drivers license* and S stands for any song. The top 5 songs with the least value of $Similarity_{DL-S}$ will be the most similar to *drivers lincense* and should be recommended.

Let us subset the columns that we need to use to compute the Euclidean distance.

```
audio_features = spotify_data[['danceability', 'energy', 'key', 'loudness','mode','speechines
                                'acousticness', 'instrumentalness', 'liveness','valence', 'tem
```

```
audio_features.head()
```

	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	liveness	v
0	0.673	0.529	0	-7.226	1	0.3060	0.0769	0.000338	0.0856	0
1	0.511	0.566	6	-7.230	0	0.2000	0.3490	0.000000	0.3400	0
2	0.699	0.687	7	-3.997	0	0.1060	0.3080	0.000036	0.1210	0
3	0.708	0.690	2	-5.181	1	0.0442	0.3480	0.000000	0.2220	0
4	0.753	0.597	8	-8.469	1	0.2920	0.0477	0.000000	0.1970	0

```
#Distribution of values of audio_features
audio_features.describe()
```

	danceability	energy	key	loudness	mode	speechiness	ac
count	243190.000000	243190.000000	243190.000000	243190.000000	243190.000000	243190.000000	24
mean	0.568357	0.580633	5.240326	-9.432548	0.670928	0.111984	0.
std	0.159444	0.236631	3.532546	4.449731	0.469877	0.198068	0.
min	0.000000	0.000000	0.000000	-60.000000	0.000000	0.000000	0.
25%	0.462000	0.405000	2.000000	-11.990000	0.000000	0.033200	0.
50%	0.579000	0.591000	5.000000	-8.645000	1.000000	0.043100	0.
75%	0.685000	0.776000	8.000000	-6.131000	1.000000	0.075300	0.
max	0.988000	1.000000	11.000000	3.744000	1.000000	0.969000	0.

Note that the audio features differ in terms of scale. Some features like *key* have a wide range of [0,11], while others like *danceability* have a very narrow range of [0,0.988]. If we use them directly, features like *danceability* will have a much higher influence on $Similarity_{DL-S}$ as compared to features like *key*. Assuming we wish all the features to have equal weight in quantifying a song's similarity to *drivers license*, we should scale the features, so that their values are comparable.

Let us scale the value of each column to a standard uniform distribution: $U[0, 1]$.

For scaling the values of a column to $U[0, 1]$, we need to subtract the minimum value of the column from each value, and divide by the range of values of the column. For example, *danceability* can be standardized as follows:

```
#Scaling danceability to U[0,1]
danceability_value_range = audio_features.danceability.max()-audio_features.danceability.min
danceability_std = (audio_features.danceability-audio_features.danceability.min())/danceabil
danceability_std
```

```
0         0.681174
1         0.517206
2         0.707490
3         0.716599
4         0.762146
            ...
243185    0.621457
243186    0.797571
243187    0.533401
243188    0.565789
243189    0.750000
Name: danceability, Length: 243190, dtype: float64
```

However, it will be cumbersome to repeat the above code for each audio feature. We can instead write a function that scales values of a column to $U[0, 1]$, and apply the function on all the audio features.

```
#Function to scale a column to U[0,1]
def scale_uniform(x):
    return (x-x.min())/(x.max()-x.min())
```

We will use the Pandas function apply() to apply the above function to the DataFrame `audio_features`.

```
#Scaling all audio features to U[0,1]
audio_features_scaled = audio_features.apply(scale_uniform)
```

The above two blocks of code can be concisely written with the `lambda` function as:

```
audio_features_scaled = audio_features.apply(lambda x: (x-x.min())/(x.max()-x.min()))
```

```
#All the audio features are scaled to U[0,1]
audio_features_scaled.describe()
```

	danceability	energy	key	loudness	mode	speechiness	ac
count	243190.000000	243190.000000	243190.000000	243190.000000	243190.000000	243190.000000	24
mean	0.575260	0.580633	0.476393	0.793290	0.670928	0.115566	0.
std	0.161380	0.236631	0.321141	0.069806	0.469877	0.204405	0.
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.
25%	0.467611	0.405000	0.181818	0.753169	0.000000	0.034262	0.
50%	0.586032	0.591000	0.454545	0.805644	1.000000	0.044479	0.
75%	0.693320	0.776000	0.727273	0.845083	1.000000	0.077709	0.
max	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.

Since we need to find the Euclidean distance from the song *drivers license*, let us find the index of the row containing features of *drivers license.*

```
#Index of the row consisting of drivers license can be found with the index attribute
drivers_license_index = spotify_data[spotify_data.track_name=='drivers license'].index[0]
```

Note that the object returned by the `index` attribute is of type `pandas.core.indexes.numeric.Int64Index`. The elements of this object can be retrieved like the elements of a python list. That is why the object is sliced with `[0]` to return the first element of the object. As there is only one observation with the `track_name` as *drivers license*, we sliced the first element. If there were multiple observations with `track_name` as *drivers license*, we will obtain the indices of all those observations with the `index` attribute.

```
audio_features_scaled.loc[drivers_license_index,:]
```

```
danceability        0.592105
energy              0.436000
key                 0.909091
```

```
loudness            0.803825
mode                1.000000
speechiness         0.062023
acousticness        0.723896
instrumentalness    0.000013
liveness            0.105000
valence             0.132000
tempo               0.590841
time_signature      0.800000
Name: 2398, dtype: float64
```

Now, we'll subtract the audio features of *drivers license* from all other songs (broadcasting):

```
#Audio features of drivers license are being subtracted from audio features of all songs by b
songs_minus_DL = audio_features_scaled-audio_features_scaled.loc[drivers_license_index,:]
```

Now, let us square the difference computed above. We'll use the in-built python function `pow()` to square the difference:

```
songs_minus_DL_sq = songs_minus_DL.pow(2)
songs_minus_DL_sq.head()
```

	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	live
0	0.007933	0.008649	0.826446	0.000580	0.0	0.064398	0.418204	1.055600e-07	0.00
1	0.005610	0.016900	0.132231	0.000577	1.0	0.020844	0.139498	1.716100e-10	0.05
2	0.013314	0.063001	0.074380	0.005586	1.0	0.002244	0.171942	5.382400e-10	0.00
3	0.015499	0.064516	0.528926	0.003154	0.0	0.000269	0.140249	1.716100e-10	0.01
4	0.028914	0.025921	0.033058	0.000021	0.0	0.057274	0.456981	1.716100e-10	0.00

Now, we'll sum the squares of differences from all audio features to compute the similarity of all songs to *drivers license*.

```
distance_squared = songs_minus_DL_sq.sum(axis = 1)
distance_squared.head()
```

```
0    1.337163
1    1.438935
2    1.516317
3    1.004043
4    0.920316
dtype: float64
```

Now, we'll sort these distances to find the top 5 songs closest to drivers's license.

```
distances_sorted = distance_squared.sort_values()
distances_sorted.head()
```

```
2398      0.000000
81844     0.008633
4397      0.011160
130789    0.015018
143744    0.015058
dtype: float64
```

Using the indices of the top 5 distances, we will identify the top 5 songs most similar to *drivers license*:

```
spotify_data.loc[distances_sorted.index[0:6],:]
```

	artist_followers	genres	artist_name	artist_popularity	track_name
2398	1444702	pop	Olivia Rodrigo	88	drivers license
81844	2264501	pop	Jay Chou	74	
4397	25457	pop	Terence Lam	60	in Bb major
130789	176266	pop	Alan Tam	54	
143744	396326	pop & rock	Laura Branigan	64	How Am I Supposed to Live W
35627	1600562	pop	Tiziano Ferro	68	Non Me Lo So Spiegare

We can see the top 5 songs most similar to *drivers license* in the *track_name* column above. Interestingly, three of the five songs are Asian! These songs indeed sound similar to *drivers license*!

7.12 Independent Practice:

7.12.1 Practice exercise 1

7.12.1.1

Read the file *Top 10 Albums By Year.csv*. This file contains the top 10 albums for each year from 1990 to 2021. Each row corresponds to a unique album.

7.12.1.2

Print the summary statistics of the data, and answer the following questions:

1. What proportion of albums have 15 or lesser tracks? Mention a range for the proportion.
2. What is the mean length of a track (in minutes)?

7.12.1.3

Why is `Worldwide Sales` not included in the summary statistics table printed in the above step?

7.12.1.4

Update the DataFrame so that `Worldwide Sales` is included in the summary statistics table. Print the summary statistics table.

Hint: Sometimes it may not be possible to convert an object to numeric(). For example, the object 'hi' cannot be converted to a numeric() by the python compiler. To avoid getting an error, use the `errors` argument of `to_numeric()` to force such conversions to NaN (missing value).

7.12.1.5

Create a new column called `mean_sales_per_year` that computes the average worldwide sales per year for each album, assuming that the worldwide sales are as of 2022. Print the first 5 rows of the updated DataFrame.

7.12.1.6

Find the album having the highest worldwide sales per year, and its artist.

7.12.1.7

Subset the data to include only Hip-Hop albums. How many Hip_Hop albums are there?

7.12.1.8

Which album amongst hip-hop has the higest mean sales per year per track, and who is its artist?

7.12.2 Practice exercise 2

7.12.2.1

Read the file *STAT303-1 survey for data analysis.csv*.

7.12.2.2

How many observations and variables are there in the data?

7.12.2.3

Rename all the columns of the data, except the first two columns, with the shorter names in the list `new_col_names` given below. The order of column names in the list is the same as the order in which the columns are to be renamed starting with the third column from the left.

7.12.2.4

Rename the following columns again:

1. Rename `do_you_smoke` to `smoke`.
2. Rename `are_you_an_introvert_or_extrovert` to `introvert_extrovert`.

Hint: Use the function rename()

7.12.2.5

Find the proportion of people going to more than 4 parties per month. Use the variable `parties_per_month`.

7.12.2.6

Among the people who go to more than 4 parties a month, what proportion of them are introverts?

7.12.2.7

Find the proportion of people in each category of the variable `how_happy`.

7.12.2.8

Among the people who go to more than 4 parties a month, what proportion of them are either `Pretty happy` or `Very happy`?

7.12.2.9

Examine the column `num_insta_followers`. Some numbers in the column contain a comma(`,`) or a tilde(`~`). Remove both these characters from the numbers in the column.

Hint: You may use the function str.replace() of the Pandas Series class.

7.12.2.10

Convert the column `num_insta_followers` to numeric. Coerce the errors.

7.12.2.11

What is the mean `internet_hours_per_day` for the top 46 people in terms of number of instagram followers?

7.12.2.12

What is the mean `internet_hours_per_day` for the remaining people?

7.12.3 Practice exercise 3

7.12.3.1

Use the updated dataset from Practice exercise 2.

The last four variables in the dataset are:

1. `cant_change_math_ability`
2. `can_change_math_ability`
3. `math_is_genetic`
4. `much_effort_is_lack_of_talent`

Each of the above variables has values - `Agree` / `Disagree`. Replace `Agree` with `1` and `Disagree` with `0`.

Hint : You can do it with any one of the following methods:

1. Use the map() function
2. Use the apply() function with the lambda function
3. Use the replace() function

Two of the above methods avoid a `for`-loop. Which ones?

8 Data Types in Pandas

8.1 Types of Data

Each column(feature) in a pandas dataframe has a datatype associated with it. Those datatypes can be grouped into **Numerical**, **Categorical**, and **Dates**.

```
<IPython.core.display.Image object>
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
movie_ratings = pd.read_csv('./datasets/movie_ratings.csv')
movie_ratings.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
0	Opal Dreams	14443	14443	9000000	Nov 22 2006	PG/PG-13
1	Major Dundee	14873	14873	3800000	Apr 07 1965	PG/PG-13
2	The Informers	315000	315000	18000000	Apr 24 2009	R
3	Buffalo Soldiers	353743	353743	15000000	Jul 25 2003	R
4	The Last Sin Eater	388390	388390	2200000	Feb 09 2007	PG/PG-13

The `dtypes` property is used to find the data types associated with each column in the DataFrame.

```
movie_ratings.dtypes
```

```
Title                 object
US Gross               int64
Worldwide Gross        int64
Production Budget      int64
Release Date          object
```

```
MPAA Rating          object
Source               object
Major Genre          object
Creative Type        object
IMDB Rating         float64
IMDB Votes            int64
dtype: object
```

8.1.1 Available Data Types and Associated Built-in Functions

In a DataFrame, columns can have different data types. Here are the common data types you'll encounter and some built-in functions associated with each type:

1. **Numerical Data (int, float)**
   - Built-in functions: `mean()`, `sum()`, `min()`, `max()`, `std()`, `median()`, `quantile()`, etc.
2. **Object Data (str or mixed types)**
   - Built-in functions: `str.contains()`, `str.startswith()`, `str.endswith()`, `str.lower()`, `str.upper()`, `str.replace()`, etc.
3. **Datetime Data (dates)**
   - Built-in functions: `dt.year`, `dt.month`, `dt.day`, `dt.strftime()`, `dt.weekday()`, `dt.hour`, etc.

These functions help in exploring and transforming the data effectively depending on the type of data in each column.

8.1.2 Data Type Filtering

We can filter the columns based on its data types

```
# select just categorical(object) columns
movie_ratings.select_dtypes(include='object').head()
```

	Title	Release Date	MPAA Rating	Source	Major Genre	Creative Ty
0	Opal Dreams	Nov 22 2006	PG/PG-13	Adapted screenplay	Drama	Fiction
1	Major Dundee	Apr 07 1965	PG/PG-13	Adapted screenplay	Western/Musical	Fiction
2	The Informers	Apr 24 2009	R	Adapted screenplay	Horror/Thriller	Fiction

	Title	Release Date	MPAA Rating	Source	Major Genre	Creative Ty
3	Buffalo Soldiers	Jul 25 2003	R	Adapted screenplay	Comedy	Fiction
4	The Last Sin Eater	Feb 09 2007	PG/PG-13	Adapted screenplay	Drama	Fiction

```
# select the numeric columns
movie_ratings.select_dtypes(include='number').head()
```

	US Gross	Worldwide Gross	Production Budget	IMDB Rating	IMDB Votes
0	14443	14443	9000000	6.5	468
1	14873	14873	3800000	6.7	2588
2	315000	315000	18000000	5.2	7595
3	353743	353743	15000000	6.9	13510
4	388390	388390	2200000	5.7	1012

8.1.3 Data Type Conversion

Often, after the inital reading, we need to convert the datatypes of some of the columns to make them suitable for analysis,as the available functions and operations depend on the column's data type. For example, the datatype of Release Date in the DataFrame `movie_ratings` is object. To perform datetime related computations on this variable, we'll need to convert it to a datatime format. We'll use the Pandas function `to_datatime()` to covert it to a datatime format. Similar functions such as `to_numeric()`, `to_string()` etc., can be used for other conversions.

In Pandas, the `errors='coerce'` parameter is often used in the context of data conversion, specifically when using the `pd.to_numeric` function. This argument tells Pandas to convert values that it can and set the ones it cannot convert to `NaN`. It's a way of gracefully handling errors without raising an exception. Read the textbook for an example

```
# check the datatype of release data column
print(movie_ratings['Release Date'].dtypes)
movie_ratings['Release Date'].head()
```

```
object
```

```
0    Nov 22 2006
1    Apr 07 1965
2    Apr 24 2009
3    Jul 25 2003
```

```
4    Feb 09 2007
Name: Release Date, dtype: object
```

Next, we'll convert the `Release Date` column in the DataFrame to the `datetime` format to facilitate further analysis.

```python
movie_ratings['Release Date'] = pd.to_datetime(movie_ratings['Release Date'])

# Let's check the datatype of release data column again
movie_ratings['Release Date'].dtypes
```

```
dtype('<M8[ns]')
```

`dtype('<M8[ns]')` means a 64-bit datetime object with nanosecond precision stored in little-endian format. This data type is commonly used to represent timestamps in high-resolution time series data.

8.1.4 Working with `datatime` Data

8.1.4.1 the `dt` accessor

The `.dt` accessor is a powerful tool in pandas that allows you to extract and manipulate components of datetime columns in a DataFrame. This is useful for analysis and feature engineering when dealing with time-related data.

You can extract various parts of a datetime column using `.dt`

Attribute	Description	Example
`.dt.year`	Extracts the year	`df['Release Date'].dt.year`
`.dt.month`	Extracts the month (1-12)	`df['Release Date'].dt.month`
`.dt.day`	Extracts the day of the month (1-31)	`df['Release Date'].dt.day`
`.dt.hour`	Extracts the hour (0-23)	`df['Release Date'].dt.hour`
`.dt.minute`	Extracts the minute	`df['Release Date'].dt.minute`
`.dt.second`	Extracts the second	`df['Release Date'].dt.second`
`.dt.weekday`	Extracts the day of the week (0=Monday, 6=Sunday)	`df['Release Date'].dt.weekday`
`.dt.dayofyear`	Extracts the day of the year (1-366)	`df['Release Date'].dt.dayofyear`
`.dt.is_leap_year`	Checks if the year is a leap year	`df['Release Date'].dt.is_leap_year`

Let's add the year for the movie_ratings dataframe next

```
# Extracting the year from the release date
movie_ratings['Release Year'] = movie_ratings['Release Date'].dt.year
movie_ratings.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
0	Opal Dreams	14443	14443	9000000	2006-11-22	PG/PG-13
1	Major Dundee	14873	14873	3800000	1965-04-07	PG/PG-13
2	The Informers	315000	315000	18000000	2009-04-24	R
3	Buffalo Soldiers	353743	353743	15000000	2003-07-25	R
4	The Last Sin Eater	388390	388390	2200000	2007-02-09	PG/PG-13

In your pandas dataframe, if having start date and end date, your can calculate the time duration between them like below

```
# let's calculate the days since release till Jan 1st 2024
movie_ratings['Days Since Release'] = (pd.Timestamp('2024-01-01') - movie_ratings['Release Da
movie_ratings.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
0	Opal Dreams	14443	14443	9000000	2006-11-22	PG/PG-13
1	Major Dundee	14873	14873	3800000	1965-04-07	PG/PG-13
2	The Informers	315000	315000	18000000	2009-04-24	R
3	Buffalo Soldiers	353743	353743	15000000	2003-07-25	R
4	The Last Sin Eater	388390	388390	2200000	2007-02-09	PG/PG-13

Filtering Data: Use extracted datetime components to filter rows. Aggregation and grouping using datetime components will be covered in future chapters.

```
# Filter rows where the release month is January
january_releases = movie_ratings[movie_ratings['Release Date'].dt.month == 1]
january_releases.head()
```

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
15	Thr3e	1008849	1060418	2400000	2007-01-05	PG/PG-13
57	Impostor	6114237	6114237	40000000	2002-01-04	PG/PG-13

	Title	US Gross	Worldwide Gross	Production Budget	Release Date	MPAA Rating
62	The Last Station	6616974	6616974	18000000	2010-01-15	R
63	The Big Bounce	6471394	6626115	50000000	2004-01-30	PG/PG-13
84	Not Easily Broken	10572742	10572742	5000000	2009-01-09	PG/PG-13

8.1.5 Working with `object` Data

In pandas, the `object` data type is a flexible data type that can store a mix of text (strings), mixed types, or arbitrary Python objects. It is commonly used for string data and is a key part of working with categorical or unstructured data in pandas.

Similar to datetime objects having a `dt` accessor, a number of specialized string methods are available when using the `str` accessor. These methods have in general matching names with the equivalent built-in string methods for single elements, but are applied element-wise on each of the values of the columns.

8.1.5.1 the `str` accessor in pandas

The `str` accessor in pandas provides a wide range of string methods that allow for efficient and convenient text processing on an entire Series of strings. Here are some commonly used str methods:

- String splitting: `str.split()`
- String joining: `str.join()`
- Substrings: `str.slice(start, stop)`, `str[0]`
- String Case Conversion: `str.lower()`, `str.upper()`, `str.capitalize()`
- Whitespace Removal: `str.strip()`, `str.lstrip()`, `str.rstrip()`
- Replacing and Removing: `str.replace('old', 'new')`
- Pattern matching and extraction: `str.contains('pattern')`, `startswith('prefix')`, `endswith('suffix')`
- String length and counting: `str.len()`, `str.count()`

Let's use the well-known titanic dataset to illustrate how to manipulate string columns in pandas dataframe next

```
titanic = pd.read_csv('./Datasets/titanic.csv')
titanic.head()
```

	PassengerId	Survived	Pclass	Name	Age	SibSp	Par
0	1	0	3	Braund, Mr. Owen Harris	22.0	1	0

	PassengerId	Survived	Pclass	Name	Age	SibSp	Par
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0
2	3	1	3	Heikkinen, Miss. Laina	26.0	0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0
4	5	0	3	Allen, Mr. William Henry	35.0	0	0

```
titanic.Name
```

```
0                                  Braund, Mr. Owen Harris
1      Cumings, Mrs. John Bradley (Florence Briggs Th...
2                                   Heikkinen, Miss. Laina
3           Futrelle, Mrs. Jacques Heath (Lily May Peel)
4                                 Allen, Mr. William Henry
                             ...
886                                  Montvila, Rev. Juozas
887                           Graham, Miss. Margaret Edith
888               Johnston, Miss. Catherine Helen "Carrie"
889                                  Behr, Mr. Karl Howell
890                                    Dooley, Mr. Patrick
Name: Name, Length: 891, dtype: object
```

The `Name` column varies in length and contains passengers' last names, titles, and first names. By extracting the title from the `Name` column, we could infer the **sex** of the passenger and add it as a new feature. This could potentially be a significant predictor of survival, as the "ladies first" principle was often applied during the Titanic evacuation. Adding this feature may enhance the model's ability to predict whether a passenger survived.

```
# Let's check the length of the name of each passenger
titanic["Name"].str.len()
```

```
0      23
1      51
2      22
3      44
4      24
       ..
886    21
887    28
888    40
889    21
```

```
890    19
Name: Name, Length: 891, dtype: int64
```

```python
# check what is the maximum length of the name
titanic.loc[titanic["Name"].str.len().idxmax(), "Name"]
```

```
'Penasco y Castellana, Mrs. Victor de Satode (Maria Josefa Perez de Soto y Vallejo)'
```

```python
# get the observations that contains the word 'Mrs'
titanic[titanic.Name.str.contains('Mrs.')]
```

	PassengerId	Survived	Pclass	Name	Age	SibSp
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	27.0	0
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	14.0	1
15	16	1	2	Hewlett, Mrs. (Mary D Kingcome)	55.0	0
...	...	...	...	...	...	...
871	872	1	1	Beckwith, Mrs. Richard Leonard (Sallie Monypeny)	47.0	1
874	875	1	2	Abelson, Mrs. Samuel (Hannah Wizosky)	28.0	1
879	880	1	1	Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)	56.0	0
880	881	1	2	Shelley, Mrs. William (Imanita Parrish Hall)	25.0	0
885	886	0	3	Rice, Mrs. William (Margaret Norton)	39.0	0

```python
# get the observations that contains the word 'Mrs'
titanic.Name.str.count('Mrs.').sum()
```

```
129
```

```python
# split the name column into two columns
titanic.Name.str.split(',', expand=True)
```

	0	1
0	Braund	Mr. Owen Harris
1	Cumings	Mrs. John Bradley (Florence Briggs Thayer)
2	Heikkinen	Miss. Laina
3	Futrelle	Mrs. Jacques Heath (Lily May Peel)

	0	1
4	Allen	Mr. William Henry
...	...	...
886	Montvila	Rev. Juozas
887	Graham	Miss. Margaret Edith
888	Johnston	Miss. Catherine Helen ”Carrie”
889	Behr	Mr. Karl Howell
890	Dooley	Mr. Patrick

```
# get the last part of the split
titanic.Name.str.split(',', expand=True).get(1)
```

```
0                                      Mr. Owen Harris
1          Mrs. John Bradley (Florence Briggs Thayer)
2                                          Miss. Laina
3                  Mrs. Jacques Heath (Lily May Peel)
4                                    Mr. William Henry
                            ...
886                                        Rev. Juozas
887                               Miss. Margaret Edith
888                     Miss. Catherine Helen "Carrie"
889                                    Mr. Karl Howell
890                                        Mr. Patrick
Name: 1, Length: 891, dtype: object
```

Create a new column `Title` that contains the title of the passengers

```
# from the name, extract the title
titanic['Title'] = titanic.Name.apply(lambda x: x.split(',')[1].split('.')[0].strip())
titanic['Title']
```

```
0         Mr
1        Mrs
2       Miss
3        Mrs
4         Mr
       ...
886      Rev
887     Miss
888     Miss
```

```
889       Mr
890       Mr
Name: Title, Length: 891, dtype: object
```

```
# get the unique titles
titanic.Title.unique()
```

```
array(['Mr', 'Mrs', 'Miss', 'Master', 'Don', 'Rev', 'Dr', 'Mme', 'Ms',
       'Major', 'Lady', 'Sir', 'Mlle', 'Col', 'Capt', 'the Countess',
       'Jonkheer'], dtype=object)
```

Let's create a mapping dictionary next

```
title_sex_mapping = {
    'Mr': 'Male',
    'Mrs': 'Female',
    'Miss': 'Female',
    'Master': 'Male',
    'Don': 'Male',
    'Rev': 'Male',
    'Dr': 'Male',  # Assumed to be Male unless you have additional context
    'Mme': 'Female',
    'Ms': 'Female',
    'Major': 'Male',
    'Lady': 'Female',
    'Sir': 'Male',
    'Mlle': 'Female',
    'Col': 'Male',
    'Capt': 'Male',
    'the Countess': 'Female',
    'Jonkheer': 'Male'
}
```

```
titanic['Sex'] = titanic['Title'].map(title_sex_mapping)
titanic.head()
```

	PassengerId	Survived	Pclass	Name	Age	SibSp	Par
0	1	0	3	Braund, Mr. Owen Harris	22.0	1	0
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0
2	3	1	3	Heikkinen, Miss. Laina	26.0	0	0

	PassengerId	Survived	Pclass	Name	Age	SibSp	Par
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0
4	5	0	3	Allen, Mr. William Henry	35.0	0	0

```
titanic.Sex.value_counts()
```

```
Sex
Male      578
Female    313
Name: count, dtype: int64
```

```
# cross tabulation
pd.crosstab(titanic.Survived, titanic.Sex)
```

Sex Survived	Female	Male
0	81	468
1	232	110

In the Titanic disaster, more women survived than men due to the social norms and evacuation protocols followed during the sinking. The principle of "women and children first" was enforced when lifeboats were being loaded. Since there were not enough lifeboats for everyone on board, priority was given to women and children, which contributed to the higher survival rate among females compared to males.

8.1.5.2 the `re` module in Python is used for regular expression

The `re` module in Python is used for **regular expressions**, which are powerful tools for text analysis and manipulation. Regular expressions allow you to search, match, and manipulate strings based on specific patterns.

Common Use Cases of `re` in String Text Analysis

- Finding patterns in text: `re.search(r'\d{4}-\d{2}-\d{2}', text)` # searches for a date in the format YYYY-MM-DD
- Extracting Specific parts of a string: `re.findall(r'\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z|a-z]` `text)` #extracts email addresses from the text.
- Replacing Parts of a String: `re.sub(r'\$\d+', '[price]', text)` #replacing any price formatted as $ with [price].

8.1.5.2.1 Commonly Used Patterns in `re`

- `\d`: Matches any digit (0-9).
- `\w`: Matches any alphanumeric character (letters and numbers).
- `\s`: Matches any whitespace character (spaces, tabs, newlines).
- `[a-z]`: Matches any lowercase letter from `a` to `z`.
- `[A-Z]`: Matches any uppercase letter from `A` to `Z`.
- `*`: Matches 0 or more occurrences of the preceding character.
- `+`: Matches 1 or more occurrences of the preceding character.
- `?`: Matches 0 or 1 occurrence of the preceding character.
- `^`: Matches the beginning of a string.
- `$`: Matches the end of a string.
- `|`: Acts as an OR operator.

```
data = pd.read_html('https://en.wikipedia.org/wiki/List_of_Chicago_Bulls_seasons')
ChicagoBulls = data[2]
ChicagoBulls.head()
```

	Season	Team	Conference	Finish	Division	Finish.1	Wins	Losses	Win%	GB	Playoffs
0	1966–67	1966–67	—	—	Western	4th	33	48	0.407	11	Lost Div
1	1967–68	1967–68	—	—	Western	4th	29	53	0.354	27	Lost Div
2	1968–69	1968–69	—	—	Western	5th	33	49	0.402	22	NaN
3	1969–70	1969–70	—	—	Western	3rd[c]	39	43	0.476	9	Lost Div
4	1970–71	1970–71	Western	3rd	Midwest[d]	2nd	51	31	0.622	2	Lost con

```
# remove all charaters between box brackets inluding the brackets themselves in the columns l
import re
def remove_brackets(x):
    return re.sub(r'\[.*?\]', '', x)

# Apply the function to each column separately using map
ChicagoBulls['Division'] = ChicagoBulls['Division'].map(remove_brackets)
ChicagoBulls['Finish'] = ChicagoBulls['Finish'].map(remove_brackets)
ChicagoBulls['Finish.1'] = ChicagoBulls['Finish.1'].map(remove_brackets)

ChicagoBulls.head()
```

	Season	Team	Conference	Finish	Division	Finish.1	Wins	Losses	Win%	GB	Playoffs
0	1966–67	1966–67	—	—	Western	4th	33	48	0.407	11	Lost Divisio

	Season	Team	Conference	Finish	Division	Finish.1	Wins	Losses	Win%	GB	Playoffs
1	1967–68	1967–68	—	—	Western	4th	29	53	0.354	27	Lost Divisio
2	1968–69	1968–69	—	—	Western	5th	33	49	0.402	22	NaN
3	1969–70	1969–70	—	—	Western	3rd	39	43	0.476	9	Lost Divisio
4	1970–71	1970–71	Western	3rd	Midwest	2nd	51	31	0.622	2	Lost confere

Another example

```
gdp_data = pd.read_html("https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_pe

gdp_data.head()
```

	Country/Territory	IMF[4][5]		World Bank[6]		United Nations[7]	
	Country/Territory	Estimate	Year	Estimate	Year	Estimate	Year
0	Monaco	—	—	240862	2022	240535	2022
1	Liechtenstein	—	—	187267	2022	197268	2022
2	Luxembourg	135321	2024	128259	2023	125897	2022
3	Bermuda	—	—	123091	2022	117568	2022
4	Switzerland	106098	2024	99995	2023	93636	2022

```
# drop the year column
gdp_data.drop(columns=["Year"], level=1, inplace=True, axis=1)

# drop the level 1 column
gdp_data = gdp_data.droplevel(1, axis=1)
```

```
column_name_cleaner = lambda x:re.split(r'\[', x)[0]

gdp_data.columns = gdp_data.columns.map(column_name_cleaner)

gdp_data.head()
```

	Country/Territory	IMF	World Bank	United Nations
0	Monaco	—	240862	240535
1	Liechtenstein	—	187267	197268
2	Luxembourg	135321	128259	125897

	Country/Territory	IMF	World Bank	United Nations
3	Bermuda	—	123091	117568
4	Switzerland	106098	99995	93636

8.1.5.3 the NLTK Library for NLP (skipped)

NLTK (Natural Language Toolkit) is a popular Python library for natural language processing (NLP) and text analysis. It provides a wide range of tools and resources for processing and analyzing human language data. NLTK is widely used in research, education, and industry for various text analysis tasks.

8.1.5.3.1 Key Features and Capabilities of NLTK

- **Tokenization**: Splits text into individual words (word tokenization) or sentences (sentence tokenization).
- **Stop Word Removal**: Provides lists of common words (like "and", "the", "is") in various languages that can be removed from text to reduce noise.
- **Stemming**: Reduces words to their root form (e.g., "running" to "run").
- **Lemmatization**: Similar to stemming, but more sophisticated. It reduces words to their dictionary form using vocabulary and morphological analysis (e.g., "better" to "good").

8.1.6 Working with `numerical` Data

8.1.6.1 Summary statistics across rows/columns in Pandas

The Pandas DataFrame class has functions such as `sum()` and `mean()` to compute sum over rows or columns of a DataFrame.

By default, functions like `mean()` and `sum()` compute the statistics for each column (i.e., all rows are aggregated) in the DataFrame. Let us compute the mean of all the numeric columns of the data:

```
movie_ratings.describe()
```

	US Gross	Worldwide Gross	Production Budget	IMDB Rating	IMDB Votes
count	2.228000e+03	2.228000e+03	2.228000e+03	2228.000000	2228.000000
mean	5.076370e+07	1.019370e+08	3.816055e+07	6.239004	33585.154847
std	6.643081e+07	1.648589e+08	3.782604e+07	1.243285	47325.651561
min	0.000000e+00	8.840000e+02	2.180000e+02	1.400000	18.000000

	US Gross	Worldwide Gross	Production Budget	IMDB Rating	IMDB Votes
25%	9.646188e+06	1.320737e+07	1.200000e+07	5.500000	6659.250000
50%	2.838649e+07	4.266892e+07	2.600000e+07	6.400000	18169.000000
75%	6.453140e+07	1.200000e+08	5.300000e+07	7.100000	40092.750000
max	7.601676e+08	2.767891e+09	3.000000e+08	9.200000	519541.000000

```
# select the numeric columns
movie_ratings.mean(numeric_only=True)
```

```
US Gross                   5.076370e+07
Worldwide Gross            1.019370e+08
Production Budget          3.816055e+07
IMDB Rating                6.239004e+00
IMDB Votes                 3.358515e+04
Release Year               2.002005e+03
ratio_wgross_by_budget     1.259483e+01
dtype: float64
```

Using the `axis` parameter:

The `axis` parameter controls whether to compute the statistic across rows or columns: * The argument `axis=0`(deafult) denotes that the mean is taken over all the rows of the DataFrame. * For computing a statistic across column the argument `axis=1` will be used.

If mean over a subset of columns is desired, then those column names can be subset from the data.

For example, let us compute the mean IMDB rating, and mean IMDB votes of all the movies:

```
movie_ratings[['IMDB Rating','IMDB Votes']].mean(axis = 0)
```

```
IMDB Rating        6.239004
IMDB Votes     33585.154847
dtype: float64
```

Pandas `sum` function

```
data = [[10, 18, 11], [13, 15, 8], [9, 20, 3]]
df = pd.DataFrame(data )
df
```

	0	1	2
0	10	18	11
1	13	15	8
2	9	20	3

```
# By default, the sum method adds values accross rows and returns the sum for each column
df.sum()
```

```
0    32
1    53
2    22
dtype: int64
```

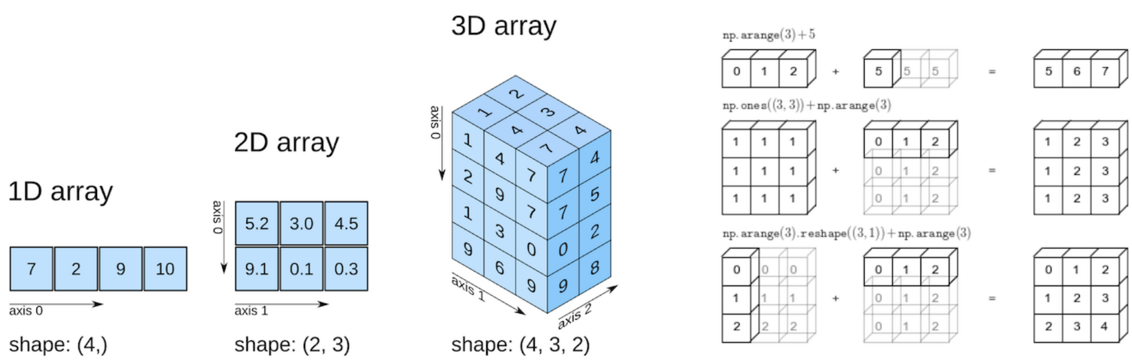
```
# By specifying the column axis (axis='columns'), the sum() method add values accross columns
df.sum(axis = 'columns')
```

```
0    39
1    36
2    32
dtype: int64
```

```
# in python, axis=1 stands for column, while axis=0 stands for rows
df.sum(axis = 1)
```

```
0    39
1    36
2    32
dtype: int64
```

9 NumPy Fundamentals



NumPy is a foundational library in Python, providing support for large, multi-dimensional arrays and matrices, along with a variety of mathematical functions. It's a critical tool in data science and machine learning because it enables efficient numerical computations, data manipulation, and linear algebra operations. Many data science packcages and machine learning algorithms rely on these operations to process data and perform complex calculations quickly. Moreover, popular libraries like `Pandas`, `SciPy`, and `TensorFlow` are built on top of NumPy, making it essential to understand for implementing and optimizing machine learning models.

9.1 Learning Objectives

By the end of this tutorial, you will be able to:

9.1.1 Foundation Skills

- **Create NumPy arrays** using various methods (`np.array()`, `np.zeros()`, `np.arange()`, etc.)
- **Understand array attributes** (`shape`, `dtype`, `size`, `ndim`) and their importance
- **Master indexing and slicing** for extracting and modifying array elements
- **Implement efficient array reshaping** and concatenation techniques

- **Choose appropriate data types** to minimize memory usage

9.2 Getting Started with NumPy

```python
# Essential imports for numerical computing
# Essential imports for numerical computing
import numpy as np
import pandas as pd
import time
import sys

# Display NumPy version and configuration
print(f"NumPy version: {np.__version__}")
print(f"NumPy is installed at: {np.__file__}")
print("\n  Ready to explore the world of numerical computing!")
```

```
NumPy version: 2.3.3
NumPy is installed at: c:\Users\lsi8012\OneDrive - Northwestern University\FA25\STAT303-1\wo

  Ready to explore the world of numerical computing!
```

If you encounter a `ModuleNotFoundError`, install NumPy using:

```
pip install numpy
```

9.3 Why NumPy?

Think of NumPy arrays as an upgrade to Python's built-in lists and tuples.
You can store data in them and perform the same kinds of computations—but with a big advantage:

- **Memory Savings**: Arrays use a fixed, compact layout in memory, reducing overhead.
- **Efficiency**: NumPy is optimized at the C level, making it far faster than pure Python structures.

- **Scalability**: Designed to handle large datasets smoothly without slowing down.

This is why NumPy is the backbone of scientific computing and data science in Python.

9.3.1 NumPy Arrays are Memory Efficient: Homogeneity & Contiguous Storage

A **NumPy array** stores elements of the **same data type** in **contiguous memory locations**. This contrasts with Python lists, which can hold mixed data types and store elements in scattered memory locations.

Because of this:
- NumPy arrays are **densely packed**, minimizing memory overhead.
- Operations can be executed much faster, since data is laid out predictably in memory.

The result: **lower memory consumption** and **higher performance** when working with large datasets.

The following example compares the memory usage of NumPy arrays with Python lists.

```
import sys

# Create a NumPy array, Python list, and tuple with the same elements
array = np.arange(1000)
py_list = list(range(1000))
py_tuple = tuple(range(1000))

# Calculate memory usage
array_memory = array.nbytes
list_memory = sys.getsizeof(py_list) + sum(sys.getsizeof(item) for item in py_list)
tuple_memory = sys.getsizeof(py_tuple) + sum(sys.getsizeof(item) for item in py_tuple)

# Display the memory usage
memory_usage = {
    "NumPy Array (in bytes)": array_memory,
    "Python List (in bytes)": list_memory,
    "Python Tuple (in bytes)": tuple_memory
}

memory_usage
```

```
{'NumPy Array (in bytes)': 8000,
 'Python List (in bytes)': 36056,
 'Python Tuple (in bytes)': 36040}
```

```
# each element in the array is a 64-bit integer
array.dtype
```

```
dtype('int64')
```

9.3.2 NumPy arrays are fast

NumPy arrays enable mathematical computations to run much faster than with native Python data structures. This speed advantage comes from three key factors:

- **Densely Packed & Homogeneous Data**
  Since NumPy arrays store elements of the same type in contiguous memory, data retrieval is faster and computations can be executed more efficiently.
- **Vectorized Computations**
  NumPy replaces slow Python `for` loops with **vectorized operations**, which are internally broken into optimized fragments and executed in parallel. This means calculations are applied to entire arrays at once, rather than element by element.
- **Integration with C/C++**
  Under the hood, NumPy is implemented in **C and C++**, which are compiled languages with very low execution overhead compared to Python. This gives NumPy both the speed of C and the usability of Python.

We'll see the faster speed on NumPy computations in the example below.

```
def my_dot(a, b):
    """
   Compute the dot product of two vectors

    Args:
      a (ndarray (n,)):  input vector
      b (ndarray (n,)):  input vector with same dimension as a

    Returns:
      x (scalar):
    """
    x=0
    for i in range(a.shape[0]):
        x = x + a[i] * b[i]
    return x
```

```
np.random.seed(1)
a = np.random.rand(10000000)  # very large arrays
b = np.random.rand(10000000)

tic = time.time()  # capture start time
c = np.dot(a, b)
toc = time.time()  # capture end time
```

```python
print(f"np.dot(a, b) =  {c:.4f}")
print(f"Vectorized version duration: {1000*(toc-tic):.4f} ms ")

tic = time.time()  # capture start time
c = my_dot(a,b)
toc = time.time()  # capture end time

print(f"my_dot(a, b) =  {c:.4f}")
print(f"loop version duration: {1000*(toc-tic):.4f} ms ")

del(a);del(b)
```

```
np.dot(a, b) =  2501072.5817
Vectorized version duration: 0.0000 ms
my_dot(a, b) =  2501072.5817
loop version duration: 1695.4978 ms
```

Both methods produce the same result, but the vectorized `np.dot()` executes almost instantly, while the manual Python loop takes over a second.

This clearly demonstrates how NumPy's **vectorization** and **C backend** make computations orders of magnitude faster than pure Python loops.

9.4 Building Blocks: NumPy Arrays Fundamentals

9.4.1 Array Creation: Your Complete Toolkit

NumPy offers multiple ways to create arrays, each optimized for different scenarios.

9.4.1.1 From Existing Data

Method	Purpose	Example Use Case
`np.array()`	Convert lists/tuples to arrays	Transform Python data structures
`np.asarray()`	Convert input to array (no copy if already array)	Ensure input is array format
`df.to_numpy()`	Convert a Pandas DataFrame	Bridge between Pandas and NumPy

```python
# 1  Creating arrays from existing data
print("1  FROM EXISTING DATA")
print("=" * 30)

# From Python lists
data_1d = [1, 2, 3, 4, 5]
arr_from_list = np.array(data_1d)
print(f"From list: {arr_from_list}")

# From nested lists (2D array)
data_2d = [[1, 2, 3], [4, 5, 6]]
arr_2d = np.array(data_2d)
print(f"2D array:\n{arr_2d}")

# From tuples
arr_from_tuple = np.array((10, 20, 30))
print(f"From tuple: {arr_from_tuple}")
```

```
1  FROM EXISTING DATA
==============================
From list: [1 2 3 4 5]
2D array:
[[1 2 3]
 [4 5 6]]
From tuple: [10 20 30]
```

9.4.1.2 Specialized Constructors

Method	Creates	When to Use
`np.zeros(shape)`	Array of zeros	Initialize arrays, placeholders
`np.ones(shape)`	Array of ones	Mathematical operations, masks
`np.full(shape, val)`	Array filled with a value	Default values, initialization
`np.eye(n)`	Identity matrix	Linear algebra operations
`np.empty(shape)`	Uninitialized array	Maximum speed (use with caution)

Pro Tip:
- Use `zeros()` for safe initialization
- Use `arange()` or `linspace()` for sequences
- Use `random` methods for simulations

```python
print("\n2  SPECIALIZED CONSTRUCTORS")
print("=" * 30)
# Arrays of zeros and ones
zeros_3x3 = np.zeros((3, 3))
ones_2x4 = np.ones((2, 4))
full_matrix = np.full((2, 3), 7)  # Fill with custom value

print(f"Zeros (3x3):\n{zeros_3x3}")
print(f"Ones (2x4):\n{ones_2x4}")
print(f"Full of 7s:\n{full_matrix}")

# Identity matrix
identity = np.eye(3)
print(f"Identity matrix:\n{identity}")
```

```
2  SPECIALIZED CONSTRUCTORS
==============================
Zeros (3x3):
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
Ones (2x4):
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]]
Full of 7s:
[[7 7 7]
 [7 7 7]]
Identity matrix:
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

9.4.1.3 Sequential & Mathematical Arrays

Method	Purpose	Best For
`np.arange(start, stop, step)`	Evenly spaced integers	Index generation, iteration

Method	Purpose	Best For
`np.linspace(start, stop, num)`	Evenly spaced floats	Plotting, function evaluation
`np.logspace(start, stop, num)`	Logarithmically spaced	Scientific computing, exponential data

```python
print("\n3  SEQUENTIAL ARRAYS")
print("=" * 30)

# Using arange (like Python's range)
sequence1 = np.arange(10)           # 0 to 9
sequence2 = np.arange(5, 15)        # 5 to 14
sequence3 = np.arange(0, 10, 2)     # 0, 2, 4, 6, 8

print(f"arange(10): {sequence1}")
print(f"arange(5, 15): {sequence2}")
print(f"arange(0, 10, 2): {sequence3}")

# Using linspace (evenly spaced floats)
linear = np.linspace(0, 1, 5)       # 5 points between 0 and 1
print(f"linspace(0, 1, 5): {linear}")
```

```
3  SEQUENTIAL ARRAYS
==============================
arange(10): [0 1 2 3 4 5 6 7 8 9]
arange(5, 15): [ 5  6  7  8  9 10 11 12 13 14]
arange(0, 10, 2): [0 2 4 6 8]
linspace(0, 1, 5): [0.   0.25 0.5  0.75 1.  ]
```

9.4.1.4 Random Data Generation

Method	Distribution	Use Case
`np.random.rand()`	Uniform [0, 1)	Simulations, sampling
`np.random.randn()`	Standard normal	Statistical modeling
`np.random.randint()`	Random integers	Discrete simulations
`np.random.choice()`	Random sample from array	Bootstrapping, resampling

```python
print("\n4 RANDOM ARRAYS")
print("=" * 30)

# Set random seed for reproducibility
np.random.seed(42)

# Different types of random arrays
random_uniform = np.random.rand(3, 3)           # Uniform [0, 1)
random_normal = np.random.randn(2, 4)           # Standard normal
random_integers = np.random.randint(1, 10, (3, 3))  # Random integers

print(f"Random uniform (3x3):\n{random_uniform}")
print(f"Random normal (2x4):\n{random_normal}")
print(f"Random integers 1-9:\n{random_integers}")
```

```
4 RANDOM ARRAYS
==============================
Random uniform (3x3):
[[0.37454012 0.95071431 0.73199394]
 [0.59865848 0.15601864 0.15599452]
 [0.05808361 0.86617615 0.60111501]]
Random normal (2x4):
[[-0.58087813 -0.52516981 -0.57138017 -0.92408284]
 [-2.61254901  0.95036968  0.81644508 -1.523876  ]]
Random integers 1-9:
[[3 7 4]
 [9 3 5]
 [3 7 5]]
```

9.4.1.5 Loading from Files

Method	File Type	Features
`np.load()`	NumPy binary `.npy`	Fast saving/loading of arrays
`np.loadtxt()`	Simple text files	Fast, lightweight parsing
`np.genfromtxt()`	Complex text files	Handles missing values and mixed data types

9.4.2 Understanding Array Attributes

Let us define a NumPy array in order to access its attributes:

```
numpy_ex = np.array([[1,2,3],[4,5,6]])
numpy_ex
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
type(numpy_ex)
```

```
numpy.ndarray
```

It is an **ndarray** type.

You can explore the attributes and methods of `numpy_ex` by typing:

```
numpy_ex.
```

and then pressing the *tab* key.

Some of the basic attributes of a NumPy array are the following:

9.4.2.1 `ndim`

Shows the number of dimensions (or axes) of the array.

```
numpy_ex.ndim
```

```
2
```

Numpy arrays can have any number of dimensions and different lengths along each dimension. We can inspect the length along each dimension using the `.shape` property of an array.

9.4.2.2 `shape`

This is a tuple of integers indicating the size of the array in each dimension. For a matrix with n rows and m columns, the shape will be *(n,m)*. The length of the shape tuple is therefore the rank, or the number of dimensions, `ndim`.

```
numpy_ex.shape
```

```
(2, 3)
```

9.4.2.3 `size`

This is the total number of elements of the array, which is the product of the elements of shape.

```
numpy_ex.size
```

```
6
```

9.4.2.4 `dtype`

Unlike lists and tuples, NumPy arrays are designed to store elements of the same type, enabling more efficient memory usage and faster computations. The data type of the elements in a NumPy array can be accessed using the `.dtype` attribute

```
numpy_ex.dtype
```

```
dtype('int64')
```

9.5 Data Types and Memory Optimization

NumPy supports a wide range of data types, each with a defined memory size. These types map directly to **C data types**, since NumPy is implemented in C at its computational core. This design allows NumPy to handle array operations far more efficiently than native Python data structures.

Data Type	Memory Size

9.5.1 Common NumPy Data Types

Data Type	Memory Size
`np.int8`	1 byte
`np.int16`	2 bytes
`np.int32`	4 bytes
`np.int64`	8 bytes
`np.uint8`	1 byte
`np.uint16`	2 bytes
`np.uint32`	4 bytes
`np.uint64`	8 bytes
`np.float16`	2 bytes
`np.float32`	4 bytes
`np.float64`	8 bytes
`np.complex64`	8 bytes
`np.complex128`	16 bytes
`np.bool_`	1 byte
`np.string_`	1 byte per character
`np.unicode_`	4 bytes per character
`np.object_`	Variable (Python objects)
`np.datetime64`	8 bytes
`np.timedelta64`	8 bytes

Tip: Choosing the right data type is crucial for **memory efficiency** and **performance**. For large datasets, using smaller types (e.g., `int16` instead of `int64`) can save significant memory.

```
# Add practical examples for data type selection
print("  CHOOSING THE RIGHT DATA TYPE")
print("=" * 40)

# Memory efficiency example
large_numbers = np.array([1000, 2000, 3000], dtype=np.int16)  # Efficient for small ranges
small_numbers = np.array([1, 2, 3], dtype=np.int8)           # Very memory efficient

print(f"int16 array memory: {large_numbers.nbytes} bytes")
print(f"int8 array memory: {small_numbers.nbytes} bytes")
```

```
# Precision example
high_precision = np.array([3.14159265359], dtype=np.float64)
low_precision = np.array([3.14159265359], dtype=np.float32)
print(f"float64 precision: {high_precision}")
print(f"float32 precision: {low_precision}")
```

```
  CHOOSING THE RIGHT DATA TYPE
========================================
int16 array memory: 6 bytes
int8 array memory: 3 bytes
float64 precision: [3.14159265]
float32 precision: [3.1415927]
```

9.5.2 Upcasting

When you create a NumPy array with elements of different data types, NumPy automatically **upcasts** them to a common type that can represent all values.
This process—also called **type coercion** or **type promotion**—follows a hierarchy of data types to prevent data loss.

Below are two common cases of upcasting, with examples:

Numeric Upcasting: If you mix integers and floats, NumPy will convert the entire array to floats.

```
arr = np.array([1, 2.5, 3])
print(arr.dtype)
```

```
float64
```

String Upcasting: If you mix numbers and strings, NumPy will upcast all elements to strings.

```
arr = np.array([1, 'hello', 3.5])
print(arr.dtype)
```

```
<U32
```

9.6 Array Indexing and Slicing: Accessing Your Data

9.6.1 Array Indexing

Similar to Python lists, NumPy uses zero-based indexing, meaning the first element of an array is accessed using index `0`. You can use positive or negative indices to access elements

```
array = np.array([10, 20, 30, 40, 50])

print(array[0])
print(array[4])
print(array[-1])
print(array[-3])
```

```
10
50
50
30
```

In multi-dimensional arrays, indices are separated by commas. The first index refers to the row, and the second index refers to the column in a 2D array.

```
# 2D array (3 rows, 3 columns)
array_2d = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(array_2d)
print(array_2d[0, 1])
print(array_2d[1, -1])
print(array_2d[-1, -1])
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
2
6
9
```

9.6.2 Array Slicing

Slicing is used to extract a sub-array from an existing array.

The Syntax for slicing is ‘array[start:stop:step]

```python
array = np.array([10, 20, 30, 40, 50])

print(array[1:4])
print(array[:3])
print(array[2:])
print(array[::2])
print(array[::-1])
```

```
[20 30 40]
[10 20 30]
[30 40 50]
[10 30 50]
[50 40 30 20 10]
```

For slicing in Multi-Dimensional Arrays, use commas to separate slicing for different dimensions

```python
###  Indexing & Slicing Quick Reference

# Extract a sub-array: elements from the first two rows and the first two columns
sub_array = array_2d[:2, :2]
print(sub_array)

# Extract all rows for the second column
col = array_2d[:, 1]
print(col)

# Extract the last two rows and last two columns
sub_array = array_2d[-2:, -2:]
print(sub_array)
```

```
NameError: name 'array_2d' is not defined
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[88], line 4
      1 ###  Indexing & Slicing Quick Reference
      2
      3 # Extract a sub-array: elements from the first two rows and the first two columns
----> 4 sub_array = array_2d[:2, :2]
      5 print(sub_array)
      7 # Extract all rows for the second column
NameError: name 'array_2d' is not defined
```

The `step` parameter can be used to select elements at regular intervals.

```
###  Conditional Selection with `np.where()` and `np.select()`

print(array[1:8:2])
print(array[::-2])
```

```
[1 3 5 7]
[9 7 5 3 1]
```

9.6.3 Combining Indexing and Slicing

You can combine indexing and slicing to extract specific elements or sub-arrays

```
# Create a 3D array
array_3d = np.array([[[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]]])

# Select specific elements and slices
print(array_3d[0, :, 1])  # Output: [2 5] (second element from each row in the first sub-arra
print(array_3d[1, 1, :2])  # Output: [10 11] (first two elements in the last row of the secon
```

```
[2 5]
[10 11]
```

9.6.4 Boolean Mask: Conditional Selection

You can use boolean arrays to filter or select elements based on a condition

```
a = np.array([10, 20, 30, 40, 50])
mask = a > 25                # [False False  True  True  True]
filtered = a[mask]           # [30 40 50]
# one-liner:
a[a > 25]  # Output: [40 50]
```

```
array([30, 40, 50])
```

Use bitwise operators with parentheses: `&` (and), `|` (or), `~` (not), `^` (xor).

```
print(a[(a > 15) & (a < 45)] )
print(a[(a < 20) | (a > 40)]  )
print(a[~(a % 20 == 0)] )
```

```
[20 30 40]
[10 50]
[10 30 50]
```

9.6.5 Indexing & Slicing Quick Reference

Operation	Syntax	Example	Result
Single element	`arr[i]`	`arr[2]`	Element at index 2
Negative indexing	`arr[-i]`	`arr[-1]`	Last element
Basic slice	`arr[start:stop]`	`arr[1:4]`	Elements 1, 2, 3
Step slice	`arr[start:stop:step]`	`arr[::2]`	Every 2nd element
2D indexing	`arr[row, col]`	`arr[0, 1]`	Row 0, Column 1
Boolean mask	`arr[condition]`	`arr[arr > 5]`	Elements > 5

9.7 Advanced Selection Methods

9.7.1 Conditional Selection with `np.where()` and `np.select()`

Use these NumPy tools for fast, vectorized **if/else** logic on arrays—perfect for data cleaning, feature engineering, and conditional transforms.

What you can do - **Replace values** based on a condition
- **Choose between arrays** element-wise
- **Apply multi-way logic** (if/elif/else) without loops

When to use which - `np.where(condition, x, y)`: simple **two-way** branching (or `np.where(condition)` to get indices). - `np.select([cond1, cond2, ...], [choice1, choice2, ...], default=...)`: clean **multi-branch** logic (3+ cases).

9.7.2 `np.where()`

- **Two forms**
  - `np.where(condition)` → returns the **indices** where `condition` is `True`.

- `np.where(condition, x, y)` → **element-wise choose**: pick from `x` where `condition` is `True`, else from `y`.

- **Chainable**
  - You can **nest** multiple `np.where()` calls, but prefer `np.select` for **3+ branches**.

```python
# Let's create sample data for our examples
ages = np.array([22, 25, 19, 35, 28, 24, 31, 26, 23, 29])
print(f"Student ages: {ages}")

scores = np.array([95, 87, 76, 63, 89, 92, 45, 78, 85, 91])
print(f"Student scores: {scores}")
```

```
Student ages: [22 25 19 35 28 24 31 26 23 29]
Student scores: [95 87 76 63 89 92 45 78 85 91]
```

```python
print("=" * 50)
print("  USING np.where()")
print("=" * 50)

# 1) Simple conditional replacement: scores < 70 -> 0
pass_fail = np.where(scores >= 70, scores, 0)
print("\n1  Pass/Fail (70+ to pass):")
print(f"Original: {scores}")
print(f"Result:   {pass_fail}")

# 2) Get indices (and values) of high scorers
high_idx = np.where(scores > 85)[0]
high_vals = scores[high_idx]
print("\n2  High Scorers (>85):")
print(f"Indices: {high_idx}")
print(f"Values : {high_vals}")
```

```
==================================================
  USING np.where()
==================================================

1  Pass/Fail (70+ to pass):
Original: [95 87 76 63 89 92 45 78 85 91]
Result:   [95 87 76  0 89 92  0 78 85 91]
```

```
2  High Scorers (>85):
Indices: [0 1 4 5 9]
Values : [95 87 89 92 91]
```

```
# Conditional replacement with different values (letter grades)
letter_grades = np.where(scores >= 90, 'A',
                np.where(scores >= 80, 'B',
                np.where(scores >= 70, 'C',
                np.where(scores >= 60, 'D', 'F'))))

# Example 3: Using np.where to find indices
print("\n3  Indices of A's and F's:")
a_idx = np.where(letter_grades == 'A')[0]
f_idx = np.where(letter_grades == 'F')[0]
print(f"A indices: {a_idx}")
print(f"F indices: {f_idx}")
```

```
3  Indices of A's and F's:
A indices: [0 5 9]
F indices: [6]
```

9.7.3 `np.select`: Multi-way Conditional Logic

While you can nest multiple `np.where()` calls for complex conditions, **`np.select()`** is preferred for **3+ branches** because it's more readable and maintainable.

9.7.4 Syntax

```
np.select(conditions, choices, default=...)
```

- `conditions`: list of boolean arrays (must be same shape or broadcastable)
- `choices`: list of result values/arrays (same length as `conditions`)
- `default`: value used where no condition is `True` (optional)

9.7.4.1 Example 1 — Letter grades (cleaner than nested `np.where()`)

```python
scores = np.array([95, 87, 76, 63, 89, 92, 45, 78, 85, 91])
conds   = [scores >= 90, scores >= 80, scores >= 70, scores >= 60]
choices = ['A',          'B',          'C',          'D']
grades  = np.select(conds, choices, default='F')
print("\n4  Letter Grades with np.select():", grades)
print(f"Original Scores: {scores}")
```

```
4  Letter Grades with np.select(): ['A' 'B' 'C' 'D' 'B' 'A' 'F' 'C' 'B' 'A']
Original Scores: [95 87 76 63 89 92 45 78 85 91]
```

9.7.4.2 Example 2 — Numeric binning (labels)

```python
x = np.array([5, 12, 20, 33, 47])
conds = [x < 10, (x >= 10) & (x < 30), x >= 30]
choices = ['low_precision', 'mid_precision', 'high_precision']
buckets = np.select(conds, choices, default='unknown')  # ← Add this!

print(buckets)
```

```
['low_precision' 'mid_precision' 'mid_precision' 'high_precision'
 'high_precision']
```

`np.select` and the default dtype gotcha

If you omit `default=...` in `np.select`, NumPy uses `0` as the fallback.
When your `choices` are **strings**, mixing them with the integer `0` forces a common dtype—strings and integers have no shared dtype—so NumPy raises a `TypeError`.

Example (raises error)

```python
import numpy as np

x = np.array([5, 12, 30])
conds   = [x < 10, (x >= 10) & (x < 20)]
choices = ['low', 'mid']

np.select(conds, choices)   # default is 0 -> TypeError (mixed str + int)
```

Best Practices:

- **Use `np.where()`** for simple binary conditions
- **Use `np.select()`** for 3+ conditions or complex logic
- **Always provide a default** to handle edge cases
- **Test conditions** to ensure they're mutually exclusive when needed

9.7.5 Finding Extremes with `np.argmin()` and `np.argmax()`

Use these to get the **index position(s)** of the minimum/maximum value.
Specify an `axis` for row/column results; omit it for the global position.

```
a = np.array([4, 1, 9, 7])
i_min = np.argmin(a)   # 1
i_max = np.argmax(a)   # 2

print("min @ index", i_min, "value:", a[i_min])
print("max @ index", i_max, "value:", a[i_max])
```

```
min @ index 1 value: 1
max @ index 2 value: 9
```

```
# Create a 2D array (e.g., student test scores)
scores = np.array([
    [85, 92, 78, 95],  # Student 0
    [88, 76, 91, 82],  # Student 1
    [95, 89, 84, 90],  # Student 2
    [72, 85, 79, 88]   # Student 3
])

print("  Student Test Scores (4 students × 4 tests):")
print(scores)
print()

# ============================================
# 1  FLATTENED (no axis) - Returns single index
# ============================================
print("=" * 60)
print("1  FLATTENED VIEW (axis=None) - Single Index")
print("=" * 60)

min_idx = np.argmin(scores)
```

```python
max_idx = np.argmax(scores)

print(f"Lowest score index (flattened):  {min_idx}")
print(f"Highest score index (flattened): {max_idx}")

# Convert flat index to 2D coordinates
min_row, min_col = np.unravel_index(min_idx, scores.shape)
max_row, max_col = np.unravel_index(max_idx, scores.shape)

print(f"\n  Minimum: {scores[min_row, min_col]} at position ({min_row}, {min_col})")
print(f"    → Student {min_row}, Test {min_col}")
print(f"  Maximum: {scores[max_row, max_col]} at position ({max_row}, {max_col})")
print(f"    → Student {max_row}, Test {max_col}")
print()
```

```
 Student Test Scores (4 students × 4 tests):
[[85 92 78 95]
 [88 76 91 82]
 [95 89 84 90]
 [72 85 79 88]]

============================================================
1  FLATTENED VIEW (axis=None) - Single Index
============================================================
Lowest score index (flattened):  12
Highest score index (flattened): 3

  Minimum: 72 at position (3, 0)
    → Student 3, Test 0
  Maximum: 95 at position (0, 3)
    → Student 0, Test 3

============================================================
2  AXIS=0 (down columns) - Which STUDENT performed best/worst?
============================================================
Lowest scoring student per test:  [3 1 0 1]
Highest scoring student per test: [2 0 1 0]

Detailed breakdown:
  Test 0: Worst = Student 3 (72), Best = Student 2 (95)
  Test 1: Worst = Student 1 (76), Best = Student 0 (92)
  Test 2: Worst = Student 0 (78), Best = Student 1 (91)
```

```
  Test 3: Worst = Student 1 (82), Best = Student 0 (95)

============================================================
3  AXIS=1 (across rows) - Which TEST was easiest/hardest?
============================================================
Worst test per student:  [2 1 2 0]
Best test per student:   [3 2 0 3]

Detailed breakdown:
  Student 0: Worst = Test 2 (78), Best = Test 3 (95)
  Student 1: Worst = Test 1 (76), Best = Test 2 (91)
  Student 2: Worst = Test 2 (84), Best = Test 0 (95)
  Student 3: Worst = Test 0 (72), Best = Test 3 (88)
```

```
# ============================================
# 2  AXIS=0 (down columns) - Compare students
# ============================================
print("=" * 60)
print("2  AXIS=0 (down columns) - Which STUDENT performed best/worst?")
print("=" * 60)

min_student_per_test = np.argmin(scores, axis=0)
max_student_per_test = np.argmax(scores, axis=0)

print(f"Lowest scoring student per test:  {min_student_per_test}")
print(f"Highest scoring student per test: {max_student_per_test}")

print("\nDetailed breakdown:")
for test_num in range(scores.shape[1]):
    worst_student = min_student_per_test[test_num]
    best_student = max_student_per_test[test_num]
    print(f"  Test {test_num}: "
          f"Worst = Student {worst_student} ({scores[worst_student, test_num]}), "
          f"Best = Student {best_student} ({scores[best_student, test_num]})")
print()
# ============================================
# 3  AXIS=1 (across rows) - Compare tests
# ============================================
print("=" * 60)
print("3  AXIS=1 (across rows) - Which TEST was easiest/hardest?")
print("=" * 60)
```

```python
min_test_per_student = np.argmin(scores, axis=1)
max_test_per_student = np.argmax(scores, axis=1)

print(f"Worst test per student:  {min_test_per_student}")
print(f"Best test per student:   {max_test_per_student}")

print("\nDetailed breakdown:")
for student_num in range(scores.shape[0]):
    worst_test = min_test_per_student[student_num]
    best_test = max_test_per_student[student_num]
    print(f"  Student {student_num}: "
          f"Worst = Test {worst_test} ({scores[student_num, worst_test]}), "
          f"Best = Test {best_test} ({scores[student_num, best_test]})")
print()
```

```
============================================================
2  AXIS=0 (down columns) - Which STUDENT performed best/worst?
============================================================
Lowest scoring student per test:  [3 1 0 1]
Highest scoring student per test: [2 0 1 0]

Detailed breakdown:
  Test 0: Worst = Student 3 (72), Best = Student 2 (95)
  Test 1: Worst = Student 1 (76), Best = Student 0 (92)
  Test 2: Worst = Student 0 (78), Best = Student 1 (91)
  Test 3: Worst = Student 1 (82), Best = Student 0 (95)

============================================================
3  AXIS=1 (across rows) - Which TEST was easiest/hardest?
============================================================
Worst test per student:  [2 1 2 0]
Best test per student:   [3 2 0 3]

Detailed breakdown:
  Student 0: Worst = Test 2 (78), Best = Test 3 (95)
  Student 1: Worst = Test 1 (76), Best = Test 2 (91)
  Student 2: Worst = Test 2 (84), Best = Test 0 (95)
  Student 3: Worst = Test 0 (72), Best = Test 3 (88)
```

Tips - `argmin`/`argmax` return **indices**, not values (use them to index back into the array). - **Axis behavior** - `axis=None` (default): flattens the entire array and returns the **global** index.

- `axis=0` (2D): compares **down the rows** within each column → returns **row indices per column**. - `axis=1` (2D): compares **across columns** within each row → returns **column indices per row**. - General ND: reduces along the specified axis; the **result shape equals the input shape with that axis removed**. - For values at those positions: - Column-wise: `i = np.argmax(M, axis=0); vals = M[i, np.arange(M.shape[1])]` - Row-wise: `j = np.argmax(M, axis=1); vals = M[np.arange(M.shape[0]), j]` - Convert a flat index to coordinates with `np.unravel_index(idx, arr.shape)`.

9.8 Array Operations: Mathematical Power at Scale

NumPy arrays excel at **vectorized operations** that process entire arrays efficiently. These operations are the foundation of scientific computing and data analysis.

9.8.1 Arithmetic Operations

NumPy arrays support all standard arithmetic operators (`+`, `-`, `*`, `/`, `**`, `%`) and apply them **element-wise**. You can perform operations between:

- **Array and Scalar**: The scalar is applied to every element (broadcasting)
- **Array and Array**: Must have compatible shapes (same shape or broadcastable)

Key Advantage: These operations are **vectorized** - they execute at C speed, not Python loop speed!

9.8.1.1 Comprehensive Arithmetic Examples

Let's explore all arithmetic operations with practical examples:

```
#  Sample Data: Sales figures for 3 stores × 4 quarters
store_sales_q1_q4 = np.array([[120, 150, 180, 200],  # Store A
                              [100, 130, 160, 190],  # Store B
                              [140, 110, 150, 170]]) # Store C

# Bonus amounts for each store and quarter
bonus_amounts = np.array([[10, 15, 20, 25],
                          [12, 18, 22, 28],
                          [8, 12, 18, 22]])

print("  Quarterly Sales (in thousands):")
print("Store  Q1   Q2   Q3   Q4")
```

```python
print("A    ", store_sales_q1_q4[0])
print("B    ", store_sales_q1_q4[1])
print("C    ", store_sales_q1_q4[2])

print("\n  Quarterly Bonuses (in thousands):")
print("Store  Q1   Q2   Q3   Q4")
print("A    ", bonus_amounts[0])
print("B    ", bonus_amounts[1])
print("C    ", bonus_amounts[2])

# For compatibility with existing examples
arr1, arr2 = store_sales_q1_q4, bonus_amounts
```

```
  Quarterly Sales (in thousands):
Store  Q1   Q2   Q3   Q4
A      [120 150 180 200]
B      [100 130 160 190]
C      [140 110 150 170]

  Quarterly Bonuses (in thousands):
Store  Q1   Q2   Q3   Q4
A      [10 15 20 25]
B      [12 18 22 28]
C      [ 8 12 18 22]
```

```python
#Element-wise summation of arrays
arr1 + arr2
```

```
array([[130, 165, 200, 225],
       [112, 148, 182, 218],
       [148, 122, 168, 192]])
```

```python
# Element-wise subtraction
arr2 - arr1
```

```
array([[-110, -135, -160, -175],
       [ -88, -112, -138, -162],
       [-132,  -98, -132, -148]])
```

```python
# Adding a scalar to an array adds the scalar to each element of the array
arr1 + 3
```

```
array([[123, 153, 183, 203],
       [103, 133, 163, 193],
       [143, 113, 153, 173]])
```

```python
# Dividing an array by a scalar divides all elements of the array by the scalar
arr1 / 2
```

```
array([[ 60.,  75.,  90., 100.],
       [ 50.,  65.,  80.,  95.],
       [ 70.,  55.,  75.,  85.]])
```

```python
# Element-wise multiplication
arr1 * arr2
```

```
array([[1200, 2250, 3600, 5000],
       [1200, 2340, 3520, 5320],
       [1120, 1320, 2700, 3740]])
```

```python
# Modulus operator with scalar
arr1 % 4
```

```
array([[0, 2, 0, 0],
       [0, 2, 0, 2],
       [0, 2, 2, 2]])
```

9.8.2 Aggregate Functions: Statistical Summaries

Aggregate functions reduce array dimensions by applying statistical operations. The `axis` parameter controls the direction of aggregation.

9.8.2.1 Global Aggregation (All Elements)

Function	Purpose	Example
`np.sum(arr)`	Sum all elements	`np.sum([[1,2],[3,4]])` → 10
`np.mean(arr)`	Average of all elements	`np.mean([[1,2],[3,4]])` → 2.5
`np.min(arr)`	Minimum value	`np.min([[1,2],[3,4]])` → 1
`np.max(arr)`	Maximum value	`np.max([[1,2],[3,4]])` → 4
`np.std(arr)`	Standard deviation	`np.std([[1,2],[3,4]])` → 1.118

9.8.2.2 Axis-Specific Aggregation

Understanding Axes in 2D Arrays:

- `axis=0`: **Down the rows** (column-wise aggregation) → Result has shape (n_cols,)
- `axis=1`: **Across the columns** (row-wise aggregation) → Result has shape (n_rows,)

Memory Tip: "Axis 0 goes down, Axis 1 goes across"

```python
# Student grades: 3 students × 4 subjects (Math, Science, English, History)
# Create sample data for demonstration
array = np.array([[4, 7, 1, 3],
                  [5, 8, 2, 6],
                  [9, 3, 5, 2]])

# Display the original array
print("Original Array:\n", array)

# Calculate the sum, mean, minimum, and maximum for the entire array
total_sum = np.sum(array)
mean_value = np.mean(array)
min_value = np.min(array)
max_value = np.max(array)

print(f"\nSum of all elements: {total_sum}")
print(f"Mean of all elements: {mean_value}")
print(f"Minimum value in the array: {min_value}")
print(f"Maximum value in the array: {max_value}")
```

```python
# Calculate the sum, mean, minimum, and maximum along each row (axis=1)
row_sum = np.sum(array, axis=1)
row_mean = np.mean(array, axis=1)
row_min = np.min(array, axis=1)
row_max = np.max(array, axis=1)

print("\nSum along each row:", row_sum)
print("Mean along each row:", row_mean)
print("Minimum value along each row:", row_min)
print("Maximum value along each row:", row_max)

# Calculate the sum, mean, minimum, and maximum along each column (axis=0)
col_sum = np.sum(array, axis=0)
col_mean = np.mean(array, axis=0)
col_min = np.min(array, axis=0)
col_max = np.max(array, axis=0)

print("\nSum along each column:", col_sum)
print("Mean along each column:", col_mean)
print("Minimum value along each column:", col_min)
print("Maximum value along each column:", col_max)
```

```
Original Array:
 [[4 7 1 3]
 [5 8 2 6]
 [9 3 5 2]]

Sum of all elements: 55
Mean of all elements: 4.583333333333333
Minimum value in the array: 1
Maximum value in the array: 9

Sum along each row: [15 21 19]
Mean along each row: [3.75 5.25 4.75]
Minimum value along each row: [1 2 2]
Maximum value along each row: [7 8 9]

Sum along each column: [18 18  8 11]
Mean along each column: [6.         6.         2.66666667 3.66666667]
Minimum value along each column: [4 3 1 2]
Maximum value along each column: [9 8 5 6]
```

9.9 Array Reshaping: Transforming Data Dimensions

Array reshaping is essential for preparing data for different computational tasks. Many operations require specific array shapes:

- **Machine Learning**: Models often expect specific input dimensions (e.g., 2D for tabular data, 4D for images)
- **Matrix Operations**: Linear algebra operations require compatible shapes for multiplicationLet's explore the essential tools for transforming array dimensions:
- **Data Analysis**: Different analysis techniques may need data in specific formats
- **Broadcasting**: Reshaping enables efficient element-wise operations between arrays

Key Insight: Reshaping **never changes the data**—only how it's organized in memory!

9.9.1 Core Reshaping Methods

9.9.1.1 `reshape()`: Intelligent Dimension Control

The **`reshape()`** method creates a **new view** of the array with different dimensions. The total number of elements must remain the same.

Syntax: `array.reshape(new_shape)` or `np.reshape(array, new_shape)`

```
print("  RESHAPE() DEMONSTRATIONS")
print("=" * 40)

# Original 1D array (12 elements)
original = np.arange(1, 13)
print(f"Original (1D): {original}")
print(f"Original shape: {original.shape}")

# Reshape to 3D array (2 × 2 × 3)
array_3d = original.reshape(2, 2, 3)
print("\n3D Array (2×2×3):")
print(array_3d)
print(f"Shape: {array_3d.shape}")

# Reshape to 2D matrices
matrix_3x4 = original.reshape(3, 4)
print("\nMatrix (3×4):")
```

```python
print(matrix_3x4)
print(f"Shape: {matrix_3x4.shape}")

matrix_4x3 = original.reshape(4, 3)
print("\nMatrix (4×3):")
print(matrix_4x3)
print(f"Shape: {matrix_4x3.shape}")

matrix_2x6 = original.reshape(2, 6)
print("\nMatrix (2×6):")
print(matrix_2x6)
print(f"Shape: {matrix_2x6.shape}")
```

```
  RESHAPE() DEMONSTRATIONS
========================================
Original (1D): [ 1  2  3  4  5  6  7  8  9 10 11 12]
Original shape: (12,)

3D Array (2×2×3):
[[[ 1  2  3]
  [ 4  5  6]]

 [[ 7  8  9]
  [10 11 12]]]
Shape: (2, 2, 3)

Matrix (3×4):
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
Shape: (3, 4)

Matrix (4×3):
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
Shape: (4, 3)

Matrix (2×6):
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
```

```
Shape: (2, 6)
```

9.9.1.2 Smart Dimension Inference with -1

Use `-1` to let NumPy **automatically calculate** one dimension based on the array size and other specified dimensions.

```python
#  Automatic Dimension Calculation (-1)

data = np.arange(24)  # 24 elements
print(f"Original data: {data}")
print(f"Total elements: {data.size}")

# Use -1 to let NumPy infer exactly one dimension (only one -1 per reshape)

# 4 rows, infer columns -> 4×6
result_1 = data.reshape(4, -1)

# infer rows, 3 columns -> 8×3
result_2 = data.reshape(-1, 3)

# 2 × 3 × ? -> 2×3×4
result_3 = data.reshape(2, 3, -1)

print("\n4×? becomes: 4×{} = {}".format(result_1.shape[1], result_1.shape))
print(result_1)

print("\n?×3 becomes: {}×3 = {}".format(result_2.shape[0], result_2.shape))
print(result_2)

print("\n2×3×? becomes: 2×3×{} = {}".format(result_3.shape[2], result_3.shape))
print(result_3)
```

```
Original data: [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]
Total elements: 24

4×? becomes: 4×6 = (4, 6)
[[ 0  1  2  3  4  5]
 [ 6  7  8  9 10 11]
 [12 13 14 15 16 17]
 [18 19 20 21 22 23]]
```

```
?×3 becomes: 8×3 = (8, 3)
[[ 0  1  2]
 [ 3  4  5]
 [ 6  7  8]
 [ 9 10 11]
 [12 13 14]
 [15 16 17]
 [18 19 20]
 [21 22 23]]

2×3×? becomes: 2×3×4 = (2, 3, 4)
[[[ 0  1  2  3]
  [ 4  5  6  7]
  [ 8  9 10 11]]

 [[12 13 14 15]
  [16 17 18 19]
  [20 21 22 23]]]
```

How `-1` Works: NumPy calculates the missing dimension using the formula:

```
missing_dimension = total_elements ÷ (product_of_known_dimensions)
```

Important: You can only use `-1` for **one dimension** per reshape operation!

9.9.1.3 `flatten()` vs `ravel()`: Converting to 1D

Both convert an array to 1D, but they differ in **copy vs view**, **speed**, and **memory order** handling.

Method	Returns	Copy/View	Speed (typical)	When to Use
`flatten()`	1D `ndarray`	**Always makes a copy**	Slower	You want an independent array you can modify safely
`ravel()`	1D `ndarray`	**View if possible**, else copy (not guaranteed)	Faster	You want a 1D view for efficiency (and accept that edits may affect the original)

Reading order options (both functions support `order=`): - **`'C'` (default)**: Row-major (left→right, top→bottom) - **`'F'`**: Column-major (top→bottom, left→right) - **`'A'`**: `'F'` if the array is Fortran-contiguous, otherwise `'C'` - **`'K'`**: As close as possible to the array's **in-memory layout** (preserves current strides, useful after slicing)

> **Gotcha:** `ravel()` may return a **copy** if a view isn't possible (e.g., non-contiguous slices or mismatched `order`). If you must ensure independence, use `flatten()` or call `.copy()` on the result of `ravel()`.

```
print("  FLATTEN() vs RAVEL() COMPARISON")
print("=" * 45)

# Original 2D array
original_2d = np.array([[1, 2, 3],
                        [4, 5, 6],
                        [7, 8, 9]])

print("Original array:")
print(original_2d)

# Method 1: flatten() - always returns a COPY
flat_copy = original_2d.flatten()

# Method 2: ravel() - returns a VIEW when possible (else a copy)
flat_view = original_2d.ravel()

print("\nResults:")
print(f"flatten() result: {flat_copy}")
print(f"ravel()   result: {flat_view}")

print("\nMemory sharing checks:")
print(f"flatten() shares memory with original? {np.shares_memory(original_2d, flat_copy)}")
print(f"ravel()   shares memory with original? {np.shares_memory(original_2d, flat_view)}")
```

```
  FLATTEN() vs RAVEL() COMPARISON
=============================================
Original array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]

Results:
```

```
flatten() result: [1 2 3 4 5 6 7 8 9]
ravel()   result: [1 2 3 4 5 6 7 8 9]

Memory sharing checks:
flatten() shares memory with original? False
ravel()   shares memory with original? True
```

```
from time import perf_counter

print("\n  READING ORDER DEMONSTRATIONS")
print("=" * 40)

# Small example matrix
matrix = np.array([[1, 2, 3],
                   [4, 5, 6]])
print(f"Matrix:\n{matrix}")

# Reading order: C (row-major) vs F (column-major)
c_order = matrix.flatten(order='C')  # left→right, top→bottom
f_order = matrix.flatten(order='F')  # top→bottom, left→right

print(f"\nC order (row-major): {c_order}")   # [1 2 3 4 5 6]
print(f"F order (col-major): {f_order}")     # [1 4 2 5 3 6]

# (Optional) ravel with order behaves similarly:
# print(matrix.ravel(order='C'))
# print(matrix.ravel(order='F'))

print("\n  Performance Test (1000×1000 array):")
large_array = np.random.rand(1000, 1000)

# Simple timing helper: take the best of a few runs to reduce noise
def best_time(fn, reps=5):
    times = []
    for _ in range(reps):
        t0 = perf_counter()
        _ = fn()
        times.append(perf_counter() - t0)
    return min(times)

flatten_time = best_time(lambda: large_array.flatten(order='K'))
ravel_time   = best_time(lambda: large_array.ravel(order='K'))
```

```
print(f"flatten(): {flatten_time:.6f} seconds")
print(f"ravel()  : {ravel_time:.6f} seconds")
if ravel_time > 0:
    print(f"ravel() is ~{flatten_time / ravel_time:.2f}× {'slower' if flatten_time > ravel_ti
```

```
  READING ORDER DEMONSTRATIONS
========================================
Matrix:
[[1 2 3]
 [4 5 6]]

C order (row-major): [1 2 3 4 5 6]
F order (col-major): [1 4 2 5 3 6]

  Performance Test (1000×1000 array):
flatten(): 0.001257 seconds
ravel()  : 0.000000 seconds
ravel() is ~6284.45× slower than flatten()
```

9.9.1.4 `resize()` — Destructive Reshaping with Size Changes

`ndarray.resize(new_shape, refcheck=True)` changes an array **in place**. It's powerful, but risky:

- **Can change the total number of elements** (unlike `reshape`)
- **Shrinking truncates** data
- **Expanding fills with zeros** (for numeric dtypes)
- **In-place / destructive**: permanently modifies the original array
- **Safety check**: raises `ValueError` if the array **references/is referenced** (use `refcheck=False` to force — unsafe)
- Don't confuse with `np.resize(a, new_shape)` which **returns a new array** and **repeats elements** to fill

```
print("  RESIZE() OPERATIONS")
print("=" * 30)

# ------------------------------
# Example 1: Expanding (pads with zeros)
# ------------------------------
array1 = np.array([1, 2, 3, 4], dtype=int)
```

```python
original_id = id(array1)

print("\n[Example 1] Expand array in-place to shape (2, 4)")
print(f"Before resize: {array1}  | shape={array1.shape}  | id={original_id}")

array1.resize(2, 4)  # 4 -> 8 elements; pads with zeros for numeric dtypes
print(f"After  resize: \n{array1}  | shape={array1.shape}  | id={id(array1)}")
print(f"Same memory object? {id(array1) == original_id}")  # True

# -------------------------------
# Example 2: Shrinking (truncates data)
# -------------------------------
array2 = np.array([[10, 20, 30, 40],
                   [50, 60, 70, 80]], dtype=int)

print("\n[Example 2] Shrink array in-place to shape (1, 3) - data is truncated")
print(f"Before shrinking:\n{array2}  | shape={array2.shape}")

array2.resize(1, 3)  # 8 -> 3 elements; truncates
print(f"After  resize:\n{array2}  | shape={array2.shape}")

# -------------------------------
# Example 3: Safe alternative - reshape (non-destructive)
# -------------------------------
array3 = np.array([1, 2, 3, 4, 5, 6], dtype=int)
print("\n[Example 3] Using reshape (safe, returns a new view/copy without changing size)")
print(f"Original: {array3}  | shape={array3.shape}  | id={id(array3)}")

# Incompatible reshape (different element count) -> raises ValueError
try:
    reshaped = array3.reshape(2, 4)  # 6 -> 8 elements (invalid)
except ValueError as e:
    print(f"reshape(2, 4) error: {e}")

# Compatible reshape
safe_reshape = array3.reshape(2, 3)  # 6 -> 6 elements (valid)
print(f"Safe reshape (2, 3):\n{safe_reshape}  | shape={safe_reshape.shape}")
print(f"Original unchanged: {array3}  | shape={array3.shape}  | id={id(array3)}")

# -------------------------------
# Bonus: np.resize (returns NEW array, repeats elements if needed)
# -------------------------------
```

```python
a = np.array([1, 2, 3])
print("\n[Bonus] np.resize returns a new array (does not modify the original)")
print(f"Original a: {a}  | id={id(a)}")
b = np.resize(a, (2, 5))  # repeats [1,2,3,1,2] in each row to fill
print(f"np.resize(a, (2,5)):\n{b}  | id={id(b)}")
print(f"Original a unchanged: {a}")
```

```
 RESIZE() OPERATIONS
==============================

[Example 1] Expand array in-place to shape (2, 4)
Before resize: [1 2 3 4]  | shape=(4,)  | id=1617695703184
After  resize:
[[1 2 3 4]
 [0 0 0 0]]  | shape=(2, 4)  | id=1617695703184
Same memory object? True

[Example 2] Shrink array in-place to shape (1, 3) - data is truncated
Before shrinking:
[[10 20 30 40]
 [50 60 70 80]]  | shape=(2, 4)
After  resize:
[[10 20 30]]  | shape=(1, 3)

[Example 3] Using reshape (safe, returns a new view/copy without changing size)
Original: [1 2 3 4 5 6]  | shape=(6,)  | id=1617695704432
reshape(2, 4) error: cannot reshape array of size 6 into shape (2,4)
Safe reshape (2, 3):
[[1 2 3]
 [4 5 6]]  | shape=(2, 3)
Original unchanged: [1 2 3 4 5 6]  | shape=(6,)  | id=1617695704432

[Bonus] np.resize returns a new array (does not modify the original)
Original a: [1 2 3]  | id=1617695704528
np.resize(a, (2,5)):
[[1 2 3 1 2]
 [3 1 2 3 1]]  | id=1617695710000
Original a unchanged: [1 2 3]
```

9.9.1.5 `transpose()` and `.T`: Matrix Transposition

Transposition flips an array along its diagonal—**rows become columns** (and vice versa). It's fundamental for linear algebra and data manipulation.

Ways to transpose - `.T`: Shorthand for transposing a 2D array. For N-D arrays, it **reverses the axis order**. - **`np.transpose(a, axes=None)`**: Function form. With `axes`, you can specify any axis permutation. - **`a.transpose(*axes)`**: Method form (equivalent to `np.transpose(a, axes=...)`).

> All of the above return a **view** (no copy) when possible.

Common applications - **Matrix multiplication**: Make shapes compatible (e.g., use `A.T @ B` when `A` is `(m, n)` and `B` is `(m, k)`). - **Data analysis**: Switch between row-major and column-major organization. - **Neural networks**: Align weight tensors and activations. - **Statistics**: Work with covariance/correlation matrices.

```
print("  TRANSPOSE OPERATIONS")
print("=" * 30)

# ------------------------------
# 2D Matrix transposition
# ------------------------------
matrix = np.array([[1, 2, 3, 4],
                   [5, 6, 7, 8]])
print(f"Original (2×4):\n{matrix}")
print(f"Shape: {matrix.shape}")

# Three equivalent ways to transpose
method1 = matrix.T
method2 = matrix.transpose()
method3 = np.transpose(matrix)

print("\nTransposed (4×2) - All methods identical:")
print(method1)
print(f"Shape: {method1.shape}")
print(f"All equal? {np.array_equal(method1, method2) and np.array_equal(method2, method3)}")

# ------------------------------
# Matrix multiplication example
# ------------------------------
print("\n  Matrix Multiplication Example:")
A = np.array([[1, 2],
```

```
               [3, 4]])
B = np.array([[5, 6],
               [7, 8]])

print(f"A (2×2): \n{A}")
print(f"B (2×2): \n{B}")
print(f"A × B:\n{A @ B}")
print(f"A × B.T:\n{A @ B.T}")
print(f"A.T × B:\n{A.T @ B}")

# ------------------------------
# Real-world example: Data conversion (students   subjects)
# ------------------------------
print("\n  PRACTICAL EXAMPLE: Student Grades")

students = ["Alice", "Bob", "Charlie"]
subjects = ["Math", "Stats", "CS"]

# rows = students, cols = subjects
students_data = np.array([
    [85, 92, 78],   # Alice
    [88, 76, 90],   # Bob
    [91, 89, 84],   # Charlie
])

print("By Students (rows = students):")
for i, student in enumerate(students):
    print(f"{student}: {students_data[i]}")

# Transpose to organize by subject (rows)
subjects_data = students_data.T

print("\nBy Subjects (rows = subjects):")
for i, subject in enumerate(subjects):
    print(f"{subject}: {subjects_data[i]}")

print("=" * 40)
```

```
 TRANSPOSE OPERATIONS
==============================
Original (2×4):
[[1 2 3 4]
 [5 6 7 8]]
Shape: (2, 4)

Transposed (4×2) - All methods identical:
[[1 5]
 [2 6]
 [3 7]
 [4 8]]
Shape: (4, 2)
All equal? True

 Matrix Multiplication Example:
A (2×2):
[[1 2]
 [3 4]]
B (2×2):
[[5 6]
 [7 8]]
A × B:
[[19 22]
 [43 50]]
A × B.T:
[[17 23]
 [39 53]]
A.T × B:
[[26 30]
 [38 44]]

 PRACTICAL EXAMPLE: Student Grades
By Students (rows = students):
Alice: [85 92 78]
Bob: [88 76 90]
Charlie: [91 89 84]

By Subjects (rows = subjects):
Math: [85 88 91]
```

```
Stats: [92 76 89]
CS: [78 90 84]
========================================
```

9.9.2 Reshaping Quick Reference

Method	Purpose	Memory	Size Change	In-Place
`reshape()`	Change dimensions	View (when possible)	No	No
`flatten()`	Convert to 1D	Copy	No	No
`ravel()`	Convert to 1D	View (when possible)	No	No
`resize()`	Change dimensions & size	In-place	Yes	Yes
`transpose()` / `.T`	Flip dimensions	View	No	No

Best Practices: - Use **`reshape()`** for safe dimension changes - Use **`ravel()`** for fast 1D conversion - Use **`flatten()`** when you need an independent copy - **Avoid `resize()`** unless you specifically need to change array size - Use **`.T`** for simple 2D matrix transposition

```
#  Practice Exercise: Image Data Reshaping
print("  PRACTICE: IMAGE DATA RESHAPING")
print("=" * 40)

# Simulate image data (height × width × channels)
image_data = np.random.randint(0, 256, (64, 64, 3), dtype=np.uint8)
print(f"Original image shape: {image_data.shape} (H×W×C)")
print(f"Total pixels: {image_data.shape[0] * image_data.shape[1]}")
print(f"Memory usage: {image_data.nbytes:,} bytes")

# Common reshaping operations in computer vision
print(f"\n  Common Computer Vision Reshaping:")

# 1. Flatten for machine learning (pixels as features)
flat_features = image_data.reshape(-1, 3)  # (4096, 3) - each pixel as RGB triplet
print(f"ML features: {flat_features.shape} (pixels×RGB)")

# 2. Batch processing format (batch × height × width × channels)
```

```python
batch_format = image_data.reshape(1, 64, 64, 3)  # Add batch dimension
print(f"Batch format: {batch_format.shape} (N×H×W×C)")

# 3. Channel-first format (channels × height × width) - PyTorch style
channels_first = image_data.transpose(2, 0, 1)  # (3, 64, 64)
print(f"Channels first: {channels_first.shape} (C×H×W)")

# 4. Grayscale conversion (average RGB channels)
grayscale = np.mean(image_data, axis=2, keepdims=True)  # (64, 64, 1)
print(f"Grayscale: {grayscale.shape} (H×W×1)")

# 5. Thumbnail creation (downsample)
thumbnail = image_data[::8, ::8, :]  # Take every 8th pixel
print(f"Thumbnail: {thumbnail.shape} (downsampled)")

print(f"\n  Memory efficiency:")
print(f"Original: {image_data.nbytes:,} bytes")
print(f"Thumbnail: {thumbnail.nbytes:,} bytes")
print(f"Reduction: {image_data.nbytes / thumbnail.nbytes:.1f}× smaller")
```

```
  PRACTICE: IMAGE DATA RESHAPING
========================================
Original image shape: (64, 64, 3) (H×W×C)
Total pixels: 4096
Memory usage: 12,288 bytes

  Common Computer Vision Reshaping:
ML features: (4096, 3) (pixels×RGB)
Batch format: (1, 64, 64, 3) (N×H×W×C)
Channels first: (3, 64, 64) (C×H×W)
Grayscale: (64, 64, 1) (H×W×1)
Thumbnail: (8, 8, 3) (downsampled)

  Memory efficiency:
Original: 12,288 bytes
Thumbnail: 192 bytes
Reduction: 64.0× smaller
```

9.10 Array Concatenation: Joining Arrays Together

Array concatenation combines multiple arrays into a single array. This is essential for:

- **Data merging**: Combining datasets from different sources
- **Batch processing**: Joining results from parallel computations

- **Feature engineering**: Combining different feature sets
- **Time series**: Appending new data to existing sequences

Key Principle: Arrays must have **compatible shapes** along all axes except the concatenation axis.

9.10.1 Understanding Axes in Concatenation

Before diving into methods, let's understand how axes work:

- **axis=0**: Concatenate **vertically** (stack rows) → increases number of rows
- **axis=1**: Concatenate **horizontally** (stack columns) → increases number of columns

- **axis=2**: For 3D+ arrays, concatenate along depth dimension

Memory Tip: Think of axis numbers as the dimension that **grows** during concatenation.

9.10.2 `np.concatenate()`: The Universal Joiner

`np.concatenate()` is the **most flexible** concatenation function. It can join arrays along any axis with fine-grained control.

Syntax: `np.concatenate((array1, array2, ...), axis=0)`

Requirements: - All arrays must have the **same number of dimensions** - Shapes must match along **all axes except the concatenation axis**

Advantages: - Works with **any number of arrays** - Can specify **any axis** for concatenation - **Most memory efficient** for large operations

9.10.2.1 Comprehensive Concatenation Examples

Let's explore concatenation with practical, real-world examples:

```
#  Real-world Example: Student Test Scores
print("  STUDENT SCORE CONCATENATION")
print("=" * 40)

# Semester 1 scores (students × subjects)
semester1 = np.array([[85, 92, 78],    # Alice: Math, Science, English
```

```
                      [88, 76, 91],    # Bob: Math, Science, English
                      [95, 89, 84]])   # Carol: Math, Science, English

# Semester 2 scores (same students, same subjects)
semester2 = np.array([[87, 89, 82],    # Alice: Math, Science, English
                      [91, 78, 89],    # Bob: Math, Science, English
                      [93, 92, 88]])   # Carol: Math, Science, English

students = ['Alice', 'Bob', 'Carol']
subjects = ['Math', 'Science', 'English']

print("Semester 1 Scores:")
for i, student in enumerate(students):
    print(f"{student:6}: {semester1[i]}")

print("\nSemester 2 Scores:")
for i, student in enumerate(students):
    print(f"{student:6}: {semester2[i]}")
```

```
  STUDENT SCORE CONCATENATION
========================================
Semester 1 Scores:
Alice : [85 92 78]
Bob   : [88 76 91]
Carol : [95 89 84]

Semester 2 Scores:
Alice : [87 89 82]
Bob   : [91 78 89]
Carol : [93 92 88]
```

```
#   AXIS=0: Vertical Concatenation (More Students)
print("\n" + "=" * 50)
print("  AXIS=0 CONCATENATION (Vertical Stacking)")
print("=" * 50)

# Add new students to the class
new_students = np.array([[82, 85, 79],    # David: Math, Science, English
                         [90, 88, 93]])   # Emma: Math, Science, English

# Concatenate along axis=0 (add rows = add students)
```

```python
expanded_class = np.concatenate((semester1, new_students), axis=0)

print(f"Original class size: {semester1.shape[0]} students")
print(f"New students added: {new_students.shape[0]} students")
print(f"Total class size: {expanded_class.shape[0]} students")
print(f"\nExpanded class scores (5 students × 3 subjects):")
print(expanded_class)

all_students = students + ['David', 'Emma']
for i, student in enumerate(all_students):
    print(f"{student:6}: {expanded_class[i]}")
```

```
array([[1, 2, 3],
       [3, 4, 5],
       [2, 2, 3],
       [1, 2, 5]])
```

```python
#  AXIS=1: Horizontal Concatenation (More Subjects)
print("\n" + "=" * 50)
print(" AXIS=1 CONCATENATION (Horizontal Stacking)")
print("=" * 50)

# Add new subjects (History and Art scores)
new_subjects_scores = np.array([[80, 92],    # Alice: History, Art
                                [85, 88],    # Bob: History, Art
                                [91, 95]])   # Carol: History, Art

# Concatenate along axis=1 (add columns = add subjects)
expanded_subjects = np.concatenate((semester1, new_subjects_scores), axis=1)

print(f"Original subjects: {semester1.shape[1]} subjects")
print(f"New subjects added: {new_subjects_scores.shape[1]} subjects")
print(f"Total subjects: {expanded_subjects.shape[1]} subjects")
print(f"\nExpanded scores (3 students × 5 subjects):")

all_subjects = subjects + ['History', 'Art']
print(f"Subjects: {all_subjects}")
print(expanded_subjects)

for i, student in enumerate(students):
    print(f"{student:6}: {expanded_subjects[i]}")
```

```
array([[1, 2, 3, 2, 2, 3],
       [3, 4, 5, 1, 2, 5]])
```

9.10.2.2 Visual Understanding of Concatenation

Here's how axis=0 and axis=1 concatenation work visually:

```
AXIS=0 (Vertical - Stack Rows):

  Array 1    +    Array 2    =    Array 1
  [1,2,3]         [7,8,9]          [1,2,3]
  [4,5,6]                      [4,5,6]
                                [7,8,9]


AXIS=1 (Horizontal - Stack Columns):
        +        =
  Array 1       Array         Combined
  [1,2,3]        [7]         [1,2,3,7]
  [4,5,6]        [8]         [4,5,6,8]
```

Rule of Thumb: - **axis=0** → **More rows** (vertical growth) - **axis=1** → **More columns** (horizontal growth)

9.10.2.3 Common Concatenation Errors and Solutions

Let's explore what happens when shapes don't match and how to fix it:

```python
#   SHAPE MISMATCH EXAMPLE
print("   CONCATENATION ERROR DEMONSTRATION")
print("=" * 45)

# Original 2D array
original_2d = np.array([[1, 2, 3],
                        [4, 5, 6]])
print(f"Original 2D array shape: {original_2d.shape}")
print(f"Original 2D array:\n{original_2d}")

# Problematic 1D array
```

```python
problem_1d = np.array([7, 8, 9])
print(f"\nProblematic 1D array shape: {problem_1d.shape}")
print(f"Problematic 1D array: {problem_1d}")

print(f"\n  Why this fails:")
print(f"   2D array: {original_2d.shape} (2 dimensions)")
print(f"   1D array: {problem_1d.shape} (1 dimension)")
print(f"    Different number of dimensions!")
```

```python
# Demonstrate the actual error
print("\n  Attempting concatenation (this will fail):")
try:
    result = np.concatenate((original_2d, problem_1d), axis=0)
    print("Success!")
except ValueError as e:
    print(f"  Error: {e}")
    print("    → Arrays have different numbers of dimensions!")
```

```
ValueError: all the input arrays must have same number of dimensions, but the array at index
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[52], line 1
----> 1 np.concatenate((arr1,arr3),axis=0)
ValueError: all the input arrays must have same number of dimensions, but the array at index
```

9.10.2.4 Solutions to Shape Mismatch Problems

When arrays don't have compatible shapes, here are the most common fixes:

```python
#   SOLUTION 1: Reshape to match dimensions
print("  SOLUTION 1: Reshape 1D → 2D")
print("=" * 35)

print(f"Original problematic shape: {problem_1d.shape}")

# Method 1: Add row dimension
solution_1 = problem_1d.reshape(1, 3)  # Convert to (1, 3)
print(f"Reshaped to row: {solution_1.shape}")
print(f"Reshaped array:\n{solution_1}")

# Method 2: Add column dimension
```

```
solution_2 = problem_1d.reshape(3, 1)  # Convert to (3, 1)
print(f"\nReshaped to column: {solution_2.shape}")
print(f"Reshaped array:\n{solution_2}")

# Method 3: Using newaxis (more explicit)
solution_3 = problem_1d[np.newaxis, :]  # Same as reshape(1, 3)
solution_4 = problem_1d[:, np.newaxis]  # Same as reshape(3, 1)
print(f"\nUsing newaxis:")
print(f"Row format: {solution_3.shape} → {solution_3}")
print(f"Col format: {solution_4.shape} → \n{solution_4}")
```

```
(3,)
```

9.10.2.5 Successful Concatenation After Reshaping

```
#   Now concatenation works!
print("  SUCCESSFUL CONCATENATION")
print("=" * 30)

# Using the reshaped array
fixed_array = problem_1d.reshape(1, 3)
print(f"Original 2D: {original_2d.shape}")
print(f"Fixed 1D→2D: {fixed_array.shape}")

# Now concatenation works
success_result = np.concatenate((original_2d, fixed_array), axis=0)
print(f"\n  Concatenation successful!")
print(f"Result shape: {success_result.shape}")
print(f"Result:\n{success_result}")

print(f"\n  What happened:")
print(f"   • Started with (2,3) + (3,) →   Incompatible")
print(f"   • Reshaped to (2,3) + (1,3) →   Compatible")
print(f"   • Final result: (3,3) array")
```

```
array([[2, 2, 3]])
```

9.10.2.6 Multiple Array Concatenation

`np.concatenate()` can join more than two arrays at once:

```
#   MULTIPLE ARRAY CONCATENATION
print("  JOINING MULTIPLE ARRAYS")
print("=" * 30)

# Create multiple arrays representing quarterly sales data
q1_sales = np.array([[100, 150], [120, 180]])  # 2 stores, 2 products
q2_sales = np.array([[110, 160], [130, 190]])
q3_sales = np.array([[105, 155], [125, 185]])
q4_sales = np.array([[115, 165], [135, 195]])

print("Quarterly Sales Data (Stores × Products):")
print(f"Q1: \n{q1_sales}")
print(f"Q2: \n{q2_sales}")
print(f"Q3: \n{q3_sales}")
print(f"Q4: \n{q4_sales}")

# Method 1: Concatenate all quarters horizontally (timeline view)
yearly_timeline = np.concatenate((q1_sales, q2_sales, q3_sales, q4_sales), axis=1)
print(f"\nHorizontal Timeline (Stores × Q1-Q4 Products):")
print(f"Shape: {yearly_timeline.shape}")
print(yearly_timeline)

# Method 2: Stack all quarters vertically (easier analysis)
yearly_stack = np.concatenate((q1_sales, q2_sales, q3_sales, q4_sales), axis=0)
print(f"\nVertical Stack (All Store-Quarters × Products):")
print(f"Shape: {yearly_stack.shape}")
print(yearly_stack)

print(f"\n  Analysis:")
print(f"Timeline view: {yearly_timeline.shape[0]} stores × {yearly_timeline.shape[1]} data po
print(f"Stacked view: {yearly_stack.shape[0]} observations × {yearly_stack.shape[1]} products
```

```
array([[1, 2, 3],
       [3, 4, 5],
       [2, 2, 3]])
```

9.10.3 Specialized Stacking Functions

While `np.concatenate()` is the most flexible, NumPy provides convenient shortcuts for common operations:

Function	Equivalent	Use Case	Advantage
`np.vstack()`	`concatenate(axis=0)`	Stack vertically (rows)	Clearer intent, handles 1D arrays
`np.hstack()`	`concatenate(axis=1)`	Stack horizontally (columns)	Clearer intent, handles 1D arrays
`np.dstack()`	`concatenate(axis=2)`	Stack along depth (3D)	Easy 3D operations
`np.column_stack()`	Special case of hstack	Treat 1D as columns	Great for combining vectors
`np.row_stack()`	Same as vstack	Treat inputs as rows	Alias for vstack

When to use each: - **`vstack/hstack`**: When you want clearer, more readable code - **`concatenate`**: When you need maximum flexibility or performance

```
#  COMPREHENSIVE STACKING COMPARISON
print("  STACKING FUNCTIONS COMPARISON")
print("=" * 40)

# Sample arrays for demonstration
array_a = np.array([[1, 2], [3, 4]])
array_b = np.array([[5, 6], [7, 8]])

print(f"Array A:\n{array_a}")
print(f"Array B:\n{array_b}")

# Vertical stacking (stack rows)
print(f"\n  VERTICAL STACKING (More Rows)")
print("-" * 35)
vstack_result = np.vstack((array_a, array_b))
concat_v_result = np.concatenate((array_a, array_b), axis=0)

print(f"np.vstack():\n{vstack_result}")
print(f"Equivalent concatenate(axis=0):\n{concat_v_result}")
print(f"Results identical: {np.array_equal(vstack_result, concat_v_result)}")

# Horizontal stacking (stack columns)
```

```
print(f"\n  HORIZONTAL STACKING (More Columns)")
print("-" * 38)
hstack_result = np.hstack((array_a, array_b))
concat_h_result = np.concatenate((array_a, array_b), axis=1)

print(f"np.hstack():\n{hstack_result}")
print(f"Equivalent concatenate(axis=1):\n{concat_h_result}")
print(f"Results identical: {np.array_equal(hstack_result, concat_h_result)}")

# Depth stacking (3D)
print(f"\n DEPTH STACKING (3D Arrays)")
print("-" * 25)
dstack_result = np.dstack((array_a, array_b))
concat_d_result = np.concatenate((array_a[..., np.newaxis], array_b[..., np.newaxis]), axis=

print(f"np.dstack() shape: {dstack_result.shape}")
print(f"First 'slice':\n{dstack_result[:,:,0]}")
print(f"Second 'slice':\n{dstack_result[:,:,1]}")

# Special case: 1D arrays
print(f"\n SPECIAL: 1D Array Handling")
print("-" * 30)
vec1 = np.array([1, 2, 3])
vec2 = np.array([4, 5, 6])

print(f"Vector 1: {vec1}")
print(f"Vector 2: {vec2}")

# vstack treats 1D as rows
vstack_1d = np.vstack((vec1, vec2))
print(f"vstack (treats as rows):\n{vstack_1d}")

# hstack treats 1D as elements
hstack_1d = np.hstack((vec1, vec2))
print(f"hstack (concatenates): {hstack_1d}")

# column_stack treats 1D as columns
column_stack_1d = np.column_stack((vec1, vec2))
print(f"column_stack (treats as cols):\n{column_stack_1d}")
```

```
Vertical Stack:
```

```
 [[1 2 3]
 [3 4 5]
 [2 2 3]
 [1 2 5]]

Horizontal Stack:
 [[1 2 3 2 2 3]
 [3 4 5 1 2 5]]
```

9.10.4 Advanced Concatenation Techniques

```
#  ADVANCED TECHNIQUES
print(" PERFORMANCE & MEMORY OPTIMIZATION")
print("=" * 40)

# Performance comparison for large arrays
large_arrays = [np.random.rand(1000, 100) for _ in range(10)]

import time

# Method 1: Multiple concatenate calls (inefficient)
start_time = time.perf_counter()
result_slow = large_arrays[0]
for arr in large_arrays[1:]:
    result_slow = np.concatenate((result_slow, arr), axis=0)
slow_time = time.perf_counter() - start_time

# Method 2: Single concatenate call (efficient)
start_time = time.perf_counter()
result_fast = np.concatenate(large_arrays, axis=0)
fast_time = time.perf_counter() - start_time

print(f" Sequential concatenation: {slow_time:.4f} seconds")
print(f" Batch concatenation: {fast_time:.4f} seconds")
print(f" Speedup: {slow_time/fast_time:.1f}x faster")
print(f" Result shape: {result_fast.shape}")
print(f" Memory usage: {result_fast.nbytes / 1e6:.1f} MB")

# Memory efficiency demonstration
print(f"\n MEMORY EFFICIENCY")
print("-" * 25)
```

```python
# Pre-allocate vs concatenate
data_size = (5000, 100)
total_arrays = 20

# Method 1: Concatenation (creates multiple copies)
start_time = time.perf_counter()
concat_list = []
for i in range(total_arrays):
    concat_list.append(np.random.rand(*data_size))
concat_result = np.concatenate(concat_list, axis=0)
concat_time = time.perf_counter() - start_time

# Method 2: Pre-allocation (more memory efficient)
start_time = time.perf_counter()
preallocated = np.empty((total_arrays * data_size[0], data_size[1]))
for i in range(total_arrays):
    start_idx = i * data_size[0]
    end_idx = (i + 1) * data_size[0]
    preallocated[start_idx:end_idx] = np.random.rand(*data_size)
prealloc_time = time.perf_counter() - start_time

print(f"Concatenation method: {concat_time:.4f} seconds")
print(f"Pre-allocation method: {prealloc_time:.4f} seconds")
print(f"Pre-allocation is {concat_time/prealloc_time:.1f}x faster")
```

```
  PERFORMANCE & MEMORY OPTIMIZATION
========================================
 Sequential concatenation: 0.0135 seconds
 Batch concatenation: 0.0022 seconds
 Speedup: 6.1x faster
 Result shape: (10000, 100)
 Memory usage: 8.0 MB

  MEMORY EFFICIENCY
-------------------------
Concatenation method: 0.0755 seconds
Pre-allocation method: 0.0809 seconds
Pre-allocation is 0.9x faster
```

9.10.4.1 Real-world Application: Time Series Data

Let's see concatenation in action with a practical time series example:

```
#  TIME SERIES CONCATENATION EXAMPLE
print("  FINANCIAL DATA PROCESSING")
print("=" * 35)

# Simulate daily stock prices for different months
np.random.seed(42)
jan_prices = np.random.uniform(100, 120, (31, 4))  # 31 days, 4 stocks
feb_prices = np.random.uniform(98, 118, (28, 4))   # 28 days, 4 stocks
mar_prices = np.random.uniform(102, 122, (31, 4))  # 31 days, 4 stocks

stock_names = ['AAPL', 'GOOGL', 'MSFT', 'TSLA']
print(f"Stock data shape - Days × Stocks:")
print(f"January: {jan_prices.shape} ({jan_prices.shape[0]} days)")
print(f"February: {feb_prices.shape} ({feb_prices.shape[0]} days)")
print(f"March: {mar_prices.shape} ({mar_prices.shape[0]} days)")

# Combine quarterly data
q1_prices = np.concatenate((jan_prices, feb_prices, mar_prices), axis=0)
print(f"\nQ1 Combined: {q1_prices.shape} ({q1_prices.shape[0]} total days)")

# Calculate monthly averages
monthly_avg = np.array([
    np.mean(jan_prices, axis=0),
    np.mean(feb_prices, axis=0),
    np.mean(mar_prices, axis=0)
])

print(f"\nMonthly averages shape: {monthly_avg.shape}")
print(f"Monthly averages:")
months = ['January', 'February', 'March']
for i, month in enumerate(months):
    print(f"{month:8}: " + " | ".join([f"{stock}: ${avg:.2f}"
                                      for stock, avg in zip(stock_names, monthly_avg[i])]))

# Add new features (volume data) using horizontal concatenation
np.random.seed(42)
q1_volumes = np.random.randint(1000000, 5000000, (q1_prices.shape[0], 4))
q1_complete = np.hstack((q1_prices, q1_volumes))
```

```python
print(f"\nWith volume data: {q1_complete.shape}")
print(f"Columns: {stock_names} (prices) + {stock_names} (volumes)")
print(f"Sample day 1: Prices={q1_complete[0, :4]}")
print(f"             Volumes={q1_complete[0, 4:].astype(int)}")
```

```
 FINANCIAL DATA PROCESSING
===================================
Stock data shape - Days × Stocks:
January: (31, 4) (31 days)
February: (28, 4) (28 days)
March: (31, 4) (31 days)

Q1 Combined: (90, 4) (90 total days)

Monthly averages shape: (3, 4)
Monthly averages:
January : AAPL: $109.57 | GOOGL: $109.65 | MSFT: $108.89 | TSLA: $110.21
February: AAPL: $106.62 | GOOGL: $106.70 | MSFT: $109.63 | TSLA: $108.16
March   : AAPL: $113.36 | GOOGL: $112.84 | MSFT: $111.29 | TSLA: $110.70

With volume data: (90, 8)
Columns: ['AAPL', 'GOOGL', 'MSFT', 'TSLA'] (prices) + ['AAPL', 'GOOGL', 'MSFT', 'TSLA'] (volu
Sample day 1: Prices=[107.49080238 119.01428613 114.63987884 111.97316968]
             Volumes=[3219110 3768307 3229084 4511566]
```

9.10.5 Concatenation Quick Reference

Operation	Function	Syntax	Result
Vertical Stack	`vstack()`	`np.vstack((A, B))`	More rows
Horizontal Stack	`hstack()`	`np.hstack((A, B))`	More columns
General Concat	`concatenate()`	`np.concatenate((A, B), axis=n)`	Custom axis
3D Stack	`dstack()`	`np.dstack((A, B))`	Depth dimension
Column Combine	`column_stack()`	`np.column_stack((v1, v2))`	Vectors → columns

9.10.6 Common Pitfalls and Solutions

Problem	Cause	Solution
`shapes not aligned`	Different dimensions	Use `reshape()` or `newaxis`
`axis out of bounds`	Wrong axis number	Check array dimensions with `.ndim`
`memory error`	Too many copies	Use pre-allocation or batch operations
`1D array issues`	Mixed 1D/2D arrays	Use `vstack`/`hstack` or reshape consistently

9.10.7 Best Practices

1. **Pre-allocate when possible** for better memory efficiency
2. **Use batch concatenation** instead of iterative concatenation

3. **Choose descriptive functions** (`vstack`/`hstack`) over generic `concatenate` when appropriate
4. **Check shapes first** with `.shape` to avoid runtime errors
5. **Consider memory usage** for large datasets - sometimes lists are better for building, then concatenate once

9.11 Independent Practice

9.11.1 Practice exercise 1

9.11.1.1

Read the coordinates of the capital cities of the world from https://gist.github.com/ofou/df09a6834a8421b4f376c8
.

Task 1: Use NumPy to print the name and coordinates of the capital city closest to the US capital - Washington DC.

Note that:

1. The *Country Name* for US is given as *United States of America* in the data.
2. The 'closeness' of capital cities from the US capital is based on the Euclidean distance of their coordinates to those of the US capital.

Hints:

1. Use the *to_numpy()* function of the *Pandas DataFrame* class to convert a DataFrame to a Numpy array
2. Use *broadcasting* to compute the euclidean distance of capital cities from Washington DC.
3. Use `np.argmin` to locate the index of the minimum value in an array
4. Exclude the capital itself to avoid trivial zero values in the results.

Task 2: Use NumPy to:

1. Print the names of the countries of the top 10 capital cities closest to the US capital - Washington DC.

2. Create and print a NumPy array containing the coordinates of the top 10 cities.

Hint: Use the *concatenate()* function from the *NumPy* library to stack the coordinates of the top 10 cities.

10 NumPy Intermediate

10.1 Learning Objectives

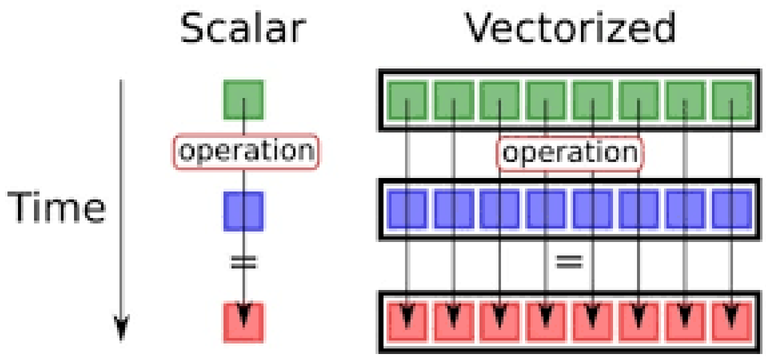
- **Apply broadcasting rules** for operations between arrays of different shapes
- **Optimize code performance** using vectorization instead of loops
- **Execute mathematical operations** (arithmetic, linear algebra, statistics)
- **Generate random data** for simulations and statistical analysis
- **Solve data science problems** using NumPy's computational power
- **Convert between NumPy and Pandas** for optimal workflow efficiency
- **Perform vectorized computations** that are 10–100x faster than Python loops

```
import numpy as np
import pandas as pd
```

10.2 Vectorization in NumPy

10.2.1 Introduction to Vectorization

Vectorization is the process of applying operations to entire arrays or matrices simultaneously rather than iterating over individual elements using loops. This approach allows NumPy to utilize highly optimized C libraries for faster computation.



Comparison between scalar (classical) computation and vectorization.

Benefits of Vectorization:

- **Performance**: Significantly faster than using for-loops in Python (often 10-100x speedup)
- **Conciseness**: Shorter and more readable code
- **Efficiency**: Reduces the overhead of Python loops and leverages lower-level optimizations
- **Parallelization**: Many vectorized operations can utilize multiple CPU cores
- **Memory Efficiency**: Better cache utilization and reduced memory overhead

10.2.2 Why Vectorization Matters

Traditional Python Loop Approach:

```
# Slow: Python loop with individual operations
result = []
for i in range(len(array1)):
    for j in range(len(array2)):
        result.append(array1[i] * array2[j] + some_value)
```

Vectorized NumPy Approach:

```
# Fast: Single vectorized operation
result = array1[:, np.newaxis] * array2 + some_value
```

The vectorized approach is faster because: 1. **Compiled C code**: NumPy operations use optimized C/Fortran libraries 2. **Reduced Python overhead**: Fewer Python function calls and type checks 3. **SIMD instructions**: Can use Single Instruction, Multiple Data operations 4. **Better memory access patterns**: More cache-friendly data access

10.2.3 Understanding Vectorized Operations with Examples

```
# Create two arrays
arr1 = np.array([1, 2, 3, 4, 5])
arr2 = np.array([10, 20, 30, 40, 50])

# Element-wise addition (vectorized operation)
sum_arr = arr1 + arr2
print("Sum Array:", sum_arr)
```

```
Sum Array: [11 22 33 44 55]
```

10.2.4 Types of Vectorized Operations

NumPy supports several categories of vectorized operations:

10.2.4.1 1. Element-wise Operations (Universal Functions - ufuncs)

These operations apply a function to each element of an array:

```
# Basic arithmetic operations (all vectorized)
arr = np.array([1, 4, 9, 16, 25])

print("Original array:", arr)
print("Square root:", np.sqrt(arr))
print("Natural log:", np.log(arr))
print("Sine:", np.sin(arr))
print("Power of 2:", np.power(arr, 2))

# Comparison operations
print("\nComparison operations:")
print("Greater than 10:", arr > 10)
print("Equal to 9:", arr == 9)
print("Between 5 and 20:", (arr >= 5) & (arr <= 20))
```

```
Original array: [ 1  4  9 16 25]
Square root: [1. 2. 3. 4. 5.]
Natural log: [0.         1.38629436 2.19722458 2.77258872 3.21887582]
```

```
Sine: [ 0.84147098 -0.7568025   0.41211849 -0.28790332 -0.13235175]
Power of 2: [  1  16  81 256 625]

Comparison operations:
Greater than 10: [False False False  True  True]
Equal to 9: [False False  True False False]
Between 5 and 20: [False False  True  True False]
```

10.2.4.2 2. Aggregation Operations

These reduce arrays to scalar values or smaller arrays:

```python
# 2D array for aggregation examples
matrix = np.array([[1, 2, 3, 4],
                   [5, 6, 7, 8],
                   [9, 10, 11, 12]])

print("Matrix:")
print(matrix)

# Aggregations across entire array
print(f"\nSum of all elements: {np.sum(matrix)}")
print(f"Mean: {np.mean(matrix)}")
print(f"Standard deviation: {np.std(matrix)}")
print(f"Min: {np.min(matrix)}, Max: {np.max(matrix)}")

# Aggregations along specific axes
print(f"\nSum along axis 0 (columns): {np.sum(matrix, axis=0)}")
print(f"Sum along axis 1 (rows): {np.sum(matrix, axis=1)}")
print(f"Mean along axis 0: {np.mean(matrix, axis=0)}")
print(f"Max along axis 1: {np.max(matrix, axis=1)}")
```

```
Matrix:
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]

Sum of all elements: 78
Mean: 6.5
Standard deviation: 3.452052529534663
Min: 1, Max: 12
```

```
Sum along axis 0 (columns): [15 18 21 24]
Sum along axis 1 (rows): [10 26 42]
Mean along axis 0: [5. 6. 7. 8.]
Max along axis 1: [ 4  8 12]
```

10.2.4.3 3. Boolean Operations and Fancy Indexing

Vectorized selection and filtering operations:

```python
# Boolean indexing - vectorized filtering
data = np.array([1, 15, 8, 23, 4, 16, 42, 3, 19, 7])

# Find elements meeting multiple conditions (vectorized)
mask = (data > 5) & (data < 20)
filtered_data = data[mask]
print(f"Original data: {data}")
print(f"Elements between 5 and 20: {filtered_data}")

# Using np.where for conditional replacement (vectorized)
# Replace values > 15 with -1, others with original value
result = np.where(data > 15, -1, data)
print(f"After conditional replacement: {result}")

# Fancy indexing - select multiple elements at once
indices = [0, 2, 4, 7]
selected = data[indices]
print(f"Elements at indices {indices}: {selected}")

# Count how many elements meet condition (vectorized)
count_large = np.sum(data > 10)
print(f"Count of elements > 10: {count_large}")
```

```
Original data: [ 1 15  8 23  4 16 42  3 19  7]
Elements between 5 and 20: [15  8 16 19  7]
After conditional replacement: [ 1 15  8 -1  4 -1 -1  3 -1  7]
Elements at indices [0, 2, 4, 7]: [1 8 4 3]
Count of elements > 10: 5
```

10.2.5 Performance Comparison: NumPy Vectorization Vs Python Loops

Understanding the performance benefits of vectorization is crucial for writing efficient scientific computing code. Let's compare different scenarios to see when and why vectorization matters.

10.2.5.1 Scenario 1: Basic Mathematical Operations

`np.dot` is a vectorized operation in NumPy that performs matrix multiplication or dot product between arrays. It efficiently computes the element-wise multiplications and then sums them up. To better understand its efficiency, let's first implement the dot product using for loops, and then compare its performance with `np.dot` to see the benefits of vectorization.

```
import time

# Function to calculate dot product using for loops
def dot_product_loops(arr1, arr2):
    result = 0
    for i in range(len(arr1)):
        result += arr1[i] * arr2[i]
    return result

# Create sample arrays of different sizes for comparison
sizes = [1000, 10000, 100000, 1000000]

print("Performance Comparison: Loops vs NumPy Vectorization")
print("=" * 60)

for size in sizes:
    # Create random arrays
    arr1 = np.random.rand(size)
    arr2 = np.random.rand(size)

    # Measure time for the loop-based implementation
    start_time = time.time()
    loop_result = dot_product_loops(arr1, arr2)
    loop_time = time.time() - start_time

    # Measure time for np.dot
    start_time = time.time()
    numpy_result = np.dot(arr1, arr2)
    numpy_time = time.time() - start_time
```

```
    # Calculate speedup
    speedup = loop_time / numpy_time if numpy_time > 0 else float('inf')

    print(f"\nArray size: {size:,}")
    print(f"Loop result: {loop_result:.5f}, Time: {loop_time:.5f}s")
    print(f"NumPy result: {numpy_result:.5f}, Time: {numpy_time:.5f}s")
    print(f"Speedup: {speedup:.1f}x faster")



print(f"\n{'='*60}")
print("Key Observations:")
print("• NumPy becomes dramatically faster as array size increases")
print("• Speedup can reach 50-100x for large arrays")
print("• NumPy overhead is minimal for small arrays but pays off quickly")
```

```
Performance Comparison: Loops vs NumPy Vectorization
============================================================

Array size: 1,000
Loop result: 234.28341, Time: 0.00000s
NumPy result: 234.28341, Time: 0.00000s
Speedup: infx faster

Array size: 10,000
Loop result: 2524.58396, Time: 0.00000s
NumPy result: 2524.58396, Time: 0.02460s
Speedup: 0.0x faster

Array size: 100,000
Loop result: 24892.29841, Time: 0.02218s
NumPy result: 24892.29841, Time: 0.00104s
Speedup: 21.4x faster

Array size: 1,000,000
Loop result: 249763.97969, Time: 0.20611s
NumPy result: 249763.97969, Time: 0.00111s
Speedup: 185.6x faster

============================================================
Key Observations:
```

- NumPy becomes dramatically faster as array size increases
- Speedup can reach 50-100x for large arrays
- NumPy overhead is minimal for small arrays but pays off quickly

10.2.5.2 Scenario 2: Complex Mathematical Operations

Let's compare more complex operations that show even greater performance differences:

```
# Complex mathematical operation: polynomial evaluation
# f(x) = 3x³ + 2x² - 5x + 1

def polynomial_loops(x_values):
    """Evaluate polynomial using loops"""
    results = []
    for x in x_values:
        result = 3*x**3 + 2*x**2 - 5*x + 1
        results.append(result)
    return results

def polynomial_vectorized(x_values):
    """Evaluate polynomial using vectorized operations"""
    return 3*x_values**3 + 2*x_values**2 - 5*x_values + 1

# Test with different sizes
size = 500000
x_data = np.random.uniform(-10, 10, size)

print("Complex Operations: Polynomial Evaluation")
print("f(x) = 3x³ + 2x² - 5x + 1")
print("-" * 40)

# Loop version
start_time = time.time()
loop_results = polynomial_loops(x_data)
loop_time = time.time() - start_time

# Vectorized version
start_time = time.time()
vector_results = polynomial_vectorized(x_data)
vector_time = time.time() - start_time

speedup = loop_time / vector_time
```

```
print(f"Array size: {size:,}")
print(f"Loop time: {loop_time:.4f}s")
print(f"Vectorized time: {vector_time:.4f}s")
print(f"Speedup: {speedup:.1f}x")

# Verify results match
print(f"Results match: {np.allclose(loop_results, vector_results)}")
```

```
Complex Operations: Polynomial Evaluation
f(x) = 3x³ + 2x² - 5x + 1
----------------------------------------
Array size: 500,000
Loop time: 0.2560s
Vectorized time: 0.0129s
Speedup: 19.9x
Results match: True
```

10.2.5.3 Performance Summary & Key Takeaways

The performance comparisons clearly demonstrate that **vectorized NumPy operations significantly outperform equivalent Python loops**:

10.2.6 Broadcasting: The Heart of NumPy Vectorization

Broadcasting is NumPy's powerful mechanism that allows operations between arrays of different shapes without explicitly reshaping them. It's the foundation that makes vectorization so flexible and intuitive.

10.2.6.1 Broadcasting Rules

NumPy broadcasting follows these rules:

1. **Start from the trailing dimension** and work backwards
2. **Dimensions are compatible** if:
    - They are equal, OR
    - One of them is 1, OR
    - One of them doesn't exist (missing dimension)
3. **Missing dimensions** are assumed to be size 1

10.2.6.2 Visual Guide to Broadcasting

```
Array 1: (3, 4)      [[1, 2, 3, 4],       Array 2: (4,)    [10, 20, 30, 40]
                      [5, 6, 7, 8],                           ↓ broadcasts to
                      [9,10,11,12]]                          [[10, 20, 30, 40],
                                                               [10, 20, 30, 40],
                                                               [10, 20, 30, 40]]
Result: (3, 4)      Element-wise addition produces (3, 4) result
```

Let's explore different broadcasting scenarios:

```python
# Comprehensive Broadcasting Examples

print("=== BROADCASTING EXAMPLES ===\n")

# Example 1: 2D array + 1D array (most common case)
arr_2d = np.array([[1, 2, 3, 4],
                   [5, 6, 7, 8],
                   [9, 10, 11, 12]])

arr_1d = np.array([10, 20, 30, 40])

print("Example 1: (3,4) + (4,) broadcasting")
print(f"2D array shape: {arr_2d.shape}")
print(f"1D array shape: {arr_1d.shape}")
result_1 = arr_2d + arr_1d
print(f"Result shape: {result_1.shape}")
print("Result:")
print(result_1)
print()

# Example 2: Broadcasting with different orientations
arr_col = np.array([[100],
                    [200],
                    [300]])  # Column vector (3,1)

print("Example 2: (3,4) + (3,1) broadcasting")
print(f"2D array shape: {arr_2d.shape}")
print(f"Column vector shape: {arr_col.shape}")
result_2 = arr_2d + arr_col
print(f"Result shape: {result_2.shape}")
print("Result:")
```

```python
print(result_2)
print()

# Example 3: Scalar broadcasting (always works)
scalar = 1000
print("Example 3: (3,4) + scalar broadcasting")
result_3 = arr_2d + scalar
print(f"Result shape: {result_3.shape}")
print("Result:")
print(result_3)
print()

# Example 4: 3D broadcasting
arr_3d = np.random.randint(1, 10, (2, 3, 4))
arr_2d_broadcast = np.array([[1, 2, 3, 4]])  # (1, 4)

print("Example 4: (2,3,4) + (1,4) broadcasting")
print(f"3D array shape: {arr_3d.shape}")
print(f"2D array shape: {arr_2d_broadcast.shape}")
result_4 = arr_3d + arr_2d_broadcast
print(f"Result shape: {result_4.shape}")
print("Broadcasting successful!")
print()
```

```
=== BROADCASTING EXAMPLES ===

Example 1: (3,4) + (4,) broadcasting
2D array shape: (3, 4)
1D array shape: (4,)
Result shape: (3, 4)
Result:
[[11 22 33 44]
 [15 26 37 48]
 [19 30 41 52]]

Example 2: (3,4) + (3,1) broadcasting
2D array shape: (3, 4)
Column vector shape: (3, 1)
Result shape: (3, 4)
Result:
[[101 102 103 104]
 [205 206 207 208]
```

```
 [309 310 311 312]]

Example 3: (3,4) + scalar broadcasting
Result shape: (3, 4)
Result:
[[1001 1002 1003 1004]
 [1005 1006 1007 1008]
 [1009 1010 1011 1012]]

Example 4: (2,3,4) + (1,4) broadcasting
3D array shape: (2, 3, 4)
2D array shape: (1, 4)
Result shape: (2, 3, 4)
Broadcasting successful!
```

```python
# Broadcasting Rules Demonstration

def check_broadcast_compatibility(shape1, shape2):
    """
    Check if two shapes are compatible for broadcasting
    """
    # Pad shorter shape with 1s on the left
    ndim1, ndim2 = len(shape1), len(shape2)
    max_ndim = max(ndim1, ndim2)

    shape1_padded = [1] * (max_ndim - ndim1) + list(shape1)
    shape2_padded = [1] * (max_ndim - ndim2) + list(shape2)

    result_shape = []
    compatible = True

    for i in range(max_ndim):
        dim1, dim2 = shape1_padded[i], shape2_padded[i]
        if dim1 == dim2:
            result_shape.append(dim1)
        elif dim1 == 1:
            result_shape.append(dim2)
        elif dim2 == 1:
            result_shape.append(dim1)
        else:
            compatible = False
            break
```

```
    return compatible, tuple(result_shape) if compatible else None

# Test different shape combinations
test_cases = [
    ((3, 4), (4,)),       # Compatible
    ((3, 4), (3, 1)),     # Compatible
    ((3, 4), (2,)),       # Incompatible
    ((2, 3, 4), (4,)),    # Compatible
    ((2, 3, 4), (3, 4)), # Compatible
    ((2, 3, 4), (2, 1, 4)), # Compatible
    ((3, 4), (2, 3)),     # Incompatible
]

print("Broadcasting Compatibility Check:")
print("-" * 50)
for shape1, shape2 in test_cases:
    compatible, result_shape = check_broadcast_compatibility(shape1, shape2)
    status = "  Compatible" if compatible else "  Incompatible"
    if compatible:
        print(f"{str(shape1):>10} + {str(shape2):>10} → {str(result_shape):>12} {status}")
    else:
        print(f"{str(shape1):>10} + {str(shape2):>10} → {'N/A':>12} {status}")
```

```
Broadcasting Compatibility Check:
--------------------------------------------------
    (3, 4) +       (4,) →       (3, 4)   Compatible
    (3, 4) +     (3, 1) →       (3, 4)   Compatible
    (3, 4) +       (2,) →          N/A   Incompatible
 (2, 3, 4) +       (4,) →    (2, 3, 4)   Compatible
 (2, 3, 4) +     (3, 4) →    (2, 3, 4)   Compatible
 (2, 3, 4) +  (2, 1, 4) →    (2, 3, 4)   Compatible
    (3, 4) +     (2, 3) →          N/A   Incompatible
```

10.2.6.3 Real-World Broadcasting Applications

Broadcasting enables elegant solutions to common data science problems:

```
# Real-world broadcasting applications

# Application 1: Data Normalization (Z-score standardization)
```

```python
print("=== APPLICATION 1: DATA NORMALIZATION ===")
# Sample data: test scores for 100 students across 5 subjects
np.random.seed(42)
scores = np.random.normal(75, 15, (100, 5))  # 100 students, 5 subjects
subjects = ['Math', 'Science', 'English', 'History', 'Art']

print(f"Original data shape: {scores.shape}")
print(f"Sample scores (first 5 students):")
for i in range(5):
    print(f"Student {i+1}: {scores[i].round(1)}")

# Calculate mean and std for each subject (broadcasting will handle the rest)
subject_means = np.mean(scores, axis=0)  # Shape: (5,)
subject_stds = np.std(scores, axis=0)    # Shape: (5,)

print(f"\nSubject means: {subject_means.round(1)}")
print(f"Subject stds: {subject_stds.round(1)}")

# Normalize using broadcasting: (100,5) - (5,) / (5,) → (100,5)
normalized_scores = (scores - subject_means) / subject_stds

print(f"\nNormalized data shape: {normalized_scores.shape}")
print(f"Normalized means (should be ~0): {np.mean(normalized_scores, axis=0).round(3)}")
print(f"Normalized stds (should be ~1): {np.std(normalized_scores, axis=0).round(3)}")
print()

# Application 2: Distance Matrix Calculation
print("=== APPLICATION 2: DISTANCE MATRIX ===")
# Calculate pairwise distances between points using broadcasting
points = np.array([[1, 2], [3, 4], [5, 6], [7, 8]])  # 4 points in 2D
n_points = points.shape[0]

print(f"Points shape: {points.shape}")
print("Points:")
for i, point in enumerate(points):
    print(f"  Point {i}: ({point[0]}, {point[1]})")

# Using broadcasting to calculate all pairwise distances efficiently
# points[:, np.newaxis, :] has shape (4, 1, 2)
# points[np.newaxis, :, :] has shape (1, 4, 2)
# Broadcasting gives us (4, 4, 2) - all pairwise differences
differences = points[:, np.newaxis, :] - points[np.newaxis, :, :]
```

```python
distances = np.sqrt(np.sum(differences**2, axis=2))

print(f"\nDistance matrix shape: {distances.shape}")
print("Distance matrix:")
print(distances.round(2))
print()

# Application 3: Image Processing with Broadcasting
print("=== APPLICATION 3: IMAGE PROCESSING ===")
# Simulate a grayscale image and apply different brightness adjustments to RGB channels
np.random.seed(42)
image_gray = np.random.randint(0, 256, (100, 100))  # 100x100 grayscale image

# Create RGB image by broadcasting
rgb_multipliers = np.array([1.0, 0.8, 0.6])  # R, G, B channel multipliers
# Reshape for broadcasting: (100, 100, 1) * (3,) → (100, 100, 3)
image_rgb = image_gray[:, :, np.newaxis] * rgb_multipliers

print(f"Original grayscale image shape: {image_gray.shape}")
print(f"RGB multipliers: {rgb_multipliers}")
print(f"Final RGB image shape: {image_rgb.shape}")
print(f"RGB image dtype: {image_rgb.dtype}")

# Show some pixel values
print(f"\nSample pixel transformation:")
print(f"Original gray value: {image_gray[50, 50]}")
print(f"RGB values: {image_rgb[50, 50].round(1)}")
print()

print("  Key Broadcasting Benefits:")
print("   • No explicit loops required")
print("   • Memory efficient (no array copying)")
print("   • Highly optimized C implementation")
print("   • Readable and concise code")
print("   • Scales to any dimension")
```

```
=== APPLICATION 1: DATA NORMALIZATION ===
Original data shape: (100, 5)
Sample scores (first 5 students):
Student 1: [82.5 72.9 84.7 97.8 71.5]
Student 2: [71.5 98.7 86.5 68.  83.1]
Student 3: [68.  68.  78.6 46.3 49.1]
```

```
Student 4: [66.6 59.8 79.7 61.4 53.8]
Student 5: [97.  71.6 76.  53.6 66.8]

Subject means: [74.3 76.2 73.6 76.9 74.5]
Subject stds: [13.4 14.7 14.7 14.3 16. ]

Normalized data shape: (100, 5)
Normalized means (should be ~0): [ 0.  0. -0. -0. -0.]
Normalized stds (should be ~1): [1. 1. 1. 1. 1.]

=== APPLICATION 2: DISTANCE MATRIX ===
Points shape: (4, 2)
Points:
  Point 0: (1, 2)
  Point 1: (3, 4)
  Point 2: (5, 6)
  Point 3: (7, 8)

Distance matrix shape: (4, 4)
Distance matrix:
[[0.   2.83 5.66 8.49]
 [2.83 0.   2.83 5.66]
 [5.66 2.83 0.   2.83]
 [8.49 5.66 2.83 0.  ]]

=== APPLICATION 3: IMAGE PROCESSING ===
Original grayscale image shape: (100, 100)
RGB multipliers: [1.  0.8 0.6]
Final RGB image shape: (100, 100, 3)
RGB image dtype: float64

Sample pixel transformation:
Original gray value: 151
RGB values: [151.  120.8  90.6]
```

Key Broadcasting Benefits:

- No explicit loops required
- Memory efficient (no array copying)
- Highly optimized C implementation
- Readable and concise code
- Scales to any dimension

```python
arr2 = np.array([[1, 2, 3, 4],
                 [5, 6, 7, 8],
                 [9, 1, 2, 3]])
```

```python
arr4 = np.array([4, 5, 6, 7])
```

```python
arr2 + arr4
```

```
array([[ 5,  7,  9, 11],
       [ 9, 11, 13, 15],
       [13,  6,  8, 10]])
```

When the expression `arr2 + arr4` is evaluated, `arr4` (which has the shape `(4,)`) is replicated three times to match the shape `(3, 4)` of `arr2`. Numpy performs the replication without actually creating three copies of the smaller dimension array, thus improving performance and using lower memory.

Broadcasting only works if one of the arrays can be replicated to match the other array's shape.

- Dimensions of size 1 will broadcast (as if the value was repeated).
- Otherwise, the dimension must have the same shape.

```python
arr5 = np.array([7, 8])
```

```python
# uncomment the code to see the error
# arr2 + arr5
```

In the above example, even if `arr5` is replicated three times, it will not match the shape of `arr2`. Hence `arr2 + arr5` cannot be evaluated successfully. See the broadcasting documentation to learn more about it.

10.2.7 Matrix Multiplication in NumPy with Vectorization

Matrix multiplication is one of the most common and computationally intensive operations in numerical computing and deep learning. NumPy offers efficient and highly optimized methods for performing matrix multiplication, which leverage vectorization to handle large matrices quickly and accurately.

Note that: NumPy matrix operations follow the standard rules of linear algebra, so it's important to ensure that the shapes of the matrices are compatible. If they are not, consider reshaping the matrices before performing multiplication

There are two commonly methods for matrix multiplication

10.2.7.1 Method 1: Matrix Multiplication Using `np.dot()`

```python
# Define two 2D arrays (matrices)
matrix1 = np.array([[1, 2, 3],
                    [4, 5, 6]])
matrix2 = np.array([[7, 8],
                    [9, 10],
                    [11, 12]])

# Matrix multiplication using np.dot
result_dot = np.dot(matrix1, matrix2)
print("Matrix Multiplication using np.dot:\n", result_dot)

# Another way to perform np.dot for matrix multiplication
result_dot2 = matrix1.dot(matrix2)
print("\nMatrix Multiplication using dot method:\n", result_dot2)
```

```
Matrix Multiplication using np.dot:
 [[ 58  64]
 [139 154]]

Matrix Multiplication using dot method:
 [[ 58  64]
 [139 154]]
```

10.2.7.2 Method 2: Matrix Multiplication using `np.matmul()` or `@`

```python
# Matrix multiplication using np.matmul or @ operator
result_matmul = np.matmul(matrix1, matrix2)
result_operator = matrix1 @ matrix2
print("\nMatrix Multiplication using np.matmul:\n", result_matmul)
print("\nMatrix Multiplication using @ operator:\n", result_operator)
```

```
Matrix Multiplication using np.matmul:
 [[ 58  64]
 [139 154]]

Matrix Multiplication using @ operator:
```

```
 [[ 58  64]
 [139 154]]
```

Note that * operator in numpy is element-wise multiplication

```
# using * operator for element-wise multiplication, please uncomment the code below to run
# element_wise = matrix1 * matrix2
# print("\nElement-wise Multiplication:\n", element_wise)
```

```
# reshape the array for element-wise multiplication
matrix2_reshaped = matrix2.reshape(2, 3)
element_wise = matrix1 * matrix2_reshaped
print("\nElement-wise Multiplication after reshaping:\n", element_wise)
```

```
Element-wise Multiplication after reshaping:
 [[ 7 16 27]
 [40 55 72]]
```

10.3 Converting Between NumPy Arrays and Pandas DataFrames

Seamless data conversion between NumPy and pandas is essential for efficient data science workflows. Each library excels in different areas, and knowing when and how to convert between them maximizes your analytical power.

10.3.1 Why Conversion Matters

NumPy and **pandas** complement each other perfectly:

Aspect	NumPy Arrays	Pandas DataFrames
Strength	Mathematical computations	Data manipulation & analysis
Speed	Faster numerical operations	Easier data exploration
Memory	More memory efficient	Rich metadata (labels, dtypes)
Use Cases	Linear algebra, statistics, ML algorithms	Data cleaning, grouping, merging
Indexing	Integer-based only	Label-based + integer-based
Missing Data	Limited NaN support	Robust NaN handling

Typical Workflow: 1. **Load data** with pandas (CSV, Excel, databases) 2. **Clean and prepare** with pandas (filtering, grouping) 3. **Convert to NumPy** for intensive computations (ML, statistics) 4. **Convert back to pandas** for results analysis and visualization

10.3.2 NumPy Array → Pandas DataFrame

When to convert: You have computational results in NumPy arrays and need pandas' data manipulation capabilities.

Common scenarios:

- Adding meaningful labels to numerical results
- Exporting to CSV, Excel, or databases
- Preparing data for visualization libraries
- Using pandas' powerful filtering and grouping

10.3.2.1 Basic Conversion Methods

```python
#   COMPREHENSIVE ARRAY → DATAFRAME CONVERSION
import numpy as np
import pandas as pd

print("  NUMPY TO PANDAS CONVERSION")
print("=" * 40)
```

```
  NUMPY TO PANDAS CONVERSION
========================================
```

```python
# Original NumPy array
scores = np.array([
    [85, 92, 78, 95],
    [88, 76, 91, 82],
    [95, 89, 84, 90]
])

students = ['Alice', 'Bob', 'Carol']
subjects = ['Math', 'Science', 'English', 'History']

print("Original NumPy array:")
print(scores)
print(f"Shape: {scores.shape}")
```

```
Original NumPy array:
[[85 92 78 95]
 [88 76 91 82]
 [95 89 84 90]]
Shape: (3, 4)
```

```
# Method 1: Basic conversion (auto index & column labels: 0..n-1)
df_basic = pd.DataFrame(scores)
print("\n  Basic DataFrame (auto index/columns):")
df_basic
```

```
  Basic DataFrame (auto index/columns):
```

	0	1	2	3
0	85	92	78	95
1	88	76	91	82
2	95	89	84	90

```
# Method 2: Custom row and column labels
df_labeled = pd.DataFrame(scores, index=students, columns=subjects)
print("\n Labeled DataFrame (custom index & columns):")
df_labeled
```

```
 Labeled DataFrame (custom index & columns):
```

	Math	Science	English	History
Alice	85	92	78	95
Bob	88	76	91	82
Carol	95	89	84	90

```
# Method 3: Add metadata (index/column names)
df_enhanced = df_labeled.copy()
df_enhanced.index.name = 'Student'
df_enhanced.columns.name = 'Subject'
```

```
print("\n  Enhanced with Metadata:")
print(f"\nIndex name: '{df_enhanced.index.name}'")
print(f"Column name: '{df_enhanced.columns.name}'")
df_enhanced
```

```
  Enhanced with Metadata:

Index name: 'Student'
Column name: 'Subject'
```

Subject Student	Math	Science	English	History
Alice	85	92	78	95
Bob	88	76	91	82
Carol	95	89	84	90

10.3.2.2 Conversion Parameters & Options

Parameter	Purpose	Example	Default
`data`	NumPy array to convert	`np.array([[1,2],[3,4]])`	Required
`columns`	Column labels	`['A', 'B']`	`[0, 1, 2, ...]`
`index`	Row labels	`['row1', 'row2']`	`[0, 1, 2, ...]`
`dtype`	Force specific data type	`dtype=float`	Inferred from array
`copy`	Create copy vs view	`copy=False`	`True` (safer)

- **Set `copy=False`** for large arrays to save memory (if you won't modify the original)

Pro Tips:- **Check data types** after conversion with `df.dtypes`

- **Always specify meaningful column names** for better code readability
- **Use `pd.Index()`** to add names to your index/columns for richer metadata

10.3.3 Pandas DataFrame → NumPy Array

When to convert: You need NumPy's computational speed and mathematical functions.

10.3.3.1 Conversion Methods & Performance

Common scenarios:

- Performance-critical computations (linear algebra, statistics)
- Integration with scientific computing libraries (SciPy, scikit-learn)
- Machine learning algorithms that expect NumPy arrays
- Mathematical operations not available in pandas

```
print(" PANDAS TO NUMPY CONVERSION")
print("=" * 40)

# Start with a DataFrame of tech stock prices
stock_data = pd.DataFrame(
    {
        "AAPL":  [150.0, 152.5, 148.2, 155.1],
        "GOOGL": [2800.0, 2750.0, 2825.0, 2900.0],
        "MSFT":  [300.0, 305.0, 298.5, 310.0],
        "TSLA":  [800.0, 795.0, 820.0, 815.0],
    },
    index=["Day 1", "Day 2", "Day 3", "Day 4"]
)

print("  Original DataFrame:")
print(stock_data)
print(f"\nShape: {stock_data.shape}")
print("Dtypes:")
print(stock_data.dtypes)
print(f"Memory usage (deep): {stock_data.memory_usage(deep=True).sum()} bytes")
```

```
 PANDAS TO NUMPY CONVERSION
========================================
  Original DataFrame:
        AAPL   GOOGL   MSFT   TSLA
Day 1  150.0  2800.0  300.0  800.0
Day 2  152.5  2750.0  305.0  795.0
Day 3  148.2  2825.0  298.5  820.0
Day 4  155.1  2900.0  310.0  815.0

Shape: (4, 4)
Dtypes:
AAPL     float64
```

```
GOOGL    float64
MSFT     float64
TSLA     float64
dtype: object
Memory usage (deep): 344 bytes
```

```python
# ----------------------------
# Method 1: .to_numpy() (recommended)
# ----------------------------
print("\n  Method 1 - .to_numpy() (recommended):")
array_modern = stock_data.to_numpy()
print(f"Shape: {array_modern.shape}")
print(f"Data type: {array_modern.dtype}")
array_modern
```

```
  Method 1 - .to_numpy() (recommended):
Shape: (4, 4)
Data type: float64

array([[ 150. , 2800. ,  300. ,  800. ],
       [ 152.5, 2750. ,  305. ,  795. ],
       [ 148.2, 2825. ,  298.5,  820. ],
       [ 155.1, 2900. ,  310. ,  815. ]])
```

```python
# ----------------------------
# Method 2: .values (legacy but still works)
# ----------------------------
print("\n   Method 2 - .values (legacy):")
array_legacy = stock_data.values
print(array_legacy)
print(f"Shape: {array_legacy.shape}")
print(f"Same result as .to_numpy()? {np.array_equal(array_modern, array_legacy)}")
```

```
   Method 2 - .values (legacy):
[[ 150.  2800.   300.   800. ]
 [ 152.5 2750.   305.   795. ]
 [ 148.2 2825.   298.5  820. ]
 [ 155.1 2900.   310.   815. ]]
Shape: (4, 4)
Same result as .to_numpy()? True
```

```python
# -----------------------------
# Method 3: Force a specific dtype (e.g., int32)
# -----------------------------
print("\n Method 3 - Force data type (int32):")
array_int = stock_data.to_numpy(dtype=np.int32)
print(array_int)
print(f"Shape: {array_int.shape}")
print(f"Data type: {array_int.dtype}")
print(f"Preview (first 2 rows):\n{array_int[:2]}")
```

```
 Method 3 - Force data type (int32):
[[ 150 2800  300  800]
 [ 152 2750  305  795]
 [ 148 2825  298  820]
 [ 155 2900  310  815]]
Shape: (4, 4)
Data type: int32
Preview (first 2 rows):
[[ 150 2800  300  800]
 [ 152 2750  305  795]]
```

```python
# -----------------------------
# Method 4: Convert specific columns only
# -----------------------------
print("\n  Method 4 - Specific columns only (AAPL, MSFT):")
tech_stocks = stock_data[["AAPL", "MSFT"]].to_numpy()
print(tech_stocks)
print(f"Shape: {tech_stocks.shape}")
print(f"Data type: {tech_stocks.dtype}")
```

```
  Method 4 - Specific columns only (AAPL, MSFT):
[[150.  300. ]
 [152.5 305. ]
 [148.2 298.5]
 [155.1 310. ]]
Shape: (4, 2)
Data type: float64
```

10.3.3.2 Method Comparison & Best Practices

Method	Recommendation	Pros	Cons
`.to_numpy()`	**Use by default**	Clear intent, stable API, future-proof	May copy data depending on dtypes/backing storage
`.to_numpy(dtype=)`	**When you need a dtype**	Explicit, consistent numeric types; easier math	Casting may be lossy or fail; extra param to manage
`.values`	**Legacy / avoid**	Works on old pandas; quick	Ambiguous with mixed/extension dtypes; not future-proof

Best practices - **Convert only needed columns** to reduce memory and copying. - **Specify `dtype`** when downstream computations require consistent numeric types. - **Check for missing values** (`NaN`, `NaT`) before casting to integer dtypes. - **Prefer `.to_numpy()` over `.values`** for clarity and forward compatibility. - Be mindful that conversions may **return a copy**, not a view—modify the DataFrame directly if needed.

10.3.4 Preserving Metadata During Conversion

Challenge: Converting a DataFrame to a NumPy array drops **column names, index labels, and dtypes**.

10.3.4.1 Practical Strategies

1) Save & restore index/columns (and dtypes)

```
df = pd.DataFrame({
    "A": pd.Series([1, 2, 3], dtype="Int64"),
    "B": pd.Categorical(["x", "y", "x"]),
}, index=["r1", "r2", "r3"])

print("Original DataFrame:")
print(df)
print(f"\nOriginal dtype of A: {df['A'].dtype}")  # Int64

# Convert to NumPy
arr = df.to_numpy()
print(f"\nNumPy array:\n{arr}")
print(f"NumPy array dtype: {arr.dtype}")  # object!
```

```
# Save metadata
cols = df.columns.copy()
index = df.index.copy()
dtypes = df.dtypes.to_dict()

# Rebuild DataFrame
df_out = pd.DataFrame(arr, index=index, columns=cols)
print(f"\nRebuilt DataFrame:")
print(df_out)
print(f"\nRebuilt dtype of A: {df_out['A'].dtype}")  # object, values are <NA>
```

```
Original DataFrame:
       A  B
r1  <NA>  x
r2  <NA>  y
r3  <NA>  x

Original dtype of A: Int64

NumPy array:
[[<NA> 'x']
 [<NA> 'y']
 [<NA> 'x']]
NumPy array dtype: object

Rebuilt DataFrame:
       A  B
r1  <NA>  x
r2  <NA>  y
r3  <NA>  x

Rebuilt dtype of A: object
```

2) Use a structured array to carry names

```
# Keeps field names (and index) inside a NumPy structured array
rec = df.to_records(index=True)               # dtype has named fields

# Back to DataFrame later
df_back = pd.DataFrame.from_records(rec).set_index("index")
df_back
```

	A	B
index		
r1	NaN	x
r2	NaN	y
r3	NaN	x

10.3.5 Comprehensive Comparison: When to Use What

Aspect	Pandas DataFrame	NumPy Array	Best Use
Data labels	Rich metadata (column names, index, dtypes)	No labels	**DataFrame** for exploration
Memory usage	Higher overhead	Minimal overhead	**Array** for large datasets
Speed	Slower (Python-level checks)	C-level, vectorized performance	**Array** for heavy compute
Missing data	Robust (`NaN`, `NaT`, nullable dtypes)	Limited (mainly float `NaN`)	**DataFrame** for dirty data
Indexing	Label & integer (`.loc`, `.iloc`)	Integer/boolean only	**DataFrame** for complex queries
Broadcasting	Limited/indirect	Full NumPy rules	**Array** for math ops
Data types	Mixed types per column	Homogeneous per array	**DataFrame** for mixed data
Plotting	Built-in `.plot()`	Manual setup	**DataFrame** for quick viz
Linear algebra	Limited	Rich (NumPy/BLAS/LAPACK)	**Array** for math/LA
Group operations	Powerful `.groupby()`, `.agg()`, `.transform()`	Manual	**DataFrame** for aggregations

10.3.5.1 Recommended Workflow

- **Exploratory analysis** → Use **Pandas**
- **Heavy computation** → Convert to **NumPy** with `df.to_numpy()`
- **Results & visualization** → Convert back to **Pandas**
- **Export & storage** → Use **Pandas** (CSV/Parquet/SQL)

Flow: Pandas (explore) → NumPy (compute) → Pandas (analyze/plot) → Pandas (export)

Tip: Convert only the needed columns and set `dtype` in `to_numpy(dtype=...)` when you need consistent numeric types. Handle missing values **before** casting.

10.4 Random Number Generation in NumPy

NumPy provides a variety of functions for generating random numbers through its numpy.random module. These functions can generate random numbers for different distributions and array shapes, making them essential for simulations, statistical sampling, and machine learning applications.

10.4.1 Key Functions for Random Number Generation in NumPy

10.4.1.1 `np.random.rand()`

It generates random numbers from a uniform distribution between 0 and 1.

```
# Generate a 2x3 array of random values between 0 and 1
rand_array = np.random.rand(2, 3)
print(rand_array)
```

```
[[0.39363552 0.47343566 0.85454739]
 [0.34000439 0.86964968 0.08813443]]
```

10.4.1.2 `np.random.randn()`

It generates random numbers from a standard normal distribution (mean = 0, standard deviation = 1). It is useful for simulating Gaussian distributed data

```
normal_array = np.random.randn(3, 3)
print(normal_array)
```

```
[[ 0.5376299   0.42818625 -0.2799933 ]
 [-1.27903074  0.51529432 -0.83428181]
 [ 2.18409673  0.57050721 -0.5806483 ]]
```

NumPy's random module can be used to generate arrays of random numbers from several different probability distributions. For example, a `3x5` array of uniformly distributed random numbers can be generated using the `uniform` function of the `random` module.

10.4.1.3 `np.random.randint()`

It generate random integers within a specific range, you can specify `low` and `high` values and the shape of the output array

```
# Generate a 4x4 matrix of random integers between 10 and 20
int_array = np.random.randint(10, 20, (4, 4))
print(int_array)
```

```
[[14 17 14 18]
 [15 17 19 17]
 [12 16 16 19]
 [15 16 11 18]]
```

10.4.1.4 `np.random.choice()`

Random selects elements from an input array

```
# Select 5 random elements from the array [1, 2, 3, 4, 5] with replacement
choice_array = np.random.choice([1, 2, 3, 4, 5], size=5, replace=True)
print(choice_array)
```

```
[4 3 5 2 2]
```

10.4.1.5 `np.random.uniform()`

It generates random floating-point numbers between a specified range (low, high) that follow uniform distribution

```
# Generate 6 random numbers from a normal distribution with mean=10 and std=2
custom_normal = np.random.normal(10, 2, size=6)
print(custom_normal)
```

```
[ 8.96337061  9.80411528  7.1623284   9.58883996 11.82330829 10.53508984]
```

```
# Generate a 1D array of 5 random numbers between -5 and 5
uniform_array = np.random.uniform(-5, 5, size=5)
print(uniform_array)
```

```
[-4.76424391 -2.90776829  4.56871733  2.3230159  -4.72083461]
```

10.4.1.6 `np.random.normal()`

It generates random numbers from a normal distribution with a specified mean and standard deviation.

```
np.random.uniform(size = (3,5))
```

```
array([[0.82164157, 0.10137984, 0.79189132, 0.811572  , 0.482274  ],
       [0.05731031, 0.17458985, 0.70742628, 0.71177794, 0.89835528],
       [0.58268615, 0.12410737, 0.12141491, 0.72113236, 0.18378467]])
```

For a full list, please check the offcial website

Random numbers can also be generated by Python's built-in random module. However, it generates one random number at a time, which makes it much slower than NumPy's random module.

10.5 Independent Study

10.5.1 Practice exercise 2:

This exercise will show vectorized computations with NumPy. Vectorized computations help perform computations more efficiently, and also make the code concise.

Q: Read the (1) quantities of roll, bun, cake and bread required by 3 people - Ben, Barbara & Beth, from *food_quantity.csv*, (2) price of these food items in two shops - Target and Kroger, from *price.csv*. Find out which shop should each person go to minimize their expenses.

```
#Reading the datasets on food quantity and price
import pandas as pd
food_qty = pd.read_csv('./datasets/food_quantity.csv',index_col=0)
price = pd.read_csv('./datasets/price.csv',index_col=0)
```

```
food_qty
```

	roll	bun	cake	bread
Person				
Ben	6	5	3	1
Barbara	3	6	2	2

Person	roll	bun	cake	bread
Beth	3	4	3	1

```
price
```

Item	Target	Kroger
roll	1.5	1.0
bun	2.0	2.5
cake	5.0	4.5
bread	16.0	17.0

First, let's start from a simple problem. We'll compute the expenses of Ben if he prefers to buy all food items from Target

```
%%time
# write a for loop to calculate the total cost of Ben's food if he shops at the Target store
total_cost = 0
for food in food_qty.columns:

    total_cost += food_qty.loc['Ben',food]*price.loc[food,'Target']
total_cost
```

```
CPU times: total: 0 ns
Wall time: 1 ms
```

```
50.0
```

```
%%time
# using numpy
total_cost = np.sum(food_qty.loc['Ben',]*price.loc[:,'Target'])
total_cost
```

```
CPU times: total: 0 ns
Wall time: 0 ns
```

```
50.0
```

Ben will spend $50 if he goes to Target

Now, let's add another layer of complication. We'll compute Ben's expenses for both stores - Target and Kroger

```
%%time
# using loops to calculate the total cost of food for Ben for all stores
total_cost = {}
for store in price.columns:
    total_cost[store] = 0
    for food in food_qty.columns:
        total_cost[store] += food_qty.loc['Ben',food]*price.loc[food,store]
total_cost
```

```
CPU times: total: 15.6 ms
Wall time: 999  s

{'Target': 50.0, 'Kroger': 49.0}
```

```
%%time
# using numpy
total_cost = np.dot(food_qty.loc['Ben',],price.loc[:,:])
total_cost
```

```
CPU times: total: 15.6 ms
Wall time: 1 ms

array([50., 49.])
```

Ben will spend $50 if he goes to Target, and $49 if he goes to Kroger. Thus, he should choose Kroger.

Now, let's add the final layer of complication, and solve the problem. We'll compute everyone's expenses for both stores - Target and Kroger

```
%%timeit
store_expense = pd.DataFrame(0.0, columns=price.columns, index = food_qty.index)
for person in store_expense.index:
    for store in store_expense.columns:
        for food in food_qty.columns:
            store_expense.loc[person, store] += food_qty.loc[person, food]*price.loc[food, s

store_expense
```

```
1.49 ms ± 22.5  s per loop (mean ± std. dev. of 7 runs, 1,000 loops each)
```

```
%%timeit
# using matrix multiplication in numpy
pd.DataFrame(np.dot(food_qty.values, price.values), columns=price.columns, index=food_qty.inc
```

```
11.3  s ± 93.2 ns per loop (mean ± std. dev. of 7 runs, 100,000 loops each)
```

Based on the above table, Ben should go to Kroger, Barbara to Target and Beth can go to either store.

As the complexity of operations increases, the number of nested for-loops tends to grow, making the code more cumbersome and difficult to manage. In contrast, leveraging NumPy arrays allows for concise and straightforward implementation, regardless of the complexity. Vectorized computations are not only cleaner but also significantly faster, offering a more efficient solution. To take advantage of this, you first need to convert a pandas DataFrame to a NumPy array for matrix multiplication. Once the computation is complete, you can convert the results back to a pandas DataFrame for further analysis or display

10.5.2 Practice exercise 3

Use matrix multiplication to find the average IMDB rating and average Rotten tomatoes rating for each genre - comedy, action, drama and horror. Use the data: *movies_cleaned.csv*. Which is the most preferred genre for IMDB users, and which is the least preferred genre for Rotten Tomatoes users?

Hint: 1. Create two matrices - one containing the IMDB and Rotten Tomatoes ratings, and the other containing the genre flags (comedy/action/drama/horror).

2. Multiply the two matrices created in 1.
3. Divide each row/column of the resulting matrix by a vector having the number of ratings in each genre to get the average rating for the genre.

Solution:

	Title	IMDB Rating	Rotten Tomatoes Rating	Running Time min	Release Date	US G
0	Broken Arrow	5.8	55	108	Feb 09 1996	7064
1	Brazil	8.0	98	136	Dec 18 1985	9929
2	The Cable Guy	5.8	52	95	Jun 14 1996	6024
3	Chain Reaction	5.2	13	106	Aug 02 1996	2122
4	Clash of the Titans	5.9	65	108	Jun 12 1981	3000

```
array([[ 5.8, 55. ],
       [ 8. , 98. ],
       [ 5.8, 52. ],
       ...,
       [ 7. , 65. ],
       [ 5.7, 26. ],
       [ 6.7, 82. ]])
```

```
array([[0, 1, 0, 0],
       [1, 0, 0, 0],
       [1, 0, 0, 0],
       ...,
       [1, 0, 0, 0],
       [0, 1, 0, 0],
       [0, 1, 0, 0]], dtype=int64)
```

We'll first find the total IMDB and Rotten tomatoes ratings for all movies of each genre, and then divide them by the number of movies of the corresponding genre to find the average rating for the genre.

For finding the total IMDB and Rotten tomatoes ratings, we'll multiply `drating_num` with `dgenre_num`. However, before multiplying, we'll check if their shapes are compatible for matrix multiplication.

```
#Shape of drating_num
drating_num.shape
```

```
(980, 2)
```

```
#Shape of dgenre_num
dgenre_num.shape
```

```
(980, 4)
```

Note that the above shapes are not compatible for matrix multiplication. We'll transpose `dgenre_num` to make the shapes compatible.

```
#Total IMDB and Rotten tomatoes ratings for each genre
ratings_sum_genre = drating_num.T.dot(dgenre_num)
ratings_sum_genre
```

```
array([[ 1785.6,  1673.1,  1630.3,   946.2],
       [14119. , 13725. , 14535. ,  6533. ]])
```

```
#Number of movies in the data will be stored in 'rows', and number of columns stored in 'cols
rows, cols = data.shape
```

```
#Getting number of movies in each genre
movies_count_genre = dgenre_num.T.dot(np.ones(rows))
movies_count_genre
```

```
array([302., 264., 239., 154.])
```

```
#Finding the average IMDB and average Rotten tomatoes ratings for each genre
ratings_sum_genre/movies_count_genre
```

```
array([[ 5.91258278,  6.3375    ,  6.82133891,  6.14415584],
       [46.75165563, 51.98863636, 60.81589958, 42.42207792]])
```

```
pd.DataFrame(ratings_sum_genre/movies_count_genre,columns = ['comedy','Action','drama','horro
             index = ['IMDB Rating','Rotten Tomatoes Rating'])
```

	comedy	Action	drama	horror
IMDB Rating	5.912583	6.337500	6.821339	6.144156
Rotten Tomatoes Rating	46.751656	51.988636	60.815900	42.422078

IMDB users prefer *drama*, and are amused the least by *comedy* movies, on an average. However, Rotten tomatoes critics would rather watch *comedy* than *horror* movies, on an average.

10.5.3 Practice exercise 4:

Random number generation using NumPy

Suppose 500 people eat at Food cart 1, and another 500 eat at Food cart 2, everyday.

The waiting time at Food cart 2 has a normal distribution with mean 8 minutes and standard deviation 3 minutes, while the waiting time at Food cart 1 has a uniform distribution with minimum 5 minutes and maximum 25 minutes.

Simulate a dataset containing waiting times for 500 ppl for 30 days in each of the food joints. Assume that the waiting times are measured simultaneously at a certain time in both places, i.e., the observations are paired.

On how many days is the average waiting time at Food cart 2 higher than that at Food cart 1?

What percentage of times the waiting time at Food cart 2 was higher than the waiting time at Food cart 1?

Try both approaches: (1) Using loops to generate data, (2) numpy array to generate data. Compare the time taken in both approaches.

```
import time as tm
```

```
#Method 1: Using loops
start_time = tm.time() #Current system time

#Initializing waiting times for 500 ppl over 30 days
waiting_times_FoodCart1 = pd.DataFrame(0,index=range(500),columns=range(30)) #FoodCart1
waiting_times_FoodCart2 = pd.DataFrame(0,index=range(500),columns=range(30)) #FoodCart2
import random as rm
for i in range(500):  #Iterating over 500 ppl
    for j in range(30): #Iterating over 30 days
        waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCa
        waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in Food
time_diff = waiting_times_FoodCart2-waiting_times_FoodCart1

print("On ",sum(time_diff.mean()>0)," days, the average waiting time at FoodCart2 higher than
print("Percentage of times waiting time at FoodCart2 was greater than that at FoodCart1 = ",
end_time = tm.time() #Current system time
print("Time taken = ", end_time-start_time)
```

```
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 for
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 for
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```

```
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
```

```
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
```

```
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
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C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:10: FutureWarning: Setting
  waiting_times_FoodCart2.iloc[i,j] = rm.gauss(8,3) #Simulating waiting time in FoodCart2 fo
C:\Users\lsi8012\AppData\Local\Temp\ipykernel_28136\4118008220.py:11: FutureWarning: Setting
  waiting_times_FoodCart1.iloc[i,j] = rm.uniform(5,25) #Simulating waiting time in FoodCart1

On  0  days, the average waiting time at FoodCart2 higher than that at FoodCart1
Percentage of times waiting time at FoodCart2 was greater than that at FoodCart1 =  16.38 %
Time taken =  3.045159339904785
```

```python
#Method 2: Using NumPy arrays
start_time = tm.time()
waiting_time_FoodCart2 = np.random.normal(8,3,size = (500,30)) #Simultaneously generating the
waiting_time_FoodCart1 = np.random.uniform(5,25,size = (500,30)) #Simultaneously generating
time_diff = waiting_time_FoodCart2-waiting_time_FoodCart1
print("On ",(time_diff.mean()>0).sum()," days, the average waiting time at FoodCart2 higher
print("Percentage of times waiting time at FoodCart2 was greater than that at FoodCart1 = ",
end_time = tm.time()
print("Time taken = ", end_time-start_time)
```

```
On  0  days, the average waiting time at FoodCart2 higher than that at FoodCart1
Percentage of times waiting time at FoodCart2 was greater than that at FoodCart1 =  16.26 %
Time taken =  0.0
```

The approach with NumPy is much faster than the one with loops.

10.5.4 Practice exercise 5

Bootstrapping: Find the 95% confidence interval of mean profit for 'Action' movies, using Bootstrapping.

Bootstrapping is a non-parametric method for obtaining confidence interval. Use the algorithm below to find the confidence interval:

1. Find the profit for each of the 'Action' movies. Suppose there are *N* such movies. We will have a *Profit* column with *N* values.

2. Randomly sample *N* values with replacement from the *Profit* column

3. Find the mean of the *N* values obtained in (b)

4. Repeat steps (b) and (c) *M=1000* times

5. The 95% Confidence interval is the range between the 2.5% and 97.5% percentile values of the 1000 means obtained in (c)
   Use the *movies_cleaned.csv* dataset.

Solution:

11 Introduction to Data Visualization

```
<IPython.core.display.Image object>
```

"One picture is worth a thousand words" - Fred R. Barnard

Visual perception offers the highest bandwidth channel, as we acquire much more information through visual perception than with all of the other channels combined, as billions of our neurons are dedicated to this task. Moreover, the processing of visual information is, at its first stages, a highly parallel process. Thus, it is generally easier for humans to comprehend information with plots, diagrams and pictures, rather than with text and numbers. This makes data visualizations a vital part of data science. Some of the key purposes of data visualization are:

1. Data visualization is the first step towards exploratory data analysis (EDA), which reveals trends, patterns, insights, or even irregularities in data.
2. Data visualization can help explain the workings of complex mathematical models.
3. Data visualization are an elegant way to summarise the findings of a data analysis project.
4. Data visualizations (especially interactive ones such as those on Tableau) may be the end-product of data analytics project, where the stakeholders make decisions based on the visualizations.

11.1 The Art of Visualization: Choosing the Right Plot Type

There are various types of plots available, and selecting the appropriate one is crucial for successful data visualization. The choice primarily depends on two factors: * The type of data you are working with, and * The role of visualization in your data analysis

11.1.1 Data Classification for Visualization

Data visualization is commonly used to plot data in a pandas DataFrame. The data can be classified into two categories:

- Numeric Data: This type of data represents quantities and can take any value within a range. Common examples include age, height, temperature, etc.

- Categorical Data: This type of data represents distinct categories or groups. It can be nominal (no inherent order, like colors or names) or ordinal (with a defined order, like ratings).

11.1.2 The Role of Visualization in Data Analysis

Data visualization is essential for effectively communicating insights derived from data analysis. By using various visualization techniques, we can **uncover patterns, and understand relationships**. Below, we discuss different types of data exploration and the relevant visualizations used for each.

11.1.2.1 Univariate Exploration

Purpose: Univariate exploration analyzes a single variable to understand its distribution, central tendency, and spread.

11.1.2.1.1 Common Visualizations:

- **Histograms**: Display the frequency distribution of a numeric variable, helping to identify the shape of the data (e.g., normal, skewed).
- **Box Plots**: Summarize key statistics of a variable, including median, quartiles, and potential outliers.
- **Bar Plots**: Show the count or proportion of categorical variables, revealing the frequency of each category.
- **Line Plots**: Used to display trends in numeric data over time, helping to visualize changes in a variable.

11.1.2.1.2 Insights Gained:

- Identify outliers and anomalies.
- Understand the range and distribution of values.
- Determine central tendency (mean, median, mode).

11.1.2.2 Bivariate Analysis

Purpose: Bivariate analysis examines the relationship between two variables, helping to understand how changes in one variable might affect another.

11.1.2.2.1 Common Visualizations:

- **Scatter Plots**: Illustrate the relationship between two numeric variables, highlighting trends and correlations.
- **Grouped Bar Plots**: Compare categorical variables against a numeric variable, revealing trends across categories.
- **Heatmaps**: Represent correlation coefficients between pairs of variables, allowing easy identification of strong correlations.

11.1.2.2.2 Insights Gained:

- Assess the strength and direction of relationships (positive, negative, or no correlation).
- Identify potential predictive relationships for further analysis.
- Discover patterns that may indicate causal relationships.

11.1.2.3 Multivariate Analysis

Purpose: Multivariate analysis investigates more than two variables simultaneously, providing a comprehensive view of complex relationships and interactions.

11.1.2.3.1 Common Visualizations:

- **Pair Plots**: Show pairwise relationships in a dataset, facilitating quick insights into correlations among multiple variables.
- **3D Scatter Plots**: Visualize the interaction between three numeric variables in a three-dimensional space.
- **Facet Grids**: Display multiple plots for different subsets of data, enabling comparisons across categories.

11.1.2.3.2 Insights Gained:

- Understand interactions and dependencies among multiple variables.
- Identify clusters or groups within the data.
- Enhance predictive modeling by considering multiple influences.

11.1.3 Summary

Choosing the appropriate plot depends on the data type and the specific analysis purpose. Numeric data typically requires plots that can handle continuous data (like line plots or histograms), while categorical data often benefits from comparisons (like bar plots or pie charts). Always consider what story you want to tell with your data and select your visualization method accordingly.

11.2 Visulization Tools

We'll use three libraries for making data visualizations - pandas, matplotlib, and seaborn.

To get started, let's import these libraries.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# let's import numpy as well
import numpy as np
```

11.2.1 Basic Plotting with Pandas

In previous chapters, we focused on using the pandas library for data reading and analysis. In addition to its powerful data manipulation capabilities, pandas also provides tools for creating basic plots, making it especially valuable for exploratory data analysis.

In this section, we will use the COVID dataset to demonstrate basic plotting techniques with pandas.

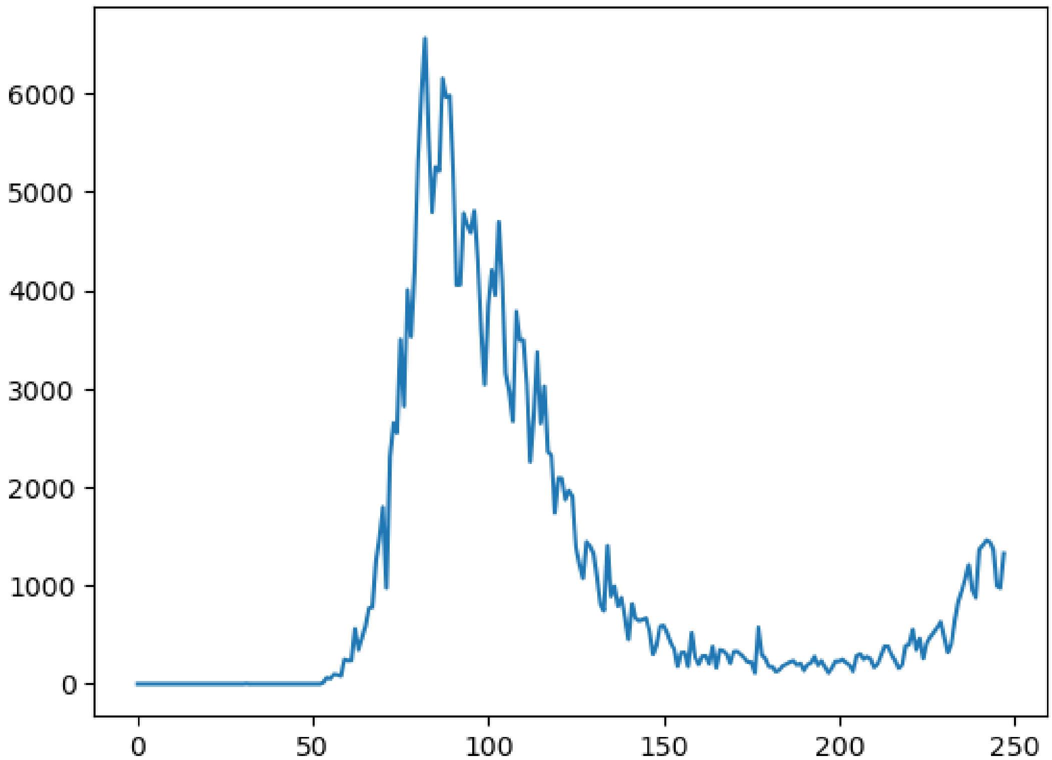
```
covid_df = pd.read_csv('./Datasets/covid.csv')
covid_df.head(5)
```

	date	new_cases	total_cases	new_deaths	total_deaths	new_tests	total_tests	cases_per_
0	2019-12-31	0.0	0.0	0.0	0.0	NaN	NaN	0.0
1	2020-01-01	0.0	0.0	0.0	0.0	NaN	NaN	0.0
2	2020-01-02	0.0	0.0	0.0	0.0	NaN	NaN	0.0
3	2020-01-03	0.0	0.0	0.0	0.0	NaN	NaN	0.0
4	2020-01-04	0.0	0.0	0.0	0.0	NaN	NaN	0.0

Let's begin by visualizing the trend of new COVID cases over time using a line plot.

A line plot is ideal for showing changes over continuous data, such as the progression of new cases over a series of dates.

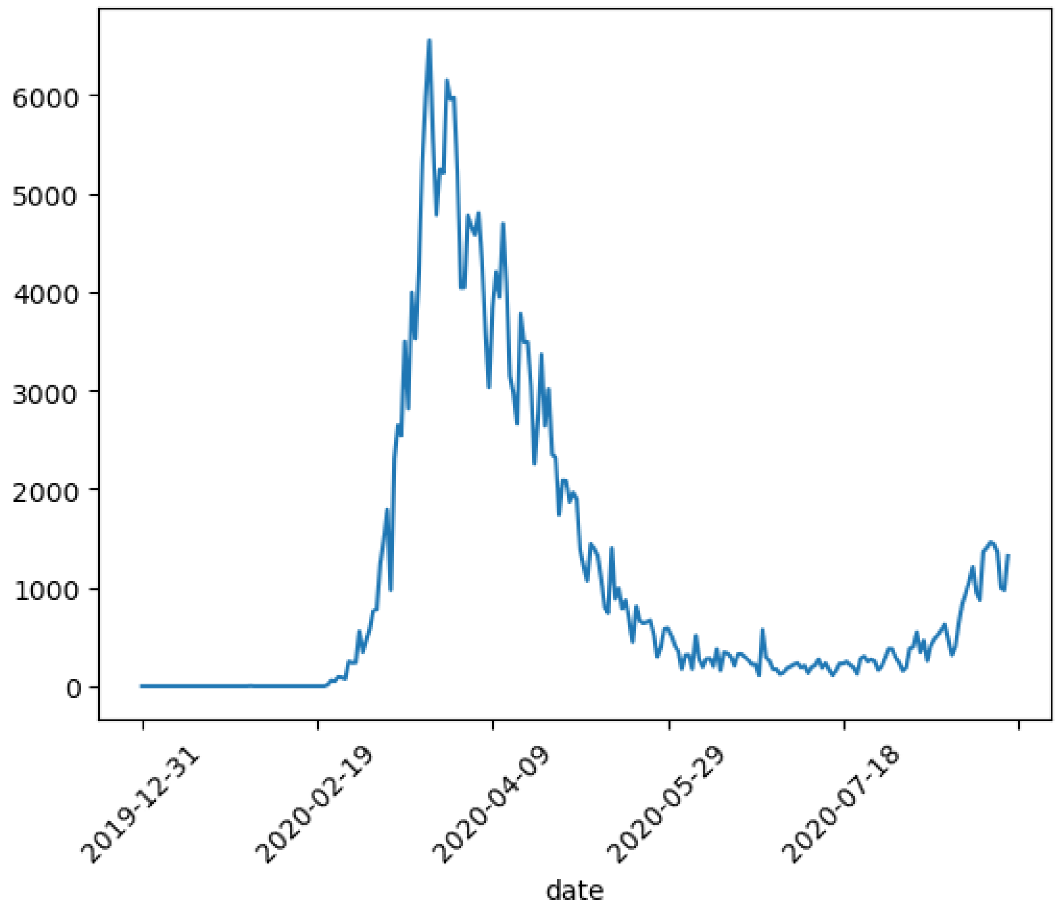
```
covid_df.new_cases.plot()
```



While this plot shows the overall trend, it's hard to tell where the peak occurred, as there are no dates on the X-axis. We can use the date column as the index for the data frame to address this issue since it is a time series dataset

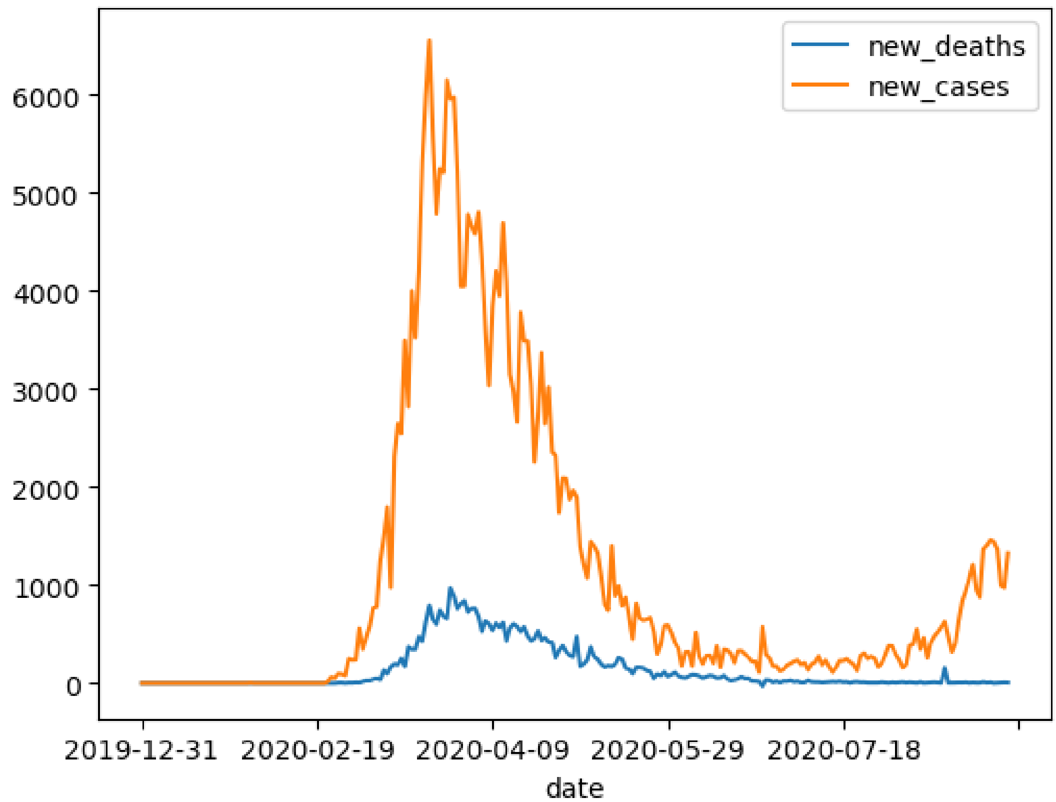
```
covid_df.set_index('date', inplace=True)
```

```
covid_df.new_cases.plot(rot=45)
```



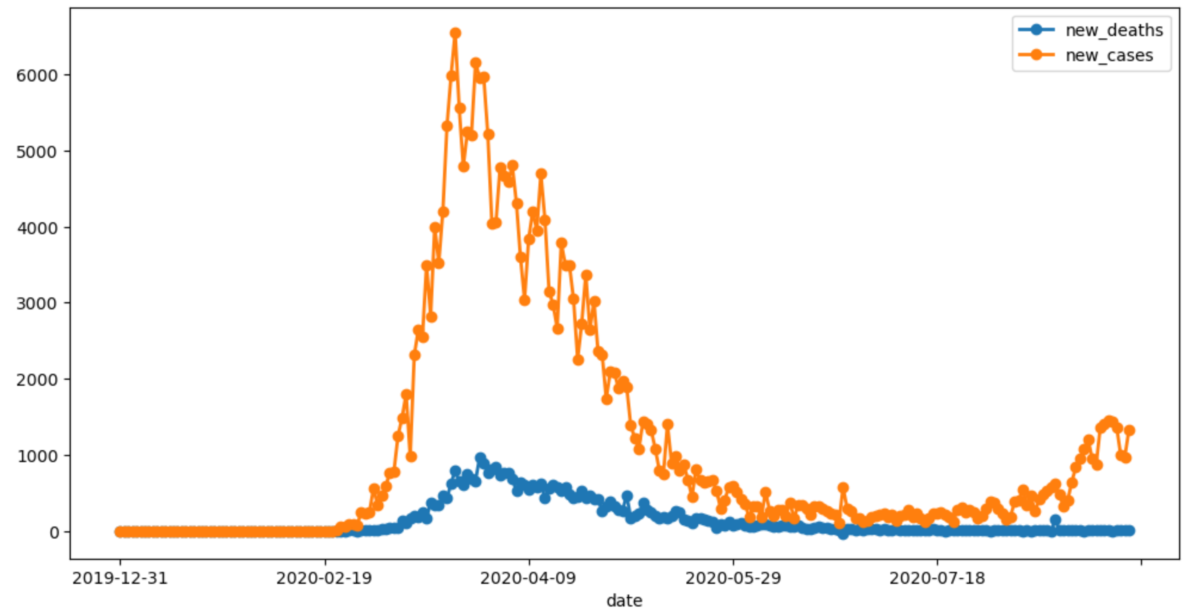
With the date set as the index, we can observe that the peak occurred around March 2020. Next, let's plot both new cases and new deaths together to compare their trends over the same time period.

```
covid_df[['new_deaths', 'new_cases']].plot()
```



By default, pandas generates line plots when using the `.plot` method. However, there are several parameters you can adjust to enhance the appearance of the line plot.

```
covid_df[['new_deaths', 'new_cases']].plot(figsize=(12, 6), linewidth=2, marker='o')
```

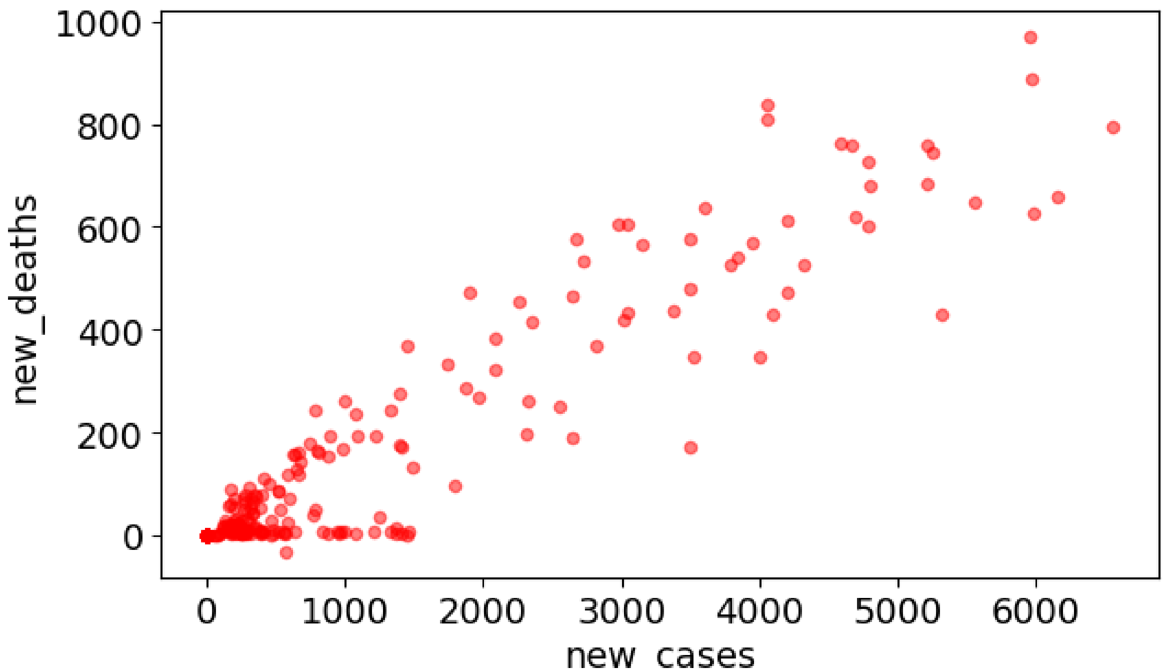


You can create other types of visualizations by setting the kind parameter in the plot function. The kind parameter accepts eleven different string values, which specify the type of plot:

- “area” is for area plots.
- “bar” is for vertical bar charts.
- “barh” is for horizontal bar charts.
- “box” is for box plots.
- “hexbin” is for hexbin plots.
- “hist” is for histograms.
- “kde” is for kernel density estimate charts.
- “density” is an alias for “kde”.
- “line” is for line graphs.
- “pie” is for pie charts.
- “scatter” is for scatter plots.

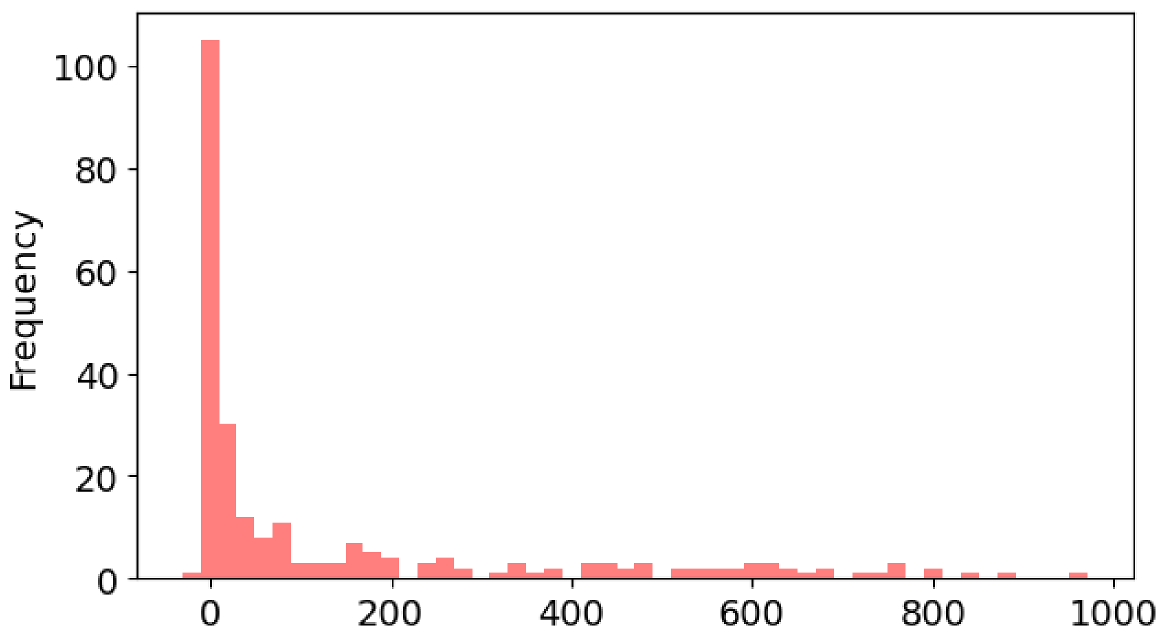
Let’s next create a scatter plot to visualize the relationship between new cases and new deaths, and explore whether there’s a correlation between them.

```
covid_df.plot(kind='scatter', x='new_cases', y='new_deaths', color='r', alpha=0.5);
```

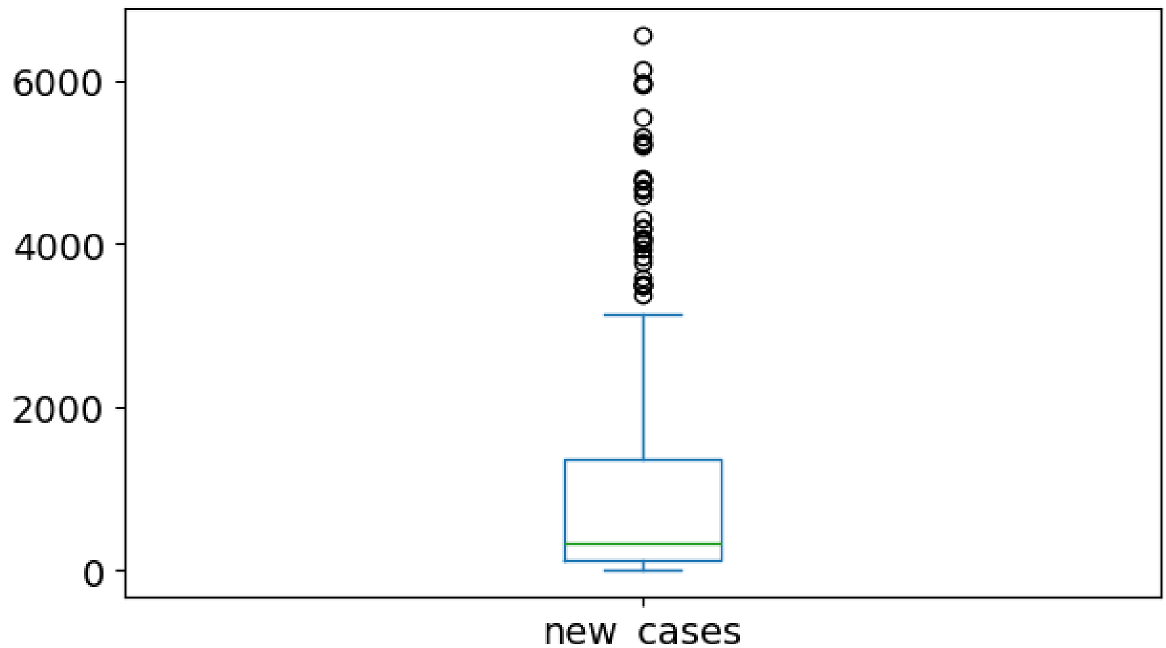


Next, let's examine the distribution of new deaths using a histogram.

```
covid_df.new_deaths.plot(kind='hist', color='r', alpha=0.5, bins=50);
```



```
covid_df.new_cases.plot(kind='box');
```



For more plot types and detailed information, refer to the official pandas documentation:

- Series.plot
- DataFrame.plot

11.2.1.1 Limitation of using pandas for plotting

While pandas provides a straightforward way to create plots, there are some limitations to be aware of:

- Customization: Pandas offers basic customization options, but it may not provide the level of detail or flexibility that you can achieve with matplotlib directly. For complex visualizations, you might need to switch to matplotlib for more control.
- Plot Types: The range of plot types available in pandas is limited compared to what you can create with matplotlib or seaborn. For instance, advanced plots like violin plots or 3D plots require switching to other libraries.
- Aesthetic Choices: The default aesthetics in pandas may not be as visually appealing as those created using seaborn or other specialized visualization libraries. For polished presentations, additional customization might be necessary.

11.2.2 Data Plotting with Matplotlib `Pyplot` Interface

Pandas data visualization is built on top of matplotlib. When you use the `.plot()` method in pandas, it internally calls matplotlib functions to create the plots.

Matplotlib is:

- a low-level graph plotting library in python that strives to emulate MATLAB,
- can be used in Python scripts, Python and IPython shells, Jupyter notebooks and web application servers.
- is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.

11.2.2.1 Matplotlib pyplot

- Matplotlib is the whole package; pyplot is a module in matplotlib
- Most of the Matplotlib utilities lies under the pyplot module, and are usually imported under the plt alias:

```
import matplotlib.pyplot as plt
```

11.2.2.2 Data Source

- Python lists, NumPy arrays as well as a pandas series
- However, all the sequences are internally converted to numpy arrays.

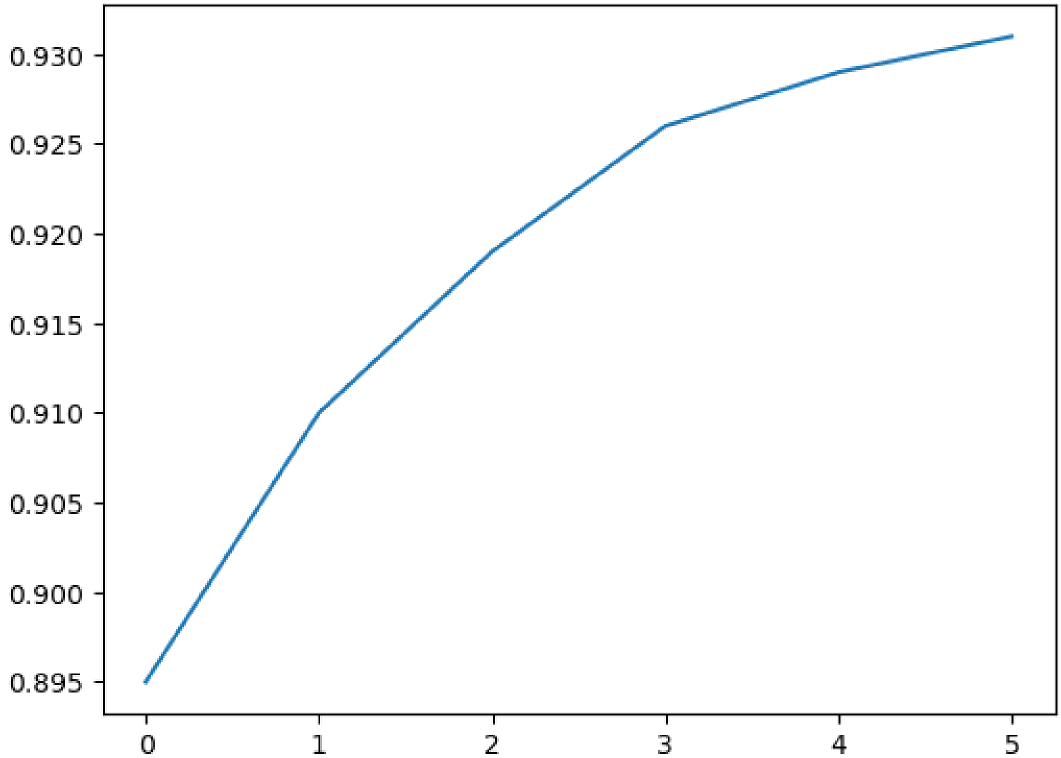
Let's create a python list to illustrate basic plotting with Matplotlib pyplot

```
yield_apples = [0.895, 0.91, 0.919, 0.926, 0.929, 0.931]
```

11.2.2.3 Basic Plotting

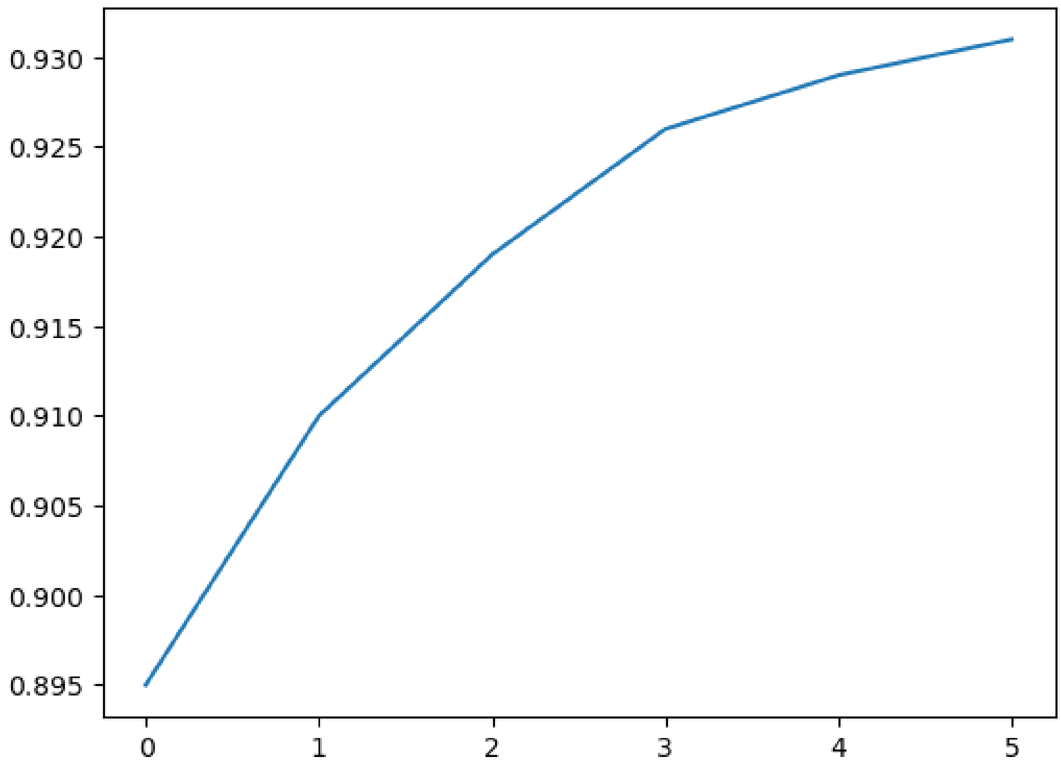
11.2.2.3.1 Plotting the overall trend

```
plt.plot(yield_apples)
```



Calling the `plt.plot` function draws the line chart as expected. It also returns a list of plots drawn `[<matplotlib.lines.Line2D at 0x2194b571df0>]`, shown within the output. We can include a semicolon (`;`) at the end of the last statement in the cell to avoiding showing the output and display just the graph.

```
plt.plot(yield_apples);
```

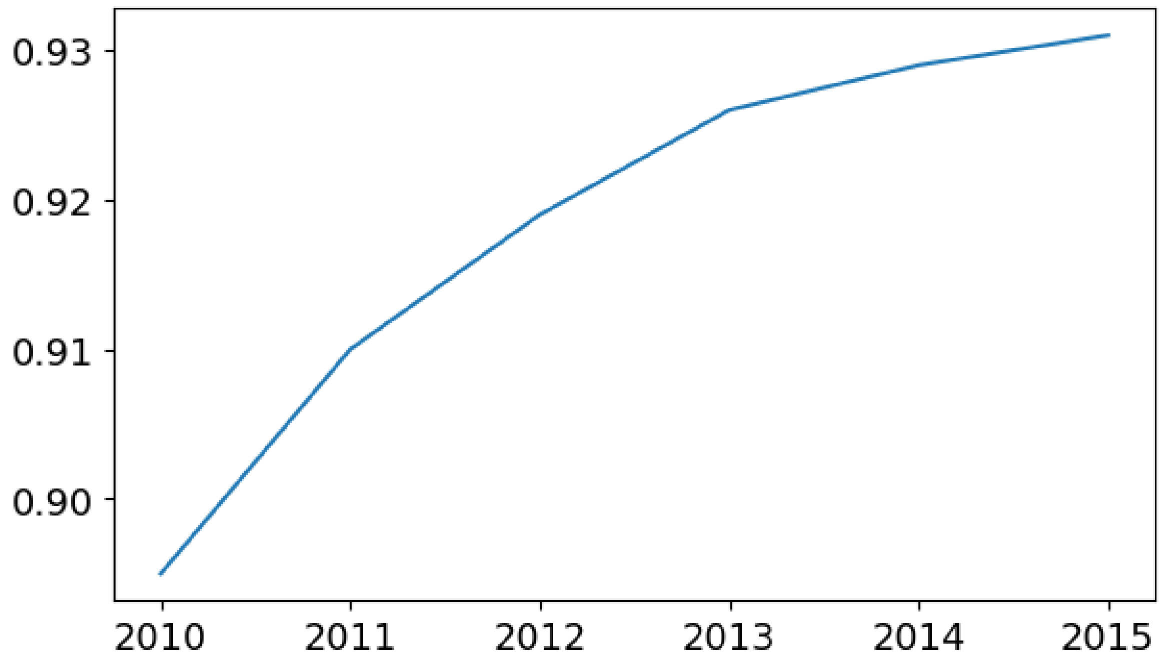


11.2.2.3.2 Customizing the X-axis

The X-axis of the plot currently shows list element indexes 0 to 5. The plot would be more informative if we could display the year for which we're plotting the data. We can do this by two arguments `plt.plot`.

```
years = [2010, 2011, 2012, 2013, 2014, 2015]
yield_apples = [0.895, 0.91, 0.919, 0.926, 0.929, 0.931]
```

```
plt.plot(years, yield_apples);
```

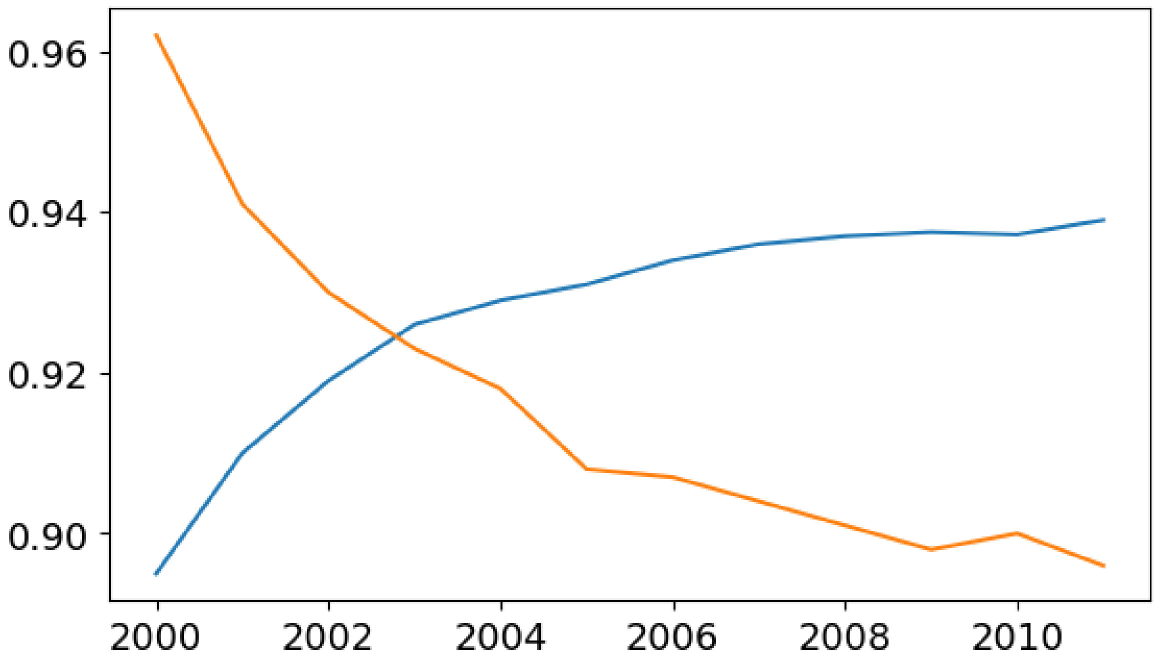


11.2.2.3.3 Ploting multiple lines

You can invoke the `plt.plot` function once for each line to plot multiple lines in the same graph. Let's compare the yields of apples vs. oranges.

```
years = range(2000, 2012)
apples = [0.895, 0.91, 0.919, 0.926, 0.929, 0.931, 0.934, 0.936, 0.937, 0.9375, 0.9372, 0.939
oranges = [0.962, 0.941, 0.930, 0.923, 0.918, 0.908, 0.907, 0.904, 0.901, 0.898, 0.9, 0.896,
```

```
plt.plot(years, apples)
plt.plot(years, oranges);
```



When plt.plot command is called without any formatting parameters, pyplot uses the following defaults:

- Figure size: 6.4 X 4.8 inches
- Plot style: solid line
- Linewidth: 1.5
- Color: Blue (code 'b', hex code: '#1f77b4')

You can also edit default styles directly by modifying the `matplotlib.rcParams` dictionary. Learn more: https://matplotlib.org/3.2.1/tutorials/introductory/customizing.html#matplotlib-rcparams .

You can customize default plot styles by directly modifying the `matplotlib.rcParams` dictionary. For more details, visit the official Matplotlib guide on customizing with rcParams.

```
import matplotlib
matplotlib.rcParams['font.size'] = 14
matplotlib.rcParams['figure.figsize'] = (7, 4)
matplotlib.rcParams['figure.facecolor'] = '#00000000'
```

Conceptual model: Plotting in Matplotlib involves multiple levels of control, from setting the figure size to customizing individual text elements. To offer complete control over the plotting process, Matplotlib provides an object-oriented interface in a hierarchical structure. This approach allows users to create and manage `Figure` and `Axes` objects, which serve as the foundation for all plotting actions. In the next chapter, you will explore how to use this object-oriented interface to gain more precise control over your plots.

11.2.2.4 Enhancing the plot

Matplotlib provides a wide range of customizable components within a figure, allowing for fine-tuned control over every aspect of the plot. These components include elements like axes, labels, ticks, legends, and the overall layout. Each can be tailored to enhance the clarity, aesthetics, and effectiveness of the visual representation, making the plot more engaging and easier to interpret.

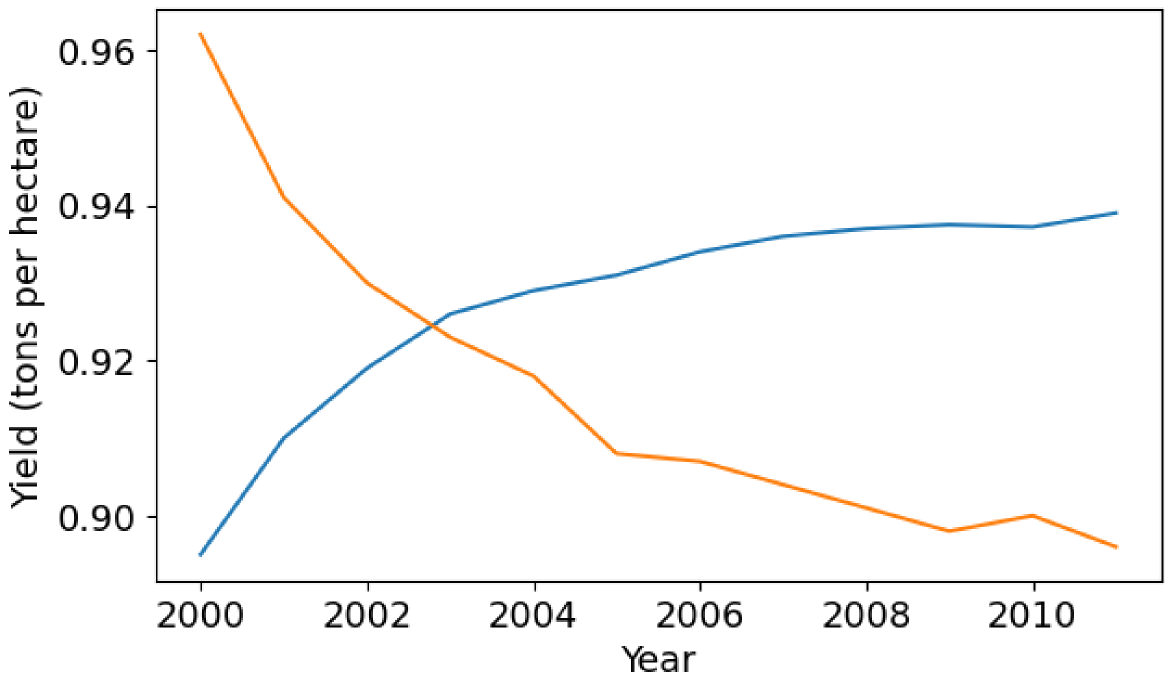
```
<IPython.core.display.Image object>
```

Figure 11.1: Matplotlib anatomy of a figure

11.2.2.4.1 Adding Axis Lables

We can add labels to the axes to show what each axis represents using the `plt.xlabel` and `plt.ylabel` methods.

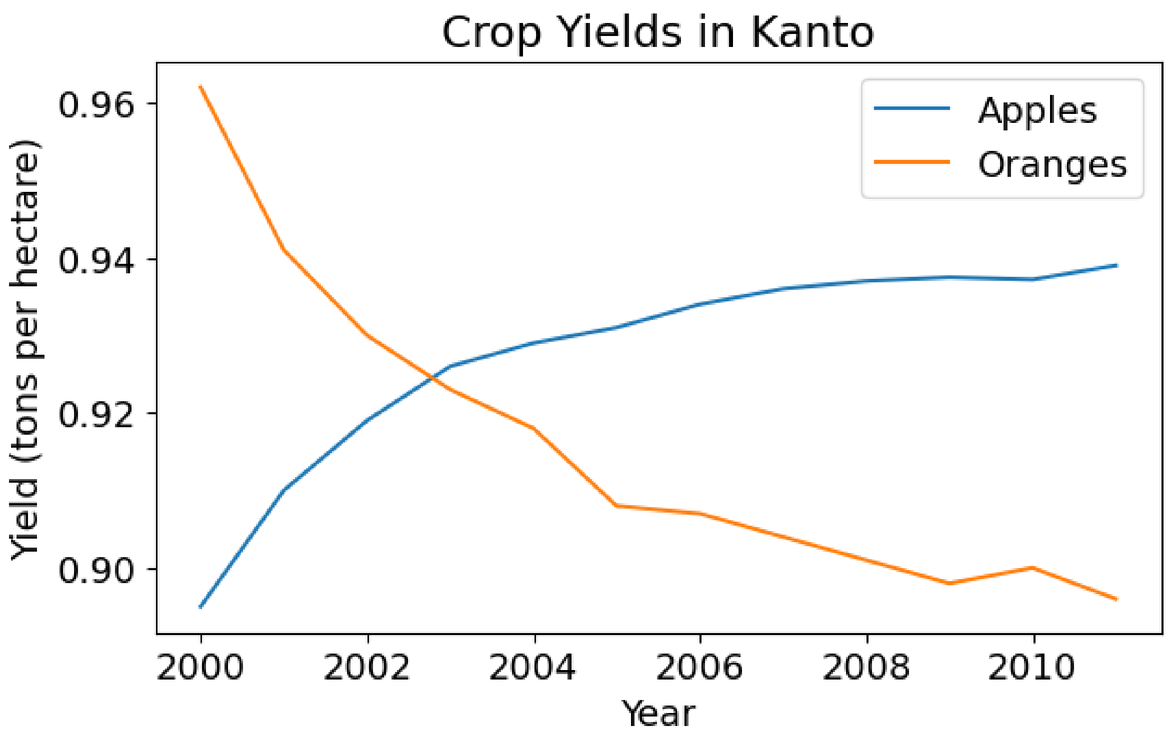
```
plt.plot(years, apples)
plt.plot(years, oranges)
plt.xlabel('Year')
plt.ylabel('Yield (tons per hectare)');
```



11.2.2.4.2 Adding Chart Title and Legend

To differentiate between multiple lines, we can include a legend within the graph using the `plt.legend` function. We can also set a title for the chart using the `plt.title` function.

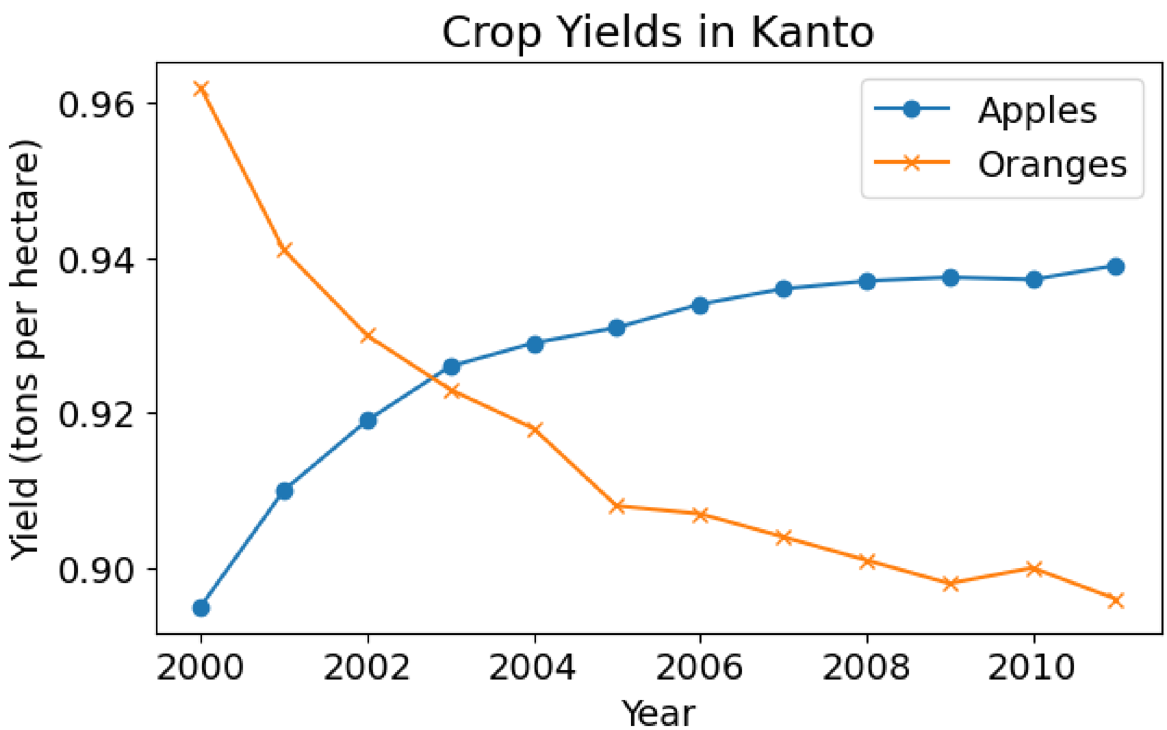
```
plt.plot(years, apples)
plt.plot(years, oranges)

plt.xlabel('Year')
plt.ylabel('Yield (tons per hectare)')

plt.title("Crop Yields in Kanto")
plt.legend(['Apples', 'Oranges']);
```



11.2.2.4.3 Adding Line Markers

We can also show markers for the data points on each line using the `marker` argument of `plt.plot`. Matplotlib provides many different markers, like a circle, cross, square, diamond, etc. You can find the full list of marker types here.

```
plt.plot(years, apples, marker='o')
plt.plot(years, oranges, marker='x')

plt.xlabel('Year')
plt.ylabel('Yield (tons per hectare)')

plt.title("Crop Yields in Kanto")
plt.legend(['Apples', 'Oranges']);
```



11.2.2.4.4 Styling Lines and Markers

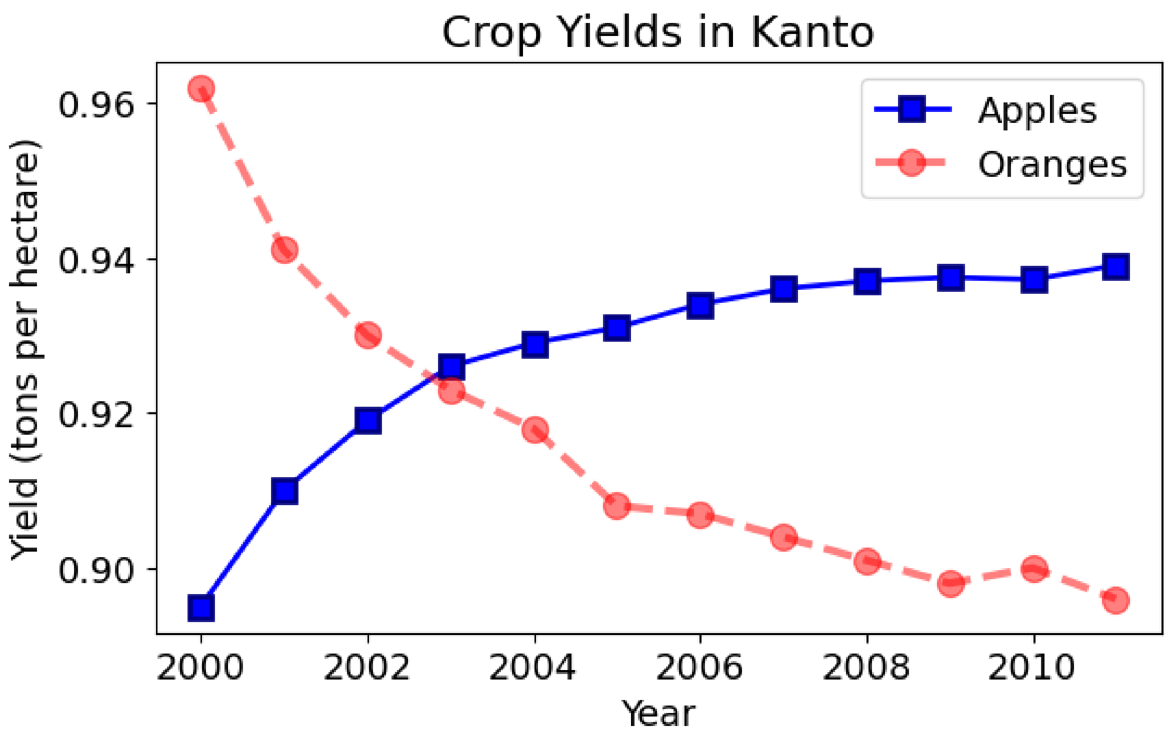
The `plt.plot` function supports many arguments for styling lines and markers:

- `color` or `c`: Set the color of the line (supported colors)
- `linestyle` or `ls`: Choose between a solid or dashed line
- `linewidth` or `lw`: Set the width of a line
- `markersize` or `ms`: Set the size of markers
- `markeredgecolor` or `mec`: Set the edge color for markers
- `markeredgewidth` or `mew`: Set the edge width for markers
- `markerfacecolor` or `mfc`: Set the fill color for markers
- `alpha`: Opacity of the plot

Check out the documentation for `plt.plot` to learn more: https://matplotlib.org/api/_as_gen/matplotlib.pyplot.plot.html#matplotlib.pyplot.plot .

```
plt.plot(years, apples, marker='s', c='b', ls='-', lw=2, ms=8, mew=2, mec='navy')
plt.plot(years, oranges, marker='o', c='r', ls='--', lw=3, ms=10, alpha=.5)

plt.xlabel('Year')
plt.ylabel('Yield (tons per hectare)')
```

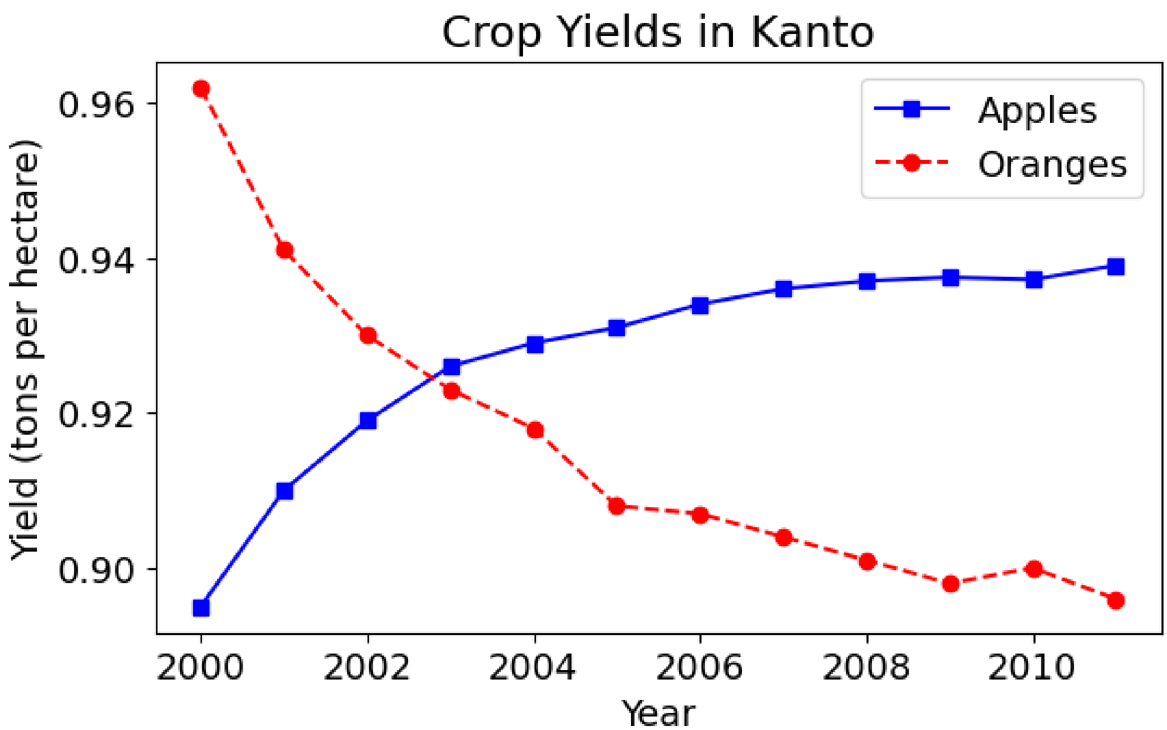
```
plt.title("Crop Yields in Kanto")
plt.legend(['Apples', 'Oranges']);
```



The `fmt` argument provides a shorthand for specifying the marker shape, line style, and line color. It can be provided as the third argument to `plt.plot`.

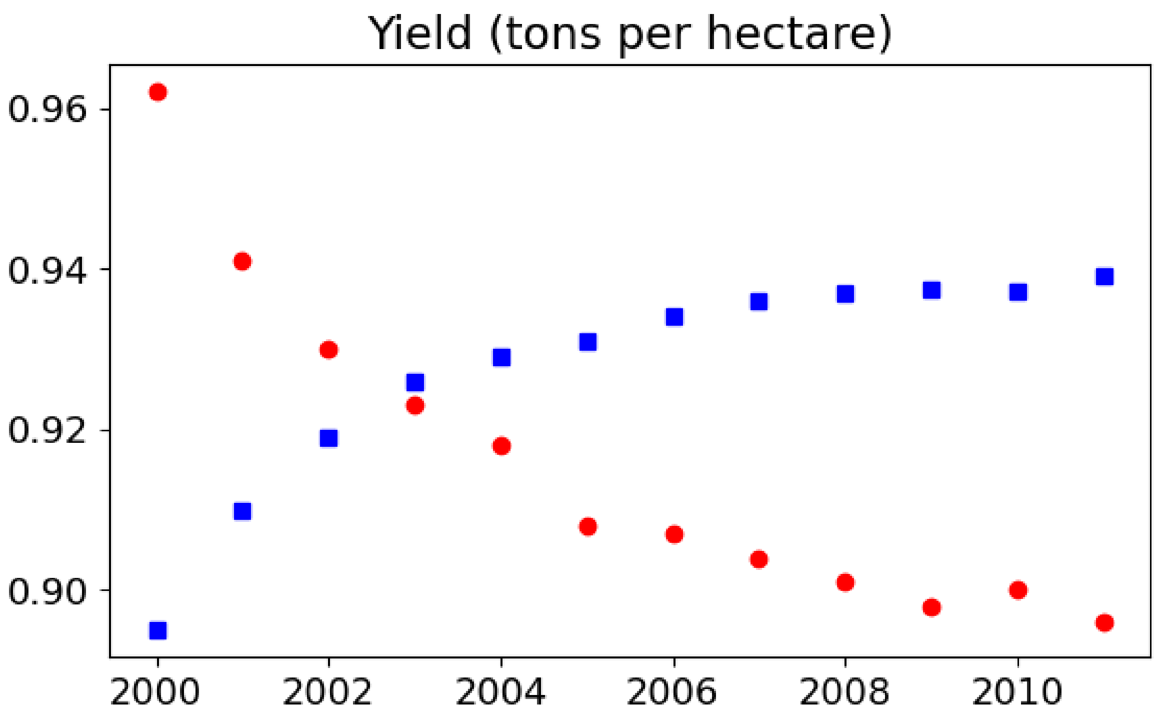
```
fmt = '[marker][line][color]'
```

```
plt.plot(years, apples, 's-b')
plt.plot(years, oranges, 'o--r')

plt.xlabel('Year')
plt.ylabel('Yield (tons per hectare)')

plt.title("Crop Yields in Kanto")
plt.legend(['Apples', 'Oranges']);
```



If you don't specify a line style in `fmt`, only markers are drawn.

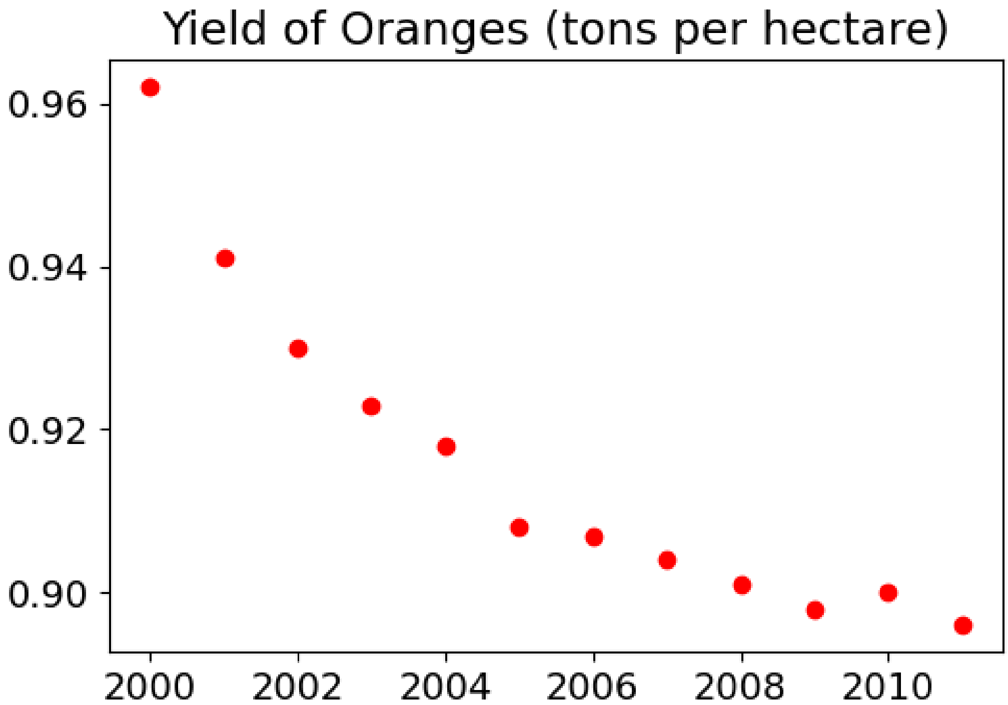
```
plt.plot(years, apples, 'sb')
plt.plot(years, oranges, 'or')
plt.title("Yield (tons per hectare)");
```



11.2.2.4.5 Changing the Figure Size

You can use the `plt.figure` function to change the size of the figure.

```
plt.figure(figsize=(6, 4))

plt.plot(years, oranges, 'or')
plt.title("Yield of Oranges (tons per hectare)");
```



11.2.2.5 Plotting other types of plots with matplotlib `pyplot`

Let's read the *fifa_data.csv* as our data source

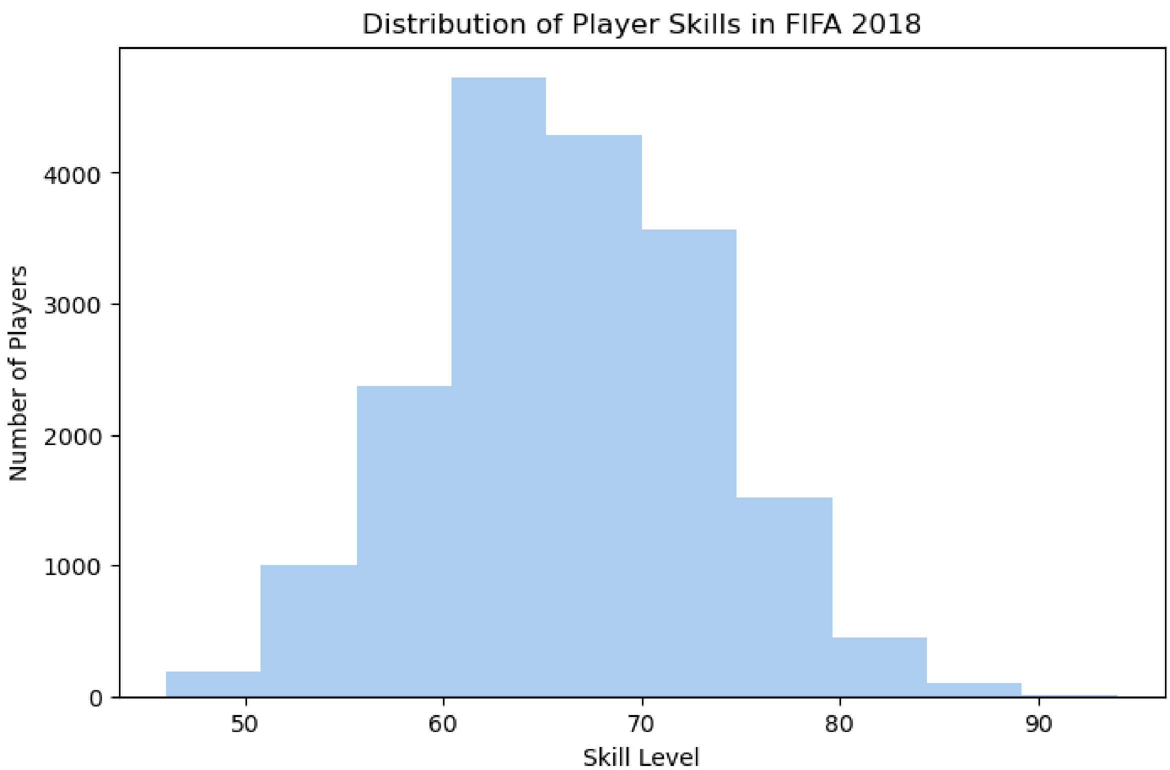
```
fifa = pd.read_csv('./Datasets/fifa_data.csv')
fifa.head(5)
```

	Unnamed: 0	ID	Name	Age	Photo	Natio
0	0	158023	L. Messi	31	https://cdn.sofifa.org/players/4/19/158023.png	Argen
1	1	20801	Cristiano Ronaldo	33	https://cdn.sofifa.org/players/4/19/20801.png	Portu
2	2	190871	Neymar Jr	26	https://cdn.sofifa.org/players/4/19/190871.png	Brazil
3	3	193080	De Gea	27	https://cdn.sofifa.org/players/4/19/193080.png	Spain
4	4	192985	K. De Bruyne	27	https://cdn.sofifa.org/players/4/19/192985.png	Belgiu

11.2.2.5.1 Histgram

```
plt.figure(figsize=(8,5))

plt.hist(fifa.Overall, color='#abcdef')

plt.ylabel('Number of Players')
plt.xlabel('Skill Level')
plt.title('Distribution of Player Skills in FIFA 2018')
```

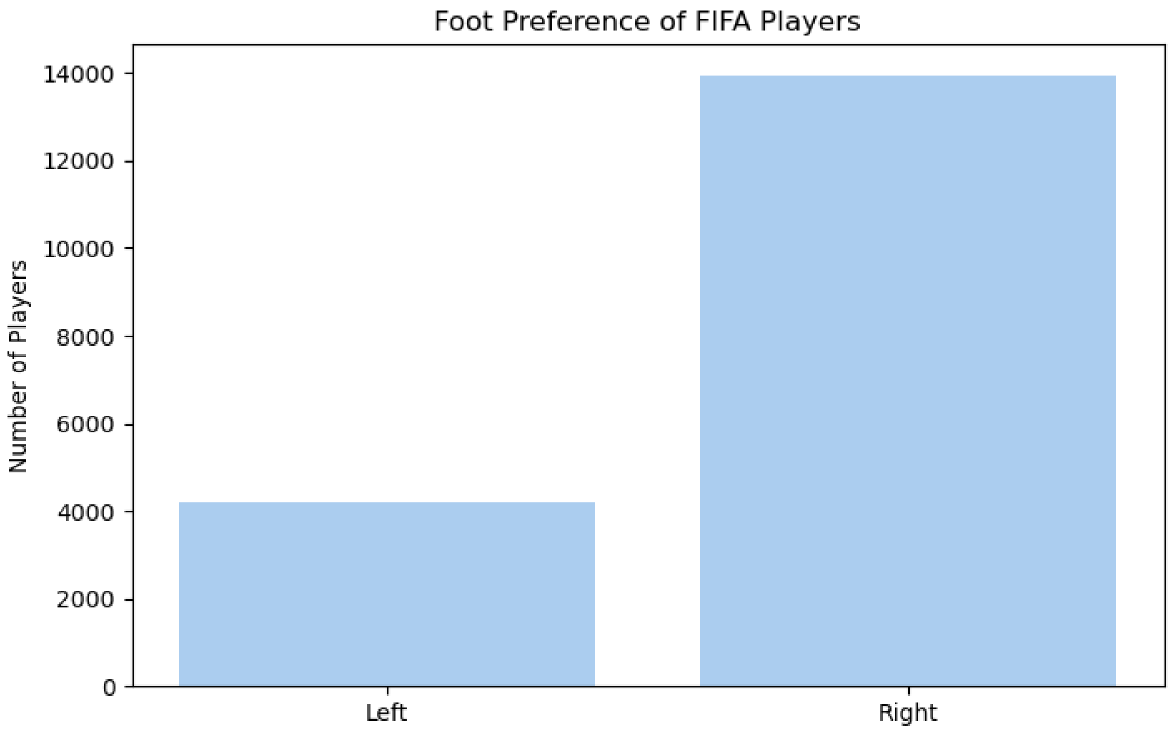
```
Text(0.5, 1.0, 'Distribution of Player Skills in FIFA 2018')
```



11.2.2.5.2 Bar chart

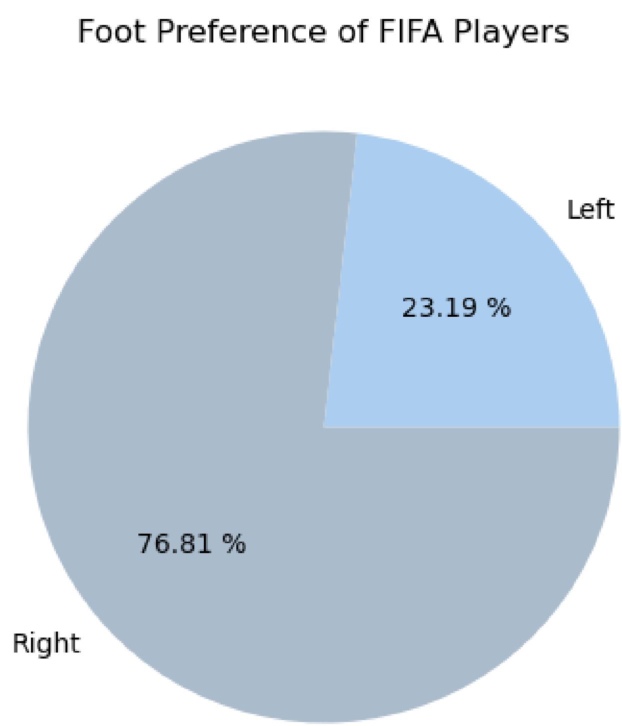
```
# plotting bar chart for the best players
plt.figure(figsize=(8,5))

foot_preference = fifa['Preferred Foot'].value_counts()
```

```
plt.bar(['Left', 'Right'], [foot_preference.iloc[1], foot_preference.iloc[0]], color='#abcde

plt.ylabel('Number of Players')
plt.title('Foot Preference of FIFA Players');
```



11.2.2.5.3 Pie chart

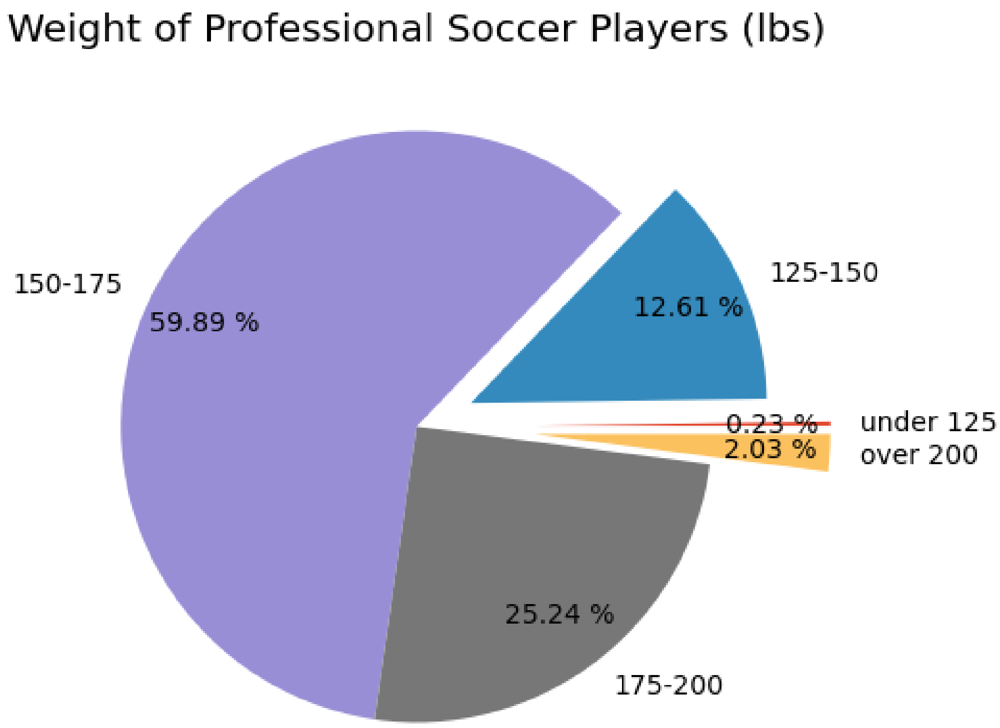
```
left = fifa.loc[fifa['Preferred Foot'] == 'Left'].count().iloc[0]
right = fifa.loc[fifa['Preferred Foot'] == 'Right'].count().iloc[0]

plt.figure(figsize=(8,5))

labels = ['Left', 'Right']
colors = ['#abcdef', '#aabbcc']

plt.pie([left, right], labels = labels, colors=colors, autopct='%.2f %%')

plt.title('Foot Preference of FIFA Players');
```



Another Pie Chart on wight of players

```
plt.figure(figsize=(8,5), dpi=100)

plt.style.use('ggplot')

fifa.Weight = [int(x.strip('lbs')) if type(x)==str else x for x in fifa.Weight]

light = fifa.loc[fifa.Weight < 125].count().iloc[0]
light_medium = fifa[(fifa.Weight >= 125) & (fifa.Weight < 150)].count().iloc[0]
medium = fifa[(fifa.Weight >= 150) & (fifa.Weight < 175)].count().iloc[0]
medium_heavy = fifa[(fifa.Weight >= 175) & (fifa.Weight < 200)].count().iloc[0]
heavy = fifa[fifa.Weight >= 200].count().iloc[0]

weights = [light,light_medium, medium, medium_heavy, heavy]
label = ['under 125', '125-150', '150-175', '175-200', 'over 200']
explode = (.4,.2,0,0,.4)

plt.title('Weight of Professional Soccer Players (lbs)')
```

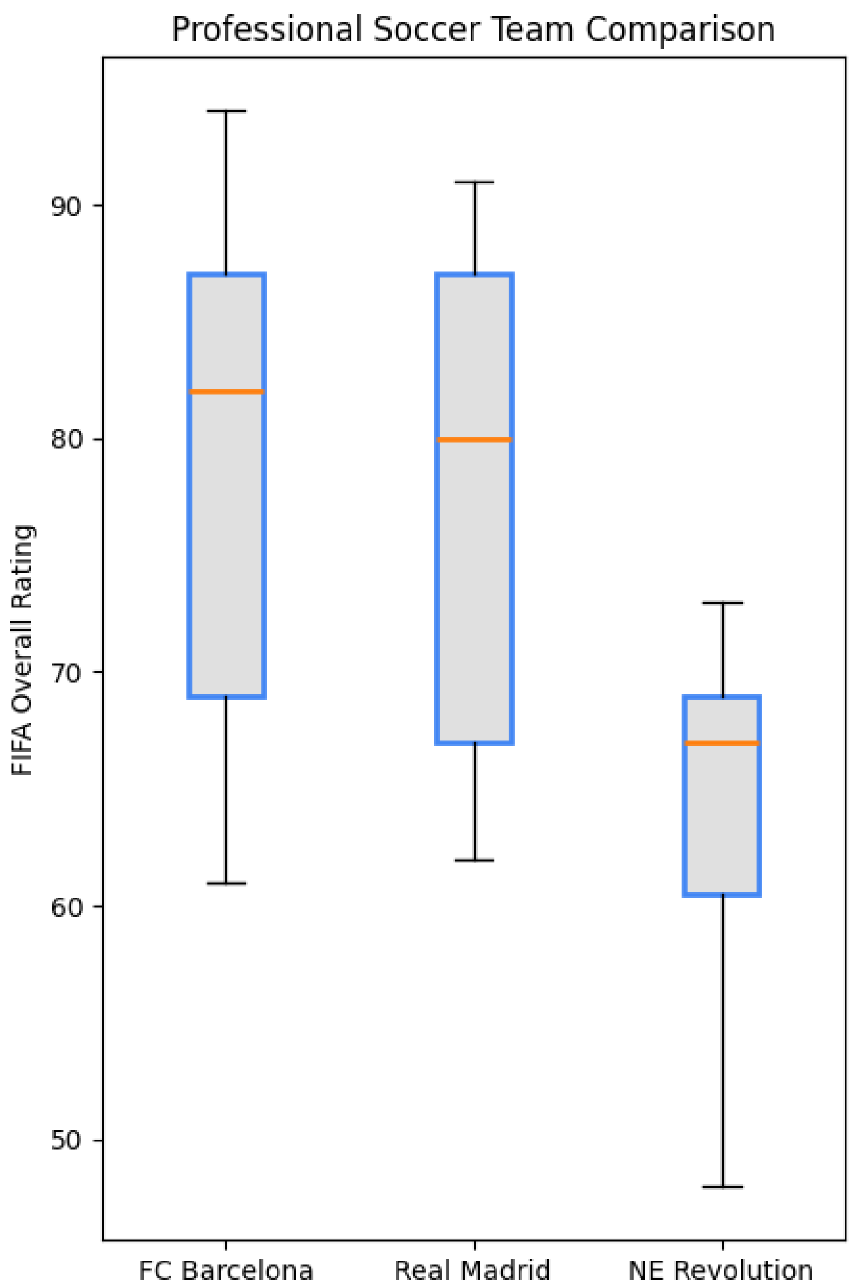
```
plt.pie(weights, labels=label, explode=explode, pctdistance=0.8,autopct='%.2f %%');
```



11.2.2.5.4 Box and Whiskers Chart

```
plt.figure(figsize=(5,8), dpi=100)

plt.style.use('default')

barcelona = fifa.loc[fifa.Club == "FC Barcelona"]['Overall']
madrid = fifa.loc[fifa.Club == "Real Madrid"]['Overall']
revs = fifa.loc[fifa.Club == "New England Revolution"]['Overall']

#bp = plt.boxplot([barcelona, madrid, revs], labels=['a','b','c'], boxprops=dict(facecolor='
bp = plt.boxplot([barcelona, madrid, revs], tick_labels=['FC Barcelona','Real Madrid','NE Rev

plt.title('Professional Soccer Team Comparison')
plt.ylabel('FIFA Overall Rating')
```

```python
for box in bp['boxes']:
    # change outline color
    box.set(color='#4286f4', linewidth=2)
    # change fill color
    box.set(facecolor = '#e0e0e0' )
    # change hatch
    #box.set(hatch = '/')
```



You've already learned how to create plots using Matplotlib's `Pyplot`. Next, you'll explore how to simplify and enhance your visualizations by using its powerful wrappers: Pandas and Seaborn.

11.2.2.6 Limitations of Using Matplotlib for Plotting

Matplotlib is a popular visualization library, but it has flaws.

- Defaults are not ideal (gridlines, background, etc need to be configured.)
- Library is low-level (doing anything complicated takes quite a bit of codes)
- Lack of integration with pandas data structures

To address these challenges, Seaborn act as higher-level interfaces to Matplotlib, offering better defaults, simpler syntax, and seamless integration with DataFrames.

11.2.3 Plotting with Seaborn

Seaborn is a powerful data visualization library built on top of Matplotlib, designed to make statistical plots easier and more attractive. It integrates seamlessly with Pandas, making it an excellent tool for plotting data from DataFrames. Seaborn comes with better default aesthetics and provides more specialized plots that are easy to implement.

Some of the key advantages of using Seaborn include:

- **Simpler syntax**: Seaborn simplifies the process of creating complex plots with just a few lines of code.
- **Beautiful default styles**: Seaborn's default plots are more polished and aesthetically pleasing compared to Matplotlib's defaults.
- **Seamless Pandas integration**: You can directly pass Pandas DataFrames to Seaborn, and it understands column names for axis labels and plot elements.

`seaborn` comes with 17 built-in datasets. That means you don't have to spend a whole lot of your time finding the right dataset and cleaning it up to make Seaborn-ready; rather you will focus on the core features of Seaborn visualization techniques to solve problems.

```python
import seaborn as sns
# get names of the builtin dataset
sns.get_dataset_names()
```

```
['anagrams',
 'anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
```

```
'exercise',
'flights',
'fmri',
'geyser',
'glue',
'healthexp',
'iris',
'mpg',
'penguins',
'planets',
'seaice',
'taxis',
'tips',
'titanic']
```

11.2.3.1 Customizing Plot Aesthetics

Seaborn provides a convenient function called `sns.set_style()` that allows users to customize the visual appearance of their plots. This function is particularly useful for enhancing the aesthetics of visualizations, making them more appealing and easier to interpret.

11.2.3.1.1 Purpose of `sns.set_style()`

The primary purpose of `sns.set_style()` is to set the visual context and style for all plots created after the call. This allows for a consistent and visually pleasing representation of data across multiple visualizations.

11.2.3.1.2 Available Style Options

Seaborn offers several built-in styles that can be set using `sns.set_style()`. The options include:

1. **`darkgrid`**:
   - A dark background with a grid overlay.
   - Ideal for visualizing data with many points or intricate details.
2. **`whitegrid`**:
   - A white background with a grid overlay.
   - Provides a clean and professional look, suitable for most types of visualizations.
3. **`dark`**:

- A dark background without gridlines.
- Useful for emphasizing data points without distractions from the grid.

4. **`white`**:

   - A simple white background without gridlines.
   - Offers a minimalist aesthetic, focusing solely on the data.

5. **`ticks`**:

   - A white background with ticks on the axes.
   - Combines the clarity of a white background with a bit of added detail for reference.

11.2.3.1.3 How to Use `sns.set_style()`

To apply a specific style, simply call **`sns.set_style()`** with the desired style name before creating your plots. Here's an example:

```
sns.set_style("whitegrid")
```

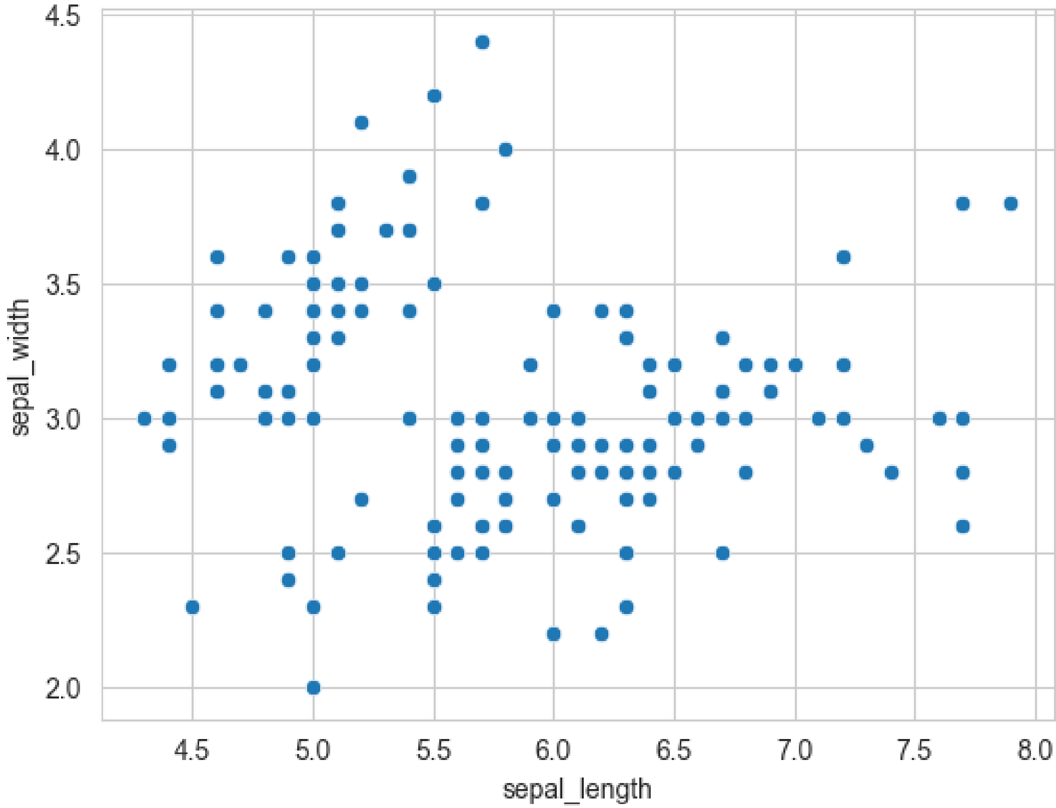
Let's do a few common plots with Seaborn tips dataset

```
# Load data into a Pandas dataframe
flowers_df = sns.load_dataset("iris")
flowers_df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

11.2.3.2 Scatterplot

```
sns.scatterplot(x=flowers_df.sepal_length, y=flowers_df.sepal_width);
```



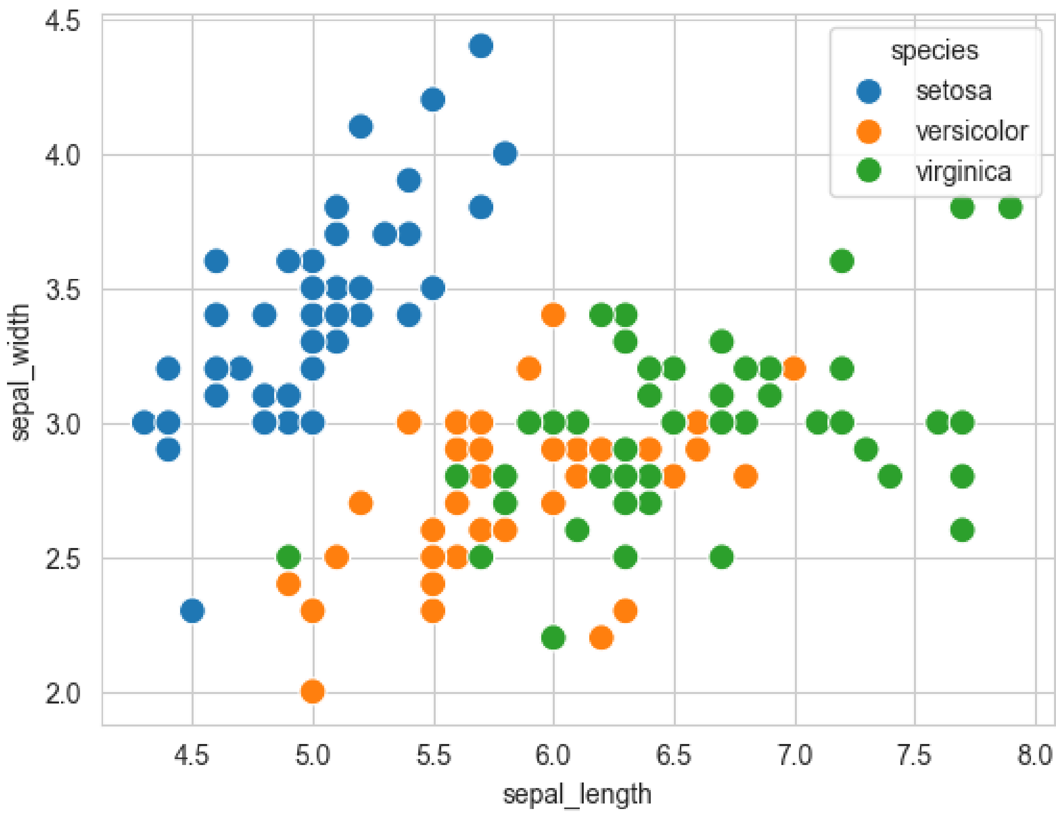
11.2.3.2.1 Adding Hues to the scatterplot

Notice how the points in the above plot seem to form distinct clusters with some outliers. We can color the dots using the flower species as a `hue`. We can also make the points larger using the `s` argument.

```
flowers_df.species.unique()
```

```
array(['setosa', 'versicolor', 'virginica'], dtype=object)
```

```
sns.scatterplot(x=flowers_df.sepal_length, y=flowers_df.sepal_width, hue=flowers_df.species,
```

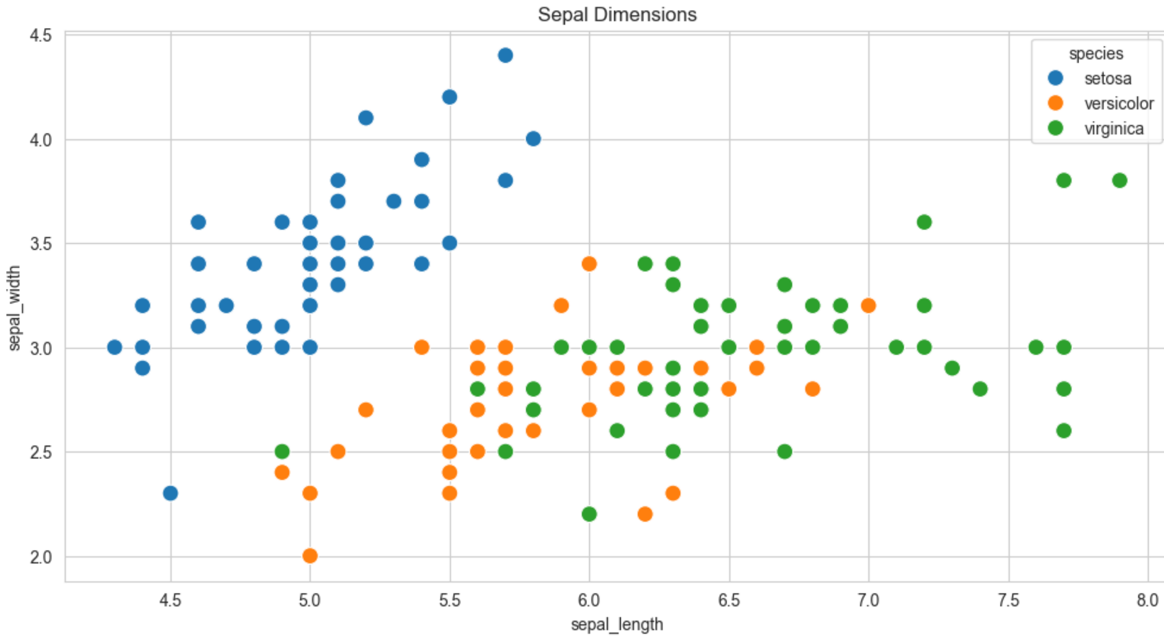


Adding hues makes the plot more informative. We can immediately tell that Setosa flowers have a smaller sepal length but higher sepal widths. In contrast, the opposite is true for Virginica flowers.

11.2.3.2.2 Customizing Seaborn Figures

Since Seaborn uses Matplotlib's plotting functions internally, we can use functions like `plt.figure` and `plt.title` to modify the figure.

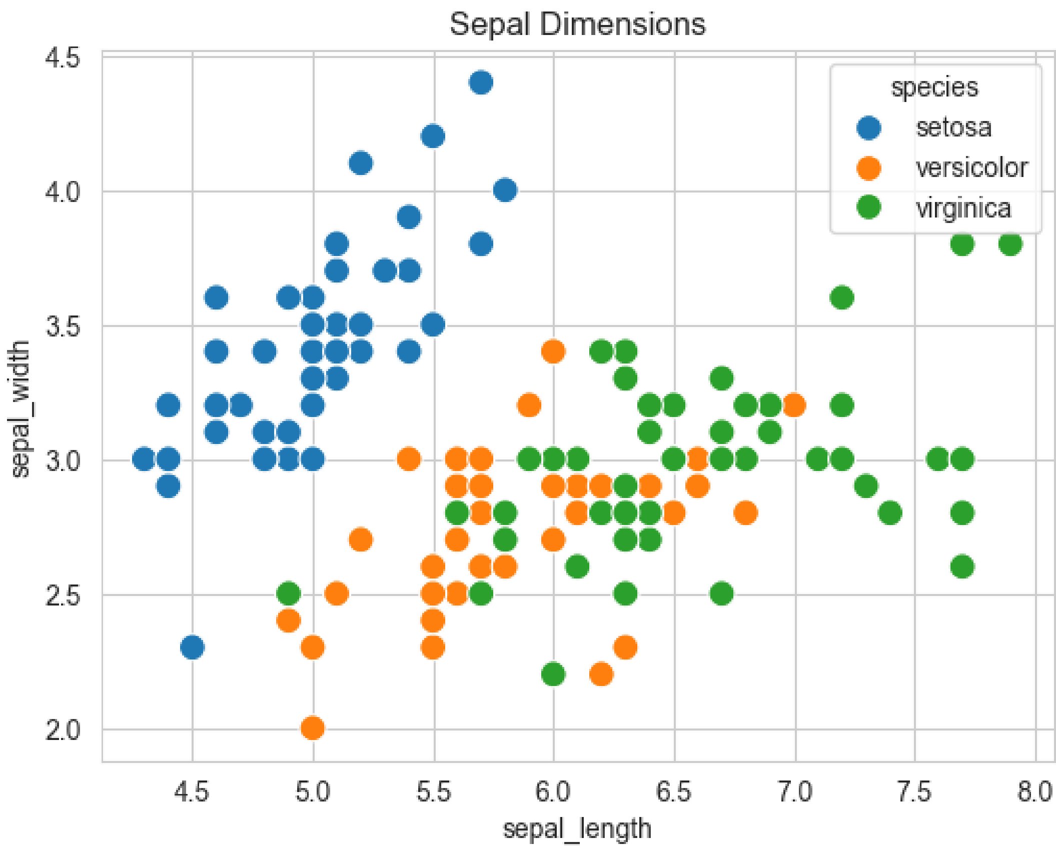
```
plt.figure(figsize=(12, 6))
plt.title('Sepal Dimensions')

sns.scatterplot(x=flowers_df.sepal_length,
                y=flowers_df.sepal_width,
                hue=flowers_df.species,
                s=100);
```



11.2.3.2.3 Integration with Pandas Data Frames

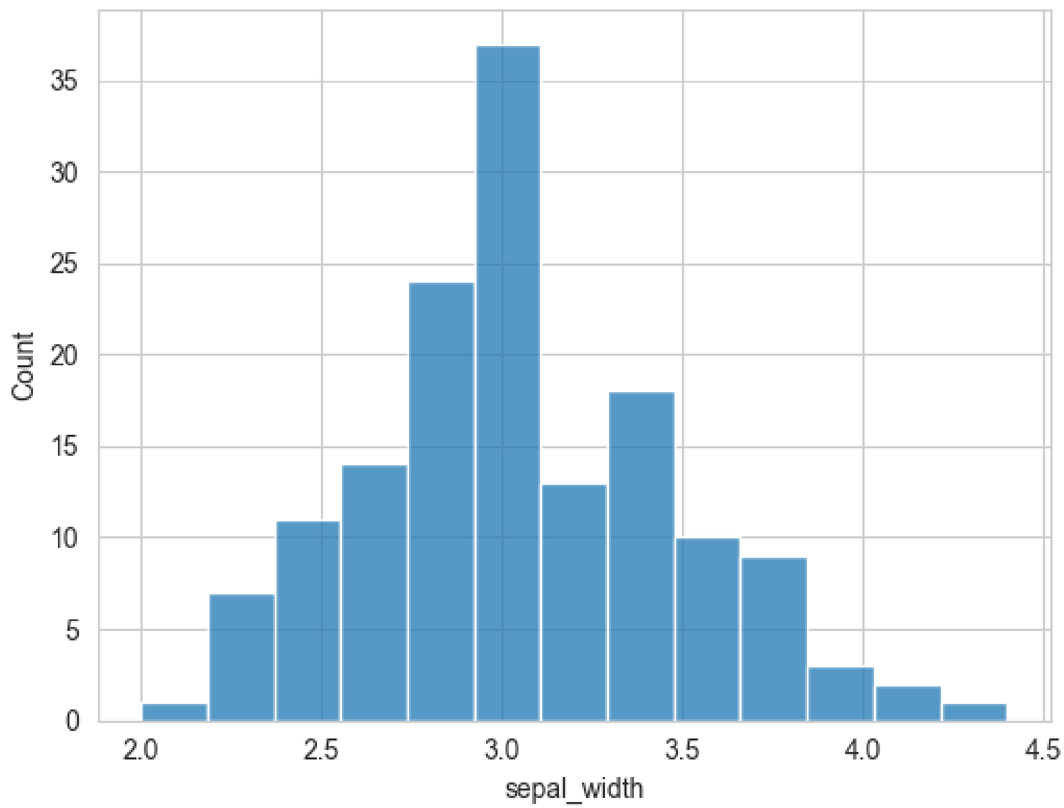
Seaborn has in-built support for Pandas data frames. Instead of passing each column as a series, you can provide column names and use the `data` argument to specify a data frame.

```
plt.title('Sepal Dimensions')
sns.scatterplot(x='sepal_length',
                y='sepal_width',
                hue='species',
                s=100,
                data=flowers_df);
```

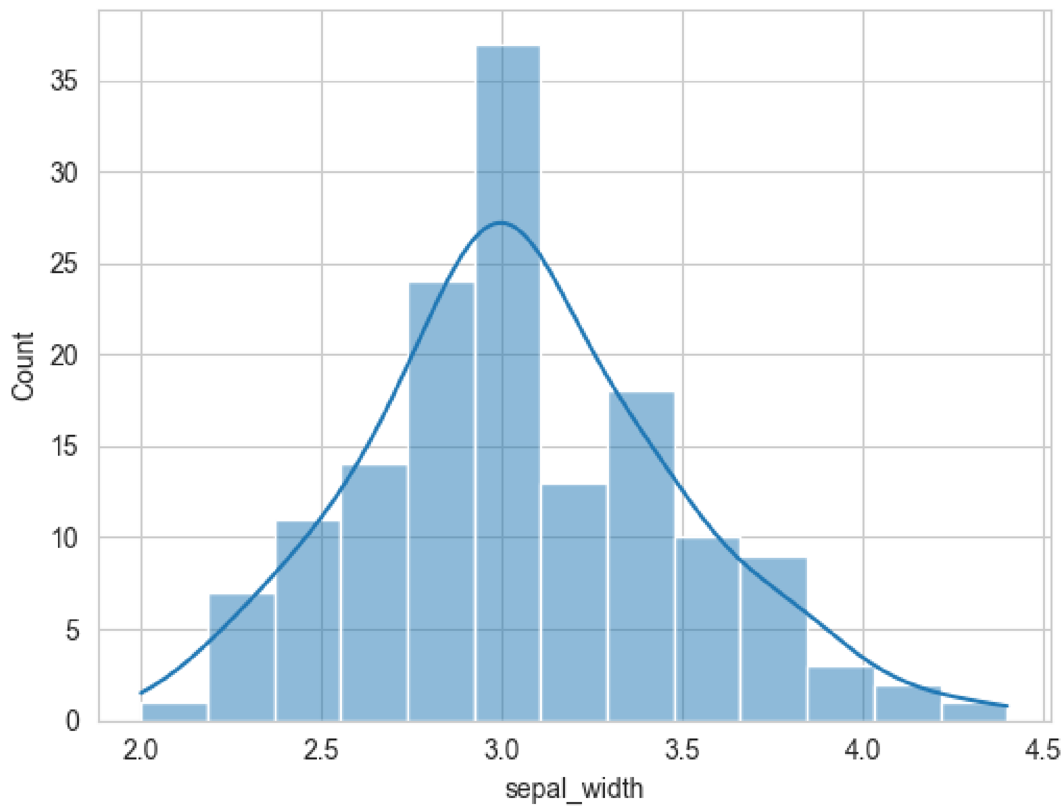


11.2.3.3 Histgram

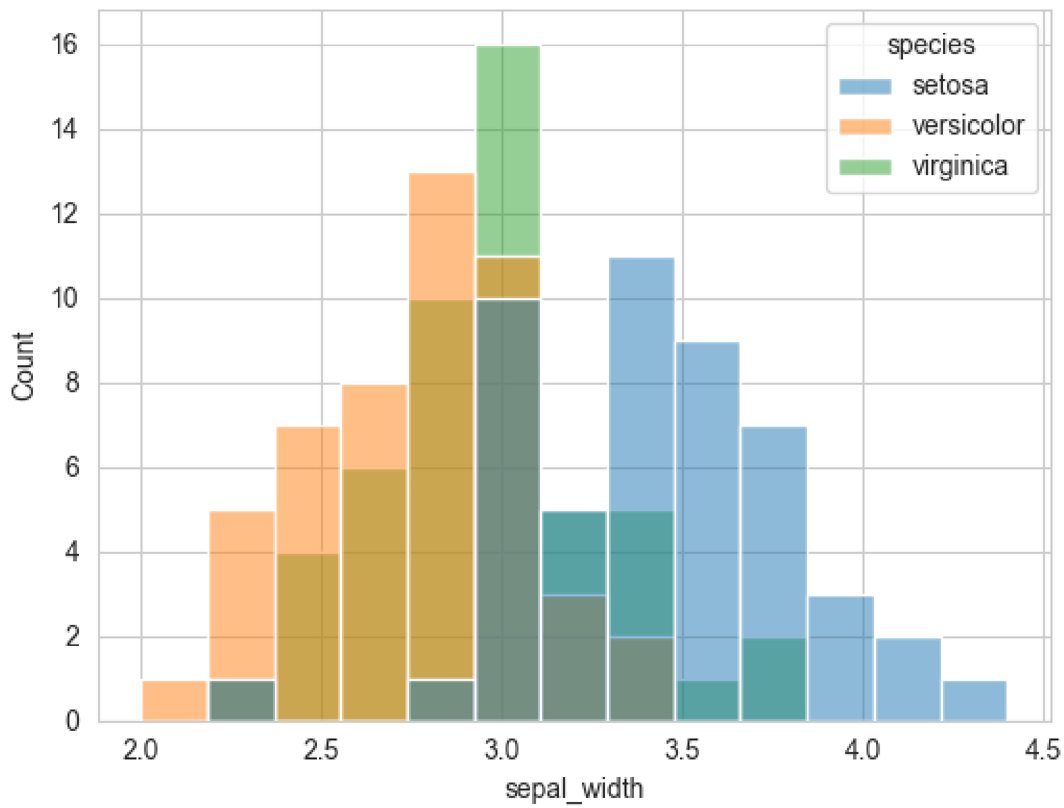
```
sns.histplot(data=flowers_df, x='sepal_width');
```




```
# show kde(kernal density estimate)
sns.histplot(data=flowers_df, x='sepal_width', kde=True);
```

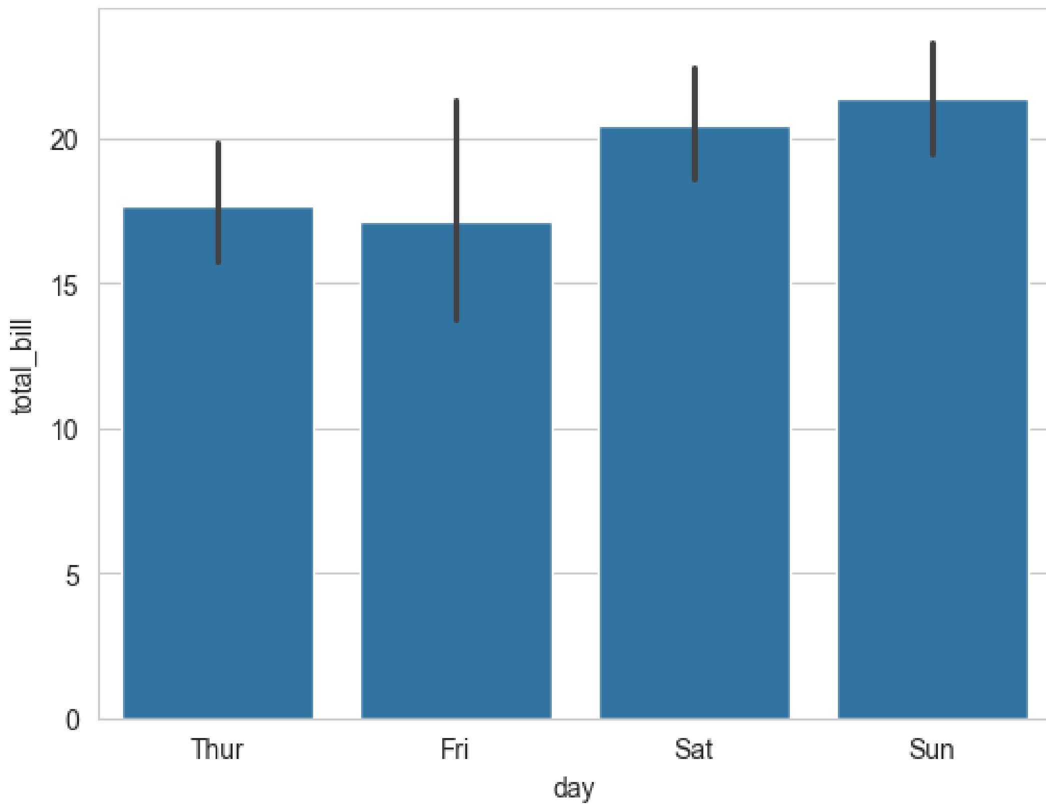



```
# adding hue
sns.histplot(data=flowers_df, x="sepal_width", hue="species");
```

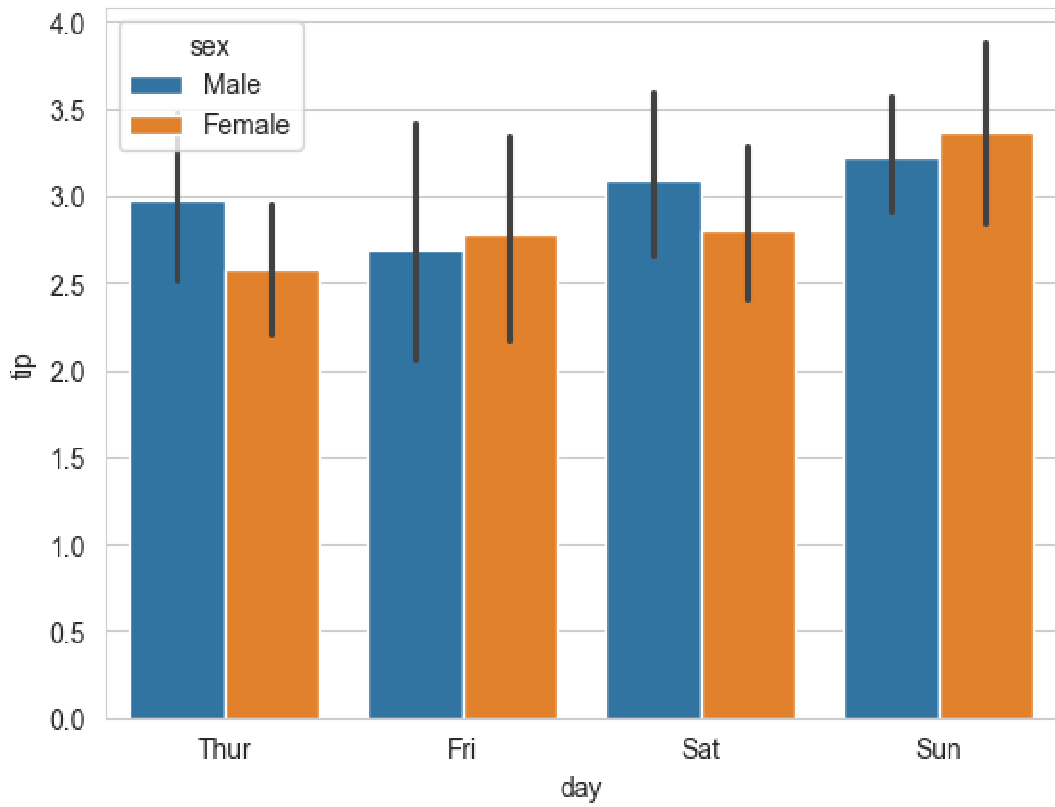


11.2.3.4 Barplot

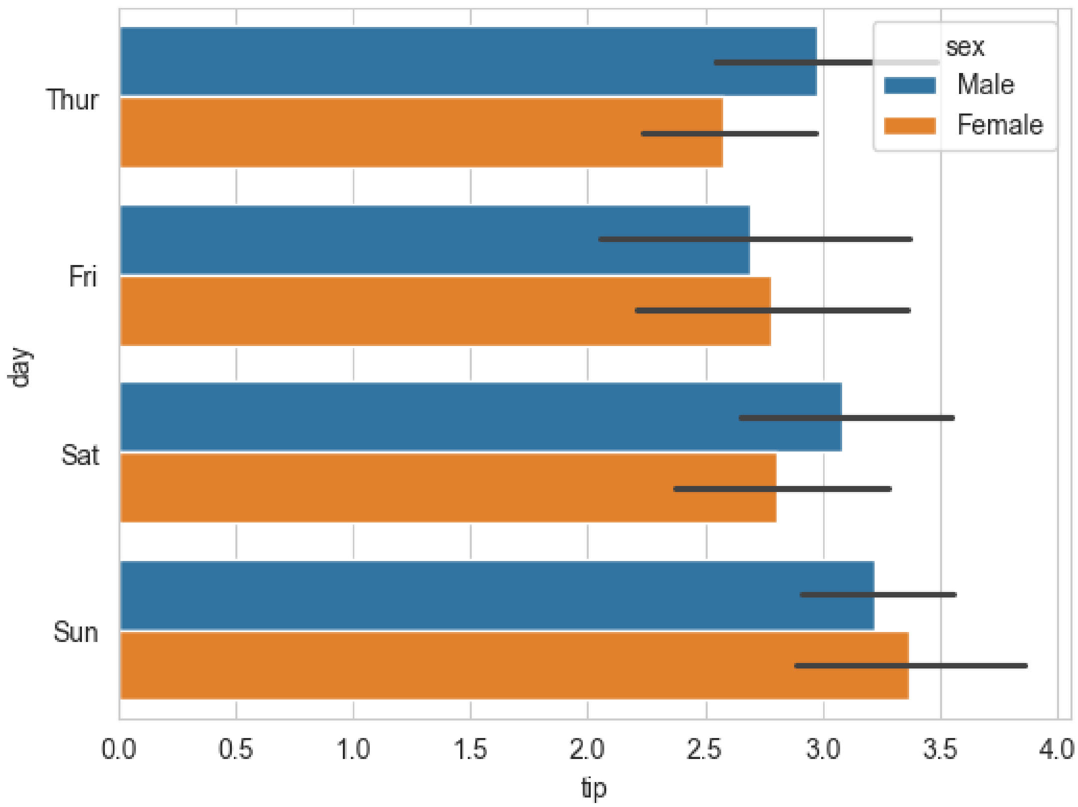
```
tips_df = sns.load_dataset("tips")
sns.barplot(x='day', y='total_bill', data=tips_df);
```




```
sns.barplot(x='day', y='tip', hue='sex', data=tips_df);
```




```
# make the bars horizontal simply by switching the axes
sns.barplot(x='tip', y='day', hue='sex', data=tips_df);
```



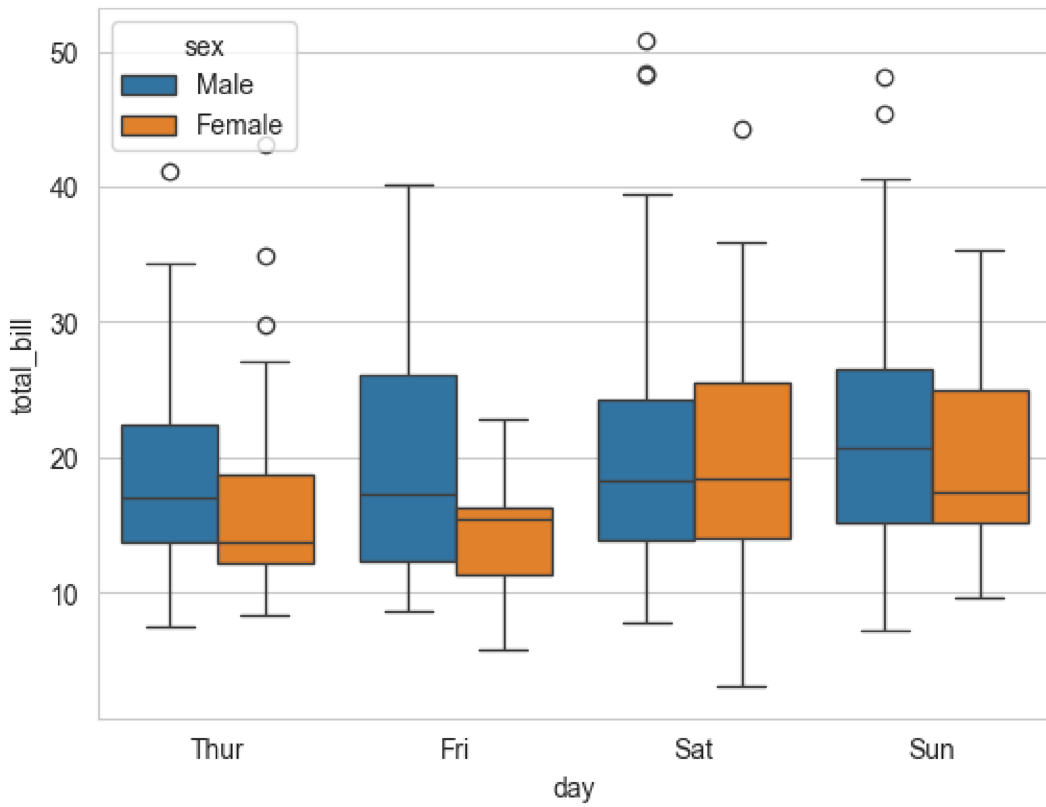
11.2.3.5 Boxplot

Purpose: Boxplots is a standardized way of visualizing the distribution of a continuous variable. They show five key metrics that describe the data distribution - median, 25th percentile value, 75th percentile value, minimum and maximum, as shown in the figure below. Note that the minimum and maximum exclude the outliers.

```
<IPython.core.display.Image object>
```

Example: Create a box plot to compare the distributions of total tips based on the day of the week, differentiating between male and female patrons.

```
sns.boxplot(data=tips_df, y='total_bill', x='day', hue='sex');
```



From the above plot, what you can observe?

11.2.3.6 Heatmap

Represent 2-dimensional data like a matrix or table using colors

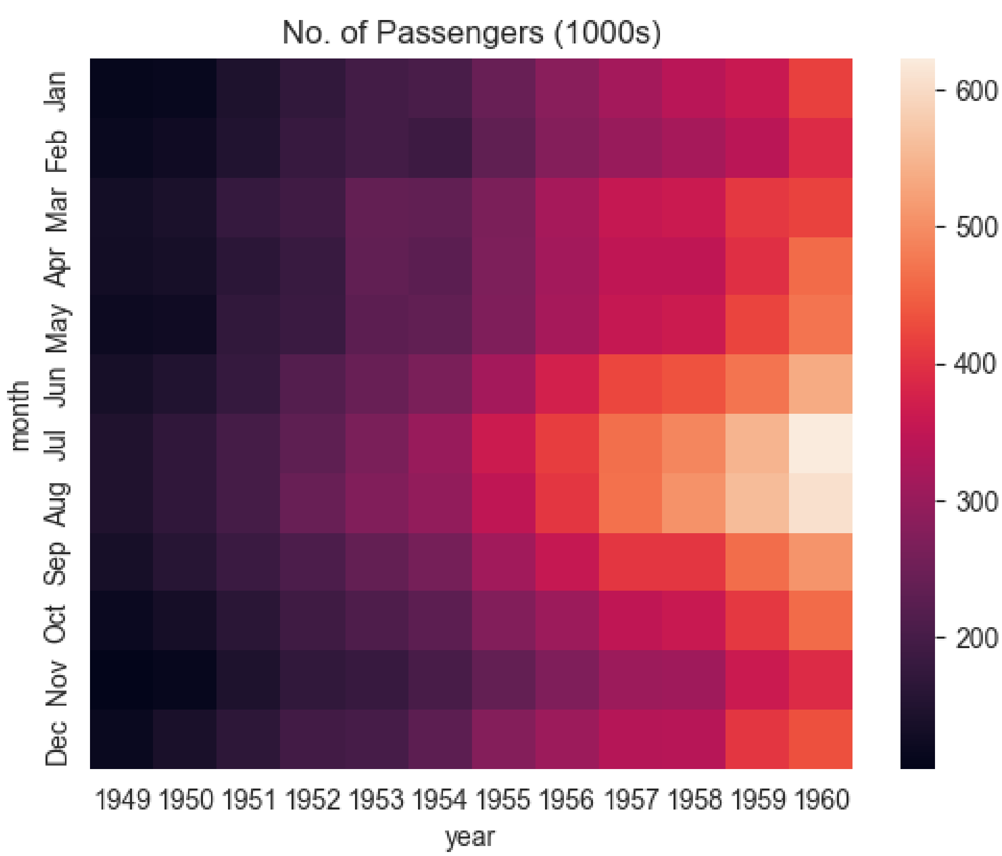
```
flights_df = sns.load_dataset("flights").pivot(index="month", columns="year", values="passeng
flights_df
# you will learn pivot in the later chapters
```

year month	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960
Jan	112	115	145	171	196	204	242	284	315	340	360	417
Feb	118	126	150	180	196	188	233	277	301	318	342	391
Mar	132	141	178	193	236	235	267	317	356	362	406	419
Apr	129	135	163	181	235	227	269	313	348	348	396	461
May	121	125	172	183	229	234	270	318	355	363	420	472

year month	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960
Jun	135	149	178	218	243	264	315	374	422	435	472	535
Jul	148	170	199	230	264	302	364	413	465	491	548	622
Aug	148	170	199	242	272	293	347	405	467	505	559	606
Sep	136	158	184	209	237	259	312	355	404	404	463	508
Oct	119	133	162	191	211	229	274	306	347	359	407	461
Nov	104	114	146	172	180	203	237	271	305	310	362	390
Dec	118	140	166	194	201	229	278	306	336	337	405	432

`flights_df` is a matrix with one row for each month and one column for each year. The values show the number of passengers (in thousands) that visited the airport in a specific month of a year. We can use the `sns.heatmap` function to visualize the footfall at the airport.

```
plt.title("No. of Passengers (1000s)")
sns.heatmap(flights_df);
```

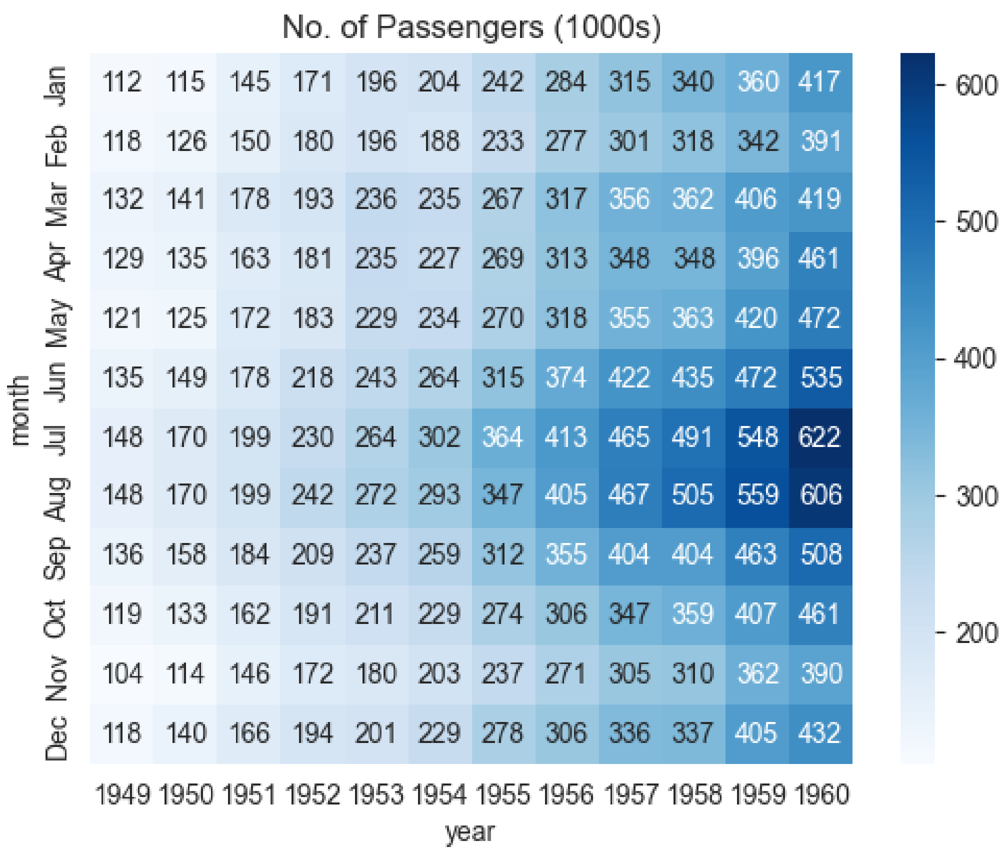


The brighter colors indicate a higher footfall at the airport. By looking at the graph, we can infer two things:

- The footfall at the airport in any given year tends to be the highest around July & August.
- The footfall at the airport in any given month tends to grow year by year.

We can also display the actual values in each block by specifying `annot=True` and using the `cmap` argument to change the color palette.

```
# fmt = "d" decimal integer. output are the number in base 10
plt.title("No. of Passengers (1000s)")
sns.heatmap(flights_df, fmt="d", annot=True, cmap='Blues');
```



11.2.3.7 Correlation Map

A correlation map is a specific type of heatmap where the values represent the correlation coefficients between variables (ranging from -1 to 1). It visually shows the strength and direction of relationships between numerical variables.

Correlation may refer to any kind of association between two random variables. However, in this book, we will always consider correlation as the linear association between two random variables, or the Pearson's correlation coefficient. Note that correlation does not imply causality and vice-versa.

The Pandas function corr() provides the pairwise correlation between all columns of a DataFrame, or between two Series. The function corrwith() provides the pairwise correlation of a DataFrame with another DataFrame or Series.

```
#Pairwise correlation amongst all columns
survey_data = pd.read_csv('./Datasets/survey_data_clean.csv')

survey_data.head()
```

	Timestamp	fav_alcohol	parties_per_month	smoke	weed
0	2022/09/13 1:43:34 pm GMT-5	I don't drink	1.0	No	Occasionally
1	2022/09/13 5:28:17 pm GMT-5	Hard liquor/Mixed drink	3.0	No	Occasionally
2	2022/09/13 7:56:38 pm GMT-5	Hard liquor/Mixed drink	3.0	No	Yes
3	2022/09/13 10:34:37 pm GMT-5	Hard liquor/Mixed drink	12.0	No	No
4	2022/09/14 4:46:19 pm GMT-5	I don't drink	1.0	No	No

```
#Pairwise correlation amongst all columns
survey_data.select_dtypes(include='number').corr()
```

	parties_per_month	love_first_sight	num_insta_followers	expected_m
parties_per_month	1.000000	0.096129	0.239705	-0.064079
love_first_sight	0.096129	1.000000	-0.024010	-0.084406
num_insta_followers	0.239705	-0.024010	1.000000	-0.130157
expected_marriage_age	-0.064079	-0.084406	-0.130157	1.000000
expected_starting_salary	0.114881	0.080138	0.127226	-0.014881
minutes_ex_per_week	0.195561	0.099244	0.099341	-0.088073
sleep_hours_per_day	-0.052542	-0.025378	-0.042421	0.182009
farthest_distance_travelled	-0.017081	-0.075539	0.011308	-0.024038
fav_number	-0.050139	0.105095	-0.124763	-0.008924
internet_hours_per_day	0.087390	-0.007652	-0.028427	-0.029772
only_child	-0.142519	0.124345	-0.152184	-0.043141
num_majors_minors	-0.073127	0.108730	0.050431	-0.055280
high_school_GPA	0.295646	0.069288	0.147402	0.017052

	parties__per__month	love__first__sight	num__insta__followers	expected__m
NU__GPA	-0.080548	-0.114041	0.004702	0.011925
age	-0.032771	0.142384	-0.230698	0.060416
height	-0.005405	0.216072	0.009318	0.044577
height__father	0.126741	0.029419	0.179684	0.026949
height__mother	0.079121	0.082684	0.129716	0.075316
procrastinator	-0.056871	0.033951	-0.089871	-0.020012
num__clubs	-0.010514	0.083342	0.265958	-0.137069
student__athlete	0.290830	0.014595	0.044807	-0.036122
AP__stats	-0.013222	-0.062992	0.005947	0.010447
used__python__before	-0.040033	-0.034692	-0.016201	0.052727
childhood__in__US	0.081905	-0.118260	0.072622	0.053759
cant__change__math__ability	-0.052912	0.005254	-0.150658	-0.072163
can__change__math__ability	0.055575	0.020758	0.130774	0.087633
math__is__genetic	-0.013374	-0.003710	-0.018411	-0.086898
much__effort__is__lack__of__talent	-0.029838	0.013376	-0.165899	0.052813

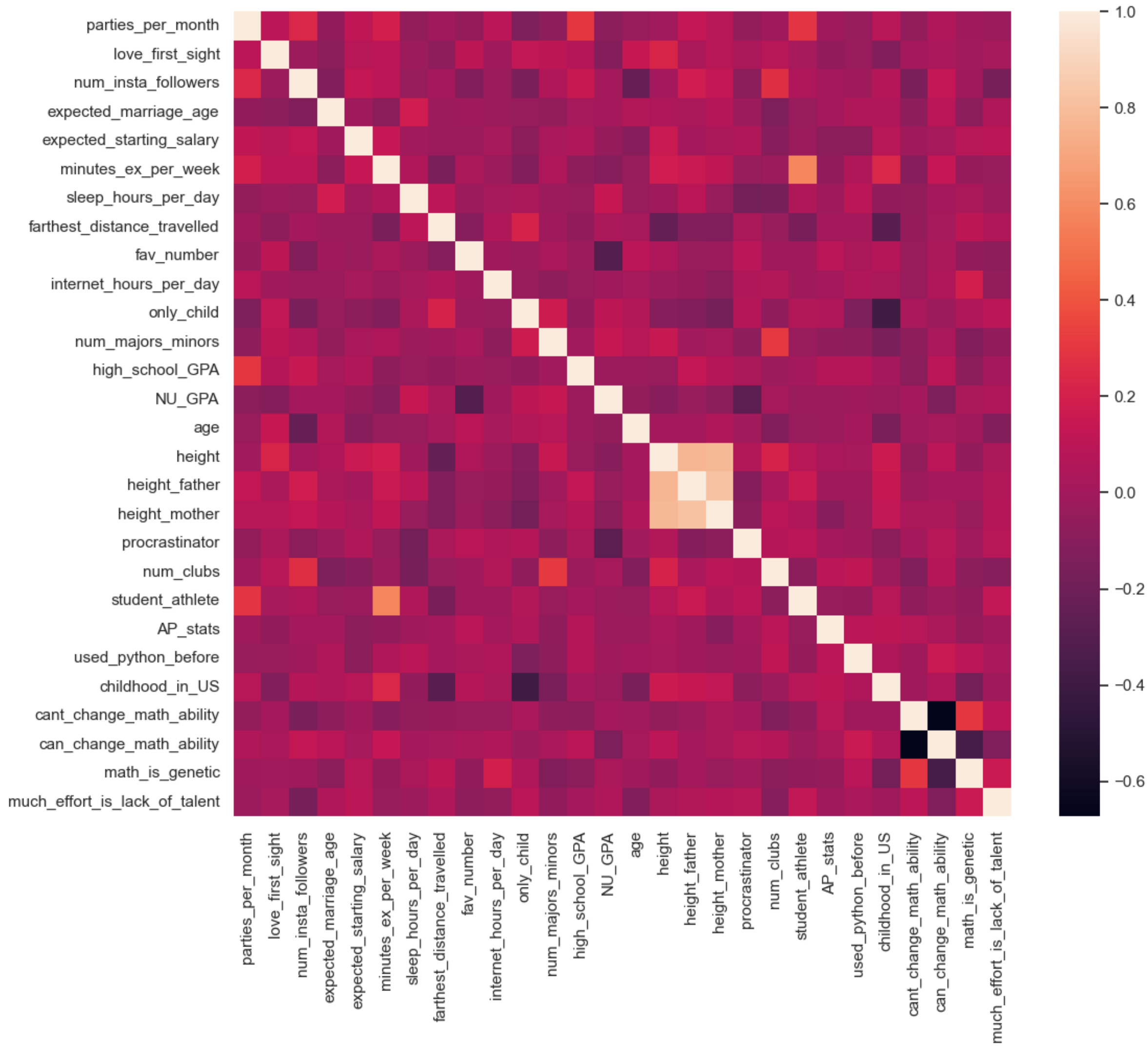
Q: Which feature is the most correlated with *NU_GPA*?

```
survey_data.select_dtypes(include='number').corrwith(survey_data.NU_GPA).sort_values(ascendi
```

```
NU_GPA                          1.000000
sleep_hours_per_day             0.143997
num_majors_minors               0.141988
only_child                      0.106440
much_effort_is_lack_of_talent   0.047840
farthest_distance_travelled     0.038238
math_is_genetic                 0.036731
num_clubs                       0.016724
expected_marriage_age           0.011925
num_insta_followers             0.004702
cant_change_math_ability        0.002094
used_python_before             -0.008536
internet_hours_per_day         -0.014531
AP_stats                       -0.026544
student_athlete                -0.027378
childhood_in_US                -0.028968
high_school_GPA                -0.030883
height_father                  -0.040120
expected_starting_salary       -0.048069
age                            -0.052039
```

```
height_mother                   -0.079276
parties_per_month               -0.080548
height                          -0.099082
minutes_ex_per_week             -0.108177
love_first_sight                -0.114041
can_change_math_ability         -0.137330
procrastinator                  -0.269552
fav_number                      -0.307656
dtype: float64
```

```
sns.set(rc={'figure.figsize':(12,10)})
sns.heatmap(survey_data.select_dtypes(include='number').corr());
```



From the above map, we can see that:

- `student athlete` is strongly postively correlated with `minutes_ex_per_week`
- `procrastinator` is strongly negatively correlated with `NU_GPA`

11.3 Independent Study

11.3.1 Practice exercise 1

Read the *gas_price.csv* file and plot the trend for each country over the time using matplotlib `pyplot`

11.3.2 Practice exercise 2

11.3.2.1

Is `NU_GPA` associated with `parties_per_month`? Analyze the association separately for *Sophomores, Juniors, and Seniors* (categories of the variable `school_year`).

Make scatterplots of `NU_GPA` vs `parties_per_month` in a 1 x 3 grid, where each grid is for a distinct `school_year`. Plot the trendline as well for each scatterplot. Use the file *survey_data_clean.csv*.

11.3.2.2

Capping the the values of `parties_per_month` to 30, and make the visualizations again.

11.3.3 Practice exercise 3

How does the expected marriage age of the people of STAT303-1 depend on their characteristics? We'll use visualizations to answer this question. Use data from the file *survey_data_clean.csv.* Proceed as follows:

11.3.3.1

Make a visualization that compares the mean `expected_marriage_age` of introverts and extroverts *(use the variable `introvert_extrovert`)*. What insights do you obtain?

11.3.3.2

Does the mean `expected_marriage_age` of introverts and extroverts depend on whether they believe in love in first sight *(variable name: `love_first_sight`)*? Update the previous visualization to answer the question.

11.3.3.3

In addition to `love_first_sight`, does the mean `expected_marriage_age` of introverts and extroverts depend on whether they are a procrastinator *(variable name: `procrastinator`)*? Update the previous visualization to answer the question.

11.3.3.4

Is there any critical information missing in the above visualizations that, if revealed, may cast doubts on the patterns observed in them?

11.3.4 Practice exercise 4

Read *Australia_weather.csv*,

11.3.4.1

Create a histogram showing the distributions of maximum temperature in Sydney, Canberra and Melbourne.

11.3.4.2

Make a density plot showing the distributions of maximum temperature in Sydney, Canberra and Melbourne.

11.3.4.3

Show the distributions of the maximum and minimum temperatures in a single plot.

11.3.4.4

Create a scatter plot with a trendline for `MinTemp` and `MaxTemp`, including a confidence interval.

Hint: Using Seaborn, the `regplot()` function enables us to overlay a trendline on the scatter plot, complete with a 95% confidence interval for the trendline

12 Advanced Data Visualization

In the previous chapter, we explored basic plotting techniques using Pandas, Seaborn, and Matplotlib to create visualizations. Now, we'll elevate our skills by diving into more advanced topics, such as crafting complex subplots and visualizing geospatial data, enabling us to build richer and more insightful plots.

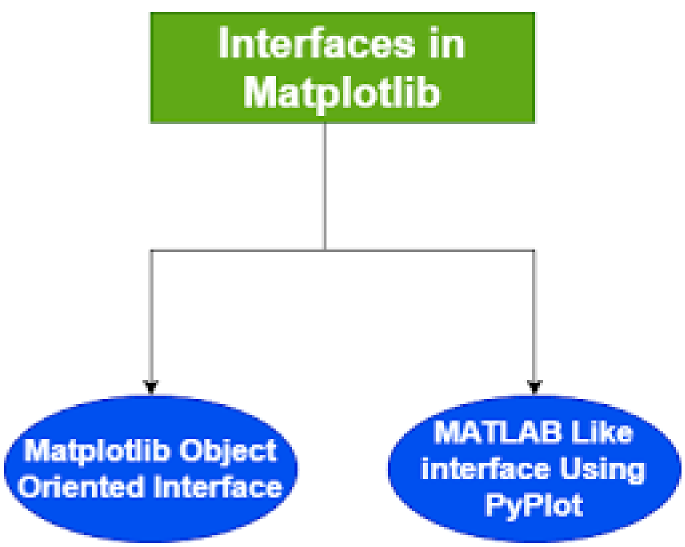
To get started, let's import necessary libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

12.1 Matplotlib Plotting Interfaces

12.1.1 Pyplot interface and OOP Interface

There are two types of interfaces in Matplotlib for visualization and these are given below:

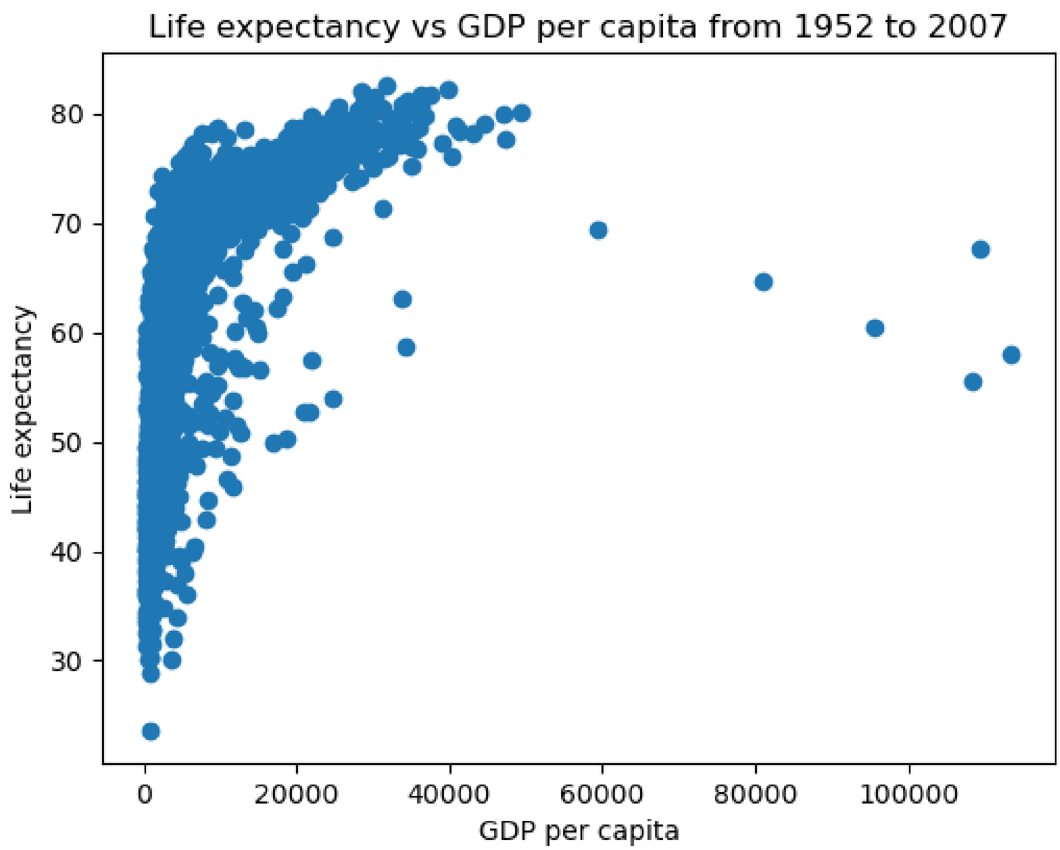


12.1.2 Plot a simple figure using two interfaces

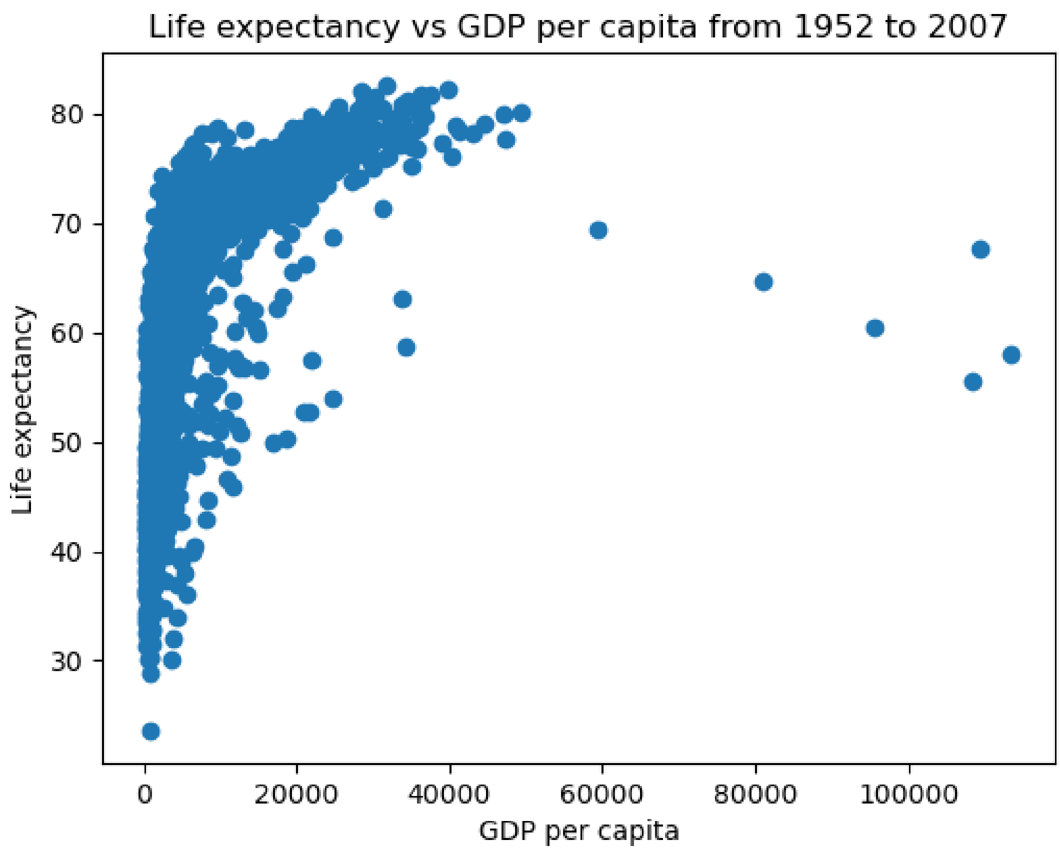
```
gdp_data = pd.read_csv('datasets/gdp_lifeExpectancy.csv')
gdp_data.head()
```

	country	continent	year	lifeExp	pop	gdpPercap
0	Afghanistan	Asia	1952	28.801	8425333	779.445314
1	Afghanistan	Asia	1957	30.332	9240934	820.853030
2	Afghanistan	Asia	1962	31.997	10267083	853.100710
3	Afghanistan	Asia	1967	34.020	11537966	836.197138
4	Afghanistan	Asia	1972	36.088	13079460	739.981106

```
#Object-oriented interface
fig, ax = plt.subplots() #Create a figure and an axes
x = gdp_data.gdpPercap
y = gdp_data.lifeExp
ax.plot(x,y,'o')   #Plot data on the axes
ax.set_xlabel('GDP per capita')    #Add an x-label to the axes
ax.set_ylabel('Life expectancy')   #Add a y-label to the axes
ax.set_title('Life expectancy vs GDP per capita from 1952 to 2007');
```




```
#pyplot interface
x = gdp_data.gdpPercap
y = gdp_data.lifeExp
plt.plot(x,y,'o') #By default, the plot() function makes a lineplot. The 'o' arguments speci
plt.xlabel('GDP per capita')  #Labelling the horizontal X-axis
plt.ylabel('Life expectancy') #Labelling the verical Y-axis
plt.title('Life expectancy vs GDP per capita from 1952 to 2007');
```



12.1.3 Pyplot Interface

In the previous chapter, our plotting is completely based on pyplot interface of Matplotlib, which is just for basic plotting. You can easily generate plots using pyplot module in the matplotlib library just by importing `matplotlib.pyplot` module.

- pyplot interface is a **state-based interface.** It implicitly tracks the plot that it wants to reference
- Simple functions are used to **add plot elements and/or modify the plot** as we need, whenever we need it.
- The Pyplot interface **shares a lot of similarities in syntax and methodology with MATLAB.**

However, The `pyplot` interface has limited functionality in these two cases:

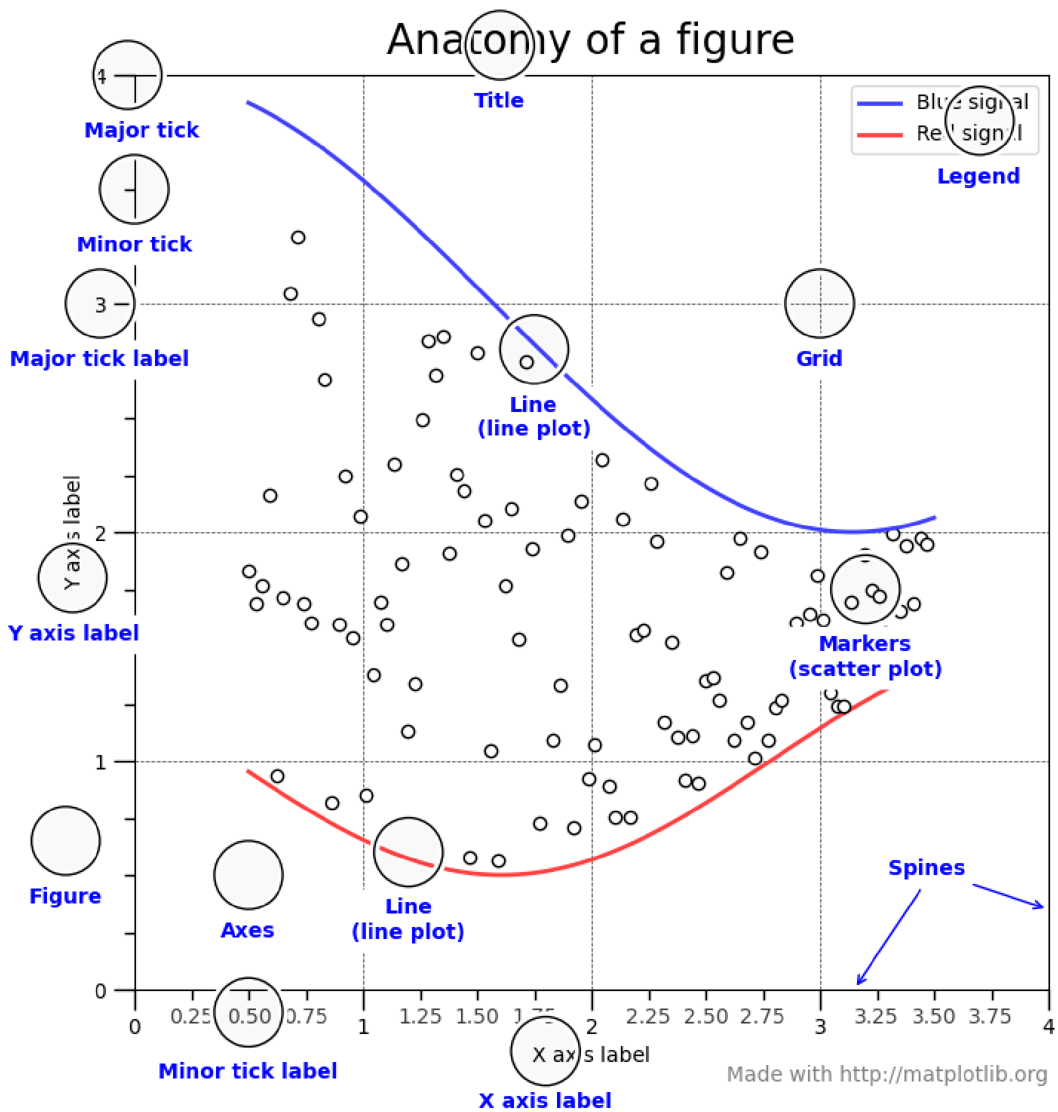
- when there is a need to make multiple plots
- when we have to make plots that require a lot of customization.

For more advanced plotting with Matplotlib, you have to learn Object-Oriented Interface.

12.2 Plotting with Object-Oriented Interface of Matplotlib

12.2.1 Matplotlib Object Anatomy

To use Matplotlib's object-oriented interface effectively, it's important to understand the anatomy of a plot—how different visual elements are structured within a figure:



12.2.2 Matplotlib Object Hierarchy

Matplotlib plots follow a **hierarchical structure**, with two core components:

- Figure: The overall container for one or more plots.
- Axes: The individual plot area where data is visualized.

Each Axes contains further elements such as:

- `Axis` (x-axis and y-axis)
- `Title`, `Labels`, `Ticks`
- Drawable objects like `Lines`, `Text`, and `Patches` (collectively called **Artists**)

Here's a simplified view of this hierarchy:

```
Figure
   Axes (1 or more)
        Axis (XAxis, YAxis)
        Title
        Label
        Tick
        Artist objects (Lines, Text, Patches, etc.)
```

Understanding this structure is key to using Matplotlib's object-oriented interface, as it allows you to **access and customize each component directly**.

12.2.2.1 Figure

The outermost object is the figure which is an instance of `figure.Figure`. It is the **top level container** for all the plot elements. The Figure is the final image that may contain one or more Axes and it keeps track of all the Axes. Figure is only a container and you can not plot data on figure.

```
# To begin with, we create a fiture instance which provides an empty canvas
fig = plt.figure()
```

```
<Figure size 640x480 with 0 Axes>
```

Figure is just a container, note that creating figure using `plt.figure()` does not automatically create an axes and hence all you can see is the figure object.

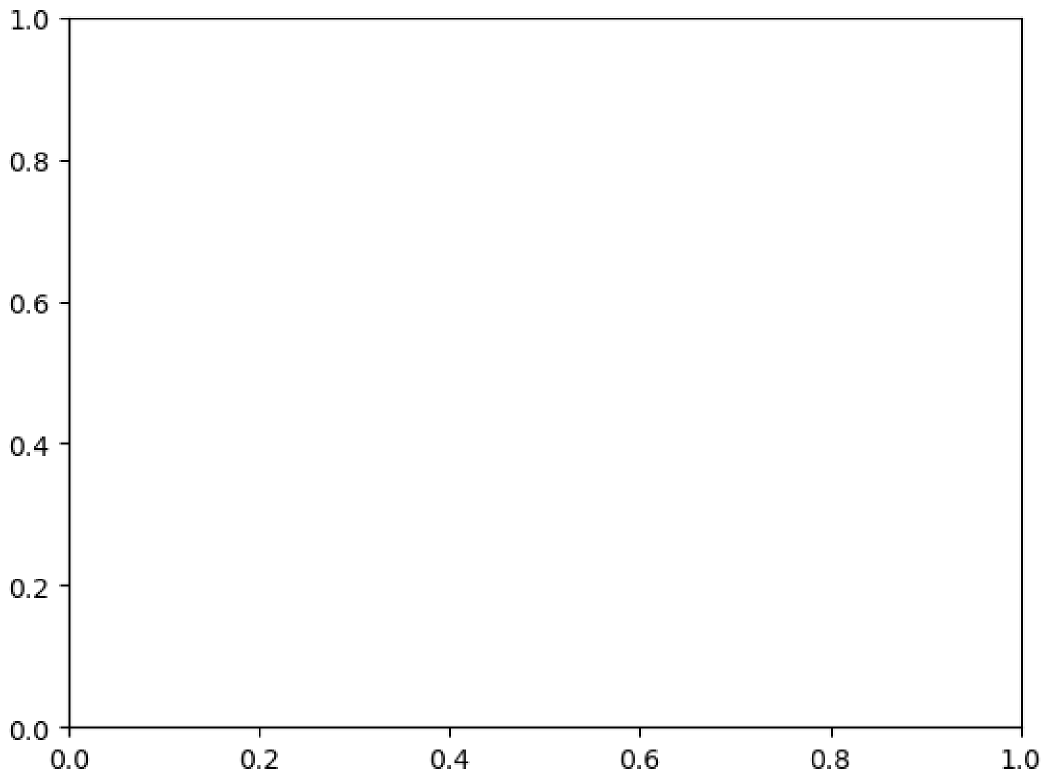
12.2.2.2 Axes

`Axes` is the instance of `matplotlib.axes.Axes`. Axes **is the area on which data are plotted**. A given figure can contain many Axes, but a given Axes object can only be in one Figure.

12.2.2.2.1 Creating an Axes Object Explicitly

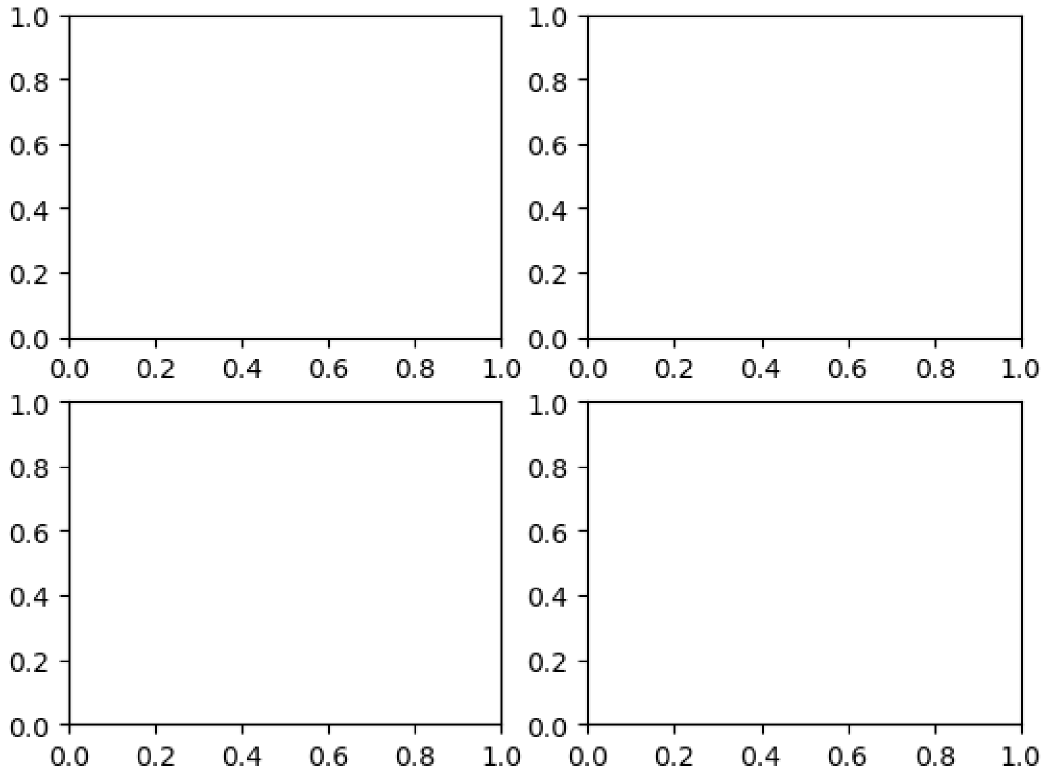
In the OOP interface, Axes objects are usually created using `plt.subplots()` or `plt.figure().add_subplot()`.

```
# A figure only contains one axes by default
fig = plt.figure()
# add axes to the figure
ax = fig.add_subplot()
```



We create a blank axes (area) for plotting, if we need to add more axes to the figure

```
# create a figure with 4 axes, arranged in 2 rows and 2 columns
fig, axes = plt.subplots(2, 2)
```



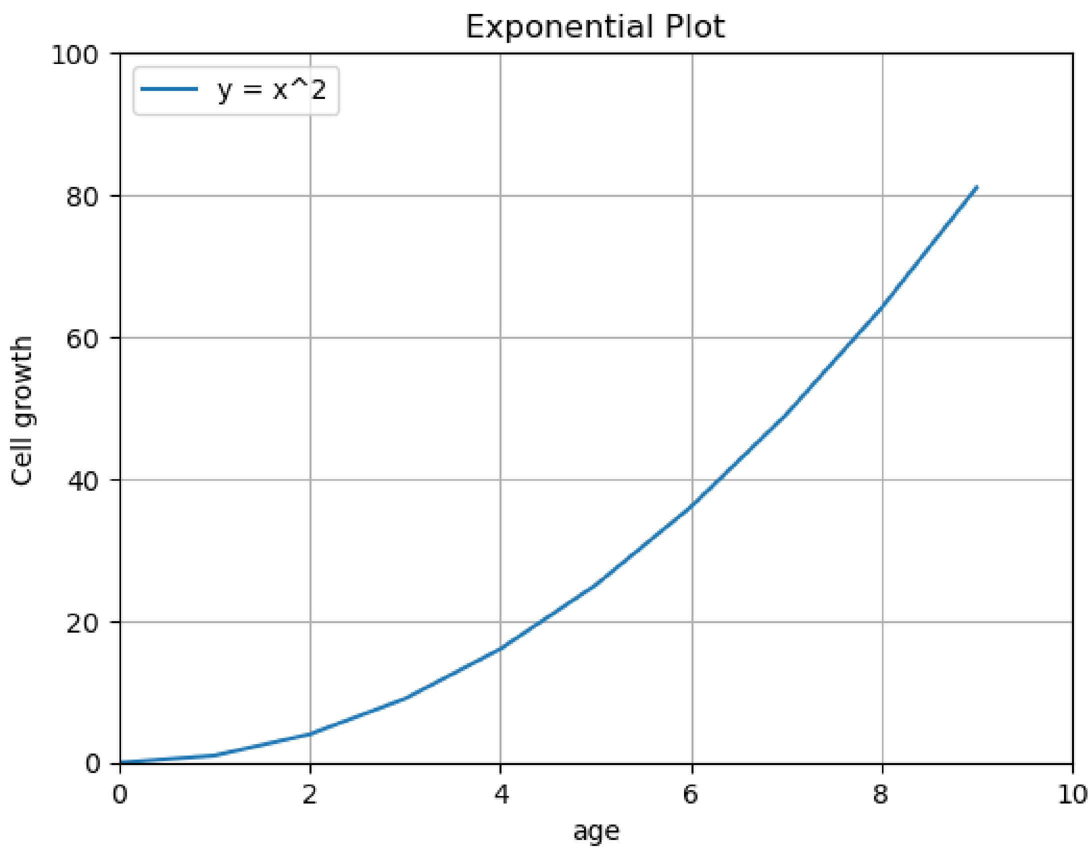
12.2.2.2.2 Components of an Axes Object

The `Axes` object contains several elements that make up the plot

- Data plotting area: Contains the data (lines, bars, points, etc.).
- X-axis and Y-axis: Controls the axis limits, labels, and ticks.
- Title and Labels: The overall title and labels for each axis.
- Gridlines: Optional lines to help align the data visually.
- Spines: The borders around the plot.
- Legend: An optional component to explain the data series.
- Annotations: Text or arrows highlighting points of interest.

This is what makes the `Axes` object central to any plot in the OOP interface of Matplotlib.

```
# create data for plotting
x = np.arange(10)
y = x**2

# create a figure and axes
fig,ax = plt.subplots(1,1)

# plot the data
ax.plot(x,y)

# set the title
ax.set_title("Exponential Plot")

# set the labels of x and y axes
ax.set_xlabel("age")
ax.set_ylabel("Cell growth")

# set the limits of x and y axes
ax.set_xlim([0, 10])
ax.set_ylim([0, 100])

# set the ticks of x and y axes
ax.set_xticks(range(0, 11, 2))
ax.set_yticks(range(0, 101, 20))

# add grid
ax.grid(True)

# add legend
ax.legend(["y = x^2"], loc = "upper left")

# show the plot
plt.show()
```



12.2.3 Creating Complex Plots with Multiple Subplots

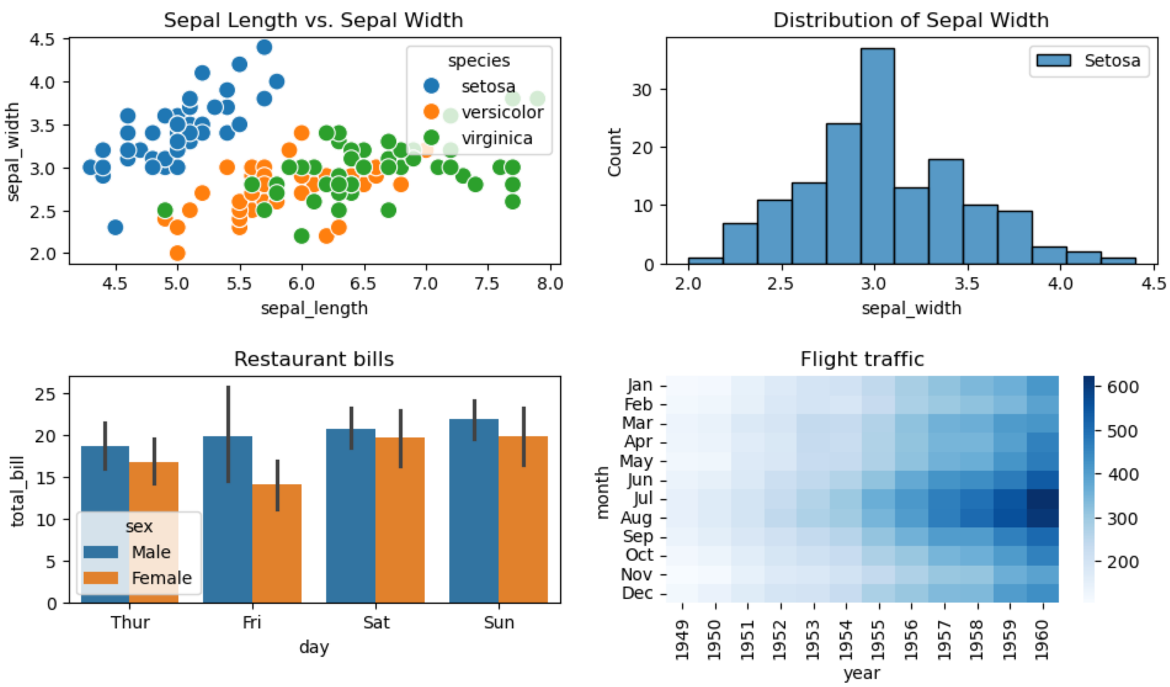
To help illustrate subplots in matplotlib, we can cover two scenarios:

- subplots that don't overlap and
- subplots inside other subplots

12.2.3.1 Plotting non-overlapped subplots

This is the most common use case, where multiple plots are placed next to each other in a grid, without overlap. The `plt.subplots()` function allows for a clean layout where each plot is contained in its own space. You can specify the number of rows and columns to create a grid of subplots.

```
flowers_df = sns.load_dataset('iris')
tips_df = sns.load_dataset('tips')
flights_df = sns.load_dataset("flights").pivot(index="month", columns="year", values="passeng


fig, axes = plt.subplots(2, 2, figsize=(10, 6))

# Pass the axes into seaborn
axes[0,0].set_title('Sepal Length vs. Sepal Width')
sns.scatterplot(x=flowers_df.sepal_length,
                y=flowers_df.sepal_width,
                hue=flowers_df.species,
                s=100,
                ax=axes[0,0])

# Use the axes for plotting
axes[0,1].set_title('Distribution of Sepal Width')
sns.histplot(flowers_df.sepal_width, ax=axes[0,1])
axes[0,1].legend(['Setosa', 'Versicolor', 'Virginica'])

# Pass the axes into seaborn
axes[1,0].set_title('Restaurant bills')
sns.barplot(x='day', y='total_bill', hue='sex', data=tips_df, ax=axes[1,0])

# Pass the axes into seaborn
axes[1,1].set_title('Flight traffic')
sns.heatmap(flights_df, cmap='Blues', ax=axes[1,1])

plt.tight_layout(pad=2)
```



- The `plt.tight_layout()` ensures the subplots do not overlap by adjusting the spacing automatically.
- Seaborn and pandas are wrappers of Matplotlib. To create a Seaborn/pandas plot in a specific Matplotlib subplot, you pass the `ax` parameter to the its plotting function. This allows you to use their visualization capabilities while fully controlling the layout of the plot using Matplotlib's `plt.subplots()`

12.2.3.2 Plotting Nested Subplots (Subplots Inside Other Subplots)

You can create a subplot inside another plot using `add_axes()` or `inset_axes()` from matplotlib's `Axes` object. This is useful for creating insets or focusing on a specific region within a larger plot.

Syntax of `add_axes()`

`ax = fig.add_axes([left, bottom, width, height])`

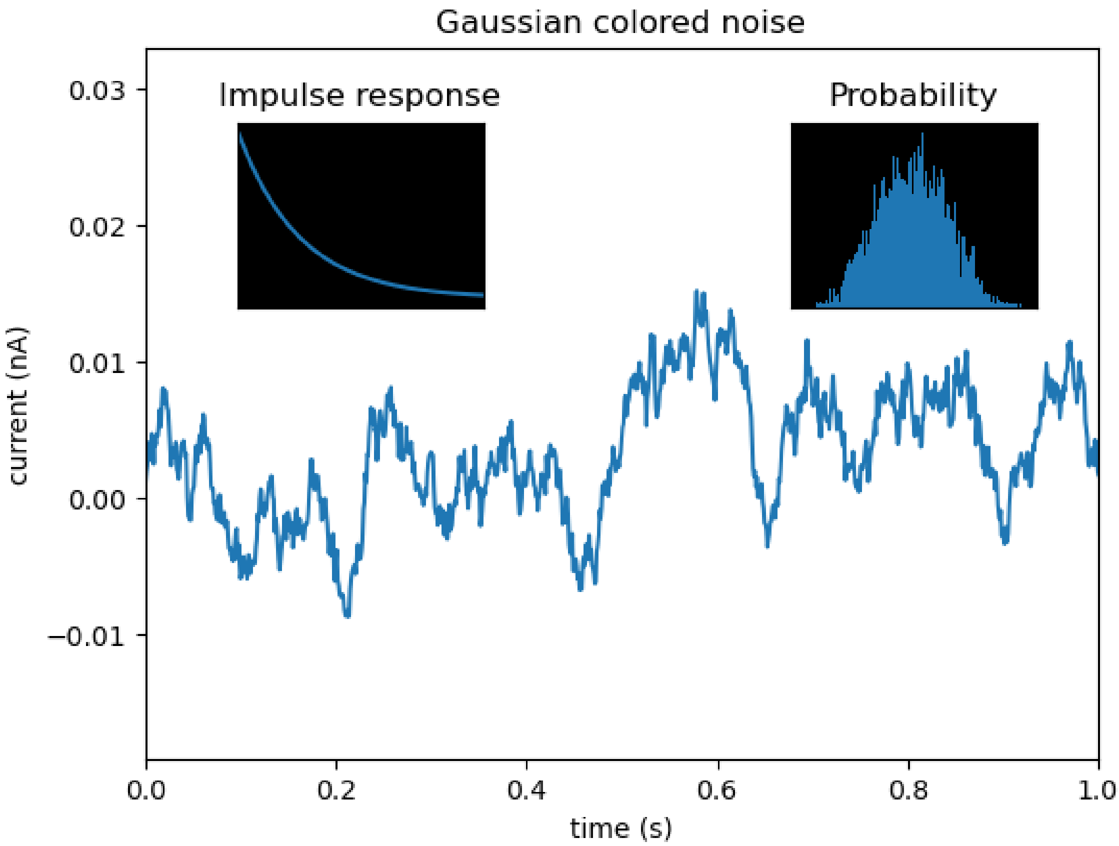
where

- left: The x-position (horizontal starting point) of the axes, as a fraction of the figure width (0 to 1).
- bottom: The y-position (vertical starting point) of the axes, as a fraction of the figure height (0 to 1).
- width: The width of the axes, as a fraction of the figure width (0 to 1).

- height: The height of the axes, as a fraction of the figure height (0 to 1).

Below is an example demonstrating the use of **add_axes()**. You can also explore the **inset_axes()** method for creating inset plots with more flexibility

```
# create inset axes within the main plot axes

np.random.seed(19680801)  # Fixing random state for reproducibility.

# create some data to use for the plot
dt = 0.001
t = np.arange(0.0, 10.0, dt)
r = np.exp(-t[:1000] / 0.05)  # impulse response
x = np.random.randn(len(t))
s = np.convolve(x, r)[:len(x)] * dt  # colored noise

fig, main_ax = plt.subplots()
main_ax.plot(t, s)
main_ax.set_xlim(0, 1)
main_ax.set_ylim(1.1 * np.min(s), 2 * np.max(s))
main_ax.set_xlabel('time (s)')
main_ax.set_ylabel('current (nA)')
main_ax.set_title('Gaussian colored noise')

# this is an inset axes over the main axes
right_inset_ax = fig.add_axes([.65, .6, .2, .2], facecolor='k')
right_inset_ax.hist(s, 400, density=True)
right_inset_ax.set(title='Probability', xticks=[], yticks=[])

# this is another inset axes over the main axes
left_inset_ax = fig.add_axes([.2, .6, .2, .2], facecolor='k')
left_inset_ax.plot(t[:len(r)], r)
left_inset_ax.set(title='Impulse response', xlim=(0, .2), xticks=[], yticks=[])

plt.show()
```



12.2.4 Advanced Customization with Matplotlib's Object-Oriented Interface

Below, we are reading the dataset of noise complaints of type Loud music/Party received the police in New York City in 2016.

```
nyc_party_complaints = pd.read_csv('datasets/party_nyc.csv')
nyc_party_complaints.head()
```

	Created Date	Closed Date	Location Type	Incident Zip	City	Borough
0	12/31/2015 0:01	12/31/2015 3:48	Store/Commercial	10034.0	NEW YORK	MANHAT
1	12/31/2015 0:02	12/31/2015 4:36	Store/Commercial	10040.0	NEW YORK	MANHAT
2	12/31/2015 0:03	12/31/2015 0:40	Residential Building/House	10026.0	NEW YORK	MANHAT
3	12/31/2015 0:03	12/31/2015 1:53	Residential Building/House	11231.0	BROOKLYN	BROOKI
4	12/31/2015 0:05	12/31/2015 3:49	Residential Building/House	10033.0	NEW YORK	MANHAT

Below, we will begin with basic plotting, utilizing Matplotlib's object-oriented interface to handle more complex tasks, such as setting the major axis formatting. When it comes to advanced customization, Matplotlib's object-oriented interface offers greater flexibility and control compared to the pyplot interface.

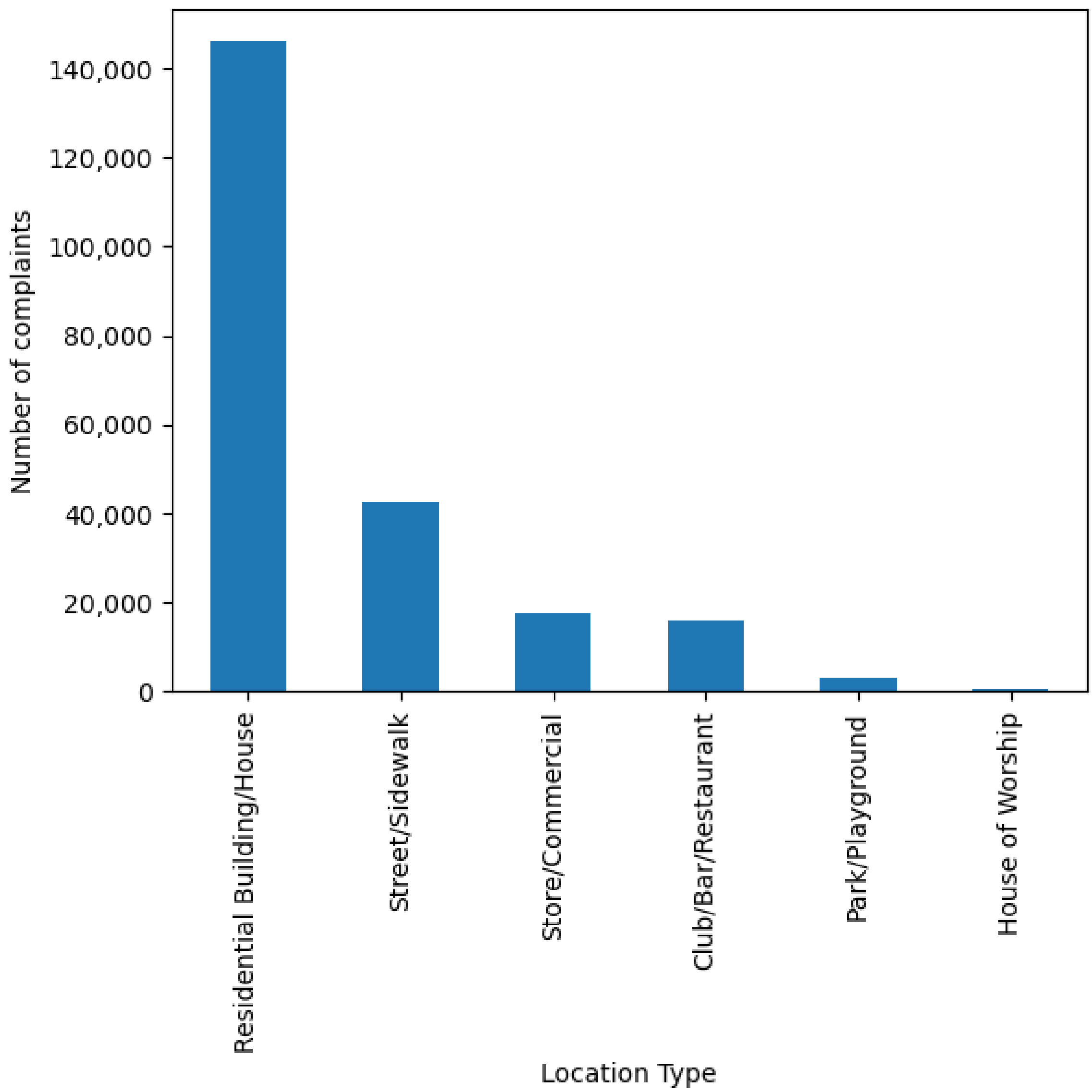
12.2.4.1 Bar plots with Pandas

Purpose of bar plots: Barplots are used to visualize any aggregate statistics of a continuous variable with respect to the categories or levels of a categorical variable.

Bar plots can be made using the pandas `bar` function with the DataFrame or Series, just like the line plots and scatterplots.

Let us visualise the locations from where the the complaints are coming.

```
ax = nyc_party_complaints['Location Type'].value_counts().plot.bar(ylabel = 'Number of compla
ax.yaxis.set_major_formatter('{x:,.0f}')
```



In the above code, we use `ax.yaxis.set_major_formatter` to format the y-axis labels in a currency style. From the above plot, we observe that most of the complaints come from residential buildings and houses, as one may expect.

For categorical variables, we can use the `.value_counts()` method to get the statistical frequency of each unique value.

```
nyc_party_complaints['Location Type'].value_counts()
```

```
Location Type
```

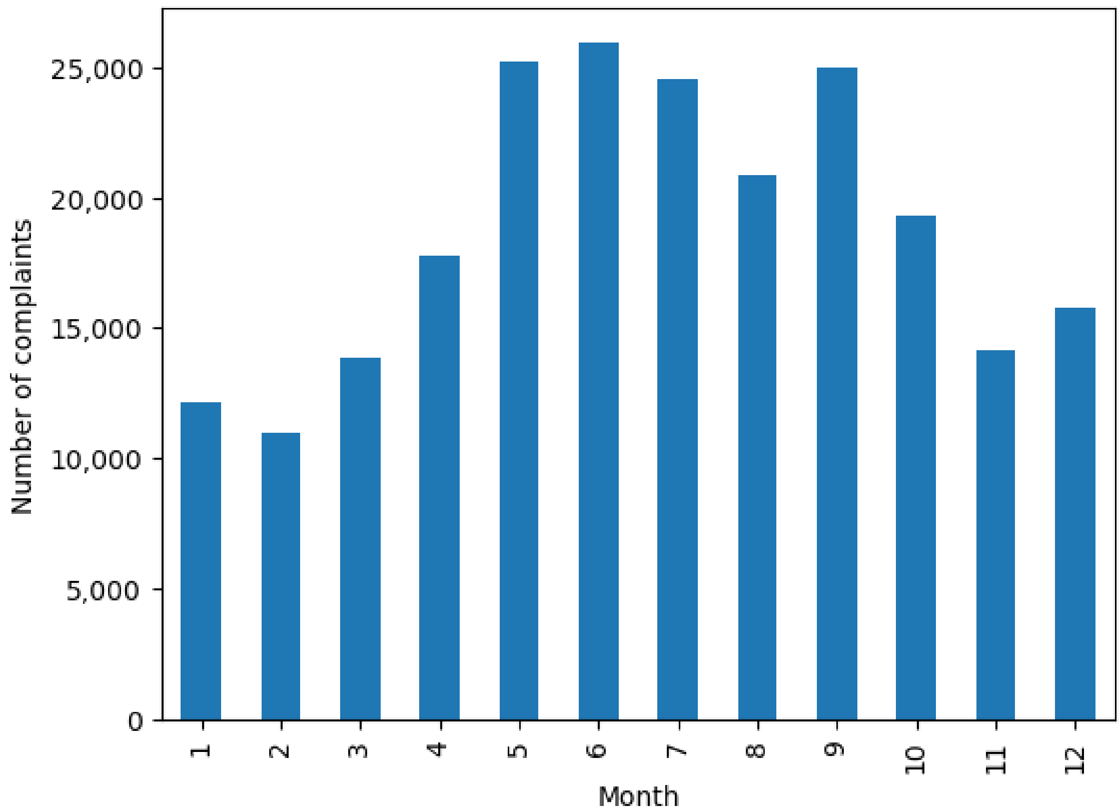
```
Residential Building/House    146040
Street/Sidewalk                42353
Store/Commercial               17617
Club/Bar/Restaurant            15766
Park/Playground                 3036
House of Worship                 602
Name: count, dtype: int64
```

Next, Let is visualize the time of the year when most complaints occur.

```
nyc_party_complaints['Month_of_the_year'].value_counts()
```

```
Month_of_the_year
6     25933
5     25192
9     25000
7     24502
8     20833
10    19332
4     17718
12    15730
11    14146
3     13880
1     12171
2     10977
Name: count, dtype: int64
```

```
#Using the pandas function bar() to create bar plot
ax = nyc_party_complaints['Month_of_the_year'].value_counts().sort_index().plot.bar(ylabel =
                                                                                     xlabel = "Montl
ax.yaxis.set_major_formatter('{x:,.0f}')
```



Try executing the code without `sort_index()` to figure out the purpose of using the function.

From the above plot, we observe that most of the complaints occur during summer and early Fall.

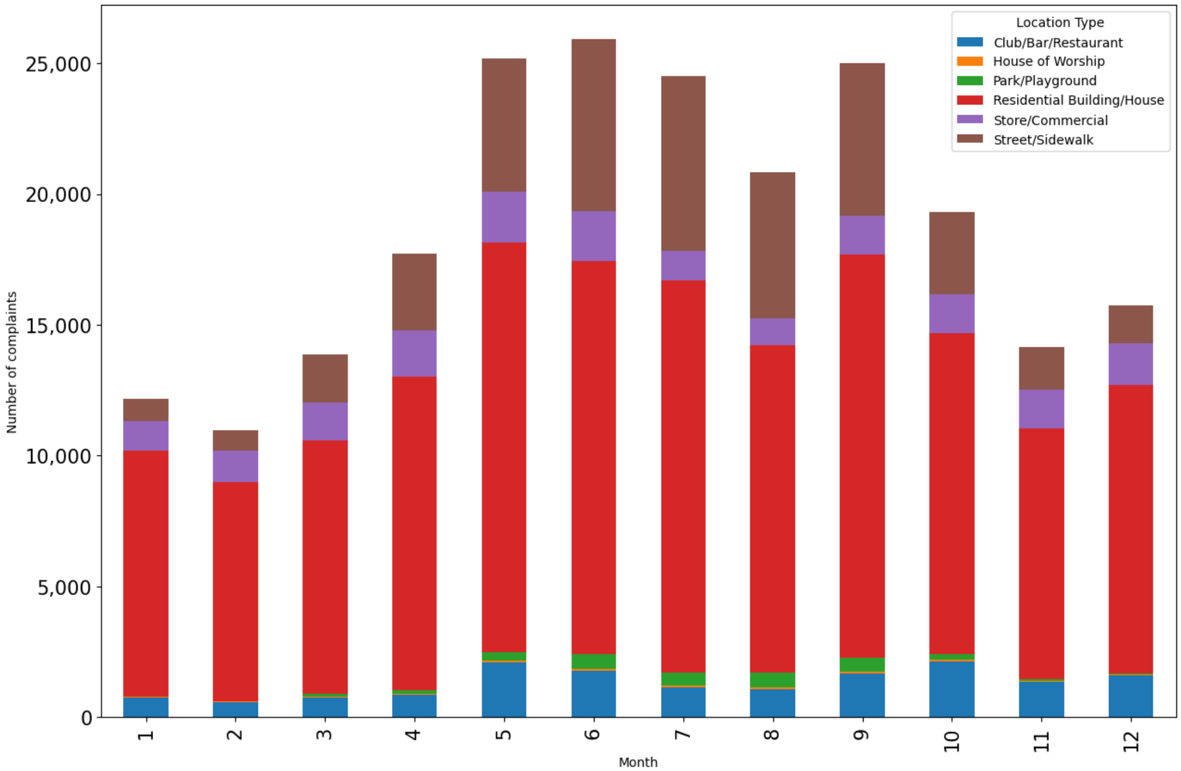
Let us create a stacked bar chart that combines both the above plots into a single plot. You may ignore the code used for re-shaping the data until Chapter 10. The purpose here is to show the utility of the pandas `bar()` function.

```
#Reshaping the data to make it suitable for a stacked barplot - ignore this code until chapte
complaints_location=pd.crosstab(nyc_party_complaints.Month_of_the_year, nyc_party_complaints
complaints_location.head()
```

Location Type Month_of_the_year	Club/Bar/Restaurant	House of Worship	Park/Playground	Residential Building/H
1	748	24	17	9393
2	570	29	16	8383
3	747	39	90	9689

Location Type Month_of_the_year	Club/Bar/Restaurant	House of Worship	Park/Playground	Residential Building/H
4	848	53	129	11984
5	2091	72	322	15676

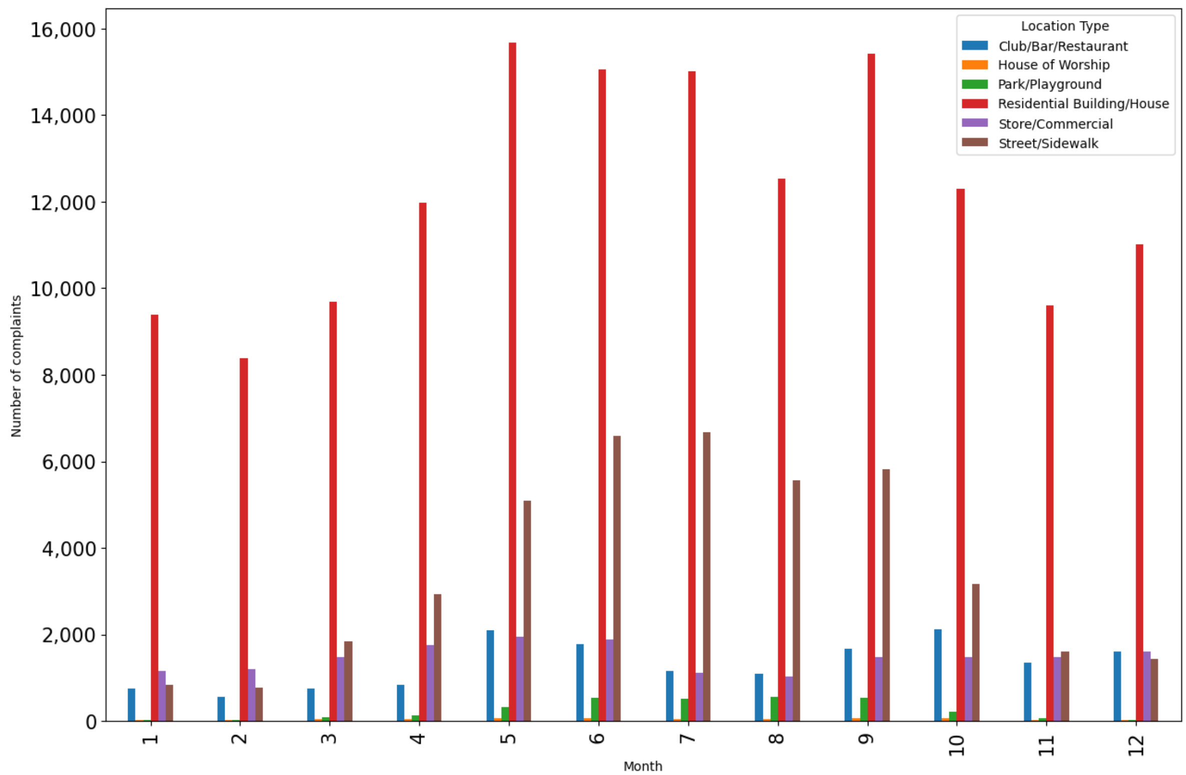
```
#Stacked bar plot showing number of complaints at different months of the year, and from dif
ax = complaints_location.plot.bar(stacked=True,ylabel = 'Number of complaints',figsize=(15, 1
ax.tick_params(axis = 'both',labelsize=15)
ax.yaxis.set_major_formatter('{x:,.0f}')
```



The above plots gives the insights about location and day of the year simultaneously that were previously separately obtained by the individual plots.

An alternative to stacked barplots are side-by-side barplots, as shown below.

```
#Side-by-side bar plot showing number of complaints at different months of the year, and fro
ax = complaints_location.plot.bar(ylabel = 'Number of complaints',figsize=(15, 10), xlabel =
ax.tick_params(axis = 'both',labelsize=15)
ax.yaxis.set_major_formatter('{x:,.0f}')
```



Question: In which scenarios should we use a stacked barplot instead of a side-by-side barplot and vice-versa?

12.2.4.2 Bar plots with confidence intervals with Seaborn

We'll group the data to obtain the total complaints for each *Location Type, Borough, Month_of_the_year, and Hour_of_the_day*. Note that you'll learn grouping data in later chapters, so you may ignore the next code block. The grouping is done to shape the data in a suitable form for visualization.

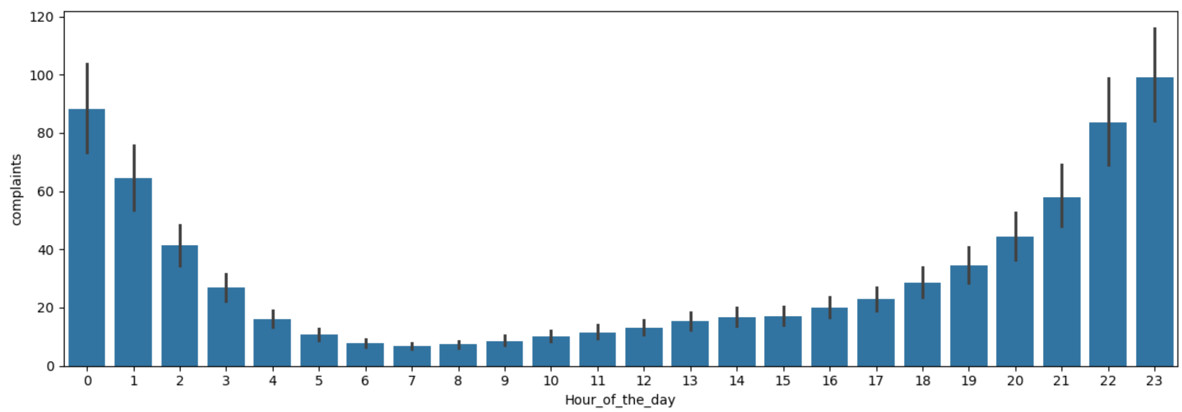
```
#Grouping the data to make it suitable for visualization using Seaborn. Ignore this code bloc
nyc_complaints_grouped = nyc_party_complaints[['Location Type','Borough','Month_of_the_year'
nyc_complaints_grouped.head()
```

	Location Type	Borough	Month_of_the_year	Hour_of_the_day	complaints
0	Club/Bar/Restaurant	BRONX	1	0	10
1	Club/Bar/Restaurant	BRONX	1	1	10
2	Club/Bar/Restaurant	BRONX	1	2	6

	Location Type	Borough	Month_of_the_year	Hour_of_the_day	complaints
3	Club/Bar/Restaurant	BRONX	1	3	6
4	Club/Bar/Restaurant	BRONX	1	4	3

Let us create a bar plot visualizing the average number of complaints with the time of the day.

```
ax = sns.barplot(x="Hour_of_the_day", y = 'complaints',  data=nyc_complaints_grouped)
ax.figure.set_figwidth(15)
```



From the above plot, we observe that most of the complaints are made around midnight. However, interestingly, there are some complaints at each hour of the day.

Note that the above barplot shows the mean number of complaints in a month at each hour of the day. The black lines are the 95% confidence intervals of the mean number of complaints.

12.2.5 `pyplot`: a convenience wrapper around the object-oriented interface

While the pyplot interface is simpler for quick, basic plots, it ultimately wraps around the object-oriented structure of Matplotlib, meaning that it's built on top of the object-oriented interface.

12.3 Creating Subplots with Seaborn

We previously demonstrated how Seaborn integrates seamlessly with Matplotlib's object-oriented interface, allowing you to pass the `ax` argument to any Seaborn function, thereby directing the plot to a specific axis within a subplot grid.

Additionally, Seaborn offers a more convenient and simplified approach to creating subplots, thanks to its high-level functions and built-in integration with Matplotlib. Here's how Seaborn makes working with subplots easier:

12.3.1 Using `Facetgrid`

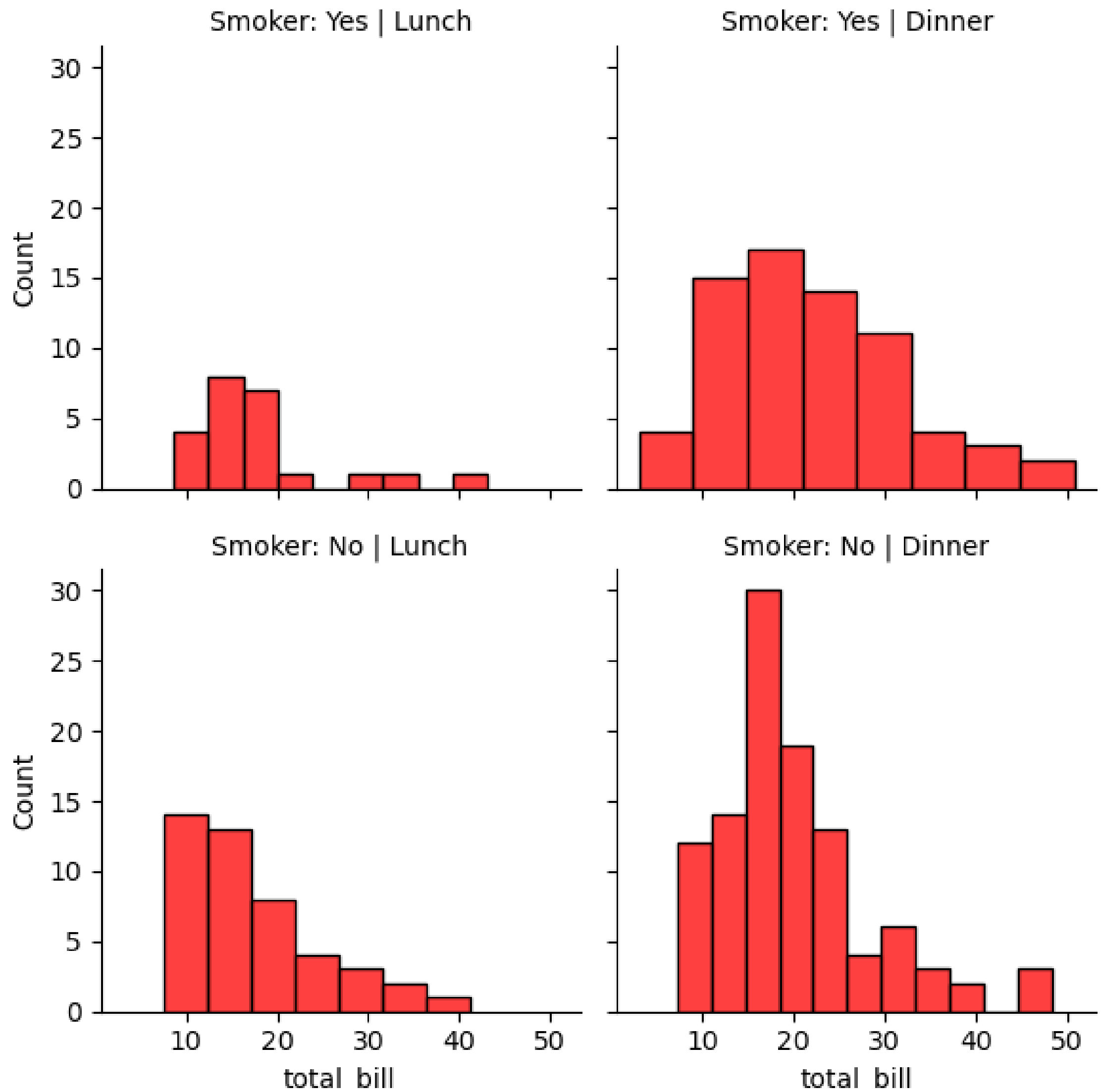
Seaborn's `FacetGrid` function make it very easy to create facet grids or subplots based on data dimensions (such as categories), which would require more manual effort with Matplotlib.

You can use the `row` and `col` parameters to control how the data is split into subplots along these dimensions.

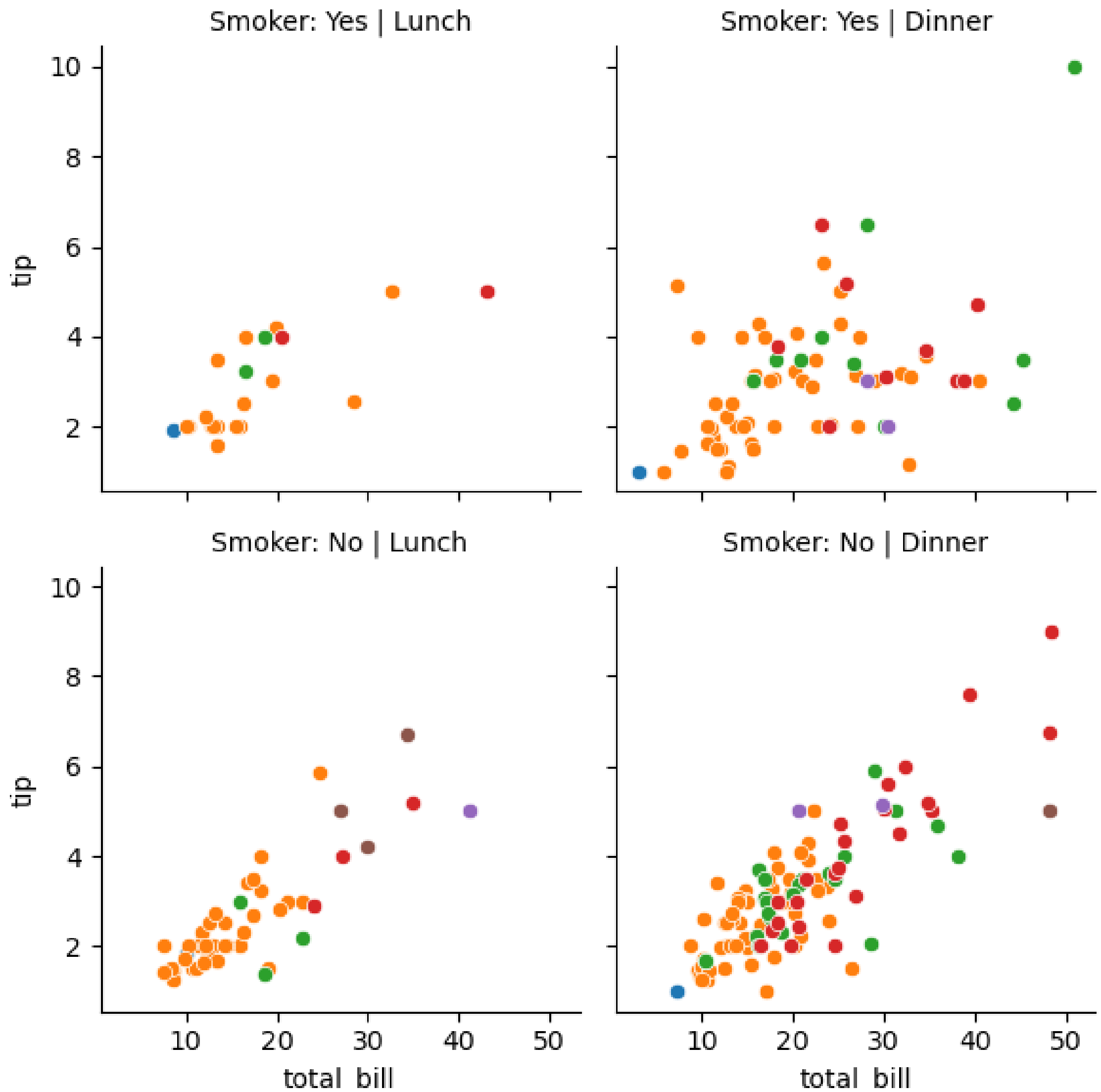
```
# Seaborn Example using FacetGrid:
tips_df = sns.load_dataset("tips")
tips_df.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

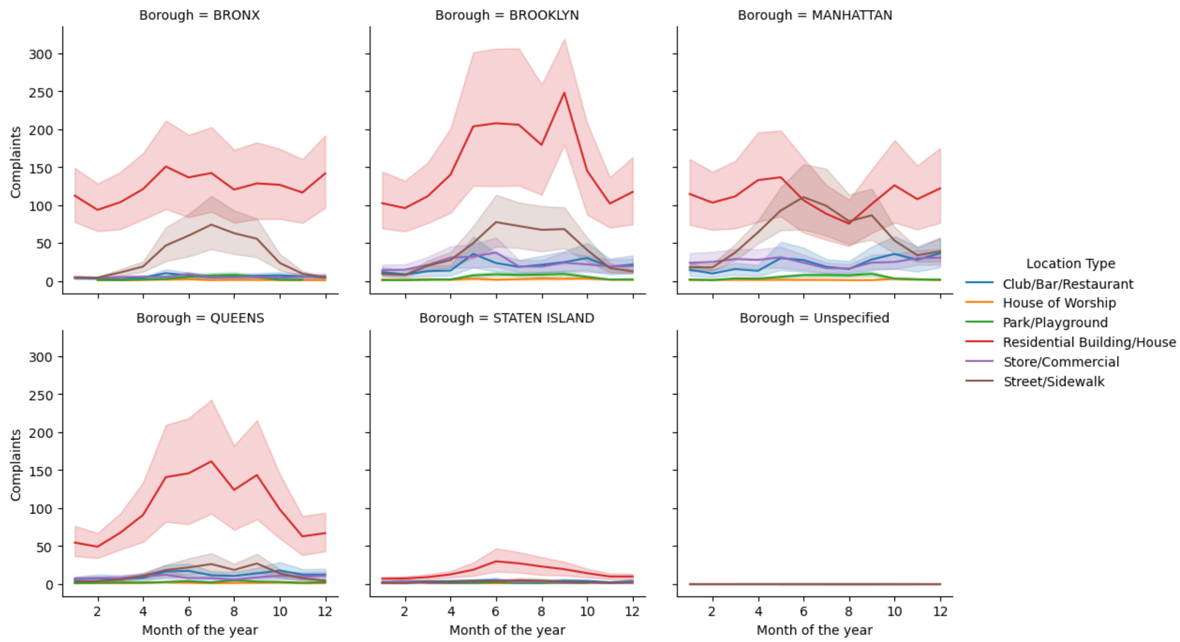
```
g = sns.FacetGrid(tips_df, col='time', row='smoker')
g.map(sns.histplot, 'total_bill', color='r')
g.set_titles(col_template="{col_name}", row_template="Smoker: {row_name}");
```




```
# adding hue to the FacetGrid
g = sns.FacetGrid(tips_df, col='time', row='smoker',hue='size')
# Plot a scatterplot of the total bill and tip for each combination of time and smoker
g.map(sns.scatterplot, 'total_bill', 'tip')
g.set_titles(col_template="{col_name}", row_template="Smoker: {row_name}");
```




```
#Visualizing the number of complaints with Month_of_the_year, Location Type, and Borough.
a = sns.FacetGrid(nyc_complaints_grouped, hue = 'Location Type', col = 'Borough',col_wrap=3,
# Plotting a lineplot to show the number of complaints with Month_of_the_year
a.map(sns.lineplot,'Month_of_the_year','complaints')
a.set_axis_labels("Month of the year", "Complaints")
a.add_legend()
```

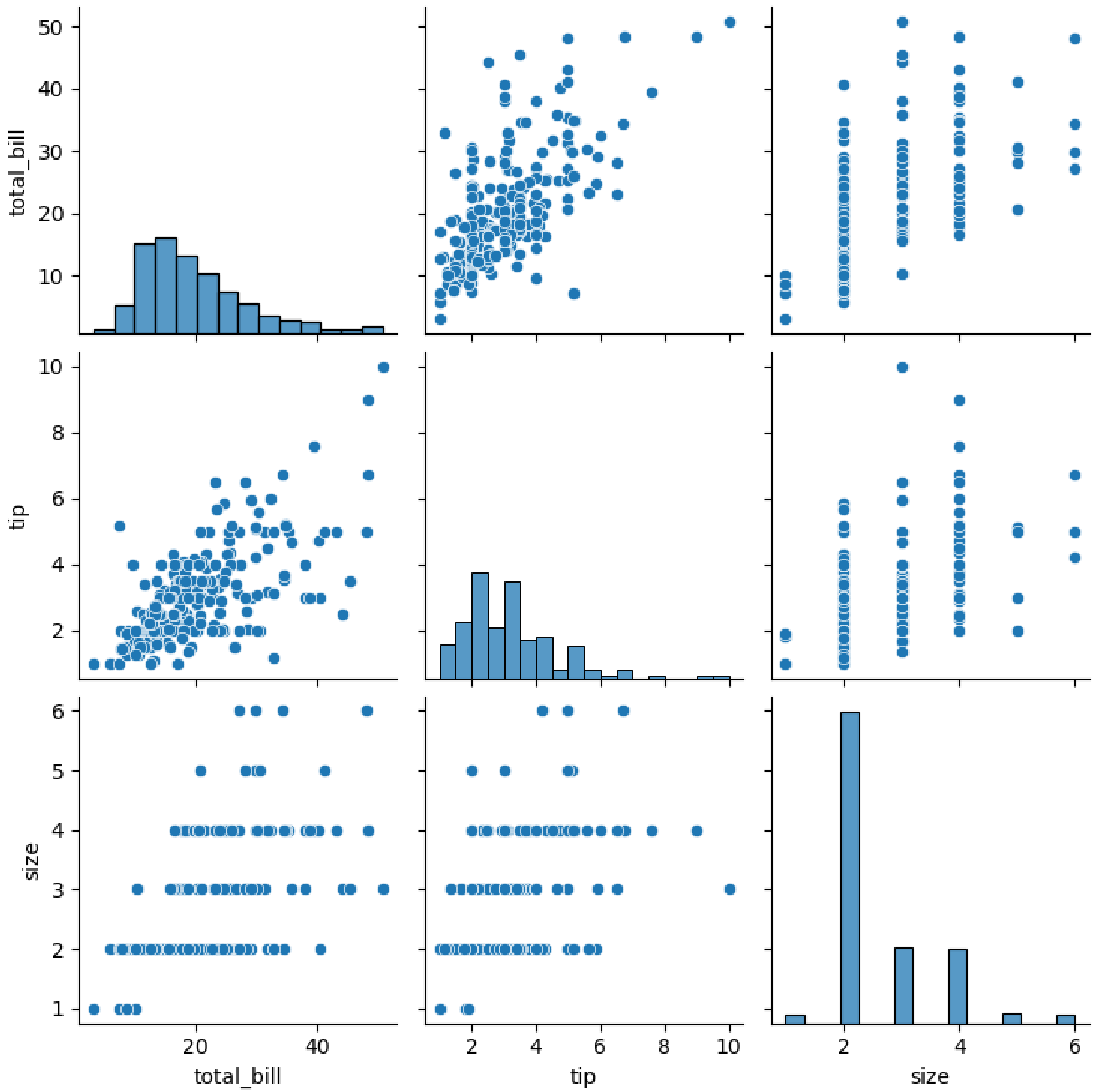


12.3.2 Using `Pairplot`

Pairplots are used to visualize the association between all variable-pairs in the data. In other words, pairplots simultaneously visualize the scatterplots between all variable-pairs.

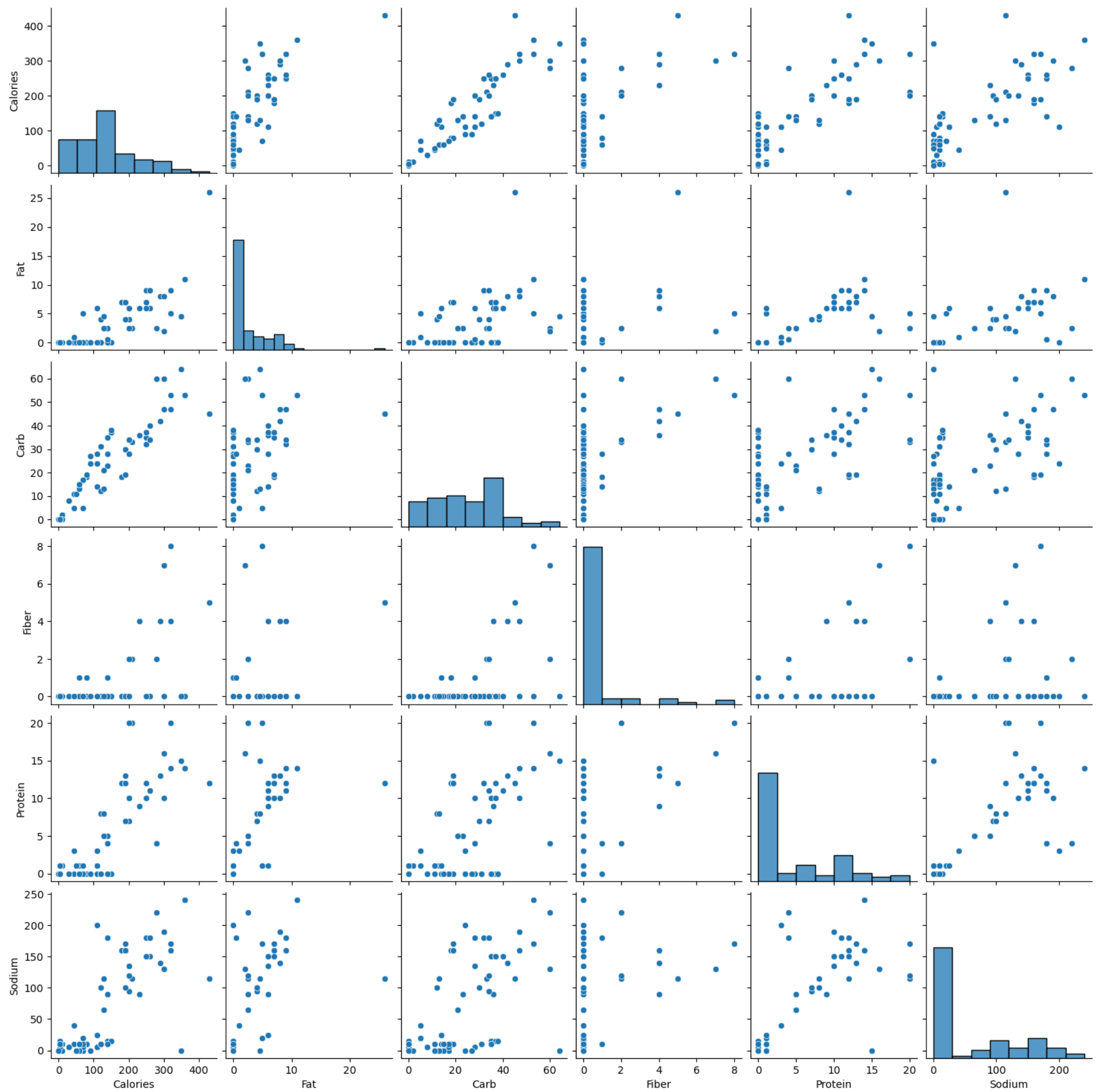
Let us visualize the pair-wise association of tips variables in the tips dataset

```
sns.pairplot(tips_df );
```



Let us visualize the pair-wise association of nutrition variables in the starbucks drinks data.

```
starbucks_drinks = pd.read_csv('datasets/starbucks-menu-nutrition-drinks.csv')
sns.pairplot(starbucks_drinks);
```



In the above pairplot, note that:

- The histograms on the diagonal of the grid show the distribution of each of the variables.
- Instead of a histogram, we can visualize the density plot with the argument kde = True.
- The scatterplots in the rest of the grid are the pair-wise plots of all the variables.

12.4 Geosptial Plotting

There are several widely used Python packages pecifically designed for working with geospatial datasets. In this lesson, we will cover:

- GeoPandas
- Folium

Let's import them

```
import geopandas as gpd
import geopandas
import folium
import geodatasets
```

```
c:\Users\lsi8012\AppData\Local\anaconda3\Lib\site-packages\paramiko\transport.py:219: Cryptog
  "class": algorithms.Blowfish,
```

12.4.1 Static Plots with GeoPandas

A **shapefile** is a widely-used format for storing geographic information system (GIS) data, specifically vector data. It contains geometries (like points, lines, and polygons) that represent features on the earth's surface, along with associated attributes for each feature, such as names, populations, or other data relevant to the feature.

12.4.1.1 Components of a Shapefile

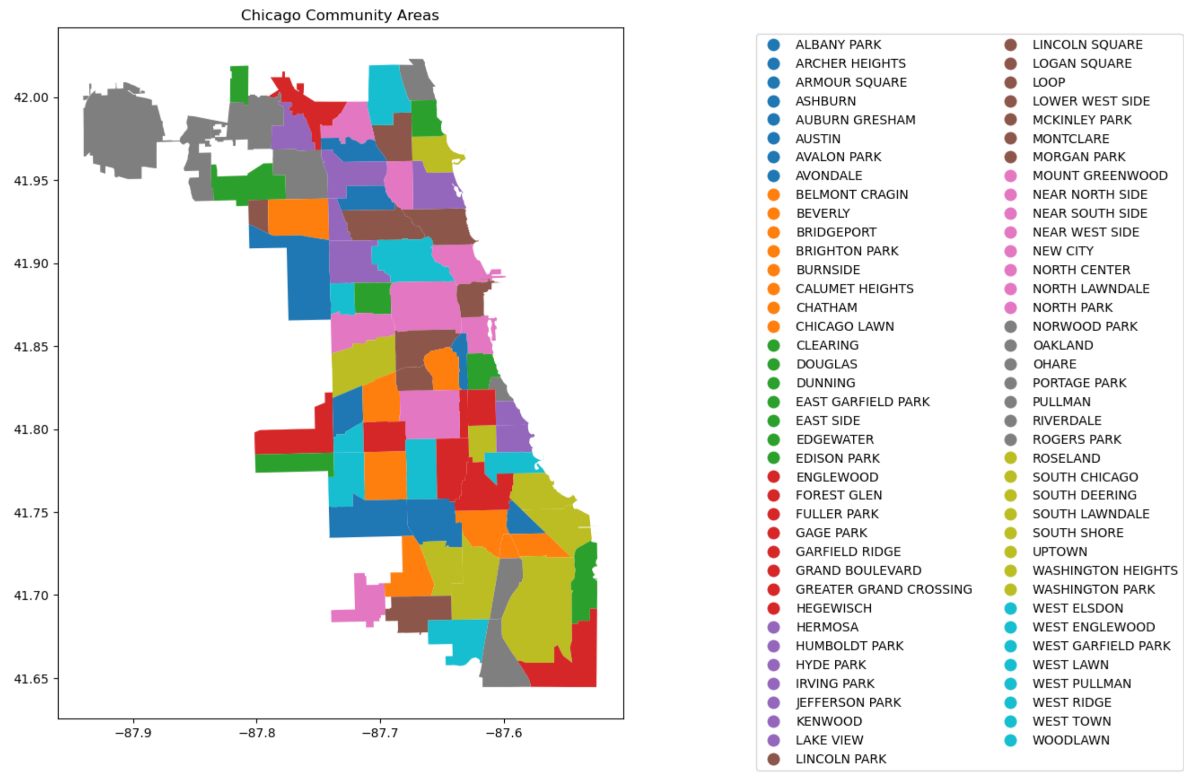
A shapefile isn't a single file but a collection of files with the same name and different extensions, which work together to store geographic and attribute data:

- `.shp`: Stores the geometry (shapes of features, like points, lines, polygons).
- `.shx`: Contains an index to quickly access geometries in the .shp file.
- `.dbf`: A table storing attributes associated with each feature.

There may also be other optional files (e.g., .prj for projection information).

```
# Create figure and axis
fig, ax = plt.subplots(figsize=(15, 10))

# Plot your GeoDataFrame
chicago = gpd.read_file(r'datasets/chicago_boundaries\geo_export_26bce2f2-c163-42a9-9329-9ca
```

```
chicago.plot(column='community', ax=ax, legend=True, legend_kwds={'ncol': 2, 'bbox_to_anchor

# Add title (optional)
plt.title('Chicago Community Areas');
```



Let’s print out the information in the shapefile

```
chicago.head()
```

	area	area_num_1	area_numbe	comarea	comarea_id	community	perimeter	shape_
0	0.0	35	35	0.0	0.0	DOUGLAS	0.0	4.6004
1	0.0	36	36	0.0	0.0	OAKLAND	0.0	1.6913
2	0.0	37	37	0.0	0.0	FULLER PARK	0.0	1.9916
3	0.0	38	38	0.0	0.0	GRAND BOULEVARD	0.0	4.8492
4	0.0	39	39	0.0	0.0	KENWOOD	0.0	2.9071

```python
chicago['geometry'].head()
```

```
0    POLYGON ((-87.60914 41.84469, -87.60915 41.844...
1    POLYGON ((-87.59215 41.81693, -87.59231 41.816...
2    POLYGON ((-87.6288 41.80189, -87.62879 41.8017...
3    POLYGON ((-87.60671 41.81681, -87.6067 41.8165...
4    POLYGON ((-87.59215 41.81693, -87.59215 41.816...
Name: geometry, dtype: geometry
```

```python
# Check the column names to see available data fields
print("Columns in the shapefile:", chicago.columns)

# Check the data types of each column
print("Data types:", chicago.dtypes)

# View the spatial extent (bounding box) of the shapes
print("Bounding box:", chicago.total_bounds)

# Check the coordinate reference system (CRS)
print("CRS:", chicago.crs)
```

```
Columns in the shapefile: Index(['area', 'area_num_1', 'area_numbe', 'comarea', 'comarea_id'
       'community', 'perimeter', 'shape_area', 'shape_len', 'geometry'],
      dtype='object')
Data types: area           float64
area_num_1      object
area_numbe      object
comarea        float64
comarea_id     float64
community       object
perimeter      float64
shape_area     float64
shape_len      float64
geometry      geometry
dtype: object
Bounding box: [-87.94011408  41.64454312 -87.5241371   42.02303859]
CRS: EPSG:4326
```

To enhance the geospatial plot, we'll use the shapefile as a background to provide context for Chicago's community areas. On top of that, we'll layer points of interest, such as restaurants, and shops, to illustrate the city's amenities. This approach will make the map more informative and visually engaging, with community boundaries as the foundation and key locations overlayed to highlight areas of interest.

Next, we will add the Divvy bicycle stations on top of the chicago shapefile

12.4.2 Dataset: Bicycle Sharing in Chicago

Divvy is Chicagoland's bike share system (in collaboration with Chicago Department of Transportation), with 6,000 bikes available at 570+ stations across Chicago and Evanston. Divvy provides residents and visitors with a convenient, fun and affordable transportation option for getting around and exploring Chicago.

Divvy, like other bike share systems, consists of a fleet of specially designed, sturdy and durable bikes that are locked into a network of docking stations throughout the region. The bikes can be unlocked from one station and returned to any other station in the system. People use bike share to explore Chicago, commute to work or school, run errands, get to appointments or social engagements, and more.

Divvy is available for use 24 hours/day, 7 days/week, 365 days/year, and riders have access to all bikes and stations across the system.

We will be using divvy trips in the year of 2013

```
# read the csv file'divvy_2013.csv' into pandas pandas dataframe
data = pd.read_csv('datasets/divvy_2013.csv')
data.head()
```

	trip_id	usertype	gender	starttime	stoptime	tripduration	from_station_id
0	3940	Subscriber	Male	2013-06-27 01:06:00	2013-06-27 09:46:00	31177	91
1	4095	Subscriber	Male	2013-06-27 12:06:00	2013-06-27 12:11:00	301	85
2	4113	Subscriber	Male	2013-06-27 11:09:00	2013-06-27 11:11:00	140	88
3	4118	Customer	NaN	2013-06-27 12:11:00	2013-06-27 12:16:00	316	85
4	4119	Subscriber	Male	2013-06-27 11:12:00	2013-06-27 11:13:00	87	88

In the Divvy dataset, each trip record includes the latitude and longitude coordinates of both the pickup and drop-off locations, which correspond to Divvy bike stations. These coordinates allow us to map the precise locations of each station, making it possible to visually display the network of Divvy stations across the city. By plotting these stations on a map, we can better understand the geographic distribution and accessibility of Divvy's bike-sharing network.

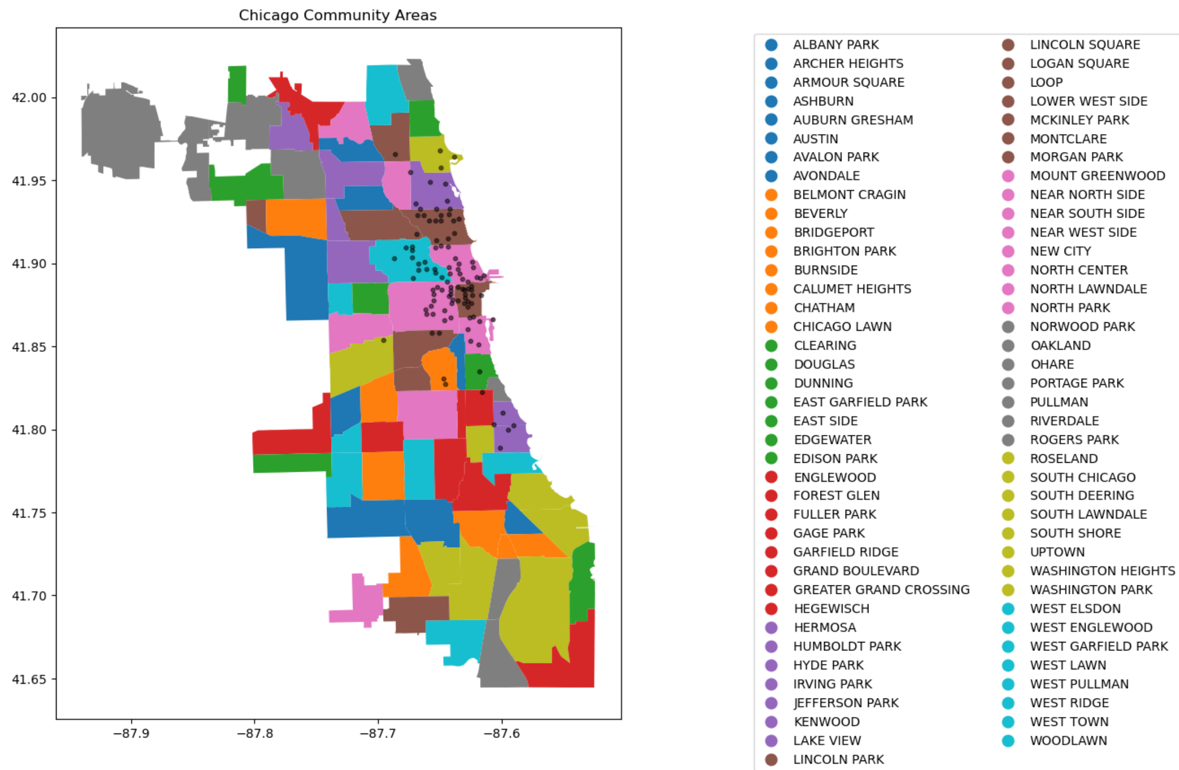
Below are the basic data cleaning steps to extract the coordinates of the Divvy stations.

```
# drop the duplicates in the column 'to_station_id', 'to_station_name', 'latitude_end', 'long
# data_station_same = data[['from_station_id', 'from_station_name', 'latitude_start', 'longi
# data_station_same.shape
```

12.4.3 Adding the divvy station to the plot

Once the coordinates are prepared, we'll add them as scatter plots on top of the Chicago shapefile

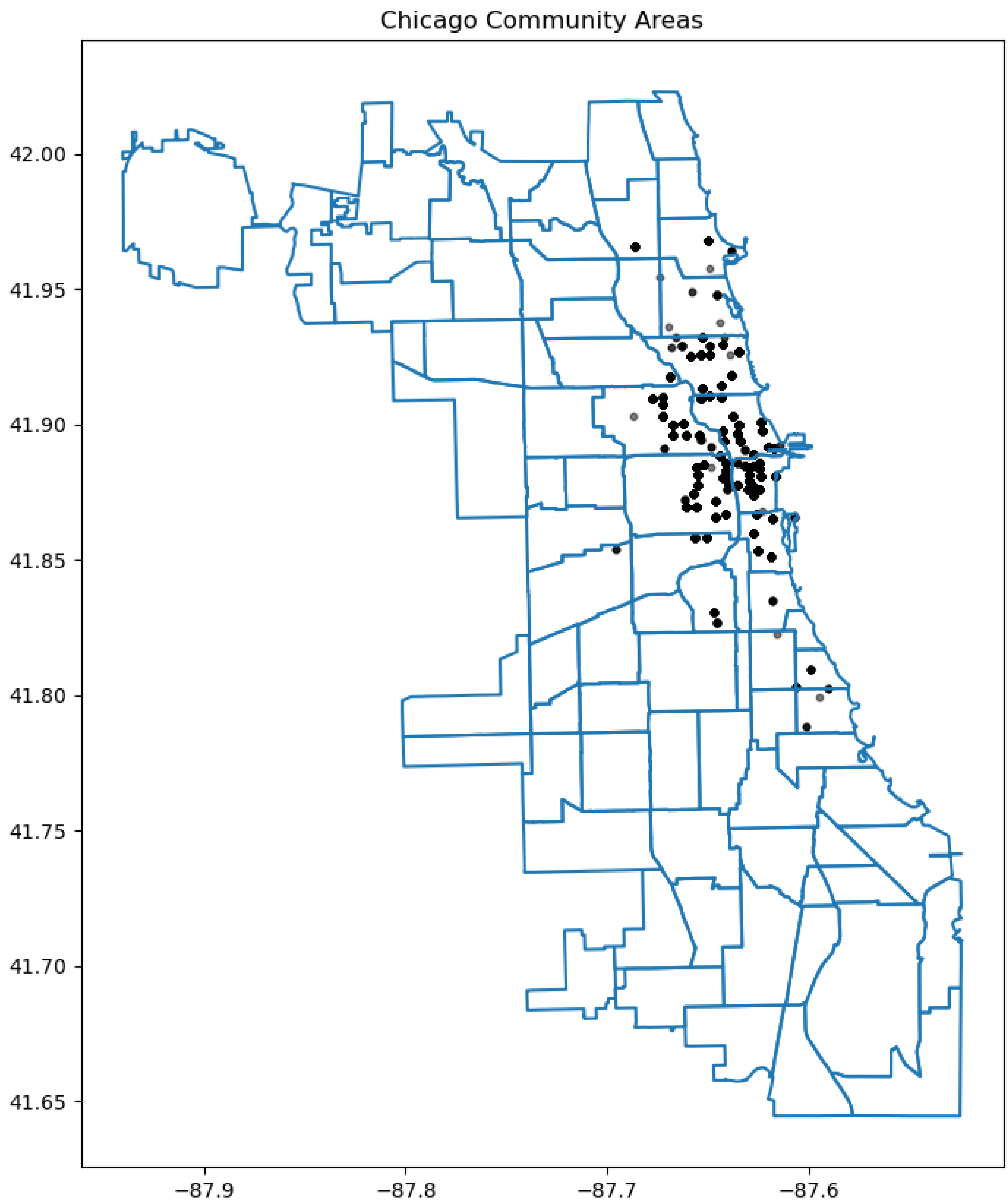
```
# Adding the stations to the plot
fig, ax = plt.subplots(figsize=(15, 10))

chicago = gpd.read_file(r'datasets/chicago_boundaries\geo_export_26bce2f2-c163-42a9-9329-9ca6
chicago.plot(column='community', ax=ax, legend=True, legend_kwds={'ncol': 2, 'bbox_to_anchor

# Plot the stations
longlat_df = data[[ 'latitude_start', 'longitude_start']].drop_duplicates()

plt.scatter(longlat_df['longitude_start'], longlat_df['latitude_start'], s=10, alpha=0.5, col


# Add title (optional)
plt.title('Chicago Community Areas');
```



12.4.4 Change the chicago shapefile

Using a different Chicago shapefile from `GeoDa` is a great way to observe how geographic boundaries or data details may vary

```
chicago = gpd.read_file(geodatasets.get_path("geoda.chicago_commpop"))

# Plot the data
fig, ax = plt.subplots(figsize=(15, 10))
chicago.boundary.plot(ax=ax)
plt.scatter(data['longitude_start'], data['latitude_start'], s=10, alpha=0.5, color='black',
plt.title('Chicago Community Areas');
```


Chicago Community Areas
42.00
41.95
41.90
41.85
41.80
41.75
41.70
41.65
−87.9
−87.8
−87.7
−87.6

12.4.5 Interactive Plotting

Alongside static plots, `geopandas` can create interactive maps based on the folium library.

Creating maps for interactive exploration mirrors the API of static plots in an explore() method of a GeoSeries or GeoDataFrame.

Here's an explanation of how `explore()` works and its key features:

Key Features of `explore()`:

1. Interactive Map Display:

- When you call explore() on a Geodataframe (gdf), it launches an interactive map widget directly within your Jupyter notebook.
- This map allows you to pan, zoom, and interact with the geometries (points, lines, polygons) in your Geodataframe.

2. Layer Control:

- explore() automatically adds the geometries from your Geodataframe as layers on the map.
- Each geometry type (points, lines, polygons) is displayed with appropriate styling and markers.

3. Tooltip Information:

- When you hover over a geometry in the map, explore() displays tooltip information that typically includes attribute data associated with that geometry.
- This feature is useful for inspecting specific details or properties of individual features in your geospatial dataset.

4. Search and Filter:

- explore() provides basic search and filter functionalities directly on the map.
- You can search for specific attribute values or filter the displayed features based on attribute criteria defined in your Geodataframe.

5. Customization:

- Although explore() provides default styling and interaction behaviors, you can customize the map further using parameters or by manipulating the Geodataframe before calling explore().

```
# use the geopandas explore default settings
chicago = gpd.read_file(geodatasets.get_path("geoda.chicago_commpop"))

chicago.explore()
```

```
<folium.folium.Map at 0x1f6a6f672c0>
```

Adding the population layer

```
# Customerize the explore settings
chicago = gpd.read_file(geodatasets.get_path("geoda.chicago_commpop"))

m = chicago.explore(
    column="POP2010",  # make choropleth based on "POP2010" column
    scheme="naturalbreaks",  # use mapclassify's natural breaks scheme
    legend=True,  # show legend
    k=10,  # use 10 bins
    tooltip=False,  # hide tooltip
    popup=["POP2010", "POP2000"],  # show popup (on-click)
    legend_kwds=dict(colorbar=False),  # do not use colorbar
    name="chicago",  # name of the layer in the map
)

m
```

```
c:\Users\lsi8012\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1446: U
  warnings.warn(
```

```
<folium.folium.Map at 0x1f6a7436e40>
```

The `explore()` method returns a `folium.Map` object, which can also be passed directly (as you do with `ax` in `plot()`). You can then use folium functionality directly on the resulting map. Next, let's add the divvy station plot.

```
type(m)
```

```
folium.folium.Map
```

12.4.6 Adding the divvy station on the interactive `Folium.Map`

We need to extract the station information from the trip dataset and add description to the station. You can skip this part

```
# Helper function for adding the description to the station
def row_to_html(row):
    row_df = pd.DataFrame(row).T
    row_df.columns = [col.capitalize() for col in row_df.columns]
    return row_df.to_html(index=False)
```

```
# Extracting the latitude, longitude, and station name for plotting, and also counting the nu
grouped_df = data.groupby(['from_station_name', 'latitude_start', 'longitude_start'])['trip_i
display(grouped_df.sort_values('trip_id', ascending=False).head())
grouped_df.rename(columns={'from_station_name':'title', 'latitude_start':'latitude', 'longitu
grouped_df['description'] = grouped_df.apply(lambda row: row_to_html(row), axis=1)
geometry = gpd.points_from_xy(grouped_df['longitude'], grouped_df['latitude'])
geo_df = gpd.GeoDataFrame(grouped_df, geometry=geometry)
# Optional: Assign Coordinate Reference System (CRS)
geo_df.crs = "EPSG:4326"  # WGS84 coordinate system
geo_df.head()
```

	from_station_name	latitude_start	longitude_start	trip_id
75	Millennium Park	41.881032	-87.624084	207
54	Lake Shore Dr & Monroe St	41.881050	-87.616970	191
72	Michigan Ave & Oak St	41.900960	-87.623777	186
68	McClurg Ct & Illinois St	41.891020	-87.617300	177
73	Michigan Ave & Pearson St	41.897660	-87.623510	127

	title	latitude	longitude	count	description
0	Aberdeen St & Jackson Blvd	41.877726	-87.654787	28	<table border=”1” class=”dataframe”>\
1	Aberdeen St & Madison St	41.881487	-87.654752	28	<table border=”1” class=”dataframe”>\
2	Adler Planetarium	41.866095	-87.607267	6	<table border=”1” class=”dataframe”>\
3	Ashland Ave & Armitage Ave	41.917859	-87.668919	20	<table border=”1” class=”dataframe”>\
4	Ashland Ave & Augusta Blvd	41.899643	-87.667700	27	<table border=”1” class=”dataframe”>\

We can add a hover tooltip (sometimes referred to as a tooltip or tooltip popup) for each point on your Folium map. This tooltip will appear when you hover over the markers on the map, providing additional information without needing to click on them. Here's how you can modify your existing code to include hover tooltips:

```
chicago = gpd.read_file(geodatasets.get_path("geoda.chicago_commpop"))

m = chicago.explore(
    column="POP2010",  # make choropleth based on "POP2010" column
    scheme="naturalbreaks",  # use mapclassify's natural breaks scheme
    legend=True,  # show legend
    k=10,  # use 10 bins
    tooltip=False,  # hide tooltip
    popup=["POP2010", "POP2000"],  # show popup (on-click)
    legend_kwds=dict(colorbar=False),  # do not use colorbar
    name="chicago",  # name of the layer in the map
)

geo_df.explore(
    m=m,  # pass the map object
    color="red",  # use red color on all points
    marker_kwds=dict(radius=5, fill=True),  # make marker radius 10px with fill
    tooltip="description",  # show "name" column in the tooltip
    tooltip_kwds=dict(labels=False),  # do not show column label in the tooltip
    name="divstation",  # name of the layer in the map
)

m
```

```
c:\Users\lsi8012\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1446: U
  warnings.warn(
```

```
<folium.folium.Map at 0x1f6a369ddf0>
```

12.5 Independent Study

12.5.1 Multiple plots in a single figure using Seaborn

Purpose: Histogram and density plots visualize the distribution of a continuous variable.

A histogram plots the number of observations occurring within discrete, evenly spaced bins of a random variable, to visualize the distribution of the variable. It may be considered a special case of a bar plot as bars are used to plot the observation counts.

A density plot uses a kernel density estimate to approximate the distribution of random variable.

Using the tips_df dataset

12.5.1.1

Make a histogram showing the distributions of total bill on each day of the week

12.5.1.2

Make a density plot showing the distributions of total bills on each day.

Part IV

From messy to insight

13 Data Cleaning and Preparation

13.1 Handling missing data

Missing values in a dataset can occur due to several reasons such as breakdown of measuring equipment, accidental removal of observations, lack of response by respondents, error on the part of the researcher, etc.

Let us read the dataset *GDP_missing_data.csv*, in which we have randomly removed some values, or put missing values in some of the columns.

We'll also read *GDP_complete_data.csv*, in which we have not removed any values. We'll use this data later to assess the accuracy of our guess or estimate of missing values in *GDP_missing_data.csv*.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import sklearn as sk
import seaborn as sns

sns.set(font_scale=1.5)

%matplotlib inline
```

```
gdp_missing_values_data = pd.read_csv('./Datasets/GDP_missing_data.csv')
gdp_complete_data = pd.read_csv('./Datasets/GDP_complete_data.csv')
```

```
gdp_missing_values_data.head()
```

	economicActivityFemale	country	lifeMale	infantMortality	gdpPerCapita	economicActivityMal
0	7.2	Afghanistan	45.0	154.0	2474.0	87.5
1	7.8	Algeria	67.5	44.0	11433.0	76.4
2	41.3	Argentina	69.6	22.0	NaN	76.2
3	52.0	Armenia	67.2	25.0	13638.0	65.0

	economicActivityFemale	country	lifeMale	infantMortality	gdpPerCapita	economicActivityMal
4	53.8	Australia	NaN	6.0	54891.0	NaN

Observe that the `gdp_missing_values_data` dataset consists of some missing values shown as NaN (Not a Number).

13.1.1 Identifying missing values in a dataframe

There are multiple ways to identify missing values in a dataframe

13.1.1.1 `describe()` Method

Note that the descriptive statistics methods associated with Pandas objects ignore missing values by default. Consider the summary statistics of `gdp_missing_values_data`:

```
gdp_missing_values_data.describe()
```

	economicActivityFemale	lifeMale	infantMortality	gdpPerCapita	economicActivityMale	illit
count	145.000000	145.000000	145.000000	145.000000	145.000000	145
mean	45.935172	65.491724	37.158621	24193.482759	76.563448	13.5
std	16.875922	9.099256	34.465699	22748.764444	7.854730	16.4
min	1.900000	36.000000	3.000000	772.000000	51.200000	0.00
25%	35.500000	62.900000	10.000000	6837.000000	72.000000	1.00
50%	47.600000	67.800000	24.000000	15184.000000	77.300000	6.60
75%	55.900000	72.400000	54.000000	35957.000000	81.600000	19.5
max	90.600000	77.400000	169.000000	122740.000000	93.000000	70.5

Observe that the `count` statistics report the number of non-missing values of each column in the data, as the number of rows in the data (see code below) is more than the number of non-missing values of all the variables in the above table. Similarly, for the rest of the statistics, such as `mean`, `std`, etc., the missing values are ignored.

```
#The dataset gdp_missing_values_data has 155 rows
gdp_missing_values_data.shape[0]
```

```
155
```

13.1.1.2 `info()` Method

Shows the count of non-null entries in each column, helping you quickly identify columns with missing values.

```
gdp_missing_values_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 155 entries, 0 to 154
Data columns (total 12 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   economicActivityFemale  145 non-null    float64
 1   country                 155 non-null    object 
 2   lifeMale                145 non-null    float64
 3   infantMortality         145 non-null    float64
 4   gdpPerCapita            145 non-null    float64
 5   economicActivityMale    145 non-null    float64
 6   illiteracyMale          145 non-null    float64
 7   illiteracyFemale        145 non-null    float64
 8   lifeFemale              145 non-null    float64
 9   geographic_location     155 non-null    object 
 10  contraception           84 non-null     float64
 11  continent               155 non-null    object 
dtypes: float64(9), object(3)
memory usage: 14.7+ KB
```

13.1.1.3 `isnull()` Method

This is one of the most direct methods. Using `df.isnull()` returns a DataFrame of Boolean values where `True` indicates a missing value. To get a summary, you can use `df.isnull().sum()` to see the count of missing values in each column.

For finding the number of missing values in each column of `gdp_missing_values_data`, we will sum up the missing values in each column of the dataset:

```
gdp_missing_values_data.isnull().sum()
```

```
economicActivityFemale    10
country                    0
lifeMale                  10
```

```
infantMortality          10
gdpPerCapita             10
economicActivityMale     10
illiteracyMale           10
illiteracyFemale         10
lifeFemale               10
geographic_location       0
contraception            71
continent                 0
dtype: int64
```

13.1.2 Types of Missing Values

In data science, missing values typically fall into three main types, each requiring different handling strategies:

13.1.2.1 Missing Completely at Random (MCAR)

- **Definition**: Missing values are entirely independent of any variables in the dataset.
- **Example**: A respondent accidentally skips a question on a survey.
- **Impact**: MCAR data can usually be ignored or imputed without biasing the analysis.
- **Handling**: Simple imputation methods, like filling with mean or median values, are often appropriate.

13.1.2.2 Missing at Random (MAR)

- **Definition**: The likelihood of a value being missing is related to other observed variables but not to the missing data itself.
- **Example**: People with higher incomes may be less likely to report their spending, but income data itself is not missing.
- **Impact**: Ignoring MAR values may bias results, so careful imputation based on related variables is recommended.
- **Handling**: More complex imputation methods, like conditional mean imputation or predictive modeling, are suitable.

13.1.2.3 Missing Not at Random (MNAR)

- **Definition**: The probability of missingness is related to the missing data itself, meaning the value is systematically missing.

- **Example**: Patients with severe health conditions might be less likely to report their health status, or students with low scores may be less likely to submit their grades.
- **Impact**: MNAR is the most challenging type, as missing values may introduce significant bias.
- **Handling**: Solutions often include sensitivity analysis, data augmentation, or modeling techniques that account for the missing mechanism, though sometimes domain-specific approaches are necessary.

Understanding the type of missing data helps in selecting the right imputation method and mitigating potential biases in the analysis.

13.1.2.4 Questions

13.1.2.4.1

Why can we ignore observations with missing values without risking skewing the analysis or trends in the case of **Missing Completely at Random (MCAR)**?

13.1.2.4.2

Why could ignoring missing values lead to biased results for **Missing at Random (MAR)** and **Missing Not at Random (MNAR)** data?

13.1.2.4.3

For the datset consisting of GDP per capita, think of hypothetical scenarios in which the missing values of GDP per capita can correspond to **MCAR / MAR / MNAR**.

13.1.3 Methods for Handling missing values

13.1.3.1 Removing Missing values

- **Row/Column Removal**: Use [`df.dropna()`] (https://pandas.pydata.org/docs/reference/api/pandas.Da
    - When to Use: if the missing values are few or the rows/columns are not critical.
    - Risks: Can reduce the dataset's size, potentially losing valuable information.

Let us drop the rows containing even a single value from `gdp_missing_values_data`.

```
gdp_no_missing_data = gdp_missing_values_data.dropna()
```

By default, `df.dropna()` will drop any row that contains at least one missing value, leaving only the rows that are completely free of missing values.

```
#Shape of gdp_no_missing_data
gdp_no_missing_data.shape
```

```
(42, 12)
```

Using `df.dropna()` to remove rows with missing values can sometimes lead to a significant reduction in data, which can be problematic if much of the data is valuable and non-missing. For example:

- **Impact of Default Behavior**: Dropping rows with even a single missing value reduced the number of rows from 155 to 42! This drastic reduction happens because, by default, dropna() removes any row with at least one missing value, keeping only rows that are completely complete.
- **Loss of Non-Missing Data**: Even though some columns may have very few missing values (e.g., at most 10), using dropna() without modification results in losing all non-missing data in affected rows. This is typically a poor choice, especially when valuable, non-missing data is removed unnecessarily.

To avoid losing too much data, you can adjust the behavior of `dropna()` using these parameters:

- `how` Parameter
  - Controls the criteria for dropping rows or columns.
  - `how='any'` (default): Drops rows or columns if any values are missing.
  - ‘how=‘all’: Drops rows or columns only if all values are missing
- `thresh` Parameter
  - Sets a minimum number of non-missing values required to retain the row or column.
  - Useful when you want to keep rows or columns with substantial, but not complete, data.

If a few values of a column are missing, we can possibly estimate them using the rest of the data, so that we can (hopefully) maximize the information that can be extracted from the data. However, if most of the values of a column are missing, it may be harder to estimate its values.

In this dataset, we see that around 50% values of the `contraception` column is missing. Thus, we’ll drop the column as it may be hard to impute its values based on a relatively small number of non-missing values.

```
#Deleting column with missing values in almost half of the observations
gdp_missing_values_data.drop(['contraception'],axis=1,inplace=True)
gdp_missing_values_data.shape
```

```
(155, 11)
```

13.1.3.2 Imputing Missing values

There are an unlimited number of ways to impute missing values. Some imputation methods are provided in the Pandas documentation.

The best way to impute them will depend on the problem:

- **For MCAR**: Simple imputation is generally acceptable.
- **For MAR**: Imputation should consider relationships with other variables, such as using conditional mean imputation.
- **For MNAR**: Imputation requires careful analysis, and domain knowledge is essential to avoid bias.

Below are some of the most common methods. Recall that we randomly introduced missing values in `gdp_missing_values_data`, while the actual values are preserved in `gdp_complete_data`.

We will apply these methods to `gdp_missing_values_data`. To evaluate each method's effectiveness in imputing missing values, we'll compare the imputed gdpPerCapita values with the actual values and calculate the `Root Mean Square Error` (RMSE).

To visualize the imputed vs actual values, let's define a helper function

```
#Index of rows with missing values for GDP per capita
null_ind_gdpPC = gdp_missing_values_data.index[gdp_missing_values_data.gdpPerCapita.isnull()]

#Defining a function to plot the imputed values vs actual values
def plot_actual_vs_predicted(y):
    fig, ax = plt.subplots(figsize=(8, 6))
    plt.rc('xtick', labelsize=15)
    plt.rc('ytick', labelsize=15)
    x = gdp_complete_data.loc[null_ind_gdpPC,'gdpPerCapita']
    # y = y.loc[null_ind_gdpPC,'gdpPerCapita']
    plt.scatter(x,y.loc[null_ind_gdpPC,'gdpPerCapita'])
    z=np.polyfit(x,y.loc[null_ind_gdpPC,'gdpPerCapita'],1)
    p=np.poly1d(z)
    plt.plot(x,x,color='orange')
```

```
    plt.xlabel('Actual GDP per capita',fontsize=15)
    plt.ylabel('Imputed GDP per capita',fontsize=15)
    ax.xaxis.set_major_formatter('${x:,.0f}')
    ax.yaxis.set_major_formatter('${x:,.0f}')
    plt.rc('axes', labelsize=10)  # Set all axis labels to fontsize 10
    # plt.title('Actual vs Imputed values for GDP per capita',fontsize=20)
    rmse = np.sqrt(np.mean((y.loc[null_ind_gdpPC,'gdpPerCapita']-x)**2))
    plt.text(10000, 50000, 'RMSE = %.2f'%rmse, fontsize=15)
```

13.1.3.2.1 Mean/Median/Mode Imputation

This approach replaces missing values with the mean or median of the column.

Let's impute missing values in the column by substituting them with the average of the non-missing values. Imputing with the mean tends to minimize the sum of squared differences between actual values and imputed values, especially in cases where data is Missing Completely at Random (MCAR). However, this might not hold true for other types of missing data, such as Missing at Random (MAR) or Missing Not at Random (MNAR).

Let us impute missing values in the column as the average of the non-missing values of the column. The sum of squared differences between actual values and the imputed values is likely to be smaller if we impute using the mean. However, this may not be true in cases other than MCAR (Missing completely at random).

```
# Extracting the columns with missing values
columns_with_missing = [col for col in gdp_missing_values_data.columns if gdp_missing_values_
columns_with_missing
```

```
['economicActivityFemale',
 'lifeMale',
 'infantMortality',
 'gdpPerCapita',
 'economicActivityMale',
 'illiteracyMale',
 'illiteracyFemale',
 'lifeFemale']
```

```
# Imputing missing values using the mean
gdp_imputed_data_mean = gdp_missing_values_data[columns_with_missing].fillna(gdp_missing_valu
```

```
plot_actual_vs_predicted(gdp_imputed_data_mean)
```

Here, since all columns with missing values are numerical, we could use the mean to impute these values. Using the mean is generally suitable only for numerical data, as it represents a central tendency specific to numbers. For categorical data, however, mean imputation would be inappropriate. Instead, the mode, which identifies the most frequently occurring value, is a more suitable choice for imputing missing values in categorical columns.

13.1.3.2.2 Conditional Imputation: Use other related variables to predict missing values

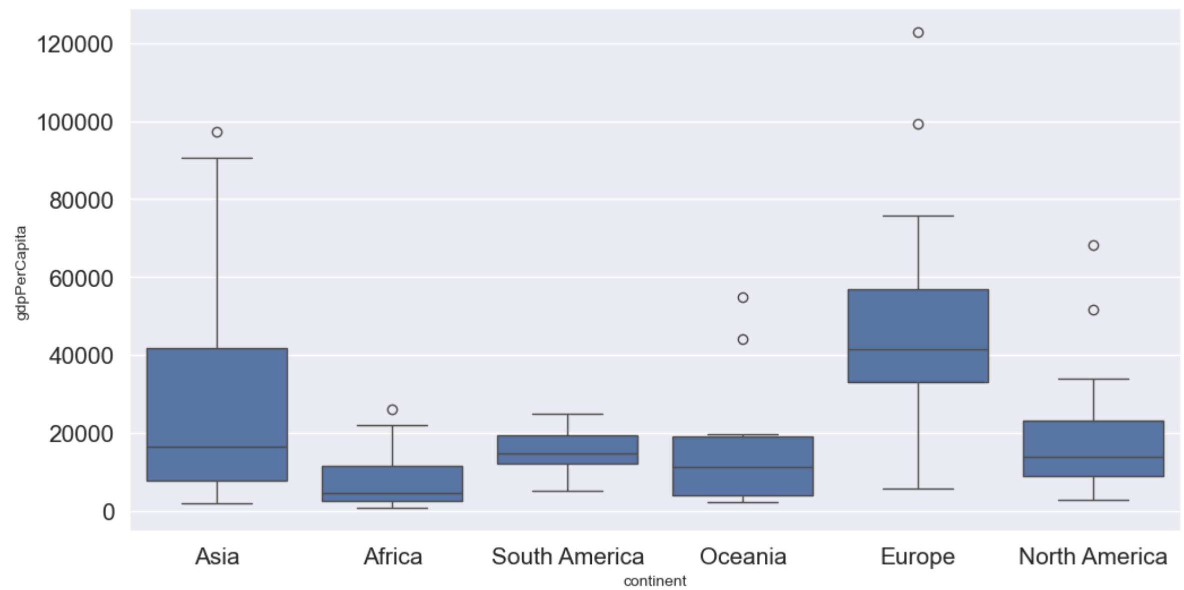
There are three approaches within this category; please see the details below.

- **Imputing missing values based on correlated variables in the data**

If a variable is highly correlated with another variable in the dataset, we can approximate its missing values using the trendline with the highly correlated variable.

Let us visualize the distribution of GDP per capita for different continents.

```
plt.rcParams["figure.figsize"] = (12,6)
sns.boxplot(x = 'continent',y='gdpPerCapita',data = gdp_missing_values_data);
```



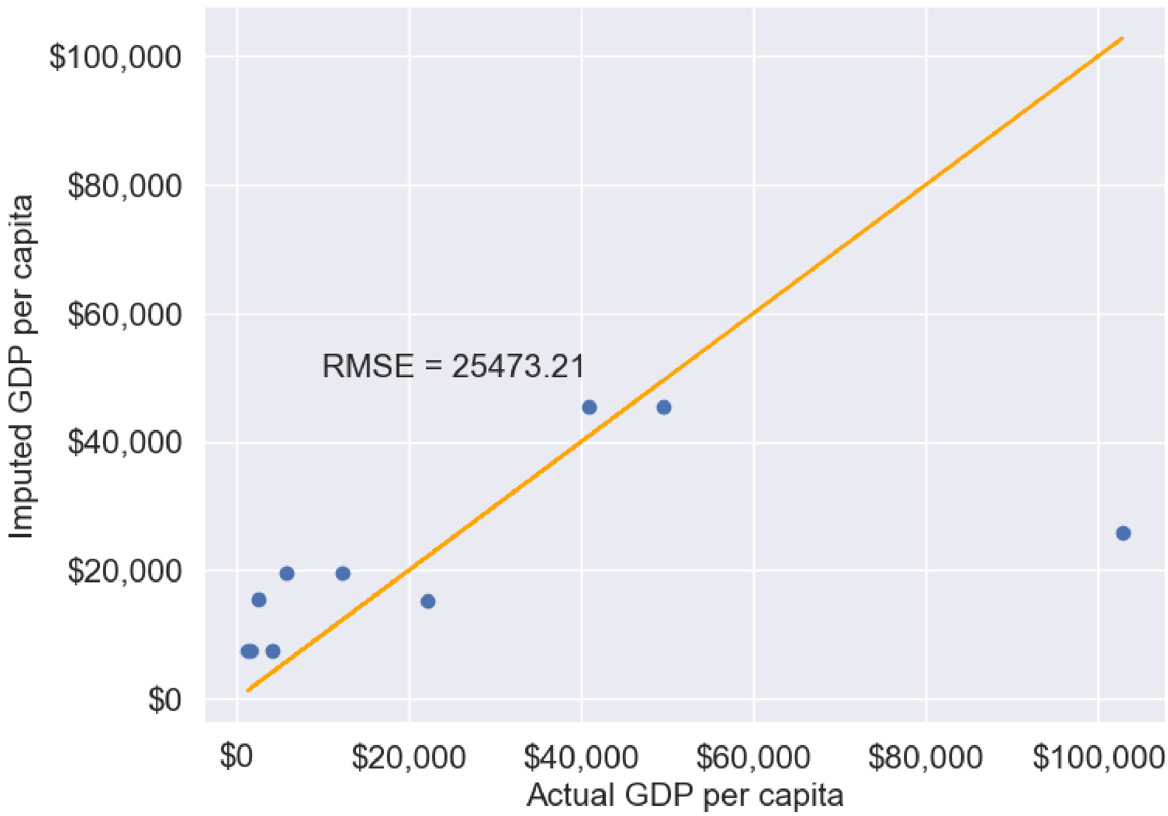
We observe that there is a distinct difference between the GDPs per capita of some of the contents. Let us impute the missing GDP per capita of a country as the mean GDP per capita of the corresponding continent. This imputation should be better than imputing the missing GDP per capita as the mean of all the non-missing values, as the GDP per capita of a country is likely to be closer to the mean GDP per capita of the continent, rather the mean GDP per capita of the whole world.

```
#Finding the mean GDP per capita of the continent - please defer the understanding of this co
avg_gdpPerCapita = gdp_missing_values_data['gdpPerCapita'].groupby(gdp_missing_values_data['c
avg_gdpPerCapita
```

```
continent
Africa            7638.178571
Asia             25922.750000
Europe           45455.303030
North America    19625.210526
Oceania          15385.857143
South America    15360.909091
Name: gdpPerCapita, dtype: float64
```

```
#Creating a copy of missing data to impute missing values
gdp_imputed_data_group_mean = gdp_missing_values_data.copy()
```

```
#Replacing missing GDP per capita with the mean GDP per capita for the corresponding contine

gdp_imputed_data_group_mean.gdpPerCapita = \
gdp_imputed_data_group_mean['gdpPerCapita'].fillna(gdp_imputed_data_group_mean['continent'].
plot_actual_vs_predicted(gdp_imputed_data_group_mean)
```



Note that the imputed values are closer to the actual values, and the RMSE has further reduced as expected.

Suppose we wish to impute the missing values of each numeric column with the average of the non-missing values of the respective column corresponding to the continent of the observation. The above logic can be extended to each column as shown in the code below.

```
all_columns_imputed_data = gdp_missing_values_data.iloc[:,[0,2,3,4,5,6,7,8]].apply(lambda x:
x.fillna(gdp_imputed_data_group_mean['continent'].map(x.groupby(gdp_missing_values_data['con
```

Using Regression

```
#Let us identify the variable highly correlated with GDP per capita.
gdp_missing_values_data.select_dtypes(include='number').corrwith(gdp_missing_values_data.gdpP
```

```
economicActivityFemale    0.078332
lifeMale                  0.579850
infantMortality          -0.572201
gdpPerCapita              1.000000
economicActivityMale     -0.134108
illiteracyMale           -0.479143
illiteracyFemale         -0.448273
lifeFemale                0.615954
dtype: float64
```

```
#The variable *lifeFemale* has the strongest correlation with GDP per capita. Let us use it t

#  Extract the variables lifeFemale and GDP per capita
x = gdp_missing_values_data.lifeFemale
y = gdp_missing_values_data.gdpPerCapita

#  Identify non-missing indices
idx_non_missing = np.isfinite(x) & np.isfinite(y)

# Fit a linear regression model (degree=1) to predict GDP per capita given lifeFemale
slope_intercept_trendline = np.polyfit(x[idx_non_missing],y[idx_non_missing],1)   #Finding th
compute_y_given_x = np.poly1d(slope_intercept_trendline)

#Creating a copy of missing data to impute missing values
gdp_imputed_data_lr = gdp_missing_values_data.copy()

#Imputing missing values of GDP per capita using the linear regression model
gdp_imputed_data_lr.loc[null_ind_gdpPC,'gdpPerCapita']=compute_y_given_x(gdp_missing_values_d
```

```
plot_actual_vs_predicted(gdp_imputed_data_lr)
```

KNN: K-nearest neighbor

In this method, we'll impute the missing value of the variable as the mean value of the K-nearest neighbors having non-missing values for that variable. The neighbors to a data-point are identified based on their Euclidean distance to the point in terms of the standardized values of rest of the variables in the data.

Let's consider a toy example to understand missing value imputation by KNN. Suppose we have to impute missing values in a toy dataset, named as `toy_data` having 4 observations and 3 variables.

```
#Toy example - A 4x3 array with missing values
nan = np.nan
toy_data = np.array([[1, 2, nan], [3, 4, 3], [nan, 6, 5], [8, 8, 7]])
toy_data
```

```
array([[ 1.,  2., nan],
       [ 3.,  4.,  3.],
       [nan,  6.,  5.],
       [ 8.,  8.,  7.]])
```

We'll use some functions from the *sklearn* library to perform the KNN imputation. It is much easier to directly use the algorithm from *sklearn*, instead of coding it from scratch.

```
#Library to compute pair-wise Euclidean distance between all observations in the data
from sklearn import metrics

#Library to impute missing values with the KNN algorithm
from sklearn import impute
```

We'll use the *sklearn* function nan_euclidean_distances() to compute the Euclidean distance between all pairs of observations in the data.

```
#This is the distance matrix containing the distance of the ith observation from the jth obse
metrics.pairwise.nan_euclidean_distances(toy_data,toy_data)
```

```
array([[ 0.        ,  3.46410162,  6.92820323, 11.29158979],
       [ 3.46410162,  0.        ,  3.46410162,  7.54983444],
       [ 6.92820323,  3.46410162,  0.        ,  3.46410162],
       [11.29158979,  7.54983444,  3.46410162,  0.        ]])
```

Note that the size of the above matrix is 4x4. This is because the $(i, j)^{th}$ element of the matrix is the distance of the i^{th} observation from the j^{th} observation. The matrix is symmetric because the distance of i^{th} observation to the j^{th} observation is the same as the distance of the j^{th} observation to the i^{th} observation.

We'll use the *sklearn* function KNNImputer() to impute the missing value of a column in `toy_data` as the mean of the values of the K nearest neighbors to the observation that have non-missing values for that column.

Let us impute the missing values in `toy_data` using the values of $K = 2$ nearest neighbors from the corresponding observation.

```
#imputing missing values with 2 nearest neighbors, where the neighbors have equal weights

#Define an object of type KNNImputer
imputer = impute.KNNImputer(n_neighbors=2)

#Use the object method 'fit_transform' to impute missing values
imputer.fit_transform(toy_data)
```

```
array([[1. , 2. , 4. ],
       [3. , 4. , 3. ],
       [5.5, 6. , 5. ],
       [8. , 8. , 7. ]])
```

The third observation was the closest to the $2nd$ and $4th$ observations based on the Euclidean distance matrix. Thus, the missing value in the $3rd$ row of the `toy_data` has been imputed as the mean of the values in the $2nd$ and $4th$ observations for the corresponding column. Similarly, the $1st$ observation is the closest to the $2nd$ and $3rd$ observations. Thus the missing value in the $1st$ row of `toy_data` has been imputed as the mean of the values in the $1st$ and $2nd$ observations for the corresponding column.

Let us use KNN to impute the missing values of `gdpPerCapita` in `gdp_missing_values_data`. We'll use only the numeric columns of the data in imputing the missing values. Also, we'll ignore `contraception` as it has a lot of missing values, and thus may not be useful.

```
#Considering numeric columns in the data to use KNN
num_cols = list(range(0,1))+list(range(2,9))
num_cols
```

```
[0, 2, 3, 4, 5, 6, 7, 8]
```

Before computing the pair-wise Euclidean distance of observations, we must standardize the data so that all columns are at the same scale. This will avoid columns with a higher magnitude of values having a higher weight in determining the Euclidean distance. Unless there is a reason to give a higher weight to a column, we assume all columns to have the same weight in the Euclidean distance computation.

We can use the code below to scale the data. However, after imputing the missing values, the data is to be scaled back to the original scale, so that each variable is in the same units as in the original dataset. However, if the code below is used, we'll lose the orginal scale of each of the columns.

```
#Scaling data to compute equally weighted distances from the 'k' nearest neighbors
scaled_data = gdp_missing_values_data.iloc[:,num_cols].apply(lambda x:(x-x.min())/(x.max()-x
```

To alleviate the problem of losing the orignial scale of the data, we'll use the MinMaxScaler object of the *sklearn* library. The object will store the original scale of the data, which will help transform the data back to the original scale once the missing values have been imputed in the standardized data.

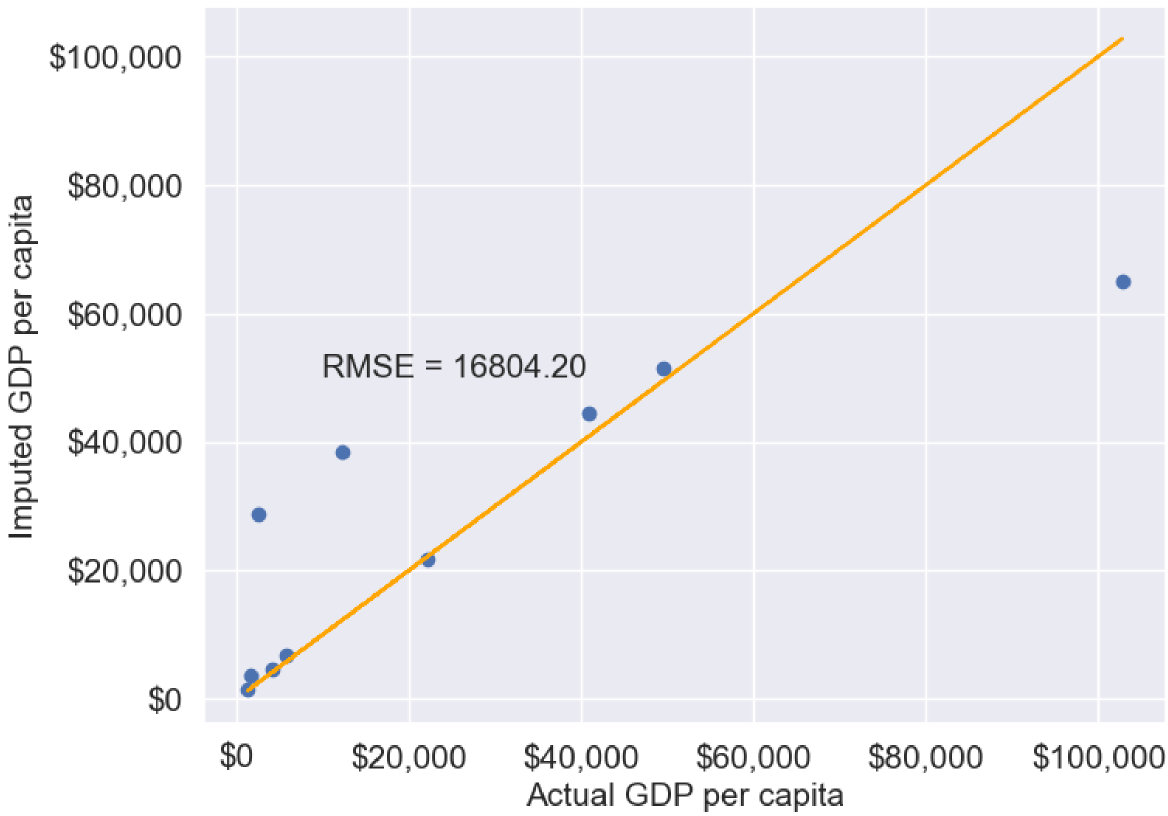
```
# Scaling data - using sklearn

#Create an object of type MinMaxScaler
scaler = sk.preprocessing.MinMaxScaler()

#Use the object method 'fit_transform' to scale the values to a standard uniform distributio
scaled_data = pd.DataFrame(scaler.fit_transform(gdp_missing_values_data.iloc[:,num_cols]))
```

```
#Imputing missing values with KNNImputer

#Define an object of type KNNImputer
imputer = impute.KNNImputer(n_neighbors=3, weights="uniform")

#Use the object method 'fit_transform' to impute missing values
imputed_arr = imputer.fit_transform(scaled_data)
```

```
#Scaling back the scaled array to obtain the data at the original scale

#Use the object method 'inverse_transform' to scale back the values to the original scale of
unscaled_data = scaler.inverse_transform(imputed_arr)
```

```
#Note the method imputes the missing value of all the columns
#However, we are interested in imputing the missing values of only the 'gdpPerCapita' column
gdp_imputed_data_knn = gdp_missing_values_data.copy()
gdp_imputed_data_knn.loc[:,'gdpPerCapita'] = unscaled_data[:,3]
```

```
#Visualizing the accuracy of missing value imputation with KNN
plot_actual_vs_predicted(gdp_imputed_data_knn)
```



Note that the RMSE is the lowest in this method. It is because this method imputes missing values as the average of the values of "similar" observations, which is smarter and more robust than the previous methods.

We chose $K = 3$ in the missing value imputation for GDP per capita. However, the value of K is typically chosen using a method known as cross validation. We'll learn about cross-validation in the next course of the sequence.

13.1.3.2.3 Forward Fill/Backward Fill

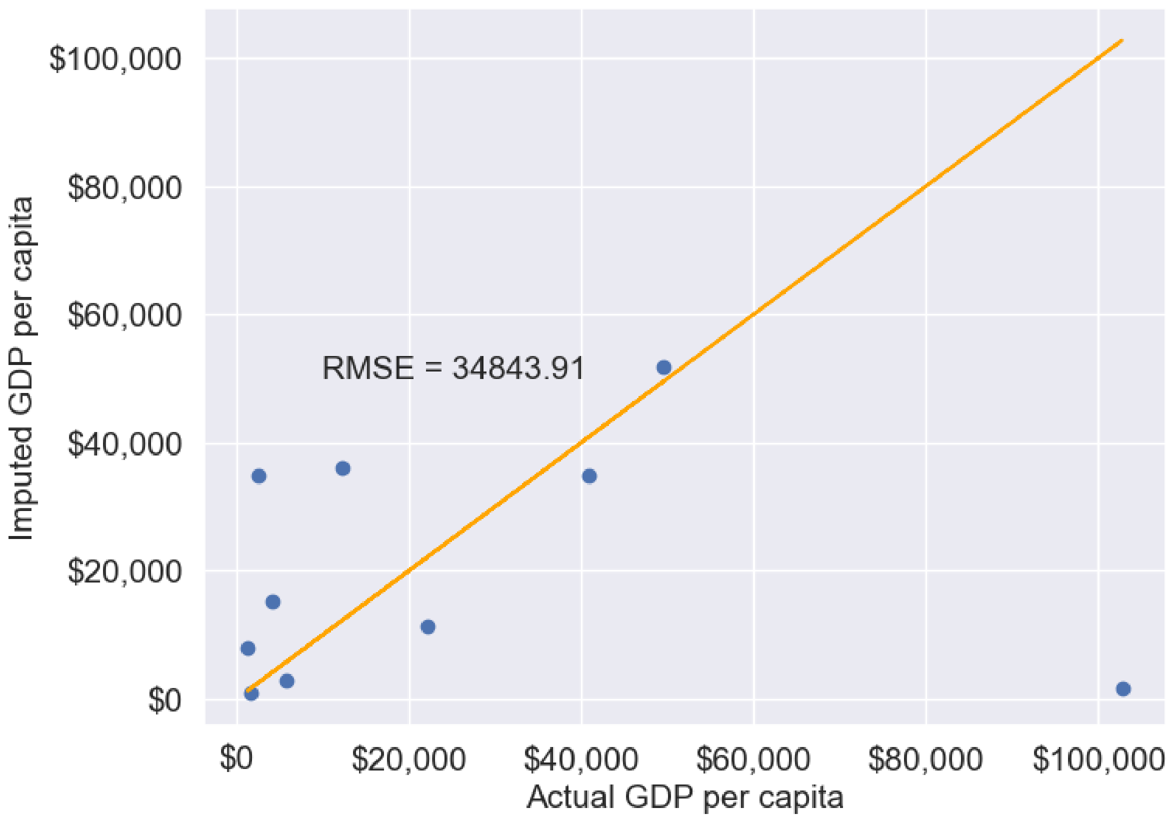
To fill missing values in a column by copying the value of the previous or next non-missing observation, we can use forward fill (propagating the last valid observation forward) or backward fill (propagating the next valid observation backward).

Below, we demonstrate forward fill. To perform backward fill instead, simply replace `ffill` with `bfill`

```
#Filling missing values: Method 1- Naive way
gdp_imputed_data_ffill = gdp_missing_values_data.ffill()
```

Let us next check how good is this method in imputing missing values.

```
plot_actual_vs_predicted(gdp_imputed_data_ffill)
```



We observe that the accuracy of imputation is poor as GDP per capita can vary a lot across countries, and the data is not sorted by GDP per capita. There is no reason why the GDP per capita of a country should be close to the GDP per capita of the country in the observation above it.

```
#Checking if any missing values are remaining
gdp_imputed_data_ffill.isnull().sum()
```

```
economicActivityFemale    0
country                   0
lifeMale                  0
infantMortality           0
gdpPerCapita              0
economicActivityMale      0
```

```
illiteracyMale          1
illiteracyFemale        0
lifeFemale              0
geographic_location     0
continent               0
dtype: int64
```

After imputing missing values, note there is still one missing value for *illiteracyMale*. Can you guess why one missing value remained?

13.2 Outlier detection and handling

An outlier is an observation that is significantly different from the rest of the data. Outlier detection and handling are important in data science because outliers can significantly impact the quality, accuracy, and interpretability of data analysis and model performance. Here are the main reasons why it's crucial to address outliers:

- **Preventing Skewed Results:** Outliers can distort statistical measures, such as the mean and standard deviation, leading to misleading interpretations of the data. For example, a few extreme values can inflate the mean, making the data seem larger than it is in reality.
- **Improving Model Accuracy:** Many machine learning algorithms are sensitive to outliers and can perform poorly if outliers are present. For instance, in linear regression, outliers can disproportionately affect the model by pulling the line of best fit toward them, reducing predictive accuracy.
- **Enhancing Robustness of Models:** Identifying and handling outliers can make models more robust and stable. By minimizing the influence of extreme values, models become less sensitive to noise, which improves generalization and reduces overfitting.

13.2.1 Outlier detection

Outlier detection is crucial for identifying unusual values in your data that may need to be investigated, removed, or transformed. Here are several popular methods for detecting outliers:

We will use *College.csv* that contains information about US universities as our dataset in this section. The description of variables of the dataset can be found on page 65 of this book.

```
# Load the College dataset
college = pd.read_csv('./Datasets/College.csv')
college.head()
```

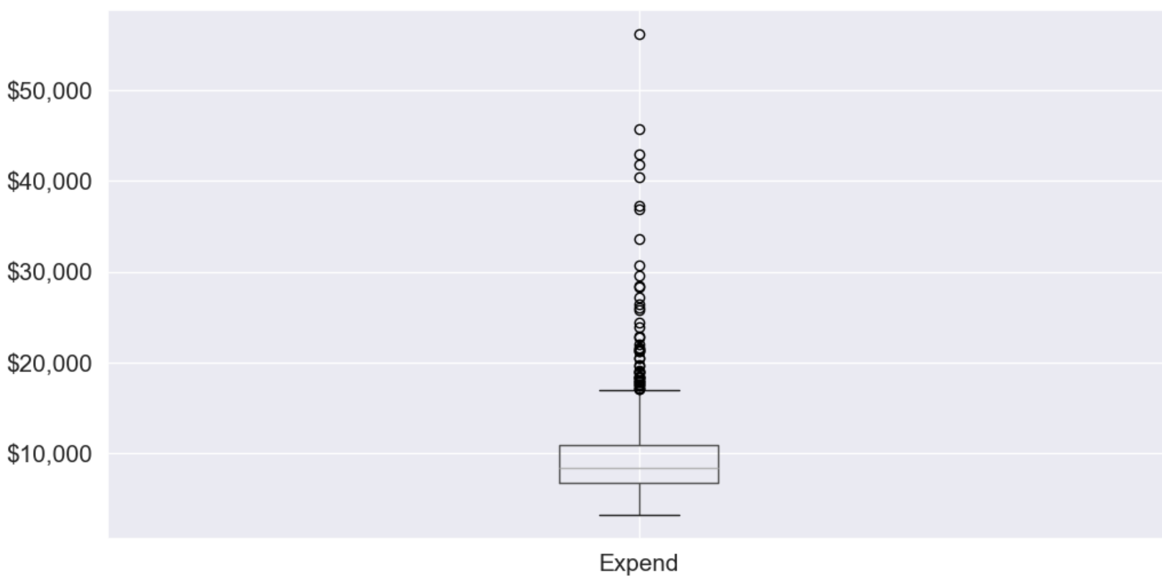
	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885
1	Adelphi University	Yes	2186	1924	512	16	29	2683
2	Adrian College	Yes	1428	1097	336	22	50	1036
3	Agnes Scott College	Yes	417	349	137	60	89	510
4	Alaska Pacific University	Yes	193	146	55	16	44	249

13.2.1.1 Visualization Methods

- **Box Plot**: A box plot is a quick and visual way to detect outliers, where points outside the "whiskers" are considered outliers.

Let us visualize outliers in average instructional expenditure per student given by the variable `Expend`.

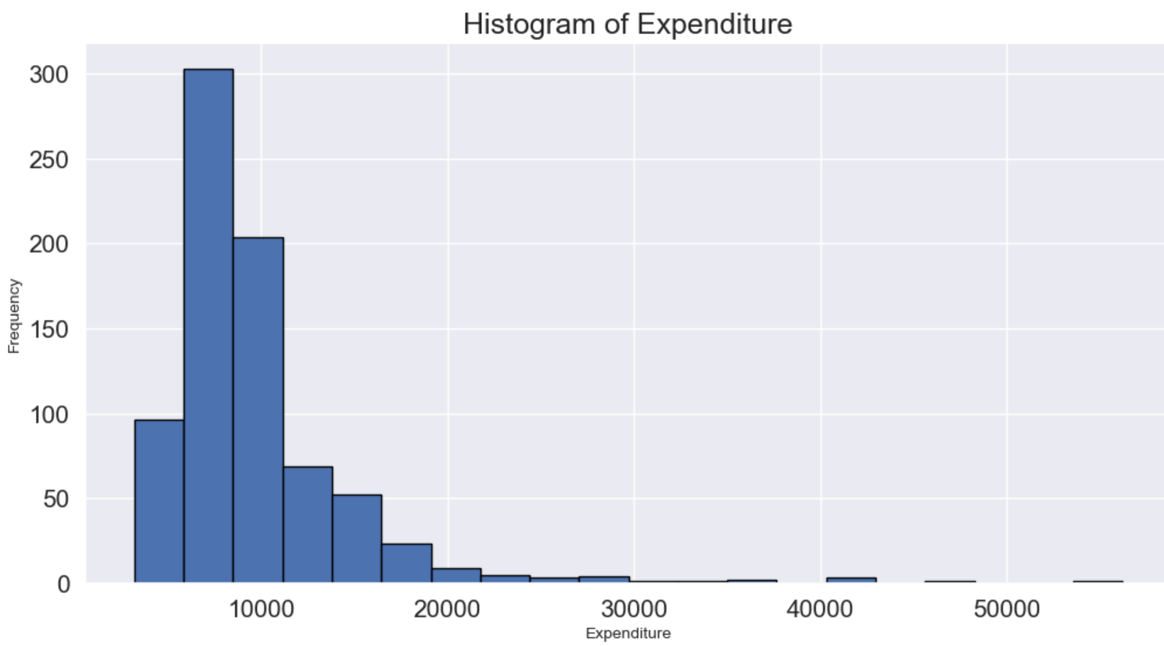
```
ax=college.boxplot(column = 'Expend')
ax.yaxis.set_major_formatter('${x:,.0f}')
```



There are several outliers (shown as circles in the above boxplot), which correspond to high values of average instructional expenditure per student.

- **Histogram**: Histograms can reveal unusual values as isolated bars, helping to identify outliers in a single feature.

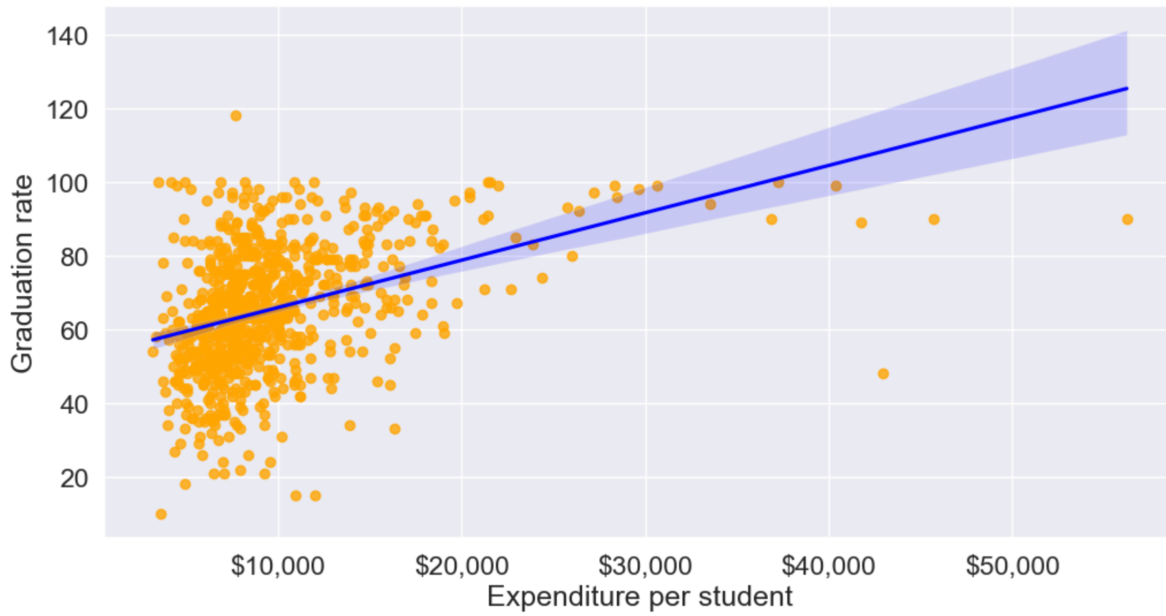
```
# create a histogram of expend
college['Expend'].plot(kind='hist', edgecolor='black', bins=20)
plt.xlabel('Expenditure')
plt.ylabel('Frequency')
plt.title('Histogram of Expenditure')
plt.show()
```



- **Scatter Plot**: A scatter plot can reveal outliers, especially in two-dimensional data. Outliers will often appear as points separated from the main cluster.

Let's make a scatterplot of 'Grad.Rate' vs 'Expend' with a trendline, to visualize potential outliers

```
sns.set(font_scale=1.5)
ax=sns.regplot(data = college, x = "Expend", y = "Grad.Rate",scatter_kws={"color": "orange"}
ax.xaxis.set_major_formatter('${x:,.0f}')
ax.set_xlabel('Expenditure per student')
ax.set_ylabel('Graduation rate')
plt.show()
```



The trendline indicates a positive correlation between `Expend` and `Grad.Rate`. However, there seems to be a lot of noise and presence of outliers in the data.

13.2.1.2 Statistical Methods

Visualization helps us identify potential outliers. To address them effectively, we need to extract these outlier instances using a defined threshold. Two commonly used statistical methods for this purpose are the Z-Score method and the Interquartile Range (IQR) (Tukey's fences).

13.2.1.2.1 Z-Score (Standard Score)

Calculates how many standard deviations a data point is from the mean. Commonly, data points with a Z-score greater than 3 (or less than -3) are considered outliers.

```
from scipy import stats

# Calculate Z-scores and identify outliers
college['z_score'] = stats.zscore(college['Expend'])
college['is_z_score_outlier'] = college['z_score'].abs() > 3

# Filter to show only the outliers
z_score_outliers = college[college['is_z_score_outlier']]
print(z_score_outliers.shape)
z_score_outliers.head()
```

(16, 21)

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Un
20	Antioch University	Yes	713	661	252	25	44	712	23
144	Columbia University	Yes	6756	1930	871	78	96	3376	55
158	Dartmouth College	Yes	8587	2273	1087	87	99	3918	32
174	Duke University	Yes	13789	3893	1583	90	98	6188	53
191	Emory University	Yes	8506	4168	1236	76	97	5544	192

Boxplot identifies outliers based on the Tukey's fences criterion:

13.2.1.2.2 IQR method (Tukey's fences)

John Tukey proposed that observations outside the range $[Q1-1.5(Q3-Q1), Q3+1.5(Q3-Q1)]$ are outliers, where $Q1$ and $Q3$ are the lower (25%) and upper (75%) quartiles respectively. Let us detect outliers based on this threshold.

```
# Calculate the first and third quartiles, and the IQR
Q1 = np.percentile(college['Expend'], 25)
Q3 = np.percentile(college['Expend'], 75)
IQR = Q3 - Q1

# Define the lower and upper bounds for outliers
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Add a new column to indicate IQR outliers
college['is_IQR_outlier'] = (college['Expend'] < lower_bound) | (college['Expend'] > upper_bo

# Filter to show only the IQR outliers
iqr_outliers = college[college['is_IQR_outlier']]
print(iqr_outliers.shape)
iqr_outliers.head()
```

(48, 22)

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Unde
3	Agnes Scott College	Yes	417	349	137	60	89	510	63

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Unde
16	Amherst College	Yes	4302	992	418	83	96	1593	5
20	Antioch University	Yes	713	661	252	25	44	712	23
60	Bowdoin College	Yes	3356	1019	418	76	100	1490	8
64	Brandeis University	Yes	4186	2743	740	48	77	2819	62

Using the IQR method, more instances in the college dataset are marked as outliers.

13.2.1.2.3 When to use each:

- Use IQR when:
  - Data is skewed
  - You don't want to assume any distribution
  - You want a more sensitive detector
- Use Z-score when:
  - Data is approximately normal
  - You want a more conservative approach
  - You need a method that's easily interpretable

The actual number of outliers marked by each method will depend on your specific dataset's distribution and characteristics.

13.2.2 Common Methods for Handling outliers

Once we identify outlier instances using statistical methods, the next step is to handle them. Below are some commonly used approaches:

- **Removing Outliers**: Discarding extreme values if they are likely due to errors or are irrelevant for the analysis.
- **Winsorizing**: Capping/flooring outliers to a specific percentile to reduce their influence without removing them completely.
- **Replacing with the median**: Replacing outlier values with the median of the remaining data. The median is less affected by extreme values than the mean, making it an ideal choice for imputation when dealing with outliers
- **Transforming Data**: Applying transformations (e.g., log, square root) to reduce the impact of outliers.

```python
def method_removal():
    """Method 1: Remove outliers"""
    data_cleaned = college[~college['is_outlier']]['Expend']
    return data_cleaned
```

```python
def method_capping():
    """Method 2: Capping (Winsorization)"""
    data_capped = college['Expend'].copy()
    data_capped[data_capped < lower_bound] = lower_bound
    data_capped[data_capped > upper_bound] = upper_bound
    return data_capped
```

```python
def method_mean_replacement():
    """Method 3: Replace with median"""
    data_median = college['Expend'].copy()
    median_value = college[~college['is_outlier']]['Expend'].median()
    data_median[college['is_outlier']] = median_value
    return data_median
```

```python
def method_log_transformation():
    """Method 4: Log transformation"""
    data_log = np.log1p(college['Expend'])
    return data_log
```

Your Practice: Try each method and compare their results to see how they differ in handling outliers.

In real-world applications, handling outliers should be approached on a case-by-case basis. However, here are some general recommendations:

Recommendations for Different Scenarios:

- Data Removal:
  - Use when: You have a large dataset and can afford to lose data
  - Pros: Simple, removes all outlier influence
  - Cons: Loss of data, potential loss of important information
- Capping (Winsorization):
  - Use when: You want to preserve data count while limiting extreme values
  - Pros: Preserves data size, reduces impact of outliers
  - Cons: Artificial boundary creation, potential loss of genuine extreme events

- Median Replacement:
  - Use when: The data is normally distributed, and outliers are verified as errors or anomalies not representing true values
  - Pros: Maintains data size, simple to implement
  - Cons: Reduces variance, may not represent the true distribution
- Log Transformation:
  - Use when: Data has a right-skewed distribution, and you want to reduce the impact of large outliers.
  - Pros: Reduces skewness, minimizes the effect of extreme values, can make data more normal-like.
  - Cons: Only applicable to positive values, may not be effective for extremely high outliers.

13.3 Data binning

Data binning is a powerful technique in data analysis where continuous numerical data is grouped into discrete intervals or "bins." Imagine it as sorting items into distinct categories or "buckets" based on their values, making it easier to observe trends, patterns, and groupings within large datasets. We encounter binning in everyday life without even realizing it.

For example, consider age groups: instead of listing every individual age, we often refer to broader categories like "under 20," "20-30," "30-40," or "40 and over." This binning approach helps simplify data interpretation and comparison across different age groups. Another common example is tax brackets. Tax rates are often applied to ranges of income, like "up to $11,000," "$11,001 to $44,725," and so forth. Here, income values are grouped into bins to determine applicable tax rates, making complex tax data easier to manage and understand.

```
<IPython.core.display.HTML object>
```

13.3.1 Key reasons for binning:

(i) Better intepretation of data

(ii) Making better recommendations

(iii) Smooth data, reduce noise

Examples:

Binning to better interpret data

1. The number of flu cases everyday may be binned to seasons such as fall, spring, winter and summer, to understand the effect of season on flu.

Binning to make recommendations:

2. A doctor may like to group patient age into bins. Grouping patient ages into categories such as Age <=12, 12<Age<=18, 18<Age<=65, Age>65 may help recommend the kind/doses of covid vaccine a patient needs.
3. A credit card company may want to bin customers based on their spend, as "High spenders", "Medium spenders" and "Low spenders". Binning will help them design customized marketing campaigns for each bin, thereby increasing customer response (or revenue). On the other hand, they use the same campaign for customers withing the same bin, thus minimizng marketing costs.

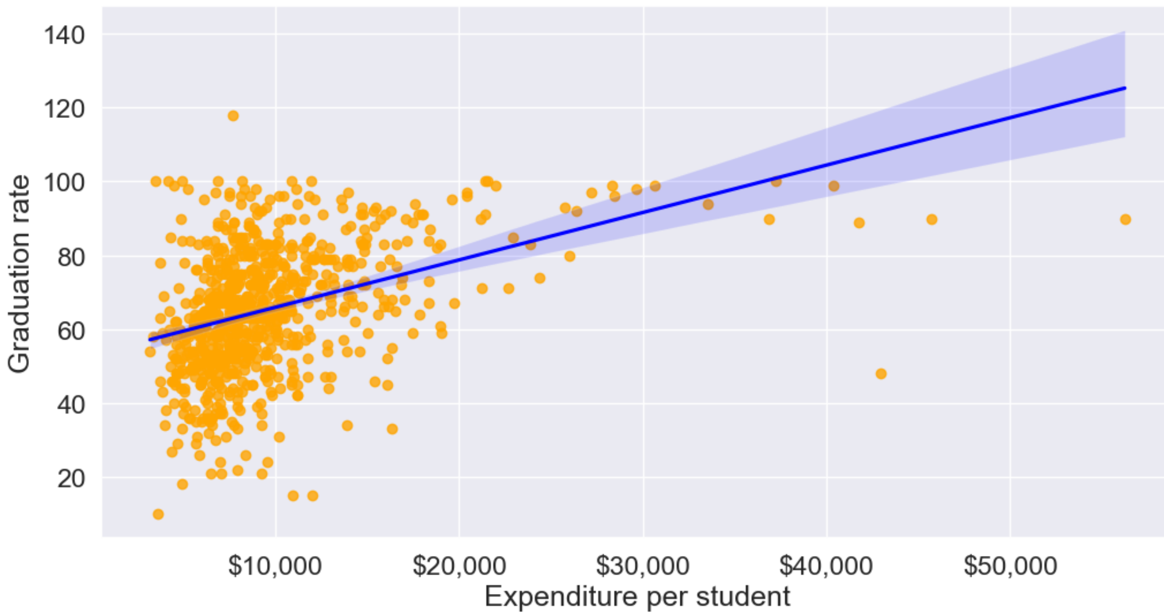
Binning to smooth data, and reduce noise

4. A sales company may want to bin their total sales to a weekly / monthly / yearly level to reduce the noise in day-to-day sales.

Using our college dataset, let's apply binning to better interpret the relationship between instructional expenditure per student (`Expend`) and graduation rate (`Grad.Rate`) for U.S. universities, and make relevant recommendations

Here is the previous scatterplot of 'Grad.Rate' vs 'Expend' with a trendline.

```
# scatter plot of Expend vs Grad.Rate
ax=sns.regplot(data = college, x = "Expend", y = "Grad.Rate",scatter_kws={"color": "orange"}
ax.xaxis.set_major_formatter('${x:,.0f}')
ax.set_xlabel('Expenditure per student')
ax.set_ylabel('Graduation rate')
plt.show()
```



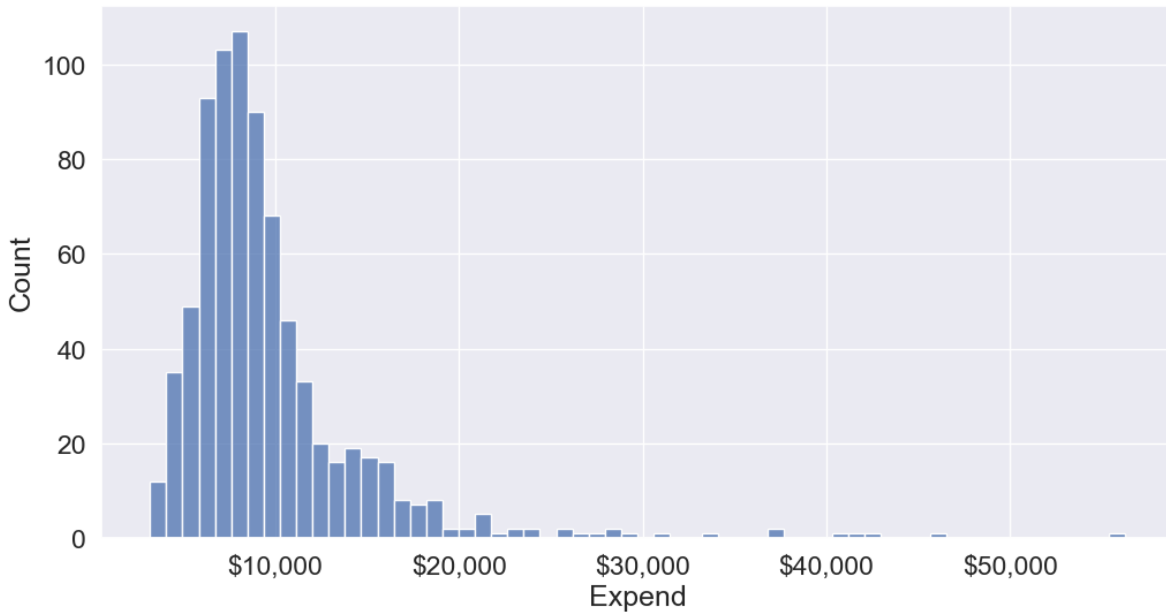
The trendline indicates a positive correlation between `Expend` and `Grad.Rate`. However, there seems to be a lot of noise and presence of outliers in the data, which makes it hard to interpret the overall trend.

13.3.2 Common binning methods:

- **Equal-width:** Divides range into equal-sized intervals
- **Equal-size:** Places equal number of observations in each bin
- **Custom:** Based on domain knowledge (like age groups above)

We'll bin `Expend` to see if we can better analyze its association with `Grad.Rate`. However, let us first visualize the distribution of `Expend`.

```
#Visualizing the distribution of expend
ax=sns.histplot(data = college, x= 'Expend')
ax.xaxis.set_major_formatter('${x:,.0f}')
```



The distribution of `Extend` is right skewed with potentially some extremely high outlying values.

13.3.2.1 Binning with equal width bins

We'll use the Pandas function cut() to bin `Expend`. This function creates bins such that all bins have the same width.

```
#Using the cut() function in Pandas to bin "Expend"
Binned_expend = pd.cut(college['Expend'],3,retbins = True)
Binned_expend
```

```
(0      (3132.953, 20868.333]
 1      (3132.953, 20868.333]
 2      (3132.953, 20868.333]
 3      (3132.953, 20868.333]
 4      (3132.953, 20868.333]
                ...
 772    (3132.953, 20868.333]
 773    (3132.953, 20868.333]
 774    (3132.953, 20868.333]
 775     (38550.667, 56233.0]
 776    (3132.953, 20868.333]
```

```
Name: Expend, Length: 777, dtype: category
Categories (3, interval[float64, right]): [(3132.953, 20868.333] < (20868.333, 38550.667] <
array([ 3132.953     , 20868.33333333, 38550.66666667, 56233.        ]))
```

The cut() function returns a tuple of length 2. The first element of the tuple are the bins, while the second element is an array containing the cut-off values for the bins.

```
type(Binned_expend)
```

```
tuple
```

```
len(Binned_expend)
```

```
2
```

Once the bins are obtained, we'll add a column in the dataset that indicates the bin for `Expend`.

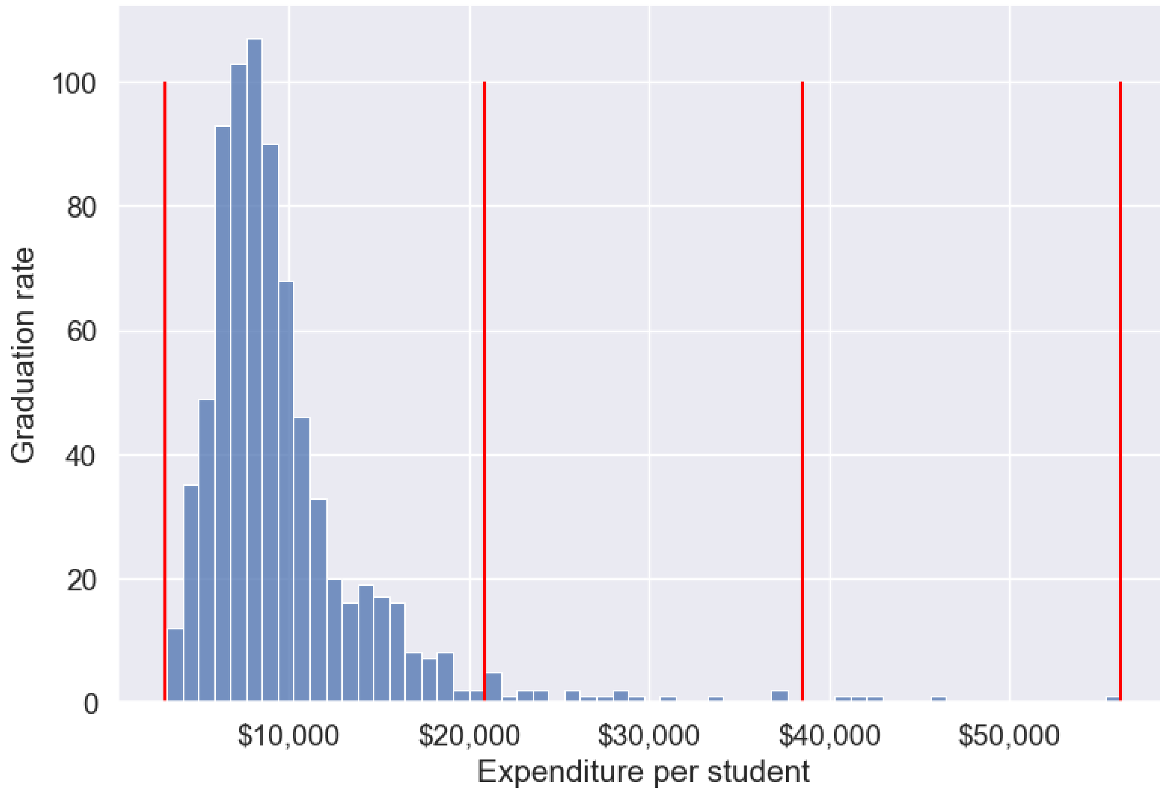
```
#Creating a categorical variable to store the level of expenditure on a student
college['Expend_bin'] = Binned_expend[0]
college.head()
```

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885
1	Adelphi University	Yes	2186	1924	512	16	29	2683
2	Adrian College	Yes	1428	1097	336	22	50	1036
3	Agnes Scott College	Yes	417	349	137	60	89	510
4	Alaska Pacific University	Yes	193	146	55	16	44	249

See the variable `Expend_bin` in the above dataset.

Let us visualize the `Expend` bins over the distribution of the `Expend` variable.

```
#Visualizing the bins for instructional expediture on a student
sns.set(font_scale=1.25)
plt.rcParams["figure.figsize"] = (9,6)
ax=sns.histplot(data = college, x= 'Expend')
plt.vlines(Binned_expend[1], 0,100,color='red')
plt.xlabel('Expenditure per student');
plt.ylabel('Graduation rate');
ax.xaxis.set_major_formatter('${x:,.0f}')
```



By default, the bins created have equal width. They are created by dividing the range between the maximum and minimum value of `Expend` into the desired number of equal-width intervals. We can label the bins as well as follows.

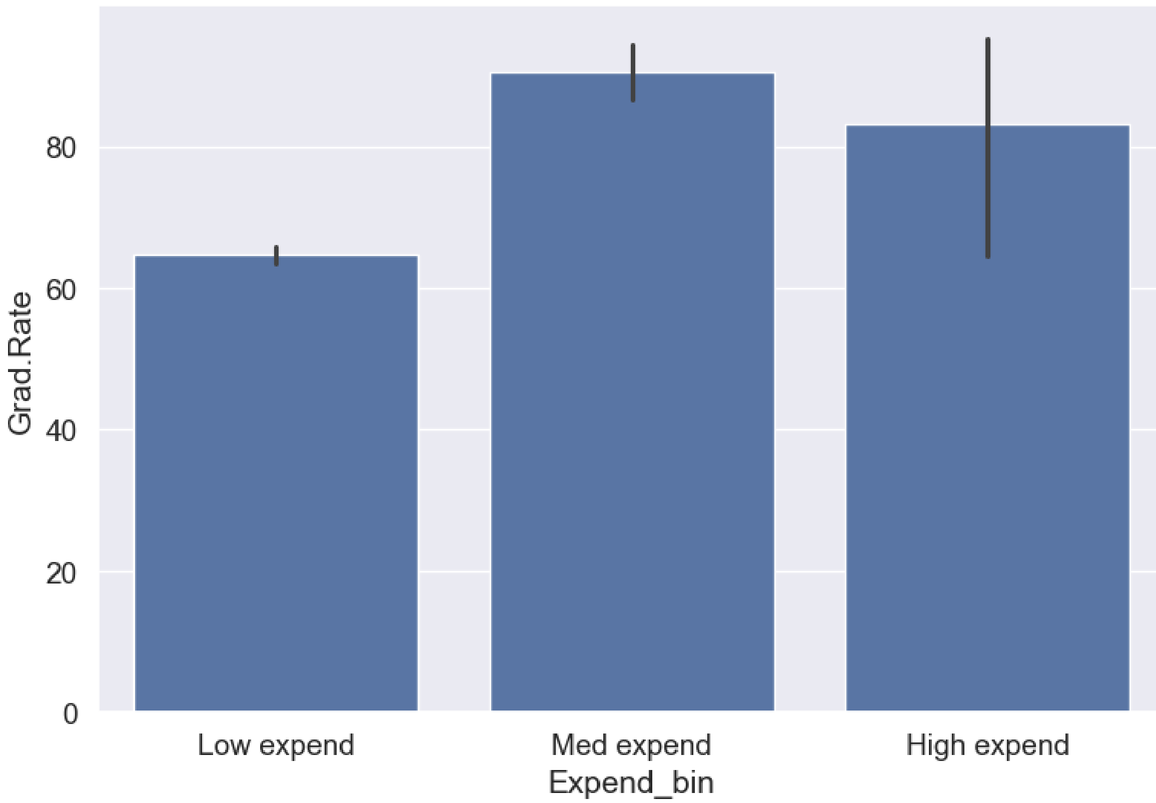
```
college['Expend_bin'] = pd.cut(college['Expend'],3,labels = ['Low expend','Med expend','High
college['Expend_bin']
```

```
0        Low expend
1        Low expend
2        Low expend
3        Low expend
4        Low expend
            ...
772      Low expend
773      Low expend
774      Low expend
775     High expend
776      Low expend
Name: Expend_bin, Length: 777, dtype: category
```

```
Categories (3, object): ['Low expend' < 'Med expend' < 'High expend']
```

Now that we have binned the variable `Expend`, let us see if we can better visualize the association of graduation rate with expenditure per student using `Expened_bin`.

```
#Visualizing average graduation rate vs categories of instructional expenditure per student
sns.barplot(x = 'Expend_bin', y = 'Grad.Rate', data = college)
```



It seems that the graduation rate is the highest for universities with medium level of expenditure per student. This is different from the trend we saw earlier in the scatter plot. Let us investigate.

Let us find the number of universities in each bin.

```
college['Expend_bin'].value_counts()
```

```
Expend_bin
Low expend     751
```

```
Med expend      21
High expend      5
Name: count, dtype: int64
```

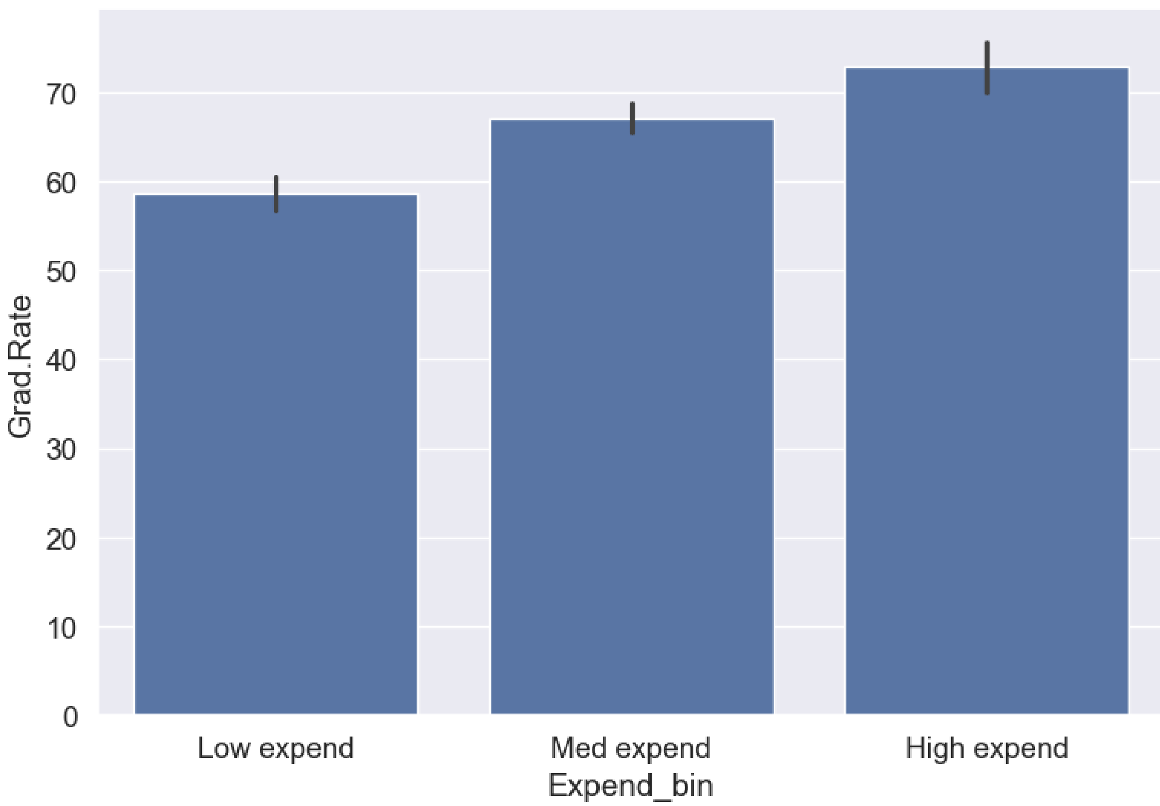
The bin `High expend` consists of only 5 universities, or 0.6% of all the universities in the dataset. These universities may be outliers that are skewing the trend (as also evident in the histogram above).

Let us see if we get the correct trend with the outliers removed from the data.

```
#Data without outliers
college_data_without_outliers = college[((college.Expend>=lower_bound) & (college.Expend<=upp
```

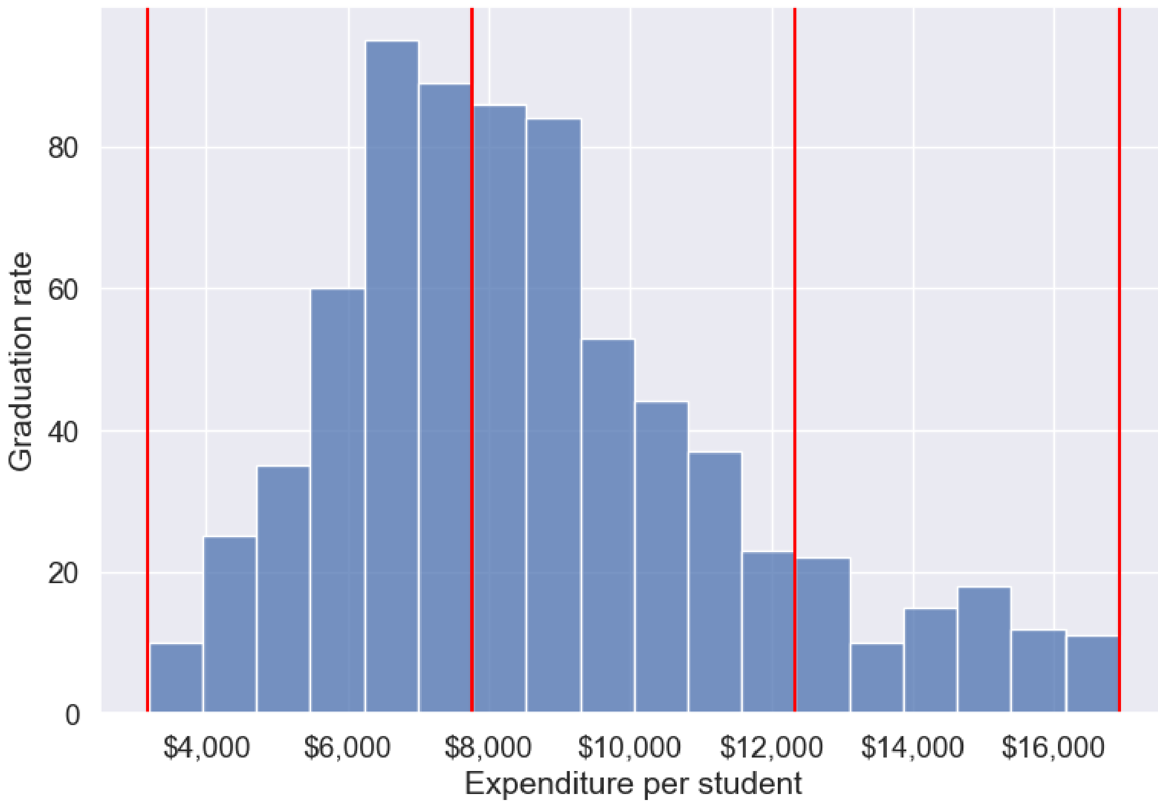
```
Binned_data = pd.cut(college_data_without_outliers['Expend'],3,labels = ['Low expend','Med ex
college_data_without_outliers.loc[:,'Expend_bin'] = Binned_data[0]
```

```
sns.barplot(x = 'Expend_bin', y = 'Grad.Rate', data = college_data_without_outliers)
```



With the outliers removed, we obtain the correct overall trend, even in the case of equal-width bins. Note that these bins have unequal number of observations as shown below.

```
ax=sns.histplot(data = college_data_without_outliers, x= 'Expend')
for i in range(4):
    plt.axvline(Binned_data[1][i], 0,100,color='red')
plt.xlabel('Expenditure per student');
plt.ylabel('Graduation rate');
ax.xaxis.set_major_formatter('${x:,.0f}')
```



Note that the right tail of the histogram has disappered since we removed outliers.

```
college_data_without_outliers['Expend_bin'].value_counts()
```

```
Expend_bin
Med expend     327
Low expend     314
High expend     88
Name: count, dtype: int64
```

Instead of removing outliers, we can address the issue by binning observations into equal-sized bins, ensuring each bin has the same number of observations. Let's explore this approach next.

13.3.2.2 Binning with equal sized bins

Let us bin the variable `Expend` such that each bin consists of the same number of observations.

We'll use the Pandas function qcut() to make equal-sized bins (in contrast to equal-width bins in the previous section).

```
#Using the Pandas function qcut() to create bins with the same number of observations
Binned_expend = pd.qcut(college['Expend'],3,retbins = True)
college['Expend_bin'] = Binned_expend[0]
```

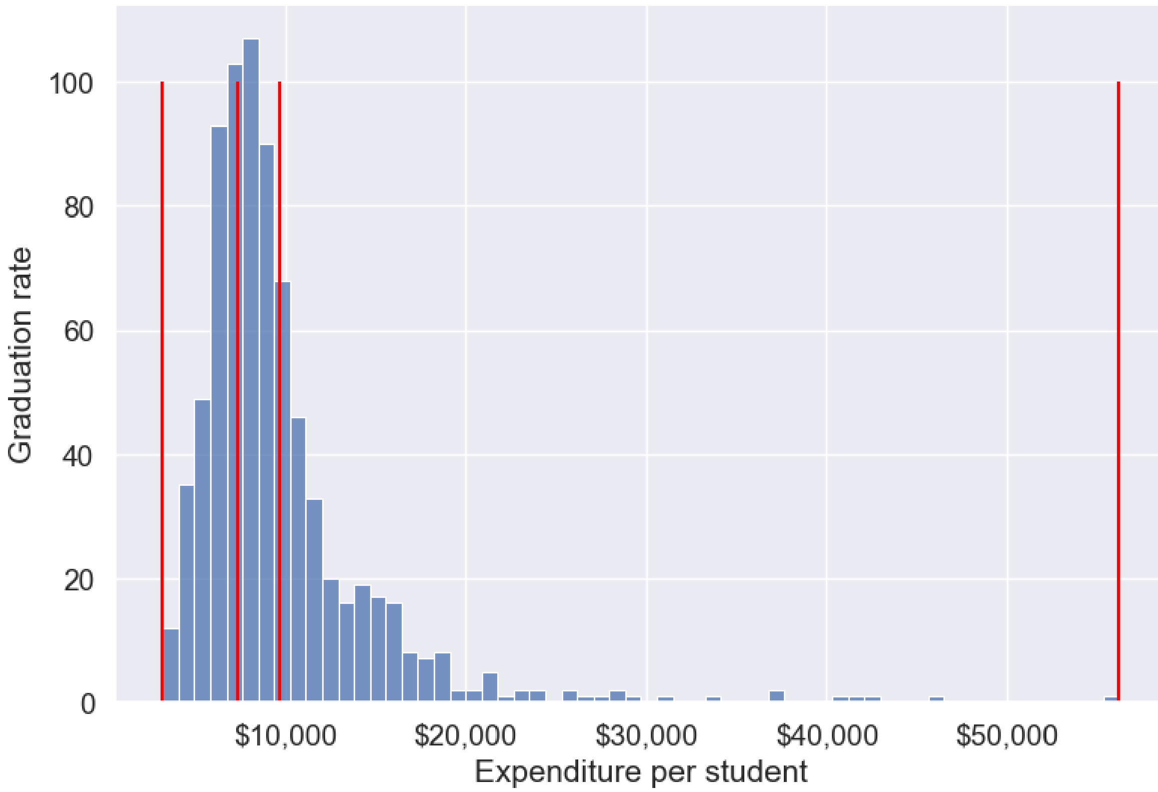
Each bin has the same number of observations with `qcut()`:

```
college['Expend_bin'].value_counts()
```

```
Expend_bin
(3185.999, 7334.333]    259
(7334.333, 9682.333]    259
(9682.333, 56233.0]     259
Name: count, dtype: int64
```

Let us visualize the `Expend` bins over the distribution of the `Expend` variable.

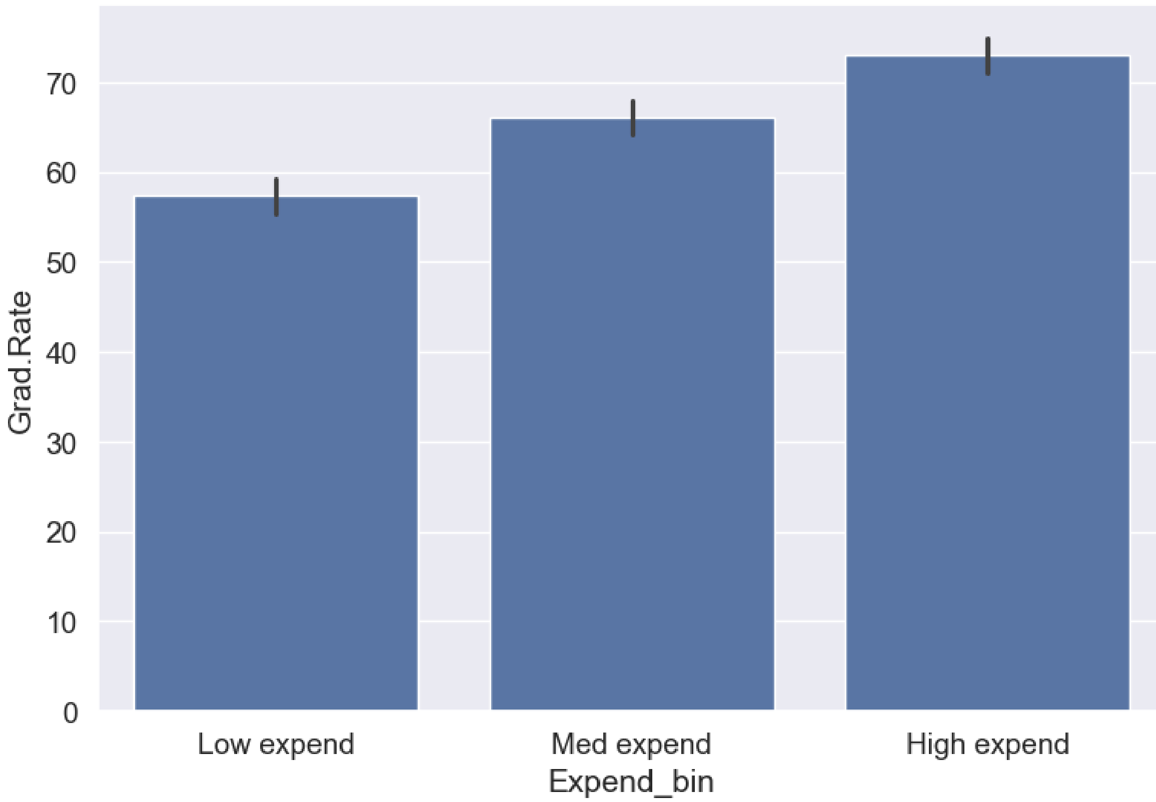
```
#Visualizing the bins for instructional expediture on a student
sns.set(font_scale=1.25)
plt.rcParams["figure.figsize"] = (9,6)
ax=sns.histplot(data = college, x= 'Expend')
plt.vlines(Binned_expend[1], 0,100,color='red')
plt.xlabel('Expenditure per student');
plt.ylabel('Graduation rate');
ax.xaxis.set_major_formatter('${x:,.0f}')
```



Note that the bin-widths have been adjusted to have the same number of observations in each bin. The bins are narrower in domains of high density, and wider in domains of sparse density.

Let us again make the barplot visualizing the average graduate rate with level of instructional expenditure per student.

```
college['Expend_bin'] = pd.qcut(college['Expend'],3,labels = ['Low expend','Med expend','High
a=sns.barplot(x = 'Expend_bin', y = 'Grad.Rate', data = college)
```



Now we see the same trend that we saw in the scatterplot, but without the noise. We have smoothed the data. Note that making equal-sized bins helps reduce the effect of outliers in the overall trend.

Suppose this analysis was done to provide recommendations to universities for increasing their graduation rate. With binning, we can can provide one recommendation to *'Low expend'* universities, and another one to *'Med expend'* universities. For example, the recommendations can be:

1. *'Low expend'* universities can expect an increase of 9 percentage points in `Grad.Rate`, if they migrate to the *'Med expend'* category.
2. *'Med expend'* universities can expect an increase of 7 percentage points in `Grad.Rate`, if they migrate to the *'High expend'* category.

The numbers in the above recommendations are based on the table below.

```
college.groupby(college.Expend_bin, observed=False)['Grad.Rate'].mean()
```

```
Expend_bin
Low expend     57.343629
```

```
Med expend     66.057915
High expend    72.988417
Name: Grad.Rate, dtype: float64
```

We can also provide recommendations based on the confidence intervals of the mean graduation rate (`Grad.Rate`). The confidence intervals, calculated below, are derived using a method known as bootstrapping. For a detailed explanation of bootstrapping, please refer to this article on Wikipedia.

```
# Bootstrapping to find 95% confidence intervals of Graduation Rate of US universities based
confidence_intervals = {}

# Loop through each expenditure bin
for expend_bin, data_sub in college.groupby('Expend_bin', observed=False):
    # Generate bootstrap samples for the graduation rate
    samples = np.random.choice(data_sub['Grad.Rate'], size=(10000, len(data_sub)), replace=T

    # Calculate the mean of each sample
    sample_means = samples.mean(axis=1)

    # Calculate the 95% confidence interval
    lower_bound = np.percentile(sample_means, 2.5)
    upper_bound = np.percentile(sample_means, 97.5)

    # Store the result in the dictionary
    confidence_intervals[expend_bin] = (round(lower_bound, 2), round(upper_bound, 2))

# Print the results
for expend_bin, ci in confidence_intervals.items():
    print(f"95% Confidence interval of Grad.Rate for {expend_bin} universities = [{ci[0]}, {
```

```
95% Confidence interval of Grad.Rate for Low expend universities = [55.36, 59.36]
95% Confidence interval of Grad.Rate for Med expend universities = [64.15, 67.95]
95% Confidence interval of Grad.Rate for High expend universities = [71.03, 74.92]
```

Apart from equal-width and equal-sized bins, custom bins can be created using the *bins* argument. Suppose, bins are to be created for `Expend` with cutoffs $\$10,000, \$20,000, \$30,000...\$60,000$. Then, we can use the *bins* argument as in the code below:

13.3.2.3 Binning with custom bins

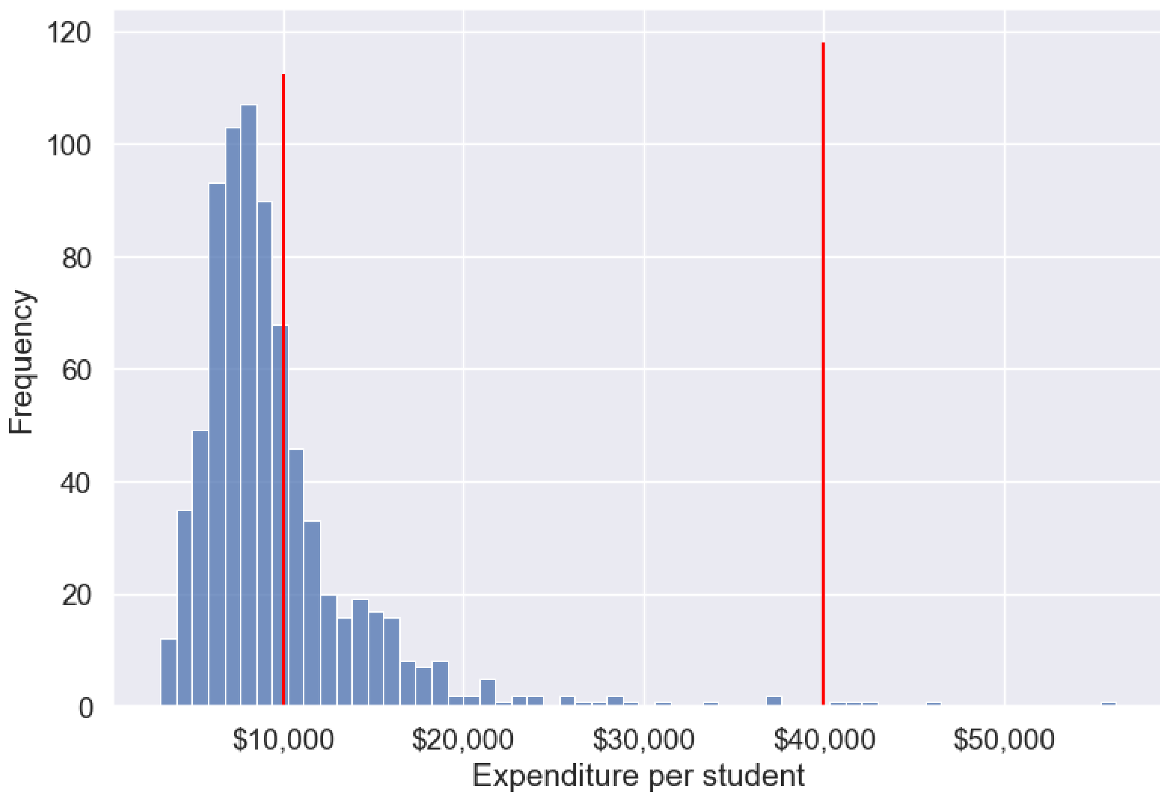
```
Binned_expend = pd.cut(college.Expend,bins = list(range(0,70000,10000)),retbins=True)
```

```
#Visualizing the bins for instructional expediture on a student
ax=sns.histplot(data = college, x= 'Expend')
plt.vlines(Binned_expend[1], 0,100,color='red')
plt.xlabel('Expenditure per student');
plt.ylabel('Graduation rate');
ax.xaxis.set_major_formatter('${x:,.0f}')
```

Next, let's create a custom bin with unequal sizes and widths.

```
# Create bins and labels
bins = [college['Expend'].min(), 10000, 40000, college['Expend'].max()]
labels = ['Low Expenditure', 'Medium Expenditure', 'High Expenditure']

# Binning the 'Expend' column
```

```python
college['Expend_Binned'] = pd.cut(college['Expend'], bins=bins, labels=labels)

# Visualizing the bins using a histogram
ax = sns.histplot(data=college, x='Expend', kde=False)

# Add vertical lines for the bin edges (the bin edges are the values in 'bins')
for bin_edge in bins[1:-1]:  # skip the first and last bins since they are the edges
    plt.vlines(bin_edge, 0, ax.get_ylim()[1], color='red')

# Add labels and formatting
plt.xlabel('Expenditure per student')
plt.ylabel('Frequency')
ax.xaxis.set_major_formatter('${x:,.0f}')

# Display the plot
plt.show()
```



As custom bin-cutoffs can be specified with the `cut()` function, custom bin quantiles can be specified with the `qcut()` function.

13.4 Dummy / Indicator variables

Dummy or indicator variables are binary variables created to represent categorical data in numerical form, typically used in regression and machine learning models. Each category is represented by a separate dummy variable with values of 0 or 1.

13.4.1 Purpose:

- **Enables categorical data** to be used in quantitative analysis: Models like linear regression and many machine learning algorithms require numerical input, making dummy variables essential for handling categorical data.
- **Preserves information**: Converts categorical information into a format that models can interpret without losing meaning.

13.4.2 When to use dummy variables

Dummy variables are especially useful in:

- **Regression models:** For categorical predictors, such as region, gender, or type of product.
- **Machine learning algorithms:** Many models, including tree-based methods, can benefit from dummy variables for categorical features.

13.4.3 How to Create Dummy variables

The pandas library in Python has a built-in function, `pd.get_dummies()`, to generate dummy variables. If a column in a DataFrame has k distinct values, we will get a DataFrame with k columns containing 0s and 1s

Let us make dummy variables with the equal-sized bins we created for the average instruction expenditure per student.

```
#Using the Pandas function qcut() to create bins with the same number of observations
Binned_expend = pd.qcut(college['Expend'],3,retbins = True,labels = ['Low_expend','Med_expen
college['Expend_bin'] = Binned_expend[0]
```

```
#Making dummy variables based on the levels (categories) of the 'Expend_bin' variable
dummy_Expend = pd.get_dummies(college['Expend_bin'], dtype=int)
```

The dummy data `dummy_Expend` has a value of 1 if the observation corresponds to the category referenced by the column name.

```
dummy_Expend.head()
```

	Low_expend	Med_expend	High_expend
0	1	0	0
1	0	0	1
2	0	1	0
3	0	0	1
4	0	0	1

Adding the dummy columns back to the original dataframe

```
#Concatenating the dummy variables to the original data
college = pd.concat([college,dummy_Expend],axis=1)
print(college.shape)
college.head()
```

```
(777, 26)
```

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885
1	Adelphi University	Yes	2186	1924	512	16	29	2683
2	Adrian College	Yes	1428	1097	336	22	50	1036
3	Agnes Scott College	Yes	417	349	137	60	89	510
4	Alaska Pacific University	Yes	193	146	55	16	44	249

We can find the correlation between the dummy variables and graduation rate to identify if any of the dummy variables will be useful to estimate graduation rate (`Grad.Rate`).

```
#Finding if dummy variables will be useful to estimate 'Grad.Rate'
dummy_Expend.corrwith(college['Grad.Rate'])
```

```
Low_expend    -0.334456
Med_expend     0.024492
High_expend    0.309964
dtype: float64
```

The dummy variables `Low expend` and `High expend` may contribute in explaining `Grad.Rate` in a regression model.

13.4.4 Using `drop_first` to Avoid Multicollinearity

In cases like linear regression, you may want to drop one dummy variable to avoid multicollinearity (perfect correlation between variables).

```
# create dummy variables, dropping the first to avoid multicollinearity
dummy_Expend = pd.get_dummies(college['Expend_bin'], dtype=int, drop_first=True)

# add the dummy variables to the original dataset
college = pd.concat([college, dummy_Expend], axis=1)

college.head()
```

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885
1	Adelphi University	Yes	2186	1924	512	16	29	2683
2	Adrian College	Yes	1428	1097	336	22	50	1036
3	Agnes Scott College	Yes	417	349	137	60	89	510
4	Alaska Pacific University	Yes	193	146	55	16	44	249

13.5 Independent Study

13.5.1 Practice exercise 1

Read *survey_data_clean.csv*. Split the columns of the dataset, such that all columns with categorical values transform into dummy variables with each category corresponding to a column of 0s and 1s. Leave the *Timestamp* column.

As all categorical columns are transformed to dummy variables, all columns have numeric values.

What is the total number of columns in the transformed data? What is the total number of columns of the original data?

Find the:

1. Top 5 variables having the highest positive correlation with `NU_GPA`.
2. Top 5 variables having the highest negative correlation with `NU_GPA`.

13.5.2 Practice exercise 2

Consider the dataset *survey_data_clean.csv* . Find the number of outliers in each column of the dataset based on both the Z-score and IQR criteria. Do not use a `for` loop.

Which column(s) have the maximum number of outliers using Z-score?

Which column(s) have the maximum number of outliers using IQR?

Do you think the outlying observations identified for those columns(s) should be considered as outliers? If not, then which type of columns should be considered when finding outliers?

14 Data Grouping and Aggregation

Grouping and aggregation are essential in data science because they enable meaningful analysis and insights from complex and large-scale datasets. Here are key reasons why they are important:

- **Data Summarization:** Grouping allows you to condense large amounts of data into concise summaries. Aggregation provides statistical summaries (e.g., mean, sum, count), making it easier to understand data trends and characteristics without needing to examine every detail.
- **Insights Across Categories:** Grouping by categories, like regions, demographics, or time periods, lets you analyze patterns within each group. For example, aggregating sales data by region can reveal which areas are performing better, or grouping customer data by age group can highlight consumer trends.
- **Time Series Analysis:** In time-series data, grouping by time intervals (e.g., day, month, quarter) allows you to observe trends over time, which can be crucial for forecasting, seasonal analysis, and trend detection.
- **Reducing Data Complexity:** By summarizing data at a higher level, you reduce complexity, making it easier to visualize and interpret, especially in large datasets where working with raw data could be overwhelming.

In this chapter, we are going to see grouping and aggregating using pandas. Grouping and aggregating will help to achieve data analysis easily using various functions. These methods will help us to the group and summarize our data and make complex analysis comparatively easy.

Througout this chapter, we will use *gdp_lifeExpectancy.csv*, let's read the csv file to pandas dataframe first

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

```
# Load the data
gdp_lifeExp_data = pd.read_csv('./Datasets/gdp_lifeExpectancy.csv')
gdp_lifeExp_data.head()
```

	country	continent	year	lifeExp	pop	gdpPercap
0	Afghanistan	Asia	1952	28.801	8425333	779.445314
1	Afghanistan	Asia	1957	30.332	9240934	820.853030
2	Afghanistan	Asia	1962	31.997	10267083	853.100710
3	Afghanistan	Asia	1967	34.020	11537966	836.197138
4	Afghanistan	Asia	1972	36.088	13079460	739.981106

14.1 Grouping by a single column

`groupby()` allows you to split a DataFrame based on the values in one or more columns. First, we'll explore grouping by a single column.

14.1.1 Syntax of `groupby()`

- `DataFrame.groupby(by="column_name")` – Groups data by the specified column.

Consider the life expectancy dataset. suppose we want to group by the observations by `continent` by passing it as an argument to the `groupby()` method.

```
#Creating a GroupBy object
grouped = gdp_lifeExp_data.groupby('continent')
#This will split the data into groups that correspond to values of the column 'continent'
```

The `groupby()` method returns a GroupBy object.

```
#A 'GroupBy' objects is created with the `groupby()` function
type(grouped)
```

```
pandas.core.groupby.generic.DataFrameGroupBy
```

The GroupBy object `grouped` contains the information of the groups in which the data is distributed. Each observation has been assigned to a specific group of the column(s) used to group the data. However, note that the dataset is not physically split into different DataFrames. For example, in the above case, each observation is assigned to a particular group depending on the value of the `continent` for that observation. However, all the observations are still in the same DataFrame `data`.

14.1.2 Attributes and methods of the *GroupBy* object

14.1.2.1 keys

The object(s) grouping the data are called *key(s)*. Here `continent` is the group key. The keys of the *GroupBy* object can be seen using Its `keys` attribute.

```
#Key(s) of the GroupBy object
grouped.keys
```

```
'continent'
```

14.1.2.2 ngroups

The number of groups in which the data is distributed based on the keys can be seen with the `ngroups` attribute.

```
#The number of groups based on the key(s)
grouped.ngroups
```

```
5
```

The group names are the *keys* of the dictionary, while the row labels are the corresponding *values*

```
#Group names
grouped.groups.keys()
```

```
dict_keys(['Africa', 'Americas', 'Asia', 'Europe', 'Oceania'])
```

14.1.2.3 groups

The `groups` attribute of the *GroupBy* object contains the group labels (or names) and the row labels of the observations in each group, as a dictionary.

```
#The groups (in the dictionary format)
grouped.groups
```

```
{'Africa': [24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
```

14.1.2.4 `groups.values`

The `groups.values` attribute of the *GroupBy* object contains the row labels of the observations in each group, as a dictionary.

```
#Group values are the row labels corresponding to a particular group
grouped.groups.values()
```

```
dict_values([Index([  24,   25,   26,   27,   28,   29,   30,   31,   32,   33,
       ...
       1694, 1695, 1696, 1697, 1698, 1699, 1700, 1701, 1702, 1703],
      dtype='int64', length=624), Index([  48,   49,   50,   51,   52,   53,   54,   55,   56
       ...
       1634, 1635, 1636, 1637, 1638, 1639, 1640, 1641, 1642, 1643],
      dtype='int64', length=300), Index([   0,    1,    2,    3,    4,    5,    6,    7,    8
       ...
       1670, 1671, 1672, 1673, 1674, 1675, 1676, 1677, 1678, 1679],
      dtype='int64', length=396), Index([  12,   13,   14,   15,   16,   17,   18,   19,   20
       ...
       1598, 1599, 1600, 1601, 1602, 1603, 1604, 1605, 1606, 1607],
      dtype='int64', length=360), Index([  60,   61,   62,   63,   64,   65,   66,   67,   68
       1092, 1093, 1094, 1095, 1096, 1097, 1098, 1099, 1100, 1101, 1102, 1103],
      dtype='int64')])
```

14.1.2.5 `size()`

The `size()` method of the *GroupBy* object returns the number of observations in each group.

```
#Number of observations in each group
grouped.size()
```

```
continent
Africa      624
Americas    300
Asia        396
Europe      360
Oceania      24
dtype: int64
```

14.1.2.6 first()

The first non missing element of each group is returned with the **first()** method of the *GroupBy* object.

```
#The first element of the group can be printed using the first() method
grouped.first()
```

	country	year	lifeExp	pop	gdpPercap
continent					
Africa	Algeria	1952	43.077	9279525	2449.008185
Americas	Argentina	1952	62.485	17876956	5911.315053
Asia	Afghanistan	1952	28.801	8425333	779.445314
Europe	Albania	1952	55.230	1282697	1601.056136
Oceania	Australia	1952	69.120	8691212	10039.595640

14.1.2.7 get_group()

This method returns the observations for a particular group of the *GroupBy* object.

```
#Observations for individual groups can be obtained using the get_group() function
grouped.get_group('Asia')
```

	country	continent	year	lifeExp	pop	gdpPercap
0	Afghanistan	Asia	1952	28.801	8425333	779.445314
1	Afghanistan	Asia	1957	30.332	9240934	820.853030
2	Afghanistan	Asia	1962	31.997	10267083	853.100710
3	Afghanistan	Asia	1967	34.020	11537966	836.197138
4	Afghanistan	Asia	1972	36.088	13079460	739.981106
...	...	...	...	...	...	...
1675	Yemen, Rep.	Asia	1987	52.922	11219340	1971.741538
1676	Yemen, Rep.	Asia	1992	55.599	13367997	1879.496673
1677	Yemen, Rep.	Asia	1997	58.020	15826497	2117.484526
1678	Yemen, Rep.	Asia	2002	60.308	18701257	2234.820827
1679	Yemen, Rep.	Asia	2007	62.698	22211743	2280.769906

14.2 Data aggregation within groups

14.2.1 Common Aggregation Functions

Aggregation functions are essential for summarizing and analyzing data in pandas. These functions allow you to compute summary statistics for your data, making it easier to identify trends and patterns.

Below are some of the most commonly used aggregation functions when working with grouped data in pandas:

- **`mean()`** – Calculates the **average** value of the group.
- **`sum()`** – Computes the **total** value by summing all elements in the group.
- **`min()`** – Finds the **minimum** value in the group.
- **`max()`** – Finds the **maximum** value in the group.
- **`count()`** – Returns the **number of occurrences** or entries in the group.
- **`median()`** – Finds the **middle value** in the sorted group.
- **`std()`** – Calculates the **standard deviation**, measuring the spread or variation in the values of the group.

Each of these functions can help summarize and provide insights into different aspects of the grouped data.

Consider the life expectancy dataset, let's find the mean life expectancy, population and GDP per capita for each country during the period of 1952 -2007.

Next, we'll find the mean statistics for each group with the `mean()` method. The method will be applied on all columns of the DataFrame and all groups.

```
#Grouping the observations by 'country'
grouped_country = gdp_lifeExp_data.drop (['continent', 'year'], axis = 1).groupby('country')

#Finding the mean stastistic of all columns of the DataFrame and all groups
grouped_country.mean()
```

	lifeExp	pop	gdpPercap
country			
Afghanistan	37.478833	1.582372e+07	802.674598
Albania	68.432917	2.580249e+06	3255.366633
Algeria	59.030167	1.987541e+07	4426.025973
Angola	37.883500	7.309390e+06	3607.100529
Argentina	69.060417	2.860224e+07	8955.553783
...	...	...	...

country	lifeExp	pop	gdpPercap
Vietnam	57.479500	5.456857e+07	1017.712615
West Bank and Gaza	60.328667	1.848606e+06	3759.996781
Yemen, Rep.	46.780417	1.084319e+07	1569.274672
Zambia	45.996333	6.353805e+06	1358.199409
Zimbabwe	52.663167	7.641966e+06	635.858042

Next, we'll find the standard deviation statistics for each group with the `std()` method.

```
grouped_country.std()
```

country	lifeExp	pop	gdpPercap
Afghanistan	5.098646	7.114583e+06	108.202929
Albania	6.322911	8.285855e+05	1192.351513
Algeria	10.340069	8.613355e+06	1310.337656
Angola	4.005276	2.672281e+06	1165.900251
Argentina	4.186470	7.546609e+06	1862.583151
...	...	...	...
Vietnam	12.172331	2.052585e+07	567.482251
West Bank and Gaza	11.000069	1.023057e+06	1716.840614
Yemen, Rep.	11.019302	5.590408e+06	609.939160
Zambia	4.453246	3.096949e+06	247.494984
Zimbabwe	7.071816	3.376895e+06	133.689213

Alternatively, you can compute other statistical metrics.

14.2.2 Multiple aggregations and Custom aggregation using `agg()`

14.2.2.1 Multiple aggregations

Directly applying the aggregate methods of the *GroupBy* object such as mean, count, etc., lets us apply only one function at a time. Also, we may wish to apply an aggregate function of our own, which is not there in the set of methods of the *GroupBy* object, such as the range of values of a column.

The `agg()` function of a *GroupBy* object lets us aggregate data using:

1. Multiple aggregation functions
2. Custom aggregate functions (in addition to in-built functions like mean, std, count etc.)

Consider the life expectancy dataset, Let us use the agg() method of the GroupBy object to simultaneously find the mean and standard deviation of the *gdpPercap* for each country.

For aggregating by multiple functions, we pass a list of strings to agg(), where the strings are the function names.

```
grouped_country['gdpPercap'].agg(['mean','std']).sort_values(by = 'mean',ascending = False).h
```

	mean	std
country		
Kuwait	65332.910472	33882.139536
Switzerland	27074.334405	6886.463308
Norway	26747.306554	13421.947245
United States	26261.151347	9695.058103
Canada	22410.746340	8210.112789

14.2.2.2 Custom aggregation

In addition to the mean and standard deviation of the gdpPercap of each country, let us also include the range of gdpPercap in the table above using lambda function

```
grouped_country['gdpPercap'].agg(lambda x: x.max() - x.min())
```

```
country
Afghanistan            342.670088
Albania               4335.973390
Algeria               3774.359280
Angola                3245.635491
Argentina             6868.064587
                         ...
Vietnam               1836.509912
West Bank and Gaza    5595.075290
Yemen, Rep.           1499.052330
Zambia                 705.723500
Zimbabwe               392.478061
Name: gdpPercap, Length: 142, dtype: float64
```

```
# define a function that calculates the range
def range_func(x):
    return x.max() - x.min()
```

```
# apply the range function to the 'gdpPercap' column besides the mean and standard deviation
grouped_country['gdpPercap'].agg(['mean', 'std', range_func]).sort_values(by = 'range_func',
```

	mean	std	range_func
country			
Kuwait	65332.910472	33882.139536	85404.702920
Singapore	17425.382267	14926.147774	44828.041413
Norway	26747.306554	13421.947245	39261.768450
Hong Kong, China	16228.700865	12207.329731	36670.557461
Ireland	15758.606238	11573.311022	35465.716022
...	...	...	...
Rwanda	675.669043	142.229906	388.246772
Senegal	1533.121694	105.399353	344.572767
Afghanistan	802.674598	108.202929	342.670088
Ethiopia	509.115155	96.427627	328.659296
Burundi	471.662990	99.329720	292.403419

For aggregating by multiple functions & changing the column names resulting from those functions, we pass a list of tuples to `agg()`, where each tuple is of length two, and contains the new column name & the function to be applied.

```
#Simultaneous renaming of columns while grouping
grouped_country['gdpPercap'].agg([('Average','mean'),('Standard Deviation','std'),('90th Per
```

	Average	Standard Deviation	90th Percentile
country			
Kuwait	65332.910472	33882.139536	109251.315590
Norway	26747.306554	13421.947245	44343.894158
United States	26261.151347	9695.058103	38764.132898
Singapore	17425.382267	14926.147774	35772.742520
Switzerland	27074.334405	6886.463308	34246.394240
...	...	...	...
Liberia	604.814141	98.988329	706.275527
Malawi	575.447212	122.999953	689.590541

country	Average	Standard Deviation	90th Percentile
Burundi	471.662990	99.329720	615.597260
Myanmar	439.333333	175.401531	592.300000
Ethiopia	509.115155	96.427627	577.448804

14.2.3 Multiple aggregate functions on multiple columns

Let us find the mean and standard deviation of lifeExp and pop for each country

```
# find the meand and standard deviation of the 'lifeExp' and 'pop' column for each country
grouped_country[['lifeExp', 'pop']].agg(['mean', 'std']).sort_values(by = ('lifeExp', 'mean')
```

	lifeExp		pop	
	mean	std	mean	std
country				
Iceland	76.511417	3.026593	2.269781e+05	4.854168e+04
Sweden	76.177000	3.003990	8.220029e+06	6.365660e+05
Norway	75.843000	2.423994	4.031441e+06	4.107955e+05
Netherlands	75.648500	2.486363	1.378680e+07	2.005631e+06
Switzerland	75.565083	4.011572	6.384293e+06	8.582009e+05
...	...	...	...	...
Mozambique	40.379500	4.599184	1.204670e+07	4.457509e+06
Guinea-Bissau	39.210250	4.937369	8.820084e+05	3.132917e+05
Angola	37.883500	4.005276	7.309390e+06	2.672281e+06
Afghanistan	37.478833	5.098646	1.582372e+07	7.114583e+06
Sierra Leone	36.769167	3.937828	3.605425e+06	1.270945e+06

14.2.4 Distinct aggregate functions on multiple columns

For aggregating by multiple functions, we pass a list of strings to `agg()`, where the strings are the function names.

For aggregating by multiple functions & changing the column names resulting from those functions, we pass a list of tuples to `agg()`, where each tuple is of length two, and contains the new column name as the first object and the function to be applied as the second object of the tuple.

For aggregating by multiple functions such that a distinct set of functions is applied to each column, we pass a dictionary to `agg()`, where the keys are the column names on which the function is to be applied, and the values are the list of strings that are the function names, or a list of tuples if we also wish to name the aggregated columns.

```
# We can use a list to apply multiple aggregation functions to a single column, and a dictio
# Use string names for the aggregation functions
grouped_country.agg({"gdpPercap": ["mean", "std"], "lifeExp": ["median", "std"], "pop": ["ma
```

	gdpPercap		lifeExp		pop	
	mean	std	median	std	max	min
country						
Afghanistan	802.674598	108.202929	39.1460	5.098646	31889923	8425333
Albania	3255.366633	1192.351513	69.6750	6.322911	3600523	1282697
Algeria	4426.025973	1310.337656	59.6910	10.340069	33333216	9279525
Angola	3607.100529	1165.900251	39.6945	4.005276	12420476	4232095
Argentina	8955.553783	1862.583151	69.2115	4.186470	40301927	17876956
…	…	…	…	…	…	…
Vietnam	1017.712615	567.482251	57.2900	12.172331	85262356	26246839
West Bank and Gaza	3759.996781	1716.840614	62.5855	11.000069	4018332	1030585
Yemen, Rep.	1569.274672	609.939160	46.6440	11.019302	22211743	4963829
Zambia	1358.199409	247.494984	46.0615	4.453246	11746035	2672000
Zimbabwe	635.858042	133.689213	53.1765	7.071816	12311143	3080907

Next, for each country, find the mean and standard deviation of the `lifeExp`, and the minimum and maximum values of gdpPercap.

```
#Specifying arguments to the function as a dictionary if distinct functions are to be applie
grouped_country.agg({'lifeExp':[('Average','mean'),('Standard deviation','std')],'gdpPercap'
```

	lifeExp		gdpPercap	
	Average	Standard deviation	min	max
country				
Afghanistan	37.478833	5.098646	635.341351	978.011439
Albania	68.432917	6.322911	1601.056136	5937.029526
Algeria	59.030167	10.340069	2449.008185	6223.367465
Angola	37.883500	4.005276	2277.140884	5522.776375
Argentina	69.060417	4.186470	5911.315053	12779.379640
…	…	…	…	…
Vietnam	57.479500	12.172331	605.066492	2441.576404
West Bank and Gaza	60.328667	11.000069	1515.592329	7110.667619

country	lifeExp Average	Standard deviation	gdpPercap min	max
Yemen, Rep.	46.780417	11.019302	781.717576	2280.769906
Zambia	45.996333	4.453246	1071.353818	1777.077318
Zimbabwe	52.663167	7.071816	406.884115	799.362176

14.3 Grouping by Multiple Columns

Above, we demonstrated grouping by a single column, which is useful for summarizing data based on one categorical variable. However, in many cases, we need to group by multiple columns. Grouping by multiple columns allows us to create more detailed summaries by accounting for multiple categorical variables. This approach enables us to analyze data at a finer granularity, revealing insights that might be missed with single-column grouping alone.

14.3.1 Basic Syntax for Grouping by Multiple Columns

Use `groupby()` with a list of column names to group data by multiple columns.

- `DataFrame.groupby(by=["col1", "col2"])`

Consider the life expectancy dataset, we can group by both country and continent to analyze `gdpPercap`, `lifeExp`, and `pop` trends for each country within each continent, providing a more comprehensive view of the data.

```
#Grouping by multiple columns
grouped_continent_contry = gdp_lifeExp_data.groupby(['continent', 'country'])[ "lifeExp"].agg
```

```
grouped_continent_contry
```

continent	country
	Iceland
	Sweden
Europe	Norway
	Netherlands
	Switzerland
...	...

continent	country
	Mozambique
Africa	Guinea-Bissau
	Angola
Asia	Afghanistan
Africa	Sierra Leone

14.3.2 Understanding Hierarchical (Multi-Level) Indexing

- Grouping by multiple columns creates a hierarchical index (also called a multi-level index).
- This index allows each level (e.g., continent, country) to act as an independent category that can be accessed individually.

In the above output, `continent` and `country` form a two-level hierarchical index, allowing us to drill down from continent-level to country-level summaries.

```
grouped_continent_contry.index.nlevels
```

```
2
```

```
# get the first level of the index
grouped_continent_contry.index.levels[0]
```

```
Index(['Africa', 'Americas', 'Asia', 'Europe', 'Oceania'], dtype='object', name='continent')
```

```
# get the second level of the index
grouped_continent_contry.index.levels[1]
```

```
Index(['Afghanistan', 'Albania', 'Algeria', 'Angola', 'Argentina', 'Australia',
       'Austria', 'Bahrain', 'Bangladesh', 'Belgium',
       ...
       'Uganda', 'United Kingdom', 'United States', 'Uruguay', 'Venezuela',
       'Vietnam', 'West Bank and Gaza', 'Yemen, Rep.', 'Zambia', 'Zimbabwe'],
      dtype='object', name='country', length=142)
```

14.3.3 Subsetting Data in a Hierarchical Index

`grouped_continent_country` is still a DataFrame with hierarchical indexing. You can use `.loc[]` for subsetting, just as you would with a single-level index.

```
# get the observations for the 'Americas' continent
grouped_continent_contry.loc['Americas'].head()
```

	mean	std	max	min
country				
Canada	74.902750	3.952871	80.653	68.750
United States	73.478500	3.343781	78.242	68.440
Puerto Rico	72.739333	3.984267	78.746	64.280
Cuba	71.045083	6.022798	78.273	59.421
Uruguay	70.781583	3.342937	76.384	66.071

```
# get the mean life expectancy for the 'Americas' continent
grouped_continent_contry.loc['Americas']['mean'].head()
```

```
country
Canada           74.902750
United States    73.478500
Puerto Rico      72.739333
Cuba             71.045083
Uruguay          70.781583
Name: mean, dtype: float64
```

```
# another way to get the mean life expectancy for the 'Americas' continent
grouped_continent_contry.loc['Americas', 'mean'].head()
```

```
country
Canada           74.902750
United States    73.478500
Puerto Rico      72.739333
Cuba             71.045083
Uruguay          70.781583
Name: mean, dtype: float64
```

You can use a tuple to access data for specific levels in a multi-level index.

```python
# get the observations for the 'United States' country
grouped_continent_contry.loc[( 'Americas', 'United States')]
```

```
mean    73.478500
std      3.343781
max     78.242000
min     68.440000
Name: (Americas, United States), dtype: float64
```

```python
grouped_continent_contry.loc[( 'Americas', 'United States'), ['mean', 'std']]
```

```
mean    73.478500
std      3.343781
Name: (Americas, United States), dtype: float64
```

```python
gdp_lifeExp_data.columns
```

```
Index(['country', 'continent', 'year', 'lifeExp', 'pop', 'gdpPercap'], dtype='object')
```

Finally, you can use `reset_index()` to convert the hierarchical index into a regular index, allowing you to apply the standard subsetting and filtering methods covered in previous chapters

```python
grouped_continent_contry.reset_index().head()
```

	continent	country	mean	std	max	min
0	Europe	Iceland	76.511417	3.026593	81.757	72.49
1	Europe	Sweden	76.177000	3.003990	80.884	71.86
2	Europe	Norway	75.843000	2.423994	80.196	72.67
3	Europe	Netherlands	75.648500	2.486363	79.762	72.13
4	Europe	Switzerland	75.565083	4.011572	81.701	69.62

14.3.4 Grouping by multiple columns and aggregating multiple variables

```python
#Grouping by multiple columns
grouped_continent_contry_multi = gdp_lifeExp_data.groupby(['continent', 'country','year'])[
grouped_continent_contry_multi
```

continent	country
Africa	Algeria
...	...
Oceania	New Zealand

Breaking Down Grouping and Aggregation

- **Grouping by Multiple Columns**:
  In this example, we are grouping the data by three columns: `continent`, `country`, and `year`. This creates groups based on unique combinations of these columns.
- **Aggregating Multiple Variables**:
  We apply multiple aggregation functions (`mean`, `std`, `max`, and `min`) to multiple variables (`lifeExp`, `pop`, and `gdpPercap`).

This type of operation is commonly referred to as **"multi-column grouping with multiple aggregations"** in pandas. It's a powerful approach because it allows us to obtain a detailed statistical summary for each combination of grouping columns across several variables.

```
# its columns are also two levels deep
grouped_continent_contry_multi.columns.nlevels
```

```
2
```

```
# pass a tuple to the loc() method to access the values of the multi-level columns with a mul
grouped_continent_contry_multi.loc[('Americas','United States'), ('lifeExp', 'mean')]
```

```
year
1952    68.440
1957    69.490
1962    70.210
```

```
1967    70.760
1972    71.340
1977    73.380
1982    74.650
1987    75.020
1992    76.090
1997    76.810
2002    77.310
2007    78.242
Name: (lifeExp, mean), dtype: float64
```

14.4 Advanced Operations within groups: `apply()`, `transform()`, and `filter()`

14.4.1 Using `apply()` on groups

The `apply()` function applies a custom function to each group, allowing for flexible operations. The function can return either a scalar, Series, or DataFrame.

Example: Consider the life expectancy dataset, find the top 3 life expectancy values for each continent

We'll first define a function that sorts a dataset by decreasing life expectancy and returns the top 3 rows. Then, we'll apply this function on each group using the `apply()` method of the *GroupBy* object.

```
# Define a function to get the top 3 rows based on life expectancy for each group
def top_3_life_expectancy(group):
    return group.nlargest(3, 'lifeExp')
```

```
#Defining the groups in the data
grouped_gdpcapital_data = gdp_lifeExp_data.groupby('continent')
```

Now we'll use the `apply()` method to apply the `top_3_life_expectancy()` function on each group of the object `grouped_gdpcapital_data`.

```
# Apply the function to each continent group
top_life_expectancy = gdp_lifeExp_data.groupby('continent')[['continent', 'country', 'year',

# Display the result
top_life_expectancy.head()
```

	continent	country	year	lifeExp	gdpPercap
0	Africa	Reunion	2007	76.442	7670.122558
1	Africa	Reunion	2002	75.744	6316.165200
2	Africa	Reunion	1997	74.772	6071.941411
3	Americas	Canada	2007	80.653	36319.235010
4	Americas	Canada	2002	79.770	33328.965070

The `top_3_life_expectancy()` function is applied to each group, and the results are concatenated internally with the `concat()` function. The output therefore has a hierarchical index whose outer level indices are the group keys.

We can also use a lambda function instead of separately defining the function `top_3_life_expectancy()`:

```
# Use a lambda function to get the top 3 life expectancy values for each continent
top_life_expectancy = (
    gdp_lifeExp_data
    .groupby('continent')[['continent', 'country', 'year', 'lifeExp', 'gdpPercap']]  # Avoid
    .apply(lambda x: x.nlargest(3, 'lifeExp'))
    .reset_index(drop=True)
)

# Display the result
top_life_expectancy.head()
```

	continent	country	year	lifeExp	gdpPercap
0	Africa	Reunion	2007	76.442	7670.122558
1	Africa	Reunion	2002	75.744	6316.165200
2	Africa	Reunion	1997	74.772	6071.941411
3	Americas	Canada	2007	80.653	36319.235010
4	Americas	Canada	2002	79.770	33328.965070

14.4.2 Using `transform()` on Groups

The `transform()` function applies a function to each group and returns a Series aligned with the original DataFrame's index. This makes it suitable for adding or modifying columns based on group-level calculations.

Recall that in the data cleaning and preparation chapter, we imputed missing values based on correlated variables in the dataset.

In the example provided, some countries had missing values for GDP per capita. To handle this, we imputed the missing GDP per capita for each country using the average GDP per capita of its corresponding continent.

Now, we'll explore an alternative approach using `groupby()` and `transform()` to perform this imputation.

Let us read the datasets and the function that makes a visualization to compare the imputed values with the actual values.

```
#Importing data with missing values
gdp_missing_data = pd.read_csv('./Datasets/GDP_missing_data.csv')

#Importing data with all values
gdp_complete_data = pd.read_csv('./Datasets/GDP_complete_data.csv')
```

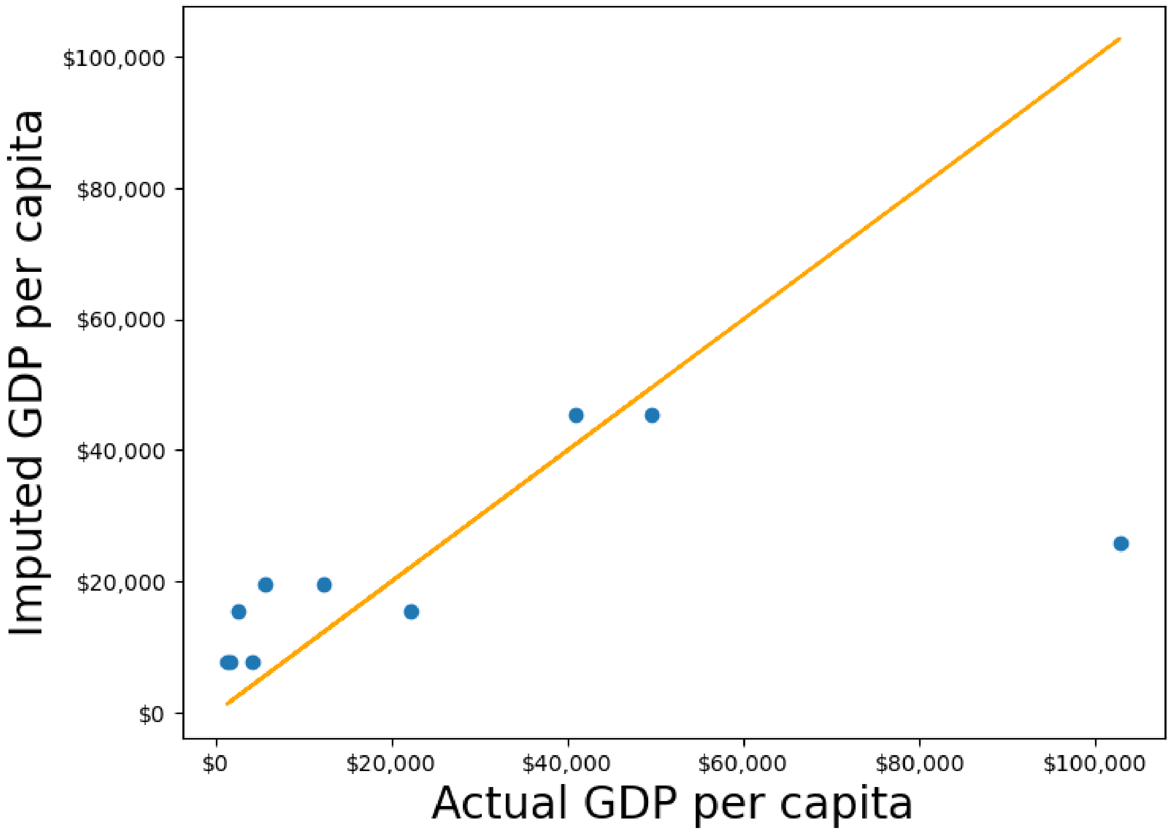
```
#Index of rows with missing values for GDP per capita
null_ind_gdpPC = gdp_missing_data.index[gdp_missing_data.gdpPerCapita.isnull()]

#Defining a function to plot the imputed values vs actual values
def plot_actual_vs_predicted():
    fig, ax = plt.subplots(figsize=(8, 6))
    plt.rc('xtick', labelsize=15)
    plt.rc('ytick', labelsize=15)
    x = gdp_complete_data.loc[null_ind_gdpPC,'gdpPerCapita']
    y = gdp_imputed_data.loc[null_ind_gdpPC,'gdpPerCapita']
    plt.scatter(x,y)
    z=np.polyfit(x,y,1)
    p=np.poly1d(z)
    plt.plot(x,x,color='orange')
    plt.xlabel('Actual GDP per capita',fontsize=20)
    plt.ylabel('Imputed GDP per capita',fontsize=20)
    ax.xaxis.set_major_formatter('${x:,.0f}')
    ax.yaxis.set_major_formatter('${x:,.0f}')
    rmse = np.sqrt(((x-y).pow(2)).mean())
    print("RMSE=",rmse)
```

Approach 1: Using the approach we used in the previous chapter

```
#Finding the mean GDP per capita of the continent
avg_gdpPerCapita = gdp_missing_data['gdpPerCapita'].groupby(gdp_missing_data['continent']).me

#Creating a copy of missing data to impute missing values
```

```
gdp_imputed_data = gdp_missing_data.copy()

#Replacing missing GDP per capita with the mean GDP per capita for the corresponding contine
for cont in avg_gdpPerCapita.index:
    gdp_imputed_data.loc[(gdp_imputed_data.continent==cont) & (gdp_imputed_data.gdpPerCapita
                     'gdpPerCapita']=avg_gdpPerCapita[cont]
plot_actual_vs_predicted()
```

```
RMSE= 25473.20645170116
```



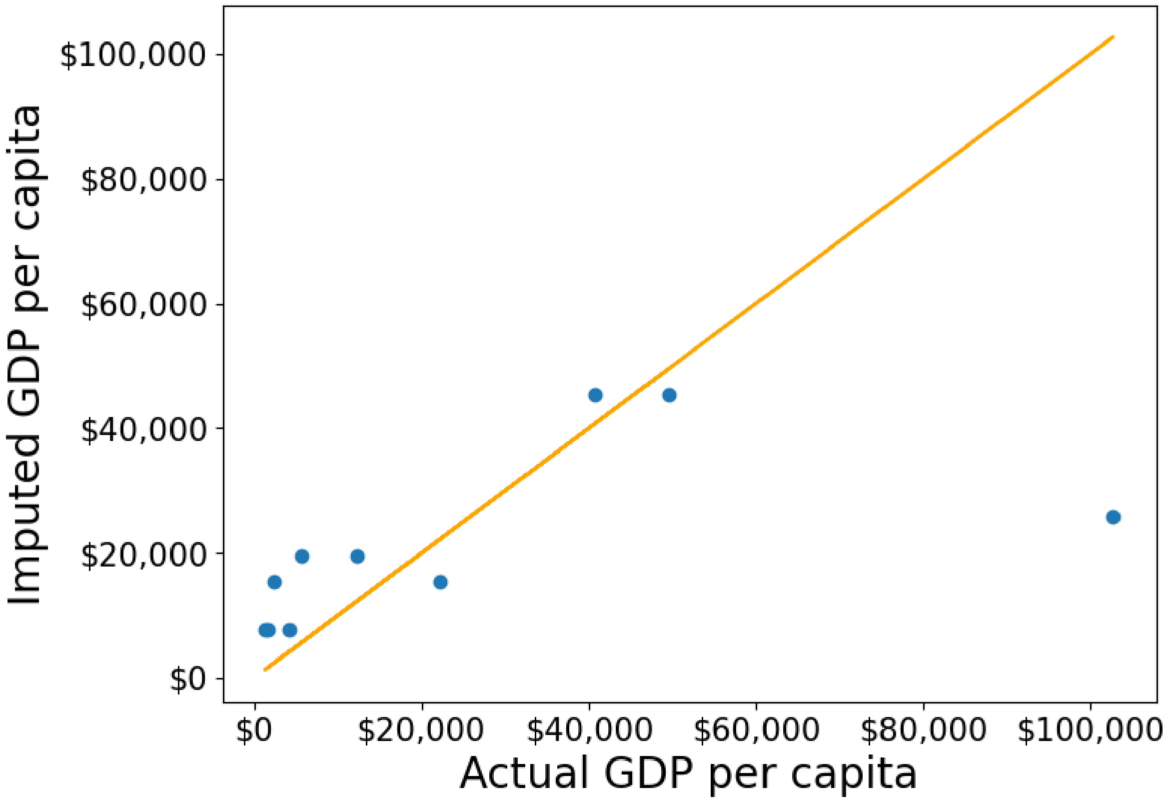
Approach 2: Using the `groupby()` and `transform()` methods.

The `transform()` function is a powerful tool for filling missing values in grouped data. It allows us to apply a function across each group and align the result back to the original DataFrame, making it perfect for filling missing values based on group statistics.

In this example, we use `transform()` to impute missing values in the `gdpPerCapita` column by filling them with the mean `gdpPerCapita` of each continent:

```
#Creating a copy of missing data to impute missing values
gdp_imputed_data = gdp_missing_data.copy()

#Grouping data by continent
grouped = gdp_missing_data.groupby('continent')

#Imputing missing values with the mean GDP per capita of the continent
gdp_imputed_data['gdpPerCapita'] = grouped['gdpPerCapita'].transform(lambda x: x.fillna(x.mea

plot_actual_vs_predicted()
```

```
RMSE= 25473.20645170116
```



Using the `transform()` function, missing values in gdpPerCapita for each group are filled with the group's mean `gdpPerCapita`. This approach is not only more convenient to write but also faster compared to using for loops. While a for loop imputes missing values one group at a time, `transform()` performs built-in operations (like mean, sum, etc.) in a way that is optimized internally, making it more efficient.

Let's use `apply()` instead of `transform()` with `groupby()`

Please copy the code below and run it in your notebook:

```
#Creating a copy of missing data to impute missing values
gdp_imputed_data = gdp_missing_data.copy()

#Grouping data by continent
grouped = gdp_missing_data.groupby('continent')

#Applying the lambda function on the 'gdpPerCapita' column of the groups
gdp_imputed_data['gdpPerCapita'] = grouped['gdpPerCapita'].apply(lambda x: x.fillna(x.mean()

plot_actual_vs_predicted()
```

Why we ran into this error? and `apply()` doesn't work?

Here's a deeper look at why `apply()` doesn't work as expected here:

14.4.2.1 Behavior of `groupby().apply()` vs. `groupby().transform()`

- **`groupby().apply()`**: This method applies a function to each group and returns the result with a hierarchical (multi-level) index by default. This hierarchical index can make it difficult to align the result back to a single column in the original DataFrame.
- **`groupby().transform()`**: In contrast, `transform()` is specifically designed to apply a function to each group and return a Series that is aligned with the original DataFrame's index. This alignment makes it directly compatible for assignment to a new or existing column in the original DataFrame.

14.4.2.2 Why `transform()` Works for Imputation

When using `transform()` to fill missing values, it applies the function (e.g., `fillna(x.mean())`) based on each group's statistics, such as the mean, while keeping the result aligned with the original DataFrame's index. This allows for smooth assignment to a column in the DataFrame without any index mismatch issues.

Additionally, `transform()` applies the function to each element in a group independently and returns a result that has the same shape as the original data, making it ideal for adding or modifying columns.

14.4.3 Using `filter()` on Groups

The `filter()` function filters entire groups based on a condition. It evaluates each group and keeps only those that meet the specified criteria.

Example: Keep only the countries where the mean life expectancy is greater than 70

```
# keep only the continent where the mean life expectancy is greater than 74
gdp_lifeExp_data.groupby('continent').filter(lambda x: x['lifeExp'].mean() > 74)['continent']
```

```
array(['Oceania'], dtype=object)
```

```
# keep only the country where the mean life expectancy is greater than 74
gdp_lifeExp_data.groupby('country').filter(lambda x: x['lifeExp'].mean() > 74)['country'].uni
```

```
array(['Australia', 'Canada', 'Denmark', 'France', 'Iceland', 'Italy',
       'Japan', 'Netherlands', 'Norway', 'Spain', 'Sweden', 'Switzerland'],
      dtype=object)
```

Using `.nunique()` get the number of countries that satisfy this condition

```
gdp_lifeExp_data.groupby('country').filter(lambda x: x['lifeExp'].mean() > 74)['country'].nun
```

```
12
```

14.5 Sampling data by group

If a dataset contains a large number of observations, operating on it can be computationally expensive. Instead, working on a sample of entire observations is a more efficient alterative. The `groupby()` method combined with `apply()` can be used for stratified random sampling from a large dataset.

Before taking the random sample, let us find the number of countries in each continent.

```
gdp_lifeExp_data.continent.value_counts()
```

```
continent
Africa      624
Asia        396
Europe      360
Americas    300
Oceania      24
Name: count, dtype: int64
```

Let us take a random sample of 650 observations from the entire dataset.

```python
sample_lifeExp_data = gdp_lifeExp_data.sample(650)
```

Now, let us see the number of countries of each continent in our sample.

```python
sample_lifeExp_data.continent.value_counts()
```

```
continent
Africa      241
Asia        149
Europe      142
Americas    109
Oceania       9
Name: count, dtype: int64
```

Some of the continent have a very low representation in the data. To rectify this, we can take a random sample of 130 observations from each of the 5 continents. In other words, we can take a random sample from each of the continent-based groups.

```python
evenly_sampled_lifeExp_data = gdp_lifeExp_data.groupby('continent').apply(lambda x:x.sample(

group_sizes = evenly_sampled_lifeExp_data.groupby(level=0).size()
print(group_sizes)
```

```
continent
Africa      130
Americas    130
Asia        130
Europe      130
Oceania     130
dtype: int64
```

The above stratified random sample equally represents all the continent.

14.6 corr(): Correlation by group

The corr() method of the *GroupBy* object returns the correlation between all pairs of columns within each group.

Example: Find the correlation between lifeExp and gdpPercap for each continent-country level combination.

```
gdp_lifeExp_data.groupby(['continent','country']).apply(lambda x:x['lifeExp'].corr(x['gdpPerc
```

```
continent  country
Africa     Algeria           0.904471
           Angola           -0.301079
           Benin             0.843949
           Botswana          0.005597
           Burkina Faso      0.881677
                                ...
Europe     Switzerland       0.980715
           Turkey            0.954455
           United Kingdom    0.989893
Oceania    Australia         0.986446
           New Zealand       0.974493
Length: 142, dtype: float64
```

Life expectancy is closely associated with GDP per capita across most continent-country combinations.

14.7 pivot_table()

The pivot_table() function in pandas is a powerful tool for performing groupwise aggregation in a structured format, similar to Excel's pivot tables. It allows you to create a summary of data by grouping and aggregating it based on specified columns. Here's an overview of how pivot_table() works for groupwise aggregation:

Note that pivot_table() is the same as pivot() except that pivot_table() aggregates the data as well in addition to re-arranging it.

Example: Consider the life expectancy dataset, calculate the average life expectancy for each country and year combination

```
pd.pivot_table(data = gdp_lifeExp_data, values = 'lifeExp',index = 'country', columns ='year
```

year country	1952	1957	1962	1967	1972	1977	1982	198
Afghanistan	28.80100	30.332000	31.997000	34.02000	36.088000	38.438000	39.854000	40.8
Albania	55.23000	59.280000	64.820000	66.22000	67.690000	68.930000	70.420000	72.0
Algeria	43.07700	45.685000	48.303000	51.40700	54.518000	58.014000	61.368000	65.7
Angola	30.01500	31.999000	34.000000	35.98500	37.928000	39.483000	39.942000	39.9
Argentina	62.48500	64.399000	65.142000	65.63400	67.065000	68.481000	69.942000	70.7
...	...	...	...	...	...	...	...	...
West Bank and Gaza	43.16000	45.671000	48.127000	51.63100	56.532000	60.765000	64.406000	67.0
Yemen, Rep.	32.54800	33.970000	35.180000	36.98400	39.848000	44.175000	49.113000	52.9
Zambia	42.03800	44.077000	46.023000	47.76800	50.107000	51.386000	51.821000	50.8
Zimbabwe	48.45100	50.469000	52.358000	53.99500	55.635000	57.674000	60.363000	62.3
All	49.05762	51.507401	53.609249	55.67829	57.647386	59.570157	61.533197	63.2

Explanation

- **values**: Specifies the column to aggregate (e.g., `lifeExp` in our example).
- **index**: Groups by the rows based on the `country` column.
- **columns**: Groups by the columns based on the `year` column.
- **aggfunc**: Uses `mean` to calculate the average life expectancy.

Common Aggregation Functions in `pivot_table()`

You can use various aggregation functions within `pivot_table()` to summarize data:

- **`mean`** – Calculates the average of values within each group.
- **`sum`** – Computes the total of values within each group.
- **`count`** – Counts the number of non-null entries within each group.
- **`min`** and **`max`** – Finds the minimum and maximum values within each group.

We can also use custom *GroupBy* aggregate functions with `pivot_table()`.

Example: Find the 90^{th} percentile of life expectancy for each country and year combination

```
pd.pivot_table(data = gdp_lifeExp_data, values = 'lifeExp',index = 'country', columns ='year
```

year country	1952	1957	1962	1967	1972	1977	1982	1987	1992	1997
Afghanistan	28.801	30.332	31.997	34.020	36.088	38.438	39.854	40.822	41.674	41.763
Albania	55.230	59.280	64.820	66.220	67.690	68.930	70.420	72.000	71.581	72.950
Algeria	43.077	45.685	48.303	51.407	54.518	58.014	61.368	65.799	67.744	69.152
Angola	30.015	31.999	34.000	35.985	37.928	39.483	39.942	39.906	40.647	40.963
Argentina	62.485	64.399	65.142	65.634	67.065	68.481	69.942	70.774	71.868	73.275
...	...	...	...	...	...	...	...	...	...	...
Vietnam	40.412	42.887	45.363	47.838	50.254	55.764	58.816	62.820	67.662	70.672
West Bank and Gaza	43.160	45.671	48.127	51.631	56.532	60.765	64.406	67.046	69.718	71.096
Yemen, Rep.	32.548	33.970	35.180	36.984	39.848	44.175	49.113	52.922	55.599	58.020
Zambia	42.038	44.077	46.023	47.768	50.107	51.386	51.821	50.821	46.100	40.238
Zimbabwe	48.451	50.469	52.358	53.995	55.635	57.674	60.363	62.351	60.377	46.809

14.8 crosstab()

14.8.1 Basic Usage of crosstab()

The `crosstab()` method is a special case of a pivot table for computing group frequncies (or size of each group).

```
# create a basic crosstab to see counts of countries in each continent
pd.crosstab(gdp_lifeExp_data['continent'], columns='count')
```

col_0 continent	count
Africa	624
Americas	300
Asia	396
Europe	360
Oceania	24

Use the `margins=True` argument to add totals (row and column margins) to the table. The `All` row and column provide totals for each age group and gender category.

```
# create a crosstab to see counts of countries in each continent and year
pd.crosstab(gdp_lifeExp_data['continent'], gdp_lifeExp_data['year'], margins=True)
```

year continent	1952	1957	1962	1967	1972	1977	1982	1987	1992	1997	2002	2007	All
Africa	52	52	52	52	52	52	52	52	52	52	52	52	624
Americas	25	25	25	25	25	25	25	25	25	25	25	25	300
Asia	33	33	33	33	33	33	33	33	33	33	33	33	396
Europe	30	30	30	30	30	30	30	30	30	30	30	30	360
Oceania	2	2	2	2	2	2	2	2	2	2	2	2	24
All	142	142	142	142	142	142	142	142	142	142	142	142	1704

This table shows the count of each year in each continent group, helping us understand the year distribution across different continent groups. We may often use it to check if the data is representative of all groups that are of interest to us.

14.8.2 Using `crosstab()` with Aggregation Functions

You can specify a `values` column and an aggregation function (`aggfunc`) to summarize numerical data for each combination of categorical variables.

Example: find the mean life expectancy for each continent and year

```
# find the mean life expectancy for each country in each continent and year
pd.crosstab(gdp_lifeExp_data['continent'], gdp_lifeExp_data['year'], values=gdp_lifeExp_data
```

year continent	1952	1957	1962	1967	1972	1977	1982	1987	1
Africa	39.135500	41.266346	43.319442	45.334538	47.450942	49.580423	51.592865	53.344788	5
Americas	53.279840	55.960280	58.398760	60.410920	62.394920	64.391560	66.228840	68.090720	6
Asia	46.314394	49.318544	51.563223	54.663640	57.319269	59.610556	62.617939	64.851182	6
Europe	64.408500	66.703067	68.539233	69.737600	70.775033	71.937767	72.806400	73.642167	7
Oceania	69.255000	70.295000	71.085000	71.310000	71.910000	72.855000	74.290000	75.320000	7

14.9 Independent Study

14.9.1 Practice exercise 1

Read the table consisting of GDP per capita of countries from the webpage: https://en.wikipedia.org/wiki/List_c
.

To only read the relevant table, read the tables that contain the word *'Country'*.

Estimate the GDP per capita of each country as the average of the estimates of the three agencies - IMF, United Nations and World Bank.

We need to do a bit of data cleaning before we could directly use the `groupby()` function. Follow the steps below:

1. Drop the "Year" column
2. Drop the level 1 column name (innermost level column)
3. Apply the following function on all the columns of the "Estimate" to convert them to numeric: `f = lambda x:pd.to_numeric(x,errors = 'coerce')`
4. Set the Country/Territory column as the index
5. Convert all other columns into numeric and coerce errors using 'errors='coerce''
6. Drop rows with NaN values
7. Find the average GDP per capital across the three agencies

```
dfs = pd.read_html('https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_per_capi
gdp_data  = dfs[0]
gdp_data.head()
```

	Country/Territory	IMF[4][5]		World Bank[6]		United Nations[7]	
	Country/Territory	Estimate	Year	Estimate	Year	Estimate	Year
0	Monaco	—	—	240862	2022	240535	2022
1	Liechtenstein	—	—	187267	2022	197268	2022
2	Luxembourg	135321	2024	128259	2023	125897	2022
3	Bermuda	—	—	123091	2022	117568	2022
4	Switzerland	106098	2024	99995	2023	93636	2022

```
# Flatten the MultiIndex to access column names
gdp_data.columns = [' '.join(col).strip() if isinstance(col, tuple) else col for col in gdp_d

# Drop all columns containing "Year"
gdp_data = gdp_data.loc[:, ~gdp_data.columns.str.contains('Year', case=False)]

gdp_data = gdp_data.rename(columns={"Country/Territory Country/Territory": "Country/Territory

gdp_data.head()
```

	Country/Territory	IMF	World Bank	United Nations
0	Monaco	—	240862	240535
1	Liechtenstein	—	187267	197268

	Country/Territory	IMF	World Bank	United Nations
2	Luxembourg	135321	128259	125897
3	Bermuda	—	123091	117568
4	Switzerland	106098	99995	93636

```
import re
column_name_cleaner = lambda x:re.split(r'\[', x)[0]

gdp_data.columns = gdp_data.columns.map(column_name_cleaner)
gdp_data.head()
```

	Country/Territory	IMF	World Bank	United Nations
0	Monaco	—	240862	240535
1	Liechtenstein	—	187267	197268
2	Luxembourg	135321	128259	125897
3	Bermuda	—	123091	117568
4	Switzerland	106098	99995	93636

```
# set the country column as the index
gdp_data.set_index('Country/Territory', inplace=True)

# convert all other columns into numeric and coerce errors
gdp_data = gdp_data.apply(pd.to_numeric, errors='coerce')

# drop rows with NaN values
gdp_data.dropna(inplace=True)

#find the average GDP per capita
gdp_data['Average GDP per capita'] = gdp_data.mean(axis=1)
gdp_data.head()
```

	IMF	World Bank	United Nations	Average GDP per capita
Country/Territory				
Luxembourg	135321.0	128259.0	125897.0	129825.666667
Switzerland	106098.0	99995.0	93636.0	99909.666667
Ireland	103500.0	103685.0	105993.0	104392.666667
Norway	90434.0	87962.0	106623.0	95006.333333
Singapore	89370.0	84734.0	78115.0	84073.000000

14.9.2 Practice exercise 2

Read the spotify dataset from `spotify_data.csv` that contains information about tracks and artists

14.9.2.1

Find the mean and standard deviation of the track popularity for each genre.

14.9.2.2

Create a new categorical column, `energy_lvl`, with two levels – 'Low energy' and 'High energy' – using equal-sized bins based on the track's energy level. Then, calculate the mean, standard deviation, and 90th percentile of track popularity for each genre and energy level combination

14.9.2.3

Find the mean and standard deviation of track popularity and danceability for each genre and energy level. What insights you can gain from the generated table

14.9.2.4

For each genre and energy level, find the mean and standard deviation of the track popularity, and the minimum and maximum values of loudness.

14.9.2.5

Find the most popular artist from each genre.

14.9.2.6

Filter the first 4 columns of the spotify dataset. Drop duplicate observartions in the resulting dataset using the Pandas DataFrame method `drop_duplicates()`. Find the top 3 most popular artists for each genre.

14.9.2.7

The spotify dataset has more than 200k observations. It may be expensive to operate with so many observations. Take a random sample of 650 observations to analyze spotify data, such that all genres are equally represented.

14.9.2.8

Find the correlation between `danceability` and `track popularity` for each genre-energy level combination.

14.9.2.9

Find the mean of track popularity for each genre-energy lvl combination such that each row corresponds to a genre, and the energy levels correspond to columns.

Hints: using `pivot_table()`

14.9.2.10

Find the 90^{th} percentile of track popularity for each genre-energy lvl combination such that each row corresponds to a genre, and the energy levels correspond to columns.

Hints: using `pivot_table()`

14.9.2.11

Find the number of observations in each group, where each groups corresponds to a distinct genre-energy lvl combination

14.9.2.12

Find the percentage of observations in each group of the above table.

14.9.2.13

What percentage of unique tracks are contributed by the top 5 artists of each genre?

Hint: Find the top 5 artists based on `artist_popularity` for each genre. Count the total number of unique tracks (`track_name`) contributed by these artists. Divide this number by the total number of unique tracks in the data. The `nunique()` function will be useful.

15 Data Reshaping and Enrichment

In this chapter, we will explore how to reshape and enrich data using pandas. Data reshaping and enriching are essential steps in data preprocessing because they help transform raw data into a more structured, useful, and meaningful format that can be effectively used for analysis or modeling.

```
<IPython.core.display.HTML object>
```

15.1 Data Reshaping

Data reshaping involves altering the structure of the data to suit the requirements of the analysis or modeling process. This is often needed when the data is in a format that is difficult to analyze or when it needs to be organized in a specific way.

15.1.1 Why Data Reshaping is Important:

- **Improves Data Organization**: Raw data may be in a long format, where each observation is recorded on a separate row, or it may be in a wide format, where the same variable is spread across multiple columns. Reshaping allows you to reorganize the data into a format that is more suitable for analysis, such as turning multiple columns into a single column (melting) or aggregating rows into summary statistics (pivoting).
- **Facilitates Analysis**: Some analysis techniques require specific data formats. For example, many machine learning algorithms require data in a matrix form, where each row represents a sample and each column represents a feature. Reshaping helps to convert data into this format.
- **Simplifies Visualization**: Visualizing data is easier when it is in the right shape. For instance, plotting categorical data across time is easier when the data is in a tidy format, rather than having multiple variables spread across columns.
- **Efficient Grouping and Aggregation**: In some cases, you may want to group data by certain categories or time periods, and reshaping can help aggregate data effectively for summarization.

15.1.2 Wide vs. Long Data Format in Data Reshaping

- **Wide Format**: Each variable in its own column; each row represents a unit, and data for multiple variables is spread across columns.
- **Long Format**: Variables are stacked into one column per variable, with additional columns used to indicate time, categories, or other distinctions, leading to a greater number of rows.

To reshape between these two formats, you can use pandas functions such as `pivot_table()`, `melt()`, `stack()`, and `unstack()`. Let's next take a look at each of these methods one by one.

Throughout this section, we will continue using the *gdp_lifeExpectancy.csv* dataset. Let's start by reading the CSV file into a pandas DataFrame.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

```
# Load the data
gdp_lifeExp_data = pd.read_csv('./Datasets/gdp_lifeExpectancy.csv')
gdp_lifeExp_data.head()
```

	country	continent	year	lifeExp	pop	gdpPercap
0	Afghanistan	Asia	1952	28.801	8425333	779.445314
1	Afghanistan	Asia	1957	30.332	9240934	820.853030
2	Afghanistan	Asia	1962	31.997	10267083	853.100710
3	Afghanistan	Asia	1967	34.020	11537966	836.197138
4	Afghanistan	Asia	1972	36.088	13079460	739.981106

15.1.3 Pivoting "long" to "wide" using `pivot_table()`

In the last chapter, we learned that `pivot_table()` is a powerful tool for performing groupwise aggregation in a structured format. The `pivot_table()` function is used to reshape data by specifying columns for the index, columns, and values.

15.1.3.1 Pivoting a single column

Consider the life expectancy dataset. Let's calculate the average life expectancy for each combination of country and year.

```
gdp_lifeExp_data_pivot = gdp_lifeExp_data.pivot_table(index=['continent','country'], columns=
gdp_lifeExp_data_pivot.head()
```

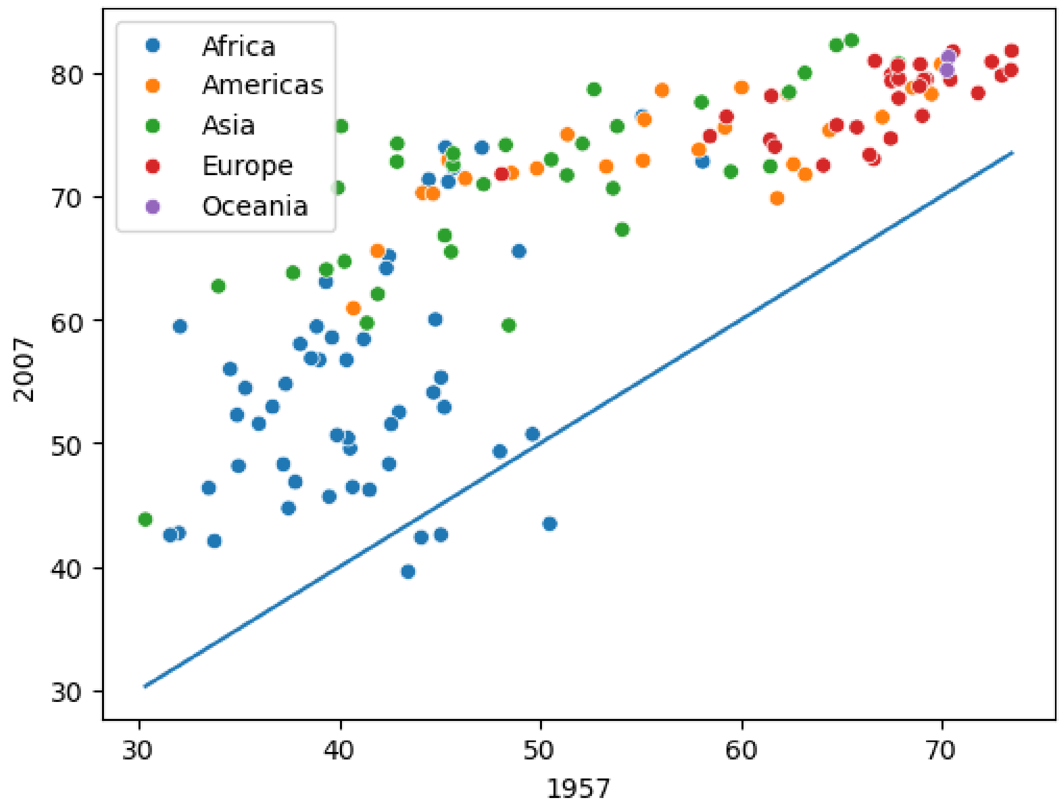
	year
continent	country
Africa	Algeria
	Angola
	Benin
	Botswana
	Burkina Faso

Explanation

- **values**: Specifies the column to aggregate (e.g., `lifeExp` in our example).
- **index**: Groups by the rows based on the `continent` and `country` column.
- **columns**: Groups by the columns based on the `year` column.
- **aggfunc**: Uses `mean` to calculate the average life expectancy.

With values of `year` as columns, it is easy to compare any metric for different years.

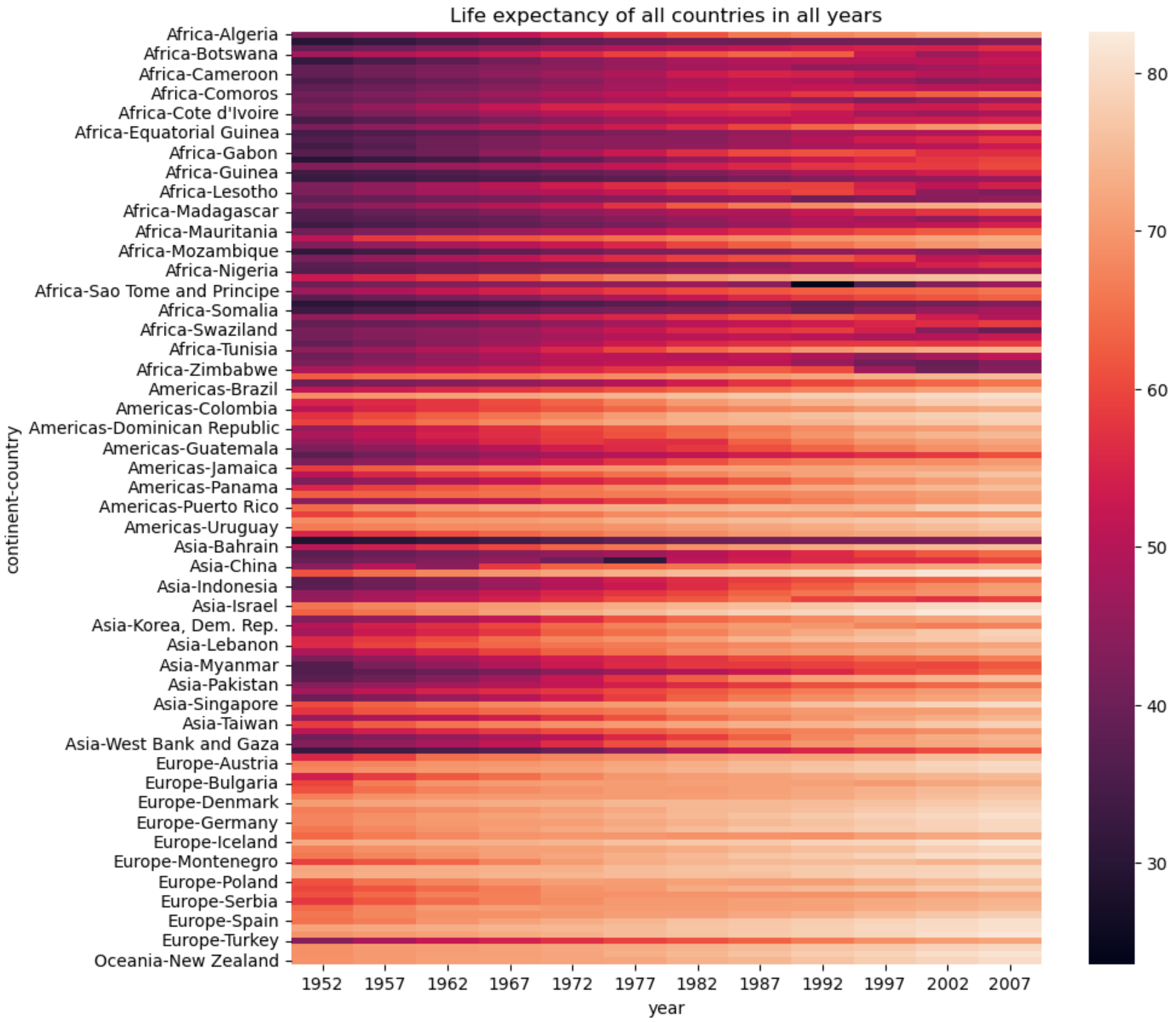
```
#visualizing the change in life expectancy of all countries in 2007 as compared to that in 19
sns.scatterplot(data = gdp_lifeExp_data_pivot, x = 1957,y=2007,hue = 'continent')
sns.lineplot(data = gdp_lifeExp_data_pivot, x = 1957,y = 1957);
```



Observe that for some African countries, the life expectancy has decreased after 50 years. It is worth investigating these countries to identify factors associated with the decrease.

Another way to visualize a pivot table is by using a heatmap, which represents the 2D table as a matrix of colors, making it easier to interpret patterns and trends.

```
# Plotting a heatmap to visualize the life expectancy of all countries in all years, figure s
plt.figure(figsize=(10,10))
plt.title("Life expectancy of all countries in all years")
sns.heatmap(gdp_lifeExp_data_pivot);
```



Common Aggregation Functions in `pivot_table()`

You can use various aggregation functions within `pivot_table()` to summarize data:

- **`mean`** – Calculates the average of values within each group.
- **`sum`** – Computes the total of values within each group.
- **`count`** – Counts the number of non-null entries within each group.
- **`min`** and **`max`** – Finds the minimum and maximum values within each group.

We can also define our own custom aggregation function and pass it to the `aggfunc` parameter of the `pivot_table()` function

For Example: Find the 90^{th} percentile of life expectancy for each country and year combination

```
pd.pivot_table(data = gdp_lifeExp_data, values = 'lifeExp',index = 'country', columns ='year
```

year country	1952	1957	1962	1967	1972	1977	1982	1987	1992	1997
Afghanistan	28.801	30.332	31.997	34.020	36.088	38.438	39.854	40.822	41.674	41.763
Albania	55.230	59.280	64.820	66.220	67.690	68.930	70.420	72.000	71.581	72.950
Algeria	43.077	45.685	48.303	51.407	54.518	58.014	61.368	65.799	67.744	69.152
Angola	30.015	31.999	34.000	35.985	37.928	39.483	39.942	39.906	40.647	40.963
Argentina	62.485	64.399	65.142	65.634	67.065	68.481	69.942	70.774	71.868	73.275
...	...	...	...	...	...	...	...	...	...	...
Vietnam	40.412	42.887	45.363	47.838	50.254	55.764	58.816	62.820	67.662	70.672
West Bank and Gaza	43.160	45.671	48.127	51.631	56.532	60.765	64.406	67.046	69.718	71.096
Yemen, Rep.	32.548	33.970	35.180	36.984	39.848	44.175	49.113	52.922	55.599	58.020
Zambia	42.038	44.077	46.023	47.768	50.107	51.386	51.821	50.821	46.100	40.238
Zimbabwe	48.451	50.469	52.358	53.995	55.635	57.674	60.363	62.351	60.377	46.809

15.1.4 Melting "wide" to "long" using `melt()`

Melting is the **inverse** of pivoting. It transforms columns into rows. The `melt()` function is used when you want to unpivot a DataFrame, turning multiple columns into rows.

Let's first consider *gdp_lifeExp_data_pivot* created in the previous section

```
gdp_lifeExp_data_pivot.head()
```

continent	year country
Africa	Algeria Angola Benin Botswana Burkina Faso

```
gdp_lifeExp_data_pivot.melt(value_name='lifeExp', var_name='year', ignore_index=False)
```

continent	country
Africa	Algeria
	Angola
	Benin
	Botswana
	Burkina Faso
...	...
Europe	Switzerland
	Turkey
	United Kingdo
Oceania	Australia
	New Zealand

With the above DataFrame, we can visualize the mean life expectancy against year separately for each country in each continent.

15.1.5 Stacking and Unstacking using `stack()` and `unstack()`

Stacking and unstacking are powerful techniques used to reshape DataFrames by moving data between rows and columns. These operations are particularly useful when working with **multi-level index** or **multi-level columns**, where the data is structured in a hierarchical format.

- `stack()`: Converts columns into rows, which means it "stacks" the data into a long format.
- `unstack()`: Converts rows into columns, reversing the effect of `stack()`.

Let's use the GDP per capita dataset and create a DataFrame with a multi-level index (e.g., continent, country) and multi-level columns (e.g., lifeExp , gdpPercap, and pop).

```
# calculate the average and standard deviation of life expectancy, population, and GDP per ca

gdp_lifeExp_multilevel_data = gdp_lifeExp_data.groupby(['continent', 'country'])[['lifeExp',
gdp_lifeExp_multilevel_data
```

continent	country
Africa	Algeria
	Angola

continent	country
	Benin
	Botswana
	Burkina Faso
...	...
	Switzerland
Europe	Turkey
	United Kingdo
Oceania	Australia
	New Zealand

The resulting DataFrame has two levels of index and two levels of columns, as shown below:

```
# check the level of row and column index
gdp_lifeExp_multilevel_data.index.nlevels
```

```
2
```

```
gdp_lifeExp_multilevel_data.columns.nlevels
```

```
2
```

if you want to see the data along with the index and columns (in a more accessible way), you can directly print the `.index` and `.columns` attributes.

```
gdp_lifeExp_multilevel_data.index
```

```
MultiIndex([( 'Africa',                  'Algeria'),
            ( 'Africa',                   'Angola'),
            ( 'Africa',                    'Benin'),
            ( 'Africa',                 'Botswana'),
            ( 'Africa',             'Burkina Faso'),
            ( 'Africa',                  'Burundi'),
            ( 'Africa',                 'Cameroon'),
            ( 'Africa', 'Central African Republic'),
            ( 'Africa',                     'Chad'),
            ( 'Africa',                  'Comoros'),
```

```
            ...
            ( 'Europe',                   'Serbia'),
            ( 'Europe',          'Slovak Republic'),
            ( 'Europe',                 'Slovenia'),
            ( 'Europe',                    'Spain'),
            ( 'Europe',                   'Sweden'),
            ( 'Europe',              'Switzerland'),
            ( 'Europe',                   'Turkey'),
            ( 'Europe',           'United Kingdom'),
            ('Oceania',                'Australia'),
            ('Oceania',              'New Zealand')],
           names=['continent', 'country'], length=142)
```

```
gdp_lifeExp_multilevel_data.columns
```

```
MultiIndex([(  'lifeExp', 'mean'),
            (  'lifeExp',  'std'),
            (      'pop', 'mean'),
            (      'pop',  'std'),
            ('gdpPercap', 'mean'),
            ('gdpPercap',  'std')],
           )
```

As seen, **the index and columns each contain tuples**. This is the reason that when performing data subsetting on a DataFrame with multi-level index and columns, you need to supply a tuple to specify the exact location of the data.

```
gdp_lifeExp_multilevel_data.loc[('Americas', 'United States'), ('lifeExp', 'mean')]
```

```
73.4785
```

```
gdp_lifeExp_multilevel_data.columns.values
```

```
array([('lifeExp', 'mean'), ('lifeExp', 'std'), ('pop', 'mean'),
       ('pop', 'std'), ('gdpPercap', 'mean'), ('gdpPercap', 'std')],
      dtype=object)
```

15.1.5.1 Understanding the `level` in multi-level index and columns

The `gdp_lifeExp_multilevel_data` dataframe have two levels of index and two level of columns, You can reference these levels in various operations, such as grouping, subsetting, or performing aggregations. The `level` parameter helps identify which part of the multi-level structure you're working with.

```
gdp_lifeExp_multilevel_data.index.get_level_values(0)
```

```
Index(['Africa', 'Africa', 'Africa', 'Africa', 'Africa', 'Africa', 'Africa',
       'Africa', 'Africa', 'Africa',
       ...
       'Europe', 'Europe', 'Europe', 'Europe', 'Europe', 'Europe', 'Europe',
       'Europe', 'Oceania', 'Oceania'],
      dtype='object', name='continent', length=142)
```

```
gdp_lifeExp_multilevel_data.index.get_level_values(1)
```

```
gdp_lifeExp_multilevel_data.columns.get_level_values(0)
```

```
Index(['lifeExp', 'lifeExp', 'pop', 'pop', 'gdpPercap', 'gdpPercap'], dtype='object')
```

```
gdp_lifeExp_multilevel_data.columns.get_level_values(1)
```

```
Index(['mean', 'std', 'mean', 'std', 'mean', 'std'], dtype='object')
```

```
gdp_lifeExp_multilevel_data.columns.get_level_values(-1)
```

```
Index(['mean', 'std', 'mean', 'std', 'mean', 'std'], dtype='object')
```

As seen above, the outermost level corresponds to `level=0`, and it increases as we move inward to the inner levels. The innermost level can be referred to as `-1`.

Now, let's use the level number in the `droplevel()` method to see how it works. The syntax for this method is as follows:

`DataFrame.droplevel(level, axis=0)`

```python
# drop the first level of row index
gdp_lifeExp_multilevel_data.droplevel(0, axis=0)
```

	lifeExp		pop		gdpPercap	
	mean	std	mean	std	mean	std
country						
Algeria	59.030167	10.340069	1.987541e+07	8.613355e+06	4426.025973	1310.337656
Angola	37.883500	4.005276	7.309390e+06	2.672281e+06	3607.100529	1165.900251
Benin	48.779917	6.128681	4.017497e+06	2.105002e+06	1155.395107	159.741306
Botswana	54.597500	5.929476	9.711862e+05	4.710965e+05	5031.503557	4178.136987
Burkina Faso	44.694000	6.845792	7.548677e+06	3.247808e+06	843.990665	183.430087
...	...	...	...	...	...	...
Switzerland	75.565083	4.011572	6.384293e+06	8.582009e+05	27074.334405	6886.463308
Turkey	59.696417	9.030591	4.590901e+07	1.667768e+07	4469.453380	2049.665102
United Kingdom	73.922583	3.378943	5.608780e+07	3.174339e+06	19380.472986	7388.189399
Australia	74.662917	4.147774	1.464931e+07	3.915203e+06	19980.595634	7815.405220
New Zealand	73.989500	3.559724	3.100032e+06	6.547108e+05	17262.622813	4409.009167

```python
# drop the first level of column index
gdp_lifeExp_multilevel_data.droplevel(0, axis=1)
```

continent	country
Africa	Algeria
	Angola
	Benin
	Botswana
	Burkina Faso
...	...
Europe	Switzerland
	Turkey
	United Kingdo
Oceania	Australia
	New Zealand

15.1.5.2 Stacking (Columns to Rows): `stack()`

The `stack()` function pivots the columns into rows. You can specify which level you want to stack using the `level` parameter. By default, it will stack the innermost level of the columns.

```
gdp_lifeExp_multilevel_data.stack(future_stack=True)
```

continent	country
Africa	Algeria
	Angola
	Benin
...	...
Europe	United Kingdo
Oceania	Australia
	New Zealand

Let's change the default setting by specifying the `level=0`(the outmost level)

```
gdp_lifeExp_multilevel_data.stack(level=0, future_stack=True)
```

continent	country
Africa	Algeria
	Angola
...	...
Oceania	Australia
	New Zealand

15.1.5.3 Unstacking (Rows to Columns) : `unstack()`

The `unstack()` function pivots the rows back into columns. By default, it will unstack the innermost level of the index.

Let's reverse the prevous dataframe to its original shape using `unstack()`.

```
gdp_lifeExp_multilevel_data.stack(level=0, future_stack=True).unstack()
```

continent	country
Africa	Algeria
	Angola
	Benin
	Botswana
	Burkina Faso
...	...
Europe	Switzerland
	Turkey
	United Kingdo
Oceania	Australia
	New Zealand

15.1.6 Transposing using `.T` attribute

If you simply want to swap rows and columns, you can use the `.T` attribute.

```
gdp_lifeExp_multilevel_data.T
```

	continent	Afr
	country	Alg
lifeExp	mean	5.9
	std	1.0
pop	mean	1.9
	std	8.6
gdpPercap	mean	4.4
	std	1.3

15.1.7 Summary of Common Reshaping Functions

Function	Description	Example
`pivot()`	Reshape data by creating new columns from existing ones.	Pivot data on index/columns.
`melt()`	Unpivot data, turning columns into rows.	Convert wide format to long.
`stack()`	Convert columns to rows.	Move column data to rows.
`unstack()`	Convert rows back to columns.	Move row data to columns.
`.T`	Transpose the DataFrame (swap rows and columns).	Transpose the entire DataFrame.

Practical Use Cases:

- **Pivoting:** Useful for time series analysis or when you want to group and summarize data by specific categories.
- **Melting:** Helpful when you need to reshape wide data for easier analysis or when preparing data for machine learning algorithms.
- **Stacking/Unstacking:** Useful when you are working with hierarchical index DataFrames.
- **Transpose:** Helpful for reorienting data for analysis or visualization.

15.1.8 Converting from Multi-Level Index to Single-Level Index DataFrame

If you're uncomfortable working with DataFrames that have multi-level indexes and columns, you can convert them into single-level DataFrames using the methods you learned in the previous chapter:

- `reset_index()`
- Flattening the multi-level columns

```
# reset the index to convert the multi-level index to a single-level index
gdp_lifeExp_multilevel_data.reset_index()
```

	continent	country	lifeExp		pop		gdpPercap	
			mean	std	mean	std	mean	s
0	Africa	Algeria	59.030167	10.340069	1.987541e+07	8.613355e+06	4426.025973	1
1	Africa	Angola	37.883500	4.005276	7.309390e+06	2.672281e+06	3607.100529	1
2	Africa	Benin	48.779917	6.128681	4.017497e+06	2.105002e+06	1155.395107	1
3	Africa	Botswana	54.597500	5.929476	9.711862e+05	4.710965e+05	5031.503557	4
4	Africa	Burkina Faso	44.694000	6.845792	7.548677e+06	3.247808e+06	843.990665	1
...	...	...	...	...	...	...	...	..
137	Europe	Switzerland	75.565083	4.011572	6.384293e+06	8.582009e+05	27074.334405	6

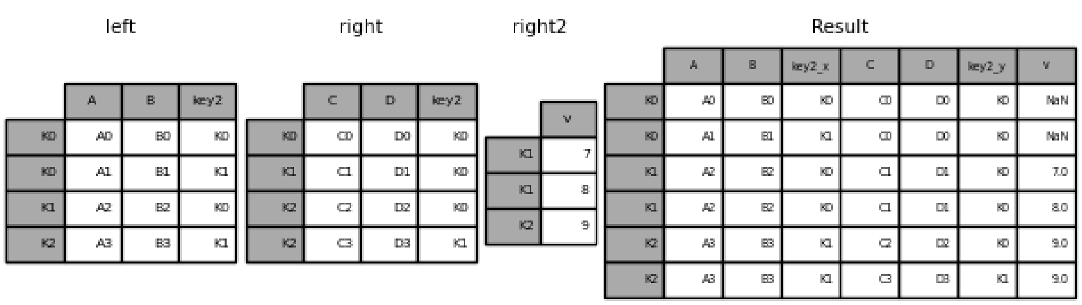
	continent	country	lifeExp		pop		gdpPercap	
			mean	std	mean	std	mean	s
138	Europe	Turkey	59.696417	9.030591	4.590901e+07	1.667768e+07	4469.453380	2
139	Europe	United Kingdom	73.922583	3.378943	5.608780e+07	3.174339e+06	19380.472986	7
140	Oceania	Australia	74.662917	4.147774	1.464931e+07	3.915203e+06	19980.595634	7
141	Oceania	New Zealand	73.989500	3.559724	3.100032e+06	6.547108e+05	17262.622813	4

```
# flatten the multi-level column
gdp_lifeExp_multilevel_data.columns = ['_'.join(col).strip() for col in gdp_lifeExp_multileve
gdp_lifeExp_multilevel_data
```

continent	country
Africa	Algeria
	Angola
	Benin
	Botswana
	Burkina Faso
...	...
Europe	Switzerland
	Turkey
	United Kingdo
Oceania	Australia
	New Zealand

15.2 Data Enriching

Data enriching involves enhancing your existing dataset by adding additional data, often from external sources, to make your dataset more insightful.



There are 3 common methods for data enriching:

- **Merging using `merge()`**: Combining datasets based on a common key.
- **Joining using `join()`**: Using indices to combine data.
- **Concatenation using `concat()`**: Stacking DataFrames vertically or horizontally.

Let's first load the existing data

```
df_existing = pd.read_csv('./datasets/LOTR.csv')
print(df_existing.shape)
df_existing.head()
```

```
(4, 3)
```

	FellowshipID	FirstName	Skills
0	1001	Frodo	Hiding
1	1002	Samwise	Gardening
2	1003	Gandalf	Spells
3	1004	Pippin	Fireworks

Let's next load the external data we want to combine to the existing data

```
df_external = pd.read_csv("./datasets/LOTR 2.csv")
print(df_external.shape)
df_external.head()
```

```
(5, 3)
```

	FellowshipID	FirstName	Age
0	1001	Frodo	50
1	1002	Samwise	39
2	1006	Legolas	2931
3	1007	Elrond	6520
4	1008	Barromir	51

15.2.1 Combining based on common keys: `merge()`

```
df_enriched_merge = pd.merge(df_existing, df_external)
print(df_enriched_merge.shape)
df_enriched_merge.head()
```

```
(2, 4)
```

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50
1	1002	Samwise	Gardening	39

You can also do this way

```
df_enriched_merge = df_existing.merge(df_external)
print(df_enriched_merge.shape)
df_enriched_merge.head()
```

```
(2, 4)
```

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50
1	1002	Samwise	Gardening	39

By default, the `merge()` function in pandas performs an **inner join**, which combines two DataFrames based on the **common keys shared by both**. Only the rows with matching keys in both DataFrames are retained in the result.

This behavior can be observed in the following code:

```
df_enriched_merge = pd.merge(df_existing, df_external, how='inner')
print(df_enriched_merge.shape)
df_enriched_merge.head()
```

```
(2, 4)
```

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50
1	1002	Samwise	Gardening	39

When you may use inner join? You should use inner join when you cannot carry out the analysis unless the observation corresponding to the *key column(s)* is present in both the tables.

Example: Suppose you wish to analyze the association between vaccinations and covid infection rate based on country-level data. In one of the datasets, you have the infection rate for each country, while in the other one you have the number of vaccinations in each country. The countries which have either the vaccination or the infection rate missing, cannot help analyze the association. In such as case you may be interested only in countries that have values for both the variables. Thus, you will use inner join to discard the countries with either value missing.

In `merge()`, the `how` parameter specifies the type of merge to be performed, the default option is `inner`, it could also be `left`, `right`, and `outer`.

These join types give you flexibility in how you want to merge and combine data from different sources.

Let's see how each of these works using our dataframes defined above

Left Join

- Returns all rows from the left DataFrame and the matching rows from the right DataFrame.
- Missing values (`NaN`) are introduced for non-matching keys from the right DataFrame.

```
df_merge_lft = pd.merge(df_existing, df_external, how='left')
print(df_merge_lft.shape)
df_merge_lft.head()
```

```
(4, 4)
```

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50.0
1	1002	Samwise	Gardening	39.0
2	1003	Gandalf	Spells	NaN
3	1004	Pippin	Fireworks	NaN

When you may use left join? You should use left join when the primary variable(s) of interest are present in the one of the datasets, and whose missing values cannot be imputed. The variable(s) in the other dataset may not be as important or it may be possible to reasonably impute their values, if missing corresponding to the observation in the primary dataset.

Examples:

1) Suppose you wish to analyze the association between the covid infection rate and the government effectiveness score (a metric used to determine the effectiveness of the government in implementing policies, upholding law and order etc.) based on the data of all countries. Let us say that one of the datasets contains the covid infection rate, while the other one contains the government effectiveness score for each country. If the infection rate for a country is missing, it might be hard to impute. However, the government effectiveness score may be easier to impute based on GDP per capita, crime rate etc. - information that is easily available online. In such a case, you may wish to use a left join where you keep all the countries for which the infection rate is known.
2) Suppose you wish to analyze the association between demographics such as age, income etc. and the amount of credit card spend. Let us say one of the datasets contains the demographic information of each customer, while the other one contains the credit card spend for the customers who made at least one purchase. In such as case, you may want to do a left join as customers not making any purchase might be absent in the card spend data. Their spend can be imputed as zero after merging the datasets.

Right join

- Returns all rows from the right DataFrame and the matching rows from the left DataFrame.
- Missing values (`NaN`) are introduced for non-matching keys from the left DataFrame.

```
df_merge_right = pd.merge(df_existing, df_external, how='right')
print(df_merge_right.shape)
df_merge_right.head()
```

```
(5, 4)
```

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50
1	1002	Samwise	Gardening	39
2	1006	Legolas	NaN	2931
3	1007	Elrond	NaN	6520
4	1008	Barromir	NaN	51

When you may use right join? You can always use a left join instead of a right join. Their purpose is the same.

15.2.1.1 outer join

- Returns all rows from both DataFrames, with `NaN` values where no match is found.

```
df_merge_outer = pd.merge(df_existing, df_external, how='outer')
print(df_merge_outer.shape)
df_merge_outer
```

```
(7, 4)
```

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50.0
1	1002	Samwise	Gardening	39.0
2	1003	Gandalf	Spells	NaN
3	1004	Pippin	Fireworks	NaN
4	1006	Legolas	NaN	2931.0
5	1007	Elrond	NaN	6520.0
6	1008	Barromir	NaN	51.0

When you may use outer join? You should use an outer join when you cannot afford to lose data present in either of the tables. All the other joins may result in loss of data.

Example: Suppose I took two course surveys for this course. If I need to analyze student sentiment during the course, I will take an outer join of both the surveys. Assume that each survey is a dataset, where each row corresponds to a unique student. Even if a student has answered one of the two surverys, it will be indicative of the sentiment, and will be useful to keep in the merged dataset.

15.2.2 Combining based on Indices: `join()`

Unlike `merge()`, `join()` in pandas combines data based on the **index**. When using `join()`, you must specify the **`suffixes`** parameter if there are overlapping column names, as it does not have default values. You can refer to the official function definition here: pandas.DataFrame.join.

Please copy the code below and run it in your notebook:

```
df_enriched_join = df_existing.join(df_external)
print(df_enriched_join.shape)
df_enriched_join.head()
```

Let's set the suffix to fix this issue

```
df_enriched_join = df_existing.join(df_external, lsuffix = '_existing',rsuffix = '_external'
print(df_enriched_join.shape)
df_enriched_join.head()
```

```
(4, 6)
```

	FellowshipID_existing	FirstName_existing	Skills	FellowshipID_external	FirstName_external
0	1001	Frodo	Hiding	1001	Frodo
1	1002	Samwise	Gardening	1002	Samwise
2	1003	Gandalf	Spells	1006	Legolas
3	1004	Pippin	Fireworks	1007	Elrond

As can be observed, **`join()` merges DataFrames using their index and performs a left join**.

Now, let's set the "FellowshipID" column as the index in both DataFrames and see what happens.

```
# set the index of the dataframes as the "FellowshipID" column
df_existing.set_index('FellowshipID', inplace=True)
df_external.set_index('FellowshipID', inplace=True)
```

```
# join the two dataframes
df_enriched_join = df_existing.join(df_external, lsuffix = '_existing',rsuffix = '_external'
print(df_enriched_join.shape)
df_enriched_join
```

```
(4, 4)
```

FellowshipID	FirstName_existing	Skills	FirstName_external	Age
1001	Frodo	Hiding	Frodo	50.0
1002	Samwise	Gardening	Samwise	39.0
1003	Gandalf	Spells	NaN	NaN
1004	Pippin	Fireworks	NaN	NaN

In this case, `df_enriched_join` will contain all rows from `df_existing` and the corresponding matching rows from `df_external`, with `NaN` values for non-matching entries.

Similar to `merge()`, you can change the type of join in `join()` by explicitly specifying the `how` parameter. However, this explanation will be skipped here, as the different join types have already been covered in the `merge()` section above.

15.2.3 Stacking vertically or horizontally: `concat()`

`concat()` is used in pandas to concatenate DataFrames along a specified axis, either rows or columns. Similar to `concat()` in NumPy, it defaults to concatenating along the rows.

```
# let's concatenate the two dataframes using default setting
df_concat = pd.concat([df_existing, df_external])
print(df_concat.shape)
df_concat
```

```
(9, 3)
```

FellowshipID	FirstName	Skills	Age
1001	Frodo	Hiding	NaN
1002	Samwise	Gardening	NaN
1003	Gandalf	Spells	NaN
1004	Pippin	Fireworks	NaN
1001	Frodo	NaN	50.0
1002	Samwise	NaN	39.0
1006	Legolas	NaN	2931.0
1007	Elrond	NaN	6520.0
1008	Barromir	NaN	51.0

```
# Concatenating along columns (horizontal)
df_concat_columns = pd.concat([df_existing, df_external], axis=1)
print(df_concat_columns.shape)
df_concat_columns
```

```
(7, 4)
```

	FirstName	Skills	FirstName	Age
FellowshipID				
1001	Frodo	Hiding	Frodo	50.0
1002	Samwise	Gardening	Samwise	39.0
1003	Gandalf	Spells	NaN	NaN
1004	Pippin	Fireworks	NaN	NaN
1006	NaN	NaN	Legolas	2931.0
1007	NaN	NaN	Elrond	6520.0
1008	NaN	NaN	Barromir	51.0

15.2.4 Missing values after data enriching

Using either of the approaches mentioned above may introduce missing values for unmatched entries. Handling these missing values is crucial for maintaining the quality and integrity of your dataset after merging or joining DataFrames. It is important to choose a method that best fits your data and analysis goals, and to document your decisions regarding missing value handling for transparency in your analysis.

Question: Read the documentations of the Pandas DataFrame methods `merge()` and `concat()`, and identify the differences. Mention examples when you can use (i) either, (ii) only `concat()`, (iii) only `merge()`

Solution:

(i) If we need to merge datasets using row indices, we can use either function.

(ii) If we need to stack datasets one on top of the other, we can only use `concat()`

(iii) If we need to merge datasets using overlapping columns we can only use `merge()`

For a comprehensive user guide, please refer to the official documentation.

15.3 Independent Study

15.3.1 Practice exercise 1

You are given the life expectancy data of each continent as a separate *.csv file.

15.3.1.1

Visualize the change of life expectancy over time for different continents.

15.3.1.2

Appending all the data files, i.e., stacking them on top of each other to form a combined datasets called "data_all_continents"

15.3.2 Practice exercise 2

15.3.2.1 Preparing GDP per capita data

Read the GDP per capita data from https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)_pe

Drop all the `Year` columns. Use the `drop()` method with the `columns`, `level` and `inplace` arguments. Print the first 2 rows of the updated DataFrame. If the first row of the DataFrame has missing values for all columns, drop that row as well.

15.3.2.2

Drop the inner level of column labels. Use the `droplevel()` method. Print the first 2 rows of the updated DataFrame.

15.3.2.3

Convert the columns consisting of GDP per capita by *IMF*, *World Bank*, and the *United Nations* to numeric. Apply a lambda function on these columns to convert them to numeric. Print the number of missing values in each column of the updated DataFrame.

Note: *Do not apply the function 3 times. Apply it once on a DataFrame consisting of these 3 columns.*

15.3.2.4

Apply the lambda function below on all the column names of the dataset obtained in the previous question to clean the column names.

```
import re
```

```
column_name_cleaner = lambda x:re.split(r'\[|/', x)[0]
```

Note: You will need to edit the parameter of the function, i.e., `x` *in the above function to make sure it is applied on column names and not columns.*

Print the first 2 rows of the updated DataFrame.

15.3.2.5

Create a new column `GDP_per_capita` that copies the GDP per capita values of the `United Nations`. If the GDP per capita is missing in the `United Nations` column, then copy it from the `World Bank` column. If the GDP per capita is missing both in the `United Nations` and the `World Bank` columns, then copy it from the `IMF` column.

Print the number of missing values in the `GDP_per_capita` column.

15.3.2.6

Drop all the columns except `Country` and `GDP_per_capita`. Print the first 2 rows of the updated DataFrame.

15.3.2.7

The country names contain some special characters *(characters other than letters)* and need to be cleaned. The following function can help clean country names:

```
import re
```

```
country_names_clean_gdp_data = lambda x: re.sub(r'[^\w\s]', '', x).strip()
```

Apply the above lambda function on the country column to clean country names. Save the cleaned dataset as `gdp_per_capita_data`. Print the first 2 rows of the updated DataFrame.

15.3.2.8 Preparing population data

Read the population data from https://en.wikipedia.org/wiki/List_of_countries_by_population_(United_Nat

- Drop all columns except `Country`, `UN Continental Region[1]`, and `Population (1 July 2023)`.
- Drop the first row since it is the total population in the world

15.3.2.9

Apply the lambda function below on all the column names of the dataset obtained in the previous question to clean the column names.

```
import re

column_name_cleaner = lambda x:re.split(r'\[|/|\(| ', x.name)[0]
```

Note: You will need to edit the parameter of the function, i.e., `x` *in the above function to make sure it is applied on column names and not columns.*

Print the first 2 rows of the updated DataFrame.

15.3.2.10

The country names contain some special characters *(characters other than letters)* and need to be cleaned. The following function can help clean country names:

```
import re

country_names_clean_population_data = lambda x: re.sub("[\(\[].*?[\)\]]", "",
x).strip()
```

Apply the above lambda function on the country column to clean country names. Save the cleaned dataset as `population_data`.

15.3.2.11 Merging GDP per capita and population datasets

Merge `gdp_per_capita_data` with `population_data` to get the population and GDP per capita of countries in a single dataset. Print the first two rows of the merged DataFrame.

Assume that:

1. We want to keep the GDP per capita of all countries in the merged dataset, even if their population in unavailable in the population dataset. For countries whose population in unavailable, their `Population` column will show `NA`.

2. We want to discard an observation of a country if its GDP per capita is unavailable.

15.3.2.12

For how many countries in `gdp_per_capita_data` does the population seem to be unavailable in `population_data`? Note that you don't need to clean country names any further than cleaned by the functions provided.

Print the observations of `gdp_per_capita_data` with missing `Population`.

15.3.3 Merging datasets with *similar* values in the *key* column

We suspect that population of more countries may be available in `population_data`. However, due to unclean country names, the observations could not merge. For example, the country *Guinea Bissau* is mentioned as *GuineaBissau* in `gdp_per_capita_data` and *Guinea-Bissau* in `population_data`. To resolve this issue, we'll use a different approach to merge datasts. We'll merge the population of a country to an observation in the GDP per capita dataset, whose name in `population_data` is the most *'similar'* to the name of the country in `gdp_per_capita_data`.

15.3.3.1

Proceed as follows:

1. For each country in `gdp_per_capita_data`, find the `country` with the most *'similar'* name in `population_data`, based on the similarity score. Use the lambda function provided below to compute the similarity score between two strings *(The higher the score, the more similar are the strings. The similarity score is* 1.0 *if two strings are exactly the same).*
2. Merge the population of the most *'similar'* country to the country in `gdp_per_capita_data`. The merged dataset must include 5 columns - the country name as it appears in `gdp_per_capita_data`, the GDP per capita, the country name of the most *'similar'* country as it appears in `population_data`, the population of that country, and the similarity score between the country names.
3. After creating the merged dataset, **print** the rows of the dataset that have similarity scores less than 1.

Use the function below to compute the similarity score between the `Country` names of the two datasets:

```
from difflib import SequenceMatcher
```

```
similar = lambda a,b: SequenceMatcher(None, a, b).ratio()
```

Note: You may use one `for` loop only for this particular question. However, it would be perfect if don't use a `for` loop

Hint:

1. Define a function that computes the index of the observation having the most *'similar'* country name in `population_data` for an observation in `gdp_per_capita_data`. The function returns a Series consisting of the most *'similar'* country name, its population, and its similarity score *(This function can be written with only one line in its body, excluding the return statement and the definition statement. However, you may use as many lines as you wish).*
2. Apply the function on the `Country` column of `gdp_per_capita_data`. A DataFrame will be obtained.
3. Concatenate the DataFrame obtained in (2) with `gdp_per_capita_data` with the pandas `concat()` function.

15.3.3.2

In the dataset obtained in the previous question, for all observations where similarity score is less than 0.8, replace the population with `Nan`.

Print the observations of the dataset having missing values of population.

A Assignment A

Python Iterables and Data Structures

Instructions

1. You may talk to a friend, discuss the questions and potential directions for solving them. However, you need to write your own solutions and code separately, and not as a group activity.
2. Write your code in the *Code* cells and your answer in the *Markdown* cells of the Jupyter notebook. Ensure that the solution is written neatly enough to understand and grade.
3. Use [Quarto](https://quarto.org) to print the *.ipynb* file as HTML. You will need to open the command prompt, navigate to the directory containing the file, and use the command: `quarto render filename.ipynb --to html`. Submit the HTML file.
4. The assignment is worth 100 points, and is due on **5th October 2025 at 11:59 pm**.
5. **Five points are properly formatting the assignment**. The breakdown is as follows:

- Must be an HTML file rendered using Quarto (2 pts).
- There aren't excessively long outputs of extraneous information (e.g. no printouts of entire data frames without good reason, there aren't long printouts of which iteration a loop is on, there aren't long sections of commented-out code, etc.) (1 pt)
- Final answers of each question are written in Markdown cells (1 pt).
- There is no piece of unnecessary / redundant code, and no unnecessary / redundant text (1 pt)

A.1 GDP per Capita (35 pts)

A.1.1 List Comprehension

USA's GDP per capita from 1960 to 2021 is given by the tuple `T` in the code cell below. The values are arranged in ascending order of the year, i.e., the first value is for 1960, the second value is for 1961, and so on.

```
T = (3007, 3067, 3244, 3375,3574, 3828, 4146, 4336, 4696, 5032,5234,5609,6094,6726,7226,7801
```

A.1.1.1

Use list comprehension to create a list of the gaps between consecutive entries in `T`, i.e, the increase in GDP per capita with respect to the previous year. The list with gaps should look like: [60, 177, …]. Let the name of this list be `GDP_increase`.

(4 points)

A.1.1.2

Use `GDP_increase` to find the maximum gap size, i.e, the maximum increase in GDP per capita.

(1 point)

A.1.2

Use list comprehension with `GDP_increase` to find the percentage of gaps that have size greater than $1000.

(3 points)

A.1.2.1

Use list comprehension with `GDP_increase` to print the list of years in which the GDP per capita increase was more than $2000.

Hint: The `enumerate()` function may help.

(4 points)

A.1.2.2

Use list comprehension to:

1. Create a list that consists of the difference between the maximum and minimum GDP per capita values for each of the 5 year-periods starting from 1976, i.e., for the periods 1976-1980, 1981-1985, 1986-1990, …, 2016-2020.
2. Find the five year period in which the difference *(between the maximum and minimum GDP per capita values)* was the least.

(4 + 2 points)

A.1.3 Dictionary

The GDP per capita of USA for most years from 1960 to 2021 is given by the dictionary `D` given in the code cell below.

```
D = {'1960':3007,'1961':3067,'1962':3244,'1963':3375,'1964':3574,'1965':3828,'1966':4146,'19
```

Find:

A.1.3.1

The GDP per capita in 2015

(2 points)

A.1.3.2

The GDP per capita of 2014 is missing. Update the dictionary to include the GDP per capita of 2014 as the average of the GDP per capita of 2013 and 2015.

(5 points)

A.1.3.3

Impute the GDP per capita of other missing years in the same manner as in (2), i.e., as the average GDP per capita of the previous year and the next year. Note that the GDP per capita is not missing for any two consecutive years.

(5 points)

A.1.3.4

Print the years and the imputed GDP per capita for the years having a missing value of GDP per capita in (3).

(5 points)

A.2 Student Survey (40 pts)

A.2.1 Marriage age responses

Below is a list consisting of responses to the question: "At what age do you think you will marry?" from students of the STAT303-1 Fall 2022 class.

```
exp_marriage_age=['24','30','28','29','30','27','26','28','30+','26','28','30','30','30','pro
```

Use list comprehension to:

A.2.1.1

Remove the elements that are not integers - such as *'probably never'*, *'30+'*, etc. What is the length of the new list?

Hint: The built-in python function of the `str` class - isdigit() may be useful to check if the string contains only digits.

(3 points)

A.2.1.2

Cap the values greater than 80 to 80, in the clean list obtained in (1). What is the mean age when people expect to marry in the new list?

(3 + 4 points)

A.2.1.3

Determine the percentage of people who expect to marry at an age of 30 or more.

(5 points)

A.2.2 Majors/Minors: Nested lists of student majors.

Below is the list consisting of the majors / minors of students of the course STAT303-1 Fall 2023. This data is a list of lists, where each sub-list *(smaller list within the outer larger list)* consists of the majors / minors of a student. Most of the students have majors / minors in one or more of these four areas:

1. Math / Statistics / Computer Science
2. Humanities / Communication
3. Social Sciences / Education
4. Physical Sciences / Natural Sciences / Engineering

There are some students having majors / minors in other areas as well.

Use list comprehension for all the questions below.

```
majors_minors = majors_minors = [['Humanities / Communications', 'Math / Statistics / Comput
```

A.2.2.1

How many students have major / minor in any three of the above mentioned four areas?

(1 point)

A.2.2.2

How many students have *Math / Statistics / Computer Science* as an area of their major / minor?

Hint: Nested list comprehension

(4 points)

A.2.2.3

How many students have *Math / Statistics / Computer Science* as the **only area** of their major / minor?

Hint: Nested list comprehension

(5 points)

A.2.2.4

How many students have *Math / Statistics / Computer Science* and *Social Sciences / Education* as a couple of areas of their major / minor?

Hint: The in-built function `all()` may be useful.

(6 points)

A.2.3 Starting salary expectations

Below is a list consisting of responses to the question: "What do you expect your starting salary to be after graduation, to the nearest thousand dollars? (ex: 47000)" from students of the STAT303-1 Fall 2023. class.

```
expected_salary = ['90000', '110000', '100000', '90k', '80000', '47000', '100000', '70000',
```

Clean `expected_salary` using list comprehensions only, and find the mean expected salary.

(5 + 4 points)

A.3 Starbucks Drinks (20 pts)

The code cell below defines an object having the nutrition information of drinks in starbucks. Assume that the manner in which the information is structured is consistent throughout the object.

```
starbucks_drinks_nutrition={'Cool Lime Starbucks Refreshers Beverage': [{'Nutrition_type': 'Calori
```

A.3.1

Use the object above to answer the following questions:

A.3.1.1

What is the datatype of the object?

(2 points)

A.3.1.2

If the object in (1) is a dictonary, what is the datatype of the values of the dictionary?

(2 points)

A.3.1.3

If the object in (1) is a dictonary, what is the datatype of the elements within the values of the dictionary?

(2 points)

A.3.1.4

How many calories are there in `Iced Coffee`?

(4 points)

A.3.1.5

Which drink(s) have the highest amount of protein in them, and what is that protein amount?

A.3.1.6

Which drink(s) have a fat content of more than 10g, and what is their fat content?

(5 points)

A.3.1.7

Use dictionary-comprehension to print the name and `carb` content of all drinks that have a `carb` content of more than 50 units.

Hint: It will be a nested dictionary comprehension.

(5 points)

A.4 AI Use Disclosure (5 pts)

In **1–3 short paragraphs**, clearly state whether **generative AI tools** were used to complete any part of this assignment.

- If **no AI was used**, write:
  "I did not use generative AI tools on this assignment."
- If **AI was used**, you must specify:
  1. **Tool(s)** used (e.g., ChatGPT, GitHub Copilot, Claude, etc.)
  2. **How** they were used (e.g., idea generation, debugging, code suggestions, writing help)
  3. **Where** they influenced your work (which questions/sections)
  4. **Edits & verification** you made (how you checked correctness, what you modified)

A.4.1 AI Use Template

- **Used AI?** Yes / No
- **Tool(s):**
- **Purpose / How used (1–3 sentences):**
- **Scope (which questions/sections):**
- **Edits & verification (what you changed and how you checked correctness):**

  Example (for illustration only):
  - Used AI? Yes
  - Tools: ChatGPT (free), GitHub Copilot
  - How used: Asked for a pandas snippet to impute missing values and for a seaborn example.
  - Scope: Q1(b) imputation, Q2(a) first draft of bar chart code.
  - Edits & verification: Rewrote code to use `groupby().transform('median')`; validated results by comparing summary stats; added axis labels manually.

Grading (5 points):
- **5 points** = complete, specific, and honest (tools named, usage described clearly)
- **3–4 points** = vague or missing some details

- **1–2 points** = minimal effort or very unclear
- **0 points** = missing or misleading disclosure

B Datasets & Templates

Datasets used in the book, and assignment / project templates can be found here